

NIMBLE User Manual

NIMBLE Development Team

Version 1.4.0



<https://r-nimble.org>
<https://github.com/nimble-dev/nimble>

Contents

I	Introduction	9
1	Welcome to NIMBLE	11
1.1	What does NIMBLE do?	11
1.2	How to use this manual	12
2	Lightning introduction	13
2.1	A brief example	13
2.2	Creating a model	13
2.3	Compiling the model	17
2.4	One-line invocation of MCMC	17
2.5	Creating, compiling and running a basic MCMC configuration	19
2.6	Customizing the MCMC	20
2.7	Running MCEM	21
2.8	Creating your own functions	23
3	More introduction	27
3.1	NIMBLE adopts and extends the BUGS language for specifying models	27
3.2	nimbleFunctions for writing algorithms	28
3.3	The NIMBLE algorithm library	29
4	Installing NIMBLE	31
4.1	Requirements to run NIMBLE	31
4.2	Installing a C++ compiler for NIMBLE to use	31
4.2.1	MacOS	31
4.2.2	Linux	32
4.2.3	Windows	32
4.3	Installing the NIMBLE package	32
4.4	Troubleshooting installation problems	33
4.5	Customizing your installation	34
4.5.1	Using your own copy of Eigen	34
4.5.2	Using libnimble	34
4.5.3	BLAS and LAPACK	35
4.5.4	Customizing compilation of the NIMBLE-generated C++	35

II	Models in NIMBLE	37
5	Writing models in NIMBLE's dialect of BUGS	39
5.1	Comparison to BUGS dialects supported by WinBUGS, OpenBUGS and JAGS . . .	39
5.1.1	Supported features of BUGS and JAGS	39
5.1.2	NIMBLE's Extensions to BUGS and JAGS	39
5.1.3	Not-supported features of BUGS and JAGS	40
5.2	Writing models	41
5.2.1	Declaring stochastic and deterministic nodes	41
5.2.2	More kinds of BUGS declarations	43
5.2.3	Vectorized versus scalar declarations	45
5.2.4	Available distributions	46
5.2.5	Available BUGS language functions	51
5.2.6	Available link functions	53
5.2.7	Truncation, censoring, and constraints	54
5.2.8	Model macros	56
6	Building and using models	61
6.1	Creating model objects	61
6.1.1	Using <i>nimbleModel</i> to create a model	61
6.1.2	Creating a model from standard BUGS and JAGS input files	66
6.1.3	Making multiple instances from the same model definition	66
6.2	NIMBLE models are objects you can query and manipulate	67
6.2.1	What are variables and nodes?	67
6.2.2	Determining the nodes and variables in a model	68
6.2.3	Accessing nodes	68
6.2.4	How nodes are named	69
6.2.5	Why use node names?	70
6.2.6	Checking if a node holds data	70
6.3	Using models in parallel	70
III	Algorithms in NIMBLE	73
7	MCMC	75
7.1	One-line invocation of MCMC: <i>nimbleMCMC</i>	76
7.2	The MCMC configuration	77
7.2.1	Default MCMC configuration	77
7.2.2	Customizing the MCMC configuration	79
7.3	Building and compiling the MCMC	85
7.4	Initializing MCMC	86
7.5	User-friendly execution of MCMC algorithms: <i>runMCMC</i>	87
7.6	Running the MCMC	88
7.6.1	Rerunning versus restarting an MCMC	88
7.6.2	Measuring sampler computation times: <i>getTimes</i>	89
7.6.3	Assessing the adaption process of <i>RW</i> and <i>RW_block</i> samplers	89
7.7	Extracting MCMC samples	89
7.8	Calculating WAIC	90

7.9	k-fold cross-validation	91
7.10	Variable selection using Reversible Jump MCMC	92
7.10.1	Using indicator variables	93
7.10.2	Without indicator variables	95
7.11	Samplers provided with NIMBLE	96
7.11.1	Conjugate ('Gibbs') samplers	96
7.11.2	Hamiltonian Monte Carlo (HMC)	98
7.11.3	Particle filter samplers	101
7.11.4	Customized log-likelihood evaluations: <i>RW_llFunction sampler</i>	101
7.12	Detailed MCMC example: <i>litters</i>	103
7.13	Comparing different MCMCs with <i>compareMCMCs</i>	105
7.14	Running MCMC chains in parallel	105
8	Particle Filters, PMCMC, and MCEM	107
8.1	Particle filters / sequential Monte Carlo and iterated filtering	107
8.1.1	Filtering algorithms	107
8.1.2	Particle MCMC (PMCMC)	112
8.2	Monte Carlo Expectation Maximization (MCEM)	114
8.2.1	Estimating the asymptotic covariance From MCEM	116
9	Laplace, AGHQ, and nested approximations	117
9.1	Laplace approximation and adaptive Gauss-Hermite quadrature	117
9.1.1	GLMM example	118
9.1.2	Using Laplace approximation	119
9.1.3	Using the Laplace approximation methods directly	121
9.1.4	Changing the optimization methods	122
9.2	Nested approximation (INLA-like) methods	123
9.2.1	Overview of the methodology	124
9.2.2	Using NIMBLE's nested approximation	127
10	Spatial models	137
10.1	Intrinsic Gaussian CAR model: <i>dcar_normal</i>	137
10.1.1	Specification and density	137
10.1.2	Example	139
10.2	Proper Gaussian CAR model: <i>dcar_proper</i>	140
10.2.1	Specification and density	140
10.2.2	Example	141
10.3	MCMC Sampling of CAR models	143
10.3.1	Initial values	143
10.3.2	Zero-neighbor regions	143
10.3.3	Zero-mean constraint	144
11	Bayesian nonparametric models	145
11.1	Bayesian nonparametric mixture models	145
11.2	Chinese Restaurant Process model	146
11.2.1	Specification and density	146
11.2.2	Example	147
11.2.3	Extensions	148

11.3	Stick-breaking model	149
11.3.1	Specification and function	149
11.3.2	Example	150
11.4	MCMC sampling of BNP models	151
11.4.1	Sampling CRP models	151
11.4.2	Sampling stick-breaking models	154
IV	Programming with NIMBLE	155
	Overview	157
12	Writing simple nimbleFunctions	159
12.1	Introduction to simple nimbleFunctions	159
12.2	R functions (or variants) implemented in NIMBLE	160
12.2.1	Finding help for NIMBLE's versions of R functions	160
12.2.2	Basic operations	160
12.2.3	Math and linear algebra	162
12.2.4	Distribution functions	165
12.2.5	Flow control: <i>if-then-else</i> , <i>for</i> , <i>while</i> , and <i>stop</i>	166
12.2.6	<i>print</i> and <i>cat</i>	166
12.2.7	Checking for user interrupts: <i>checkInterrupt</i>	166
12.2.8	Optimization: <i>optim</i> and <i>nimOptim</i>	166
12.2.9	Integration: <i>integrate</i> and <i>nimIntegrate</i>	167
12.2.10	'nim' synonyms for some functions	168
12.3	How NIMBLE handles types of variables	168
12.3.1	nimbleList data structures	169
12.3.2	How numeric types work	169
12.4	Declaring argument and return types	172
12.5	Compiled nimbleFunctions pass arguments by reference	173
12.6	Calling external compiled code	173
12.7	Calling uncompiled R functions from compiled nimbleFunctions	173
13	Creating user-defined distributions and functions for models	175
13.1	User-defined functions	175
13.2	User-defined distributions	177
13.2.1	Using <i>registerDistributions</i> for alternative parameterizations and providing other information	179
13.3	Advanced user-defined functions and distributions	180
13.4	User-defined model macros	182
14	Working with NIMBLE models	183
14.1	The variables and nodes in a NIMBLE model	183
14.1.1	Determining the nodes in a model	183
14.1.2	Understanding lifted nodes	185
14.1.3	Determining dependencies in a model	185
14.2	Accessing information about nodes and variables	186
14.2.1	Getting distributional information about a node	186

14.2.2	Getting information about a distribution	187
14.2.3	Getting distribution parameter values for a node	187
14.2.4	Getting distribution bounds for a node	188
14.3	Carrying out model calculations	189
14.3.1	Core model operations: calculation and simulation	189
14.3.2	Pre-defined nimbleFunctions for operating on model nodes: <i>simNodes</i> , <i>calcNodes</i> , and <i>getLogProbNodes</i>	191
14.3.3	Accessing log probabilities via <i>logProb</i> variables	192
15	Data structures in NIMBLE	195
15.1	The modelValues data structure	195
15.1.1	Creating modelValues objects	195
15.1.2	Accessing contents of modelValues	197
15.2	The nimbleList data structure	201
15.2.1	Pre-defined nimbleList types	203
15.2.2	Using <i>eigen</i> and <i>svd</i> in nimbleFunctions	204
16	Writing nimbleFunctions to interact with models	207
16.1	Overview	207
16.2	Using and compiling nimbleFunctions	209
16.3	Writing setup code	209
16.3.1	Useful tools for setup functions	209
16.3.2	Accessing and modifying numeric values from setup	210
16.3.3	Determining numeric types in nimbleFunctions	210
16.3.4	Control of setup outputs	210
16.4	Writing run code	211
16.4.1	Driving models: <i>calculate</i> , <i>calculateDiff</i> , <i>simulate</i> , <i>getLogProb</i>	211
16.4.2	Getting and setting variable and node values	211
16.4.3	Getting parameter values and node bounds	214
16.4.4	Using modelValues objects	214
16.4.5	Using model variables and modelValues in expressions	217
16.4.6	Including other methods in a nimbleFunction	217
16.4.7	Using other nimbleFunctions	218
16.4.8	Virtual nimbleFunctions and nimbleFunctionLists	219
16.4.9	Character objects	221
16.4.10	User-defined data structures	222
16.5	Example: writing user-defined samplers to extend NIMBLE's MCMC engine	223
16.5.1	User-defined samplers and posterior predictive nodes	225
16.6	Copying nimbleFunctions (and NIMBLE models)	226
16.7	Debugging nimbleFunctions	226
16.8	Timing nimbleFunctions with <i>run.time</i>	227
16.9	Clearing and unloading compiled objects	227
16.10	Reducing memory usage	228
V	Automatic Derivatives in NIMBLE	229
17	Automatic Derivatives	231

17.1	How to turn on derivatives in a model	232
17.2	How to support derivatives in user-defined functions and distributions	232
17.3	What operations are and aren't supported for AD	233
17.4	Basics of obtaining derivatives in <code>nimbleFunctions</code>	234
17.4.1	Checking derivatives with uncompiled execution	236
17.4.2	Holding some local variables out of derivative tracking	237
17.4.3	Using AD with multiple <code>nimbleFunctions</code>	239
17.4.4	Understanding more about how AD works: <i>taping</i> of operations	240
17.4.5	Resetting a <code>nimDerivs</code> call	241
17.4.6	A note on performance benchmarking	244
17.5	Advanced uses: double taping	244
17.6	Derivatives involving model calculations	246
17.6.1	Method 1: <code>nimDerivs</code> of <code>model\$calculate</code>	246
17.6.2	Method 2: <code>nimDerivs</code> of a method that calls <code>model\$calculate</code>	249
17.7	Parameter transformations	253
18	Example: maximum likelihood estimation using <code>optim</code> with gradients from <code>nimDerivs</code>.	255

Part I

Introduction

Chapter 1

Welcome to NIMBLE

NIMBLE is a system for building and sharing analysis methods for statistical models from R, especially for hierarchical models and computationally-intensive methods. While NIMBLE is embedded in R, it goes beyond R by supporting separate programming of models and algorithms along with compilation for fast execution.

As of version 1.4.0, NIMBLE has been around for a while and is reasonably stable, but we have a lot of plans to expand and improve it. The algorithm library provides MCMC with a lot of user control and ability to write new samplers easily. Other algorithms include Laplace approximation, adaptive Gauss-Hermite quadrature (AGHQ) approximation, INLA-like nested approximation, and Monte Carlo Expectation Maximization (MCEM). Some algorithms and model components are provided in separate packages, including nimbleHMC (Hamiltonian Monte Carlo with the no U-turn sampler), nimbleSMC (particle filtering, particle MCMC, iterated particle filtering, and ensemble Kalman Filter), nimbleEcology (ecological model components), and nimbleSCR (spatial capture-recapture).

But NIMBLE is about much more than providing an algorithm library. It provides a language for writing model-generic algorithms. We hope you will program in NIMBLE and make an R package providing your method. Of course, NIMBLE is open source, so we also hope you'll contribute to its development.

Please join the mailing lists (see R-nimble.org/more/issues-and-groups) and help improve NIMBLE by telling us what you want to do with it, what you like, and what could be better. We have a lot of ideas for how to improve it, but we want your help and ideas too. You can also follow and contribute to developer discussions on [our GitHub repository](#).

If you use NIMBLE in your work, please cite us, as this helps justify past and future funding for the development of NIMBLE. For more information, please call `citation('nimble')` in R.

1.1 What does NIMBLE do?

NIMBLE makes it easier to program statistical algorithms that will run efficiently and work on many different models from R.

You can think of NIMBLE as comprising four pieces:

1. A system for writing statistical models flexibly, which is an extension of the BUGS language¹.
2. A library of algorithms such as MCMC.
3. A language, called NIMBLE, embedded within and similar in style to R, for writing algorithms that operate on models written in BUGS.
4. A compiler that generates C++ for your models and algorithms, compiles that C++, and lets you use it seamlessly from R without knowing anything about C++.

NIMBLE stands for Numerical Inference for statistical Models for Bayesian and Likelihood Estimation.

Although NIMBLE was motivated by algorithms for hierarchical statistical models, it's useful for other goals too. You could use it for simpler models. And since NIMBLE can automatically compile R-like functions into C++ that use the Eigen library for fast linear algebra, you can use it to program fast numerical functions without any model involved².

One of the beauties of R is that many of the high-level analysis functions are themselves written in R, so it is easy to see their code and modify them. The same is true for NIMBLE: the algorithms are themselves written in the NIMBLE language.

1.2 How to use this manual

We suggest everyone start with the Lightning Introduction in Chapter 2.

Then, if you want to jump into using NIMBLE's algorithms without learning about NIMBLE's programming system, go to Part II to learn how to build your model and Part III to learn how to apply NIMBLE's built-in algorithms to your model.

If you want to learn about NIMBLE programming (`nimbleFunctions`), go to Part IV. This teaches how to program user-defined function or distributions to use in BUGS code, compile your R code for faster operations, and write algorithms with NIMBLE. These algorithms could be specific algorithms for your particular model (such as a user-defined MCMC sampler for a parameter in your model) or general algorithms you can distribute to others. In fact the algorithms provided as part of NIMBLE and described in Part III are written as `nimbleFunctions`.

¹See Chapter 5 for information about NIMBLE's version of BUGS.

²The packages `Rcpp` and `RcppEigen` provide different ways of connecting C++, the Eigen library and R. In those packages you program directly in C++, while in NIMBLE you program in R in a `nimbleFunction` and the NIMBLE compiler turns it into C++.

Chapter 2

Lightning introduction

2.1 A brief example

Here we'll give a simple example of building a model and running some algorithms on the model, as well as creating our own user-specified algorithm. The goal is to give you a sense for what one can do in the system. Later sections will provide more detail.

We'll use the *pump* model example from BUGS¹. We could load the model from the standard BUGS example file formats (Section 6.1.2), but instead we'll show how to enter it directly in R.

In this 'lightning introduction' we will:

1. Create the model for the pump example.
2. Compile the model.
3. Create a basic MCMC configuration for the pump model.
4. Compile and run the MCMC
5. Customize the MCMC configuration and compile and run that.
6. Create, compile and run a Monte Carlo Expectation Maximization (MCEM) algorithm, which illustrates some of the flexibility NIMBLE provides to combine R and NIMBLE.
7. Write a short `nimbleFunction` to generate simulations from designated nodes of any model.

2.2 Creating a model

First we define the model code, its constants, data, and initial values for MCMC.

```
pumpCode <- nimbleCode({  
  for (i in 1:N){  
    theta[i] ~ dgamma(alpha,beta)  
    lambda[i] <- theta[i]*t[i]  
    x[i] ~ dpois(lambda[i])  
  }  
  alpha ~ dexp(1.0)  
  beta ~ dgamma(0.1,1.0)  
})
```

¹The data set describes failure rates of some pumps.

```

pumpConsts <- list(N = 10,
                  t = c(94.3, 15.7, 62.9, 126, 5.24,
                        31.4, 1.05, 1.05, 2.1, 10.5))

pumpData <- list(x = c(5, 1, 5, 14, 3, 19, 1, 1, 4, 22))

pumpInits <- list(alpha = 1, beta = 1,
                  theta = rep(0.1, pumpConsts$N))

```

Here $x[i]$ is the number of failures recorded during a time duration of length $t[i]$ for the i^{th} pump. $\theta[i]$ is a failure rate, and the goal is estimate parameters α and β . Now let's create the model and look at some of its nodes.

```

pump <- nimbleModel(code = pumpCode, name = "pump", constants = pumpConsts,
                   data = pumpData, inits = pumpInits)

```

```

pump$getNodeNames()

```

```

## [1] "alpha"          "beta"           "lifted_d1_over_beta"
## [4] "theta[1]"       "theta[2]"       "theta[3]"
## [7] "theta[4]"       "theta[5]"       "theta[6]"
## [10] "theta[7]"       "theta[8]"       "theta[9]"
## [13] "theta[10]"      "lambda[1]"      "lambda[2]"
## [16] "lambda[3]"      "lambda[4]"      "lambda[5]"
## [19] "lambda[6]"      "lambda[7]"      "lambda[8]"
## [22] "lambda[9]"      "lambda[10]"     "x[1]"
## [25] "x[2]"           "x[3]"           "x[4]"
## [28] "x[5]"           "x[6]"           "x[7]"
## [31] "x[8]"           "x[9]"           "x[10]"

```

```

pump$x

```

```

## [1] 5 1 5 14 3 19 1 1 4 22

```

```

pump$logProb_x

```

```

## [1] -2.998011 -1.118924 -1.882686 -2.319466 -4.254550 -20.739651
## [7] -2.358795 -2.358795 -9.630645 -48.447798

```

```

pump$alpha

```

```

## [1] 1

```

```

pump$theta

```

```

## [1] 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1

```

```

pump$lambda

```

```

## [1] 9.430 1.570 6.290 12.600 0.524 3.140 0.105 0.105 0.210 1.050

```

Notice that in the list of nodes, NIMBLE has introduced a new node, `lifted_d1_over_beta`. We

call this a ‘lifted’ node. Like R, NIMBLE allows alternative parameterizations, such as the scale or rate parameterization of the gamma distribution. Choice of parameterization can generate a lifted node, as can using a link function or a distribution argument that is an expression. It’s helpful to know why they exist, but you shouldn’t need to worry about them.

Thanks to the plotting capabilities of the `igraph` package that NIMBLE uses to represent the directed acyclic graph, we can plot the model (Figure 2.1).

```
pump$plotGraph()
```

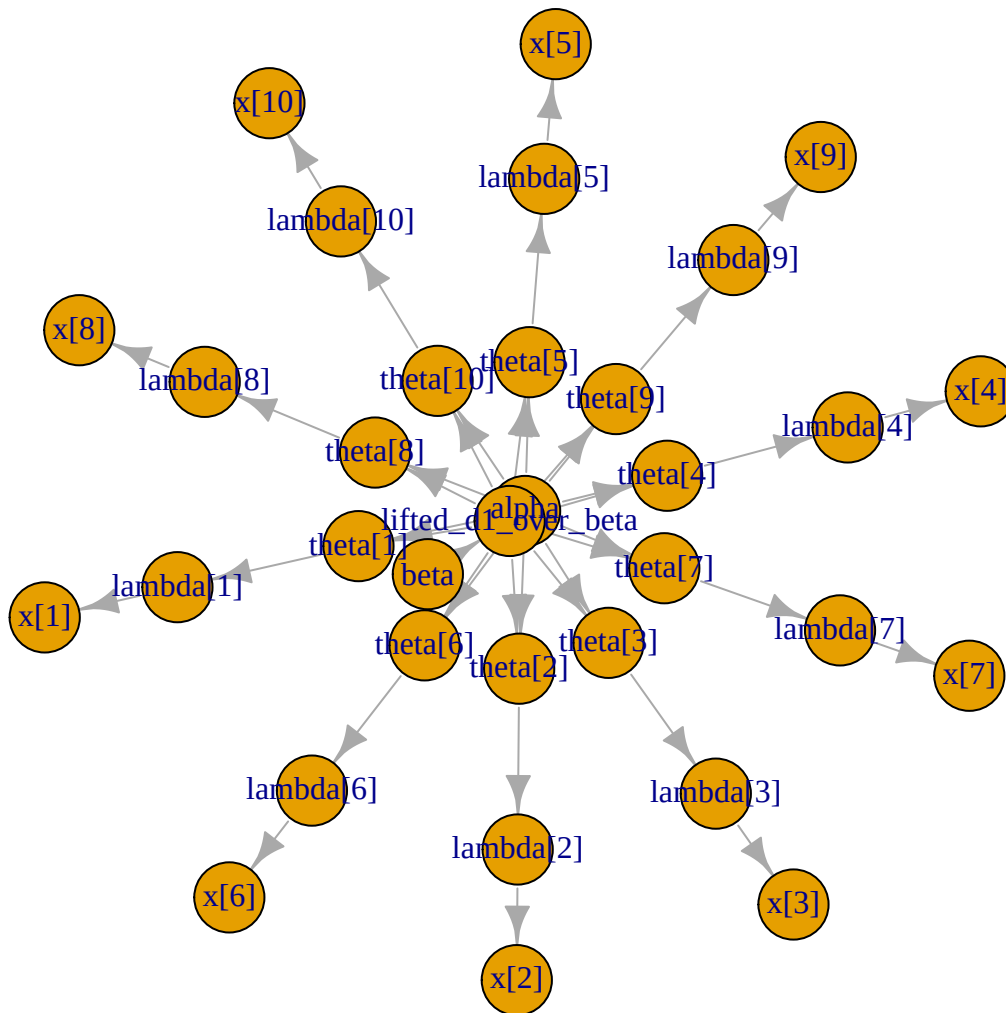


Figure 2.1: Directed Acyclic Graph plot of the pump model, thanks to the `igraph` package

You are in control of the model. By default, `nimbleModel` does its best to initialize a model, but let’s say you want to re-initialize `theta`. To simulate from the prior for `theta` (overwriting the initial values previously in the model) we first need to be sure the parent nodes of all `theta[i]` nodes are fully initialized, including any non-stochastic nodes such as lifted nodes. We then use the `simulate` function to simulate from the distribution for `theta`. Finally we use the `calculate` function to calculate the dependencies of `theta`, namely `lambda` and the log probabilities of `x` to ensure all parts of the model are up to date. First we show how to use the model’s `getDependencies` method to query information about its graph.

```
# Show all dependencies of alpha and beta terminating in stochastic nodes
pump$getDependencies(c("alpha", "beta"))
```

```
## [1] "alpha"          "beta"          "lifted_d1_over_beta"
## [4] "theta[1]"      "theta[2]"      "theta[3]"
## [7] "theta[4]"      "theta[5]"      "theta[6]"
## [10] "theta[7]"      "theta[8]"      "theta[9]"
## [13] "theta[10]"
```

```
# Now show only the deterministic dependencies
pump$getDependencies(c("alpha", "beta"), determOnly = TRUE)
```

```
## [1] "lifted_d1_over_beta"
```

```
# Check that the lifted node was initialized.
pump[["lifted_d1_over_beta"]] # It was.
```

```
## [1] 1
```

```
# Now let's simulate new theta values
set.seed(1) # This makes the simulations here reproducible
pump$simulate("theta")
pump$theta # the new theta values
```

```
## [1] 0.15514136 1.88240160 1.80451250 0.83617765 1.22254365 1.15835525
## [7] 0.99001994 0.30737332 0.09461909 0.15720154
```

```
# lambda and logProb_x haven't been re-calculated yet
pump$lambda # these are the same values as above
```

```
## [1] 9.430 1.570 6.290 12.600 0.524 3.140 0.105 0.105 0.210 1.050
```

```
pump$logProb_x
```

```
## [1] -2.998011 -1.118924 -1.882686 -2.319466 -4.254550 -20.739651
## [7] -2.358795 -2.358795 -9.630645 -48.447798
```

```
pump$getLogProb("x") # The sum of logProb_x
```

```
## [1] -96.10932
```

```
pump$calculate(pump$getDependencies(c("theta")))
```

```
## [1] -262.204
```

```
pump$lambda # Now they have.
```

```
## [1] 14.6298299 29.5537051 113.5038360 105.3583839 6.4061287 36.3723548
## [7] 1.0395209 0.3227420 0.1987001 1.6506161
```

```
pump$logProb_x
```

```
## [1] -6.002009 -26.167496 -94.632145 -65.346457 -2.626123 -7.429868
## [7] -1.000761 -1.453644 -9.840589 -39.096527
```


Notice that the first `getDependencies` call returned dependencies from `alpha` and `beta` down to the next stochastic nodes in the model. The second call requested only deterministic dependencies. The call to `pump$simulate("theta")` expands `"theta"` to include all nodes in `theta`. After simulating into `theta`, we can see that `lambda` and the log probabilities of `x` still reflect the old values of `theta`, so we calculate them and then see that they have been updated.

2.3 Compiling the model

Next we compile the model, which means generating C++ code, compiling that code, and loading it back into R with an object that can be used just like the uncompiled model. The values in the compiled model will be initialized from those of the original model in R, but the original and compiled models are distinct objects so any subsequent changes in one will not be reflected in the other.

```
Cpump <- compileNimble(pump)
Cpump$theta
```

```
## [1] 0.15514136 1.88240160 1.80451250 0.83617765 1.22254365 1.15835525
## [7] 0.99001994 0.30737332 0.09461909 0.15720154
```

Note that the compiled model is used when running any NIMBLE algorithms via C++, so the model needs to be compiled before (or at the same time as) any compilation of algorithms, such as the compilation of the MCMC done in the next section.

2.4 One-line invocation of MCMC

The most direct approach to invoking NIMBLE's MCMC engine is using the `nimbleMCMC` function. This function would generally take the code, data, constants, and initial values as input, but it can also accept the (compiled or uncompiled) model object as an argument. It provides a variety of options for executing and controlling multiple chains of NIMBLE's default MCMC algorithm, and returning posterior samples, posterior summary statistics, and/or WAIC values.

For example, to execute two MCMC chains of 10,000 samples each, and return samples, summary statistics, and WAIC values:

```
mcmc.out <- nimbleMCMC(code = pumpCode, constants = pumpConsts,
                      data = pumpData, inits = pumpInits,
                      nchains = 2, niter = 10000,
                      summary = TRUE, WAIC = TRUE,
                      monitors = c('alpha', 'beta', 'theta'))
```

```
## [Warning] There are 7 individual pWAIC values that are greater than 0.4. This may indicate
```

```
names(mcmc.out)
```

```
## [1] "samples" "summary" "WAIC"
```

```
mcmc.out$summary
```

```
## $chain1
```

```
##           Mean           Median          St.Dev.      95%CI_low 95%CI_upp
```

```

## alpha      0.69804352 0.65835063 0.27037676 0.287898244 1.3140461
## beta       0.92862598 0.82156847 0.54969128 0.183699137 2.2872696
## theta[1]   0.06019274 0.05676327 0.02544956 0.021069950 0.1199230
## theta[2]   0.10157737 0.08203988 0.07905076 0.008066869 0.3034085
## theta[3]   0.08874755 0.08396502 0.03760562 0.031186960 0.1769982
## theta[4]   0.11567784 0.11301465 0.03012598 0.064170937 0.1824525
## theta[5]   0.60382223 0.54935089 0.31219612 0.159731108 1.3640771
## theta[6]   0.61204831 0.60085518 0.13803302 0.372712375 0.9135269
## theta[7]   0.90263434 0.70803389 0.73960182 0.074122175 2.7598261
## theta[8]   0.89021051 0.70774794 0.72668155 0.072571029 2.8189252
## theta[9]   1.57678136 1.44390008 0.76825189 0.455195149 3.4297368
## theta[10]  1.98954127 1.96171250 0.42409802 1.241383787 2.9012192
##
## $chain2
##           Mean      Median    St.Dev.   95%CI_low 95%CI_upp
## alpha      0.69101961 0.65803654 0.26548378 0.277195564 1.2858148
## beta       0.91627273 0.81434426 0.53750825 0.185772263 2.2702428
## theta[1]   0.05937364 0.05611283 0.02461866 0.020956151 0.1161870
## theta[2]   0.10017726 0.08116259 0.07855024 0.008266343 0.3010355
## theta[3]   0.08908126 0.08390782 0.03704170 0.031330829 0.1736876
## theta[4]   0.11592652 0.11356920 0.03064645 0.063595333 0.1829574
## theta[5]   0.59755632 0.54329373 0.31871551 0.149286703 1.3748728
## theta[6]   0.61080189 0.59946693 0.13804343 0.371373877 0.9097319
## theta[7]   0.89902759 0.70901502 0.72930369 0.076243503 2.7441445
## theta[8]   0.89954594 0.70727079 0.73345905 0.071250926 2.8054633
## theta[9]   1.57530029 1.45005738 0.75242164 0.469959364 3.3502795
## theta[10]  1.98911473 1.96227061 0.42298189 1.246910723 2.9102326
##
## $all.chains
##           Mean      Median    St.Dev.   95%CI_low 95%CI_upp
## alpha      0.69453156 0.65803654 0.26795776 0.28329854 1.2999319
## beta       0.92244935 0.81828160 0.54365539 0.18549077 2.2785444
## theta[1]   0.05978319 0.05646474 0.02504028 0.02102807 0.1183433
## theta[2]   0.10087731 0.08162361 0.07880204 0.00811108 0.3017967
## theta[3]   0.08891440 0.08394667 0.03732417 0.03123228 0.1749967
## theta[4]   0.11580218 0.11326039 0.03038683 0.06385253 0.1827382
## theta[5]   0.60068928 0.54668011 0.31548032 0.15363752 1.3686801
## theta[6]   0.61142510 0.60015416 0.13803618 0.37203765 0.9122467
## theta[7]   0.90083096 0.70852800 0.73445465 0.07550465 2.7534885
## theta[8]   0.89487822 0.70761105 0.73007484 0.07211191 2.8067373
## theta[9]   1.57604083 1.44719278 0.76035931 0.46374515 3.3866706
## theta[10]  1.98932800 1.96195345 0.42352979 1.24334249 2.9068229

mcmc.out$WAIC

## nimbleList object of type waicNimbleList
## Field "WAIC":
## [1] 48.69896

```

```
## Field "lppd":
## [1] -19.99653
## Field "pWAIC":
## [1] 4.352945
```

See Section 7.1 or `help(nimbleMCMC)` for more details about using `nimbleMCMC`.

Note that the WAIC value varies depending on what quantities are treated as parameters. See Section 7.8 or `help(waic)` for more details.

2.5 Creating, compiling and running a basic MCMC configuration

At this point we have initial values for all of the nodes in the model, and we have both the original and compiled versions of the model. As a first algorithm to try on our model, let's use NIMBLE's default MCMC. Note that conjugate relationships are detected for all nodes except for `alpha`, on which the default sampler is a random walk Metropolis sampler.

```
pumpConf <- configureMCMC(pump, print = TRUE)
```

```
## ===== Monitors =====
## thin = 1: alpha, beta
## ===== Samplers =====
## RW sampler (1)
##   - alpha
## conjugate sampler (11)
##   - beta
##   - theta[] (10 elements)
```

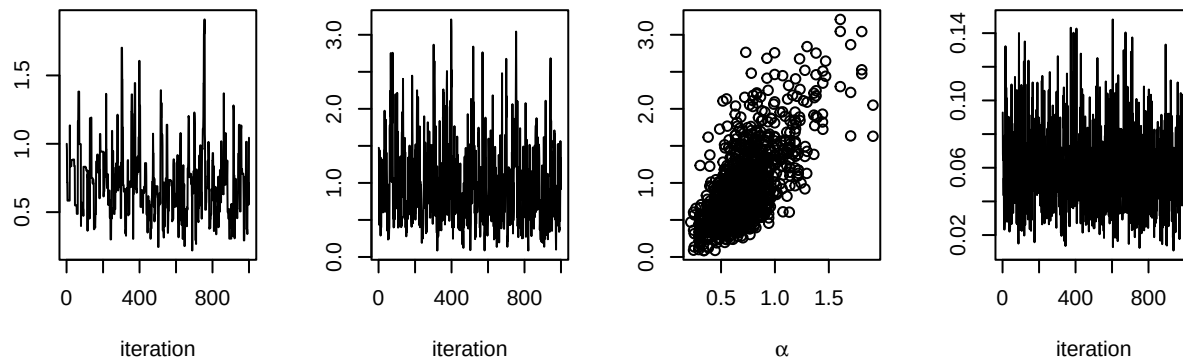
```
pumpConf$addMonitors(c("alpha", "beta", "theta"))
```

```
## thin = 1: alpha, beta, theta
```

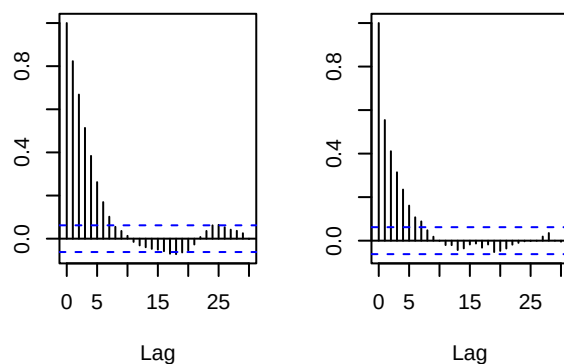
```
pumpMCMC <- buildMCMC(pumpConf)
CpumpMCMC <- compileNimble(pumpMCMC, project = pump)
```

```
niter <- 1000
set.seed(1)
samples <- runMCMC(CpumpMCMC, niter = niter)
```

```
par(mfrow = c(1, 4), mai = c(.6, .4, .1, .2))
plot(samples[, "alpha"], type = "l", xlab = "iteration",
      ylab = expression(alpha))
plot(samples[, "beta"], type = "l", xlab = "iteration",
      ylab = expression(beta))
plot(samples[, "alpha"], samples[, "beta"], xlab = expression(alpha),
      ylab = expression(beta))
plot(samples[, "theta[1]"], type = "l", xlab = "iteration",
      ylab = expression(theta[1]))
```



```
acf(samples[, "alpha"]) # plot autocorrelation of alpha sample
acf(samples[, "beta"])  # plot autocorrelation of beta sample
```



Notice the posterior correlation between `alpha` and `beta`. A measure of the mixing for each is the autocorrelation for each parameter, shown by the acf plots.

2.6 Customizing the MCMC

Let's add an adaptive block sampler on `alpha` and `beta` jointly and see if that improves the mixing.

```
pumpConf$addSampler(target = c("alpha", "beta"), type = "RW_block",
  control = list(adaptInterval = 100))
```

```
pumpMCMC2 <- buildMCMC(pumpConf)
```

```
# need to reset the nimbleFunctions in order to add the new MCMC
CpumpNewMCMC <- compileNimble(pumpMCMC2, project = pump,
  resetFunctions = TRUE)
```

```
set.seed(1)
```

```
CpumpNewMCMC$run(niter)
```

```
samplesNew <- as.matrix(CpumpNewMCMC$mvSamples)
```

```
par(mfrow = c(1, 4), mai = c(.6, .4, .1, .2))
```

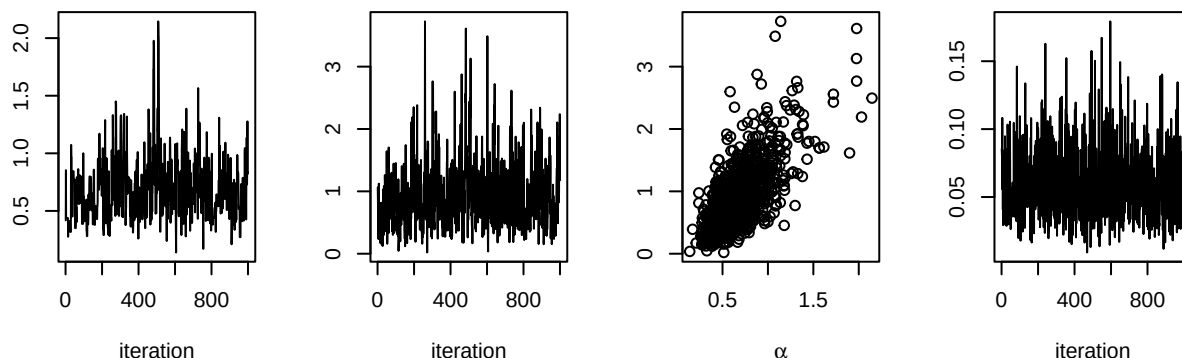
```
plot(samplesNew[, "alpha"], type = "l", xlab = "iteration",
  ylab = expression(alpha))
```

```
plot(samplesNew[, "beta"], type = "l", xlab = "iteration",
```

```

ylab = expression(beta))
plot(samplesNew[, "alpha"], samplesNew[, "beta"], xlab = expression(alpha),
      ylab = expression(beta))
plot(samplesNew[, "theta[1]"], type = "l", xlab = "iteration",
      ylab = expression(theta[1]))

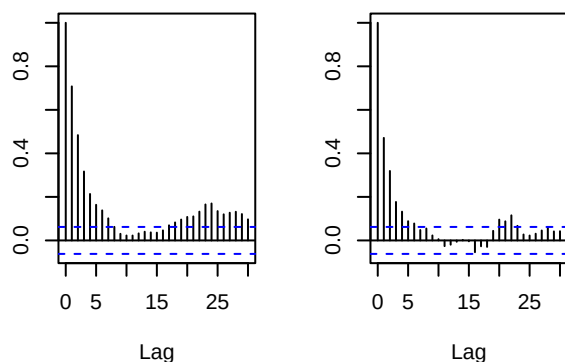
```



```

acf(samplesNew[, "alpha"]) # plot autocorrelation of alpha sample
acf(samplesNew[, "beta"]) # plot autocorrelation of beta sample

```



We can see that the block sampler has decreased the autocorrelation for both `alpha` and `beta`. Of course these are just short runs, and what we are really interested in is the effective sample size of the MCMC per computation time, but that's not the point of this example.

Once you learn the MCMC system, you can write your own samplers and include them. The entire system is written in `nimbleFunctions`.

2.7 Running MCEM

NIMBLE is a system for working with algorithms, not just an MCMC engine. So let's try maximizing the marginal likelihood for `alpha` and `beta` using Monte Carlo Expectation Maximization².

```

pump2 <- nimbleModel(code = pumpCode, name = "pump", constants = pumpConsts,
                    data = pumpData, inits = pumpInits, buildDerivs = TRUE)

pumpMCEM <- buildMCEM(model = pump2, latentNodes = "theta[1:10]")

```

²Note that for this model, one could analytically integrate over `theta` and then numerically maximize the resulting marginal likelihood.

```

Cpump <- compileNimble(pump2)
CpumpMCEM <- compileNimble(pumpMCEM, project = pump2)

set.seed(1)
pumpMLE <- CpumpMCEM$findMLE()

```

```

## [Note] Iteration Number: 1.
## [Note] Current number of MCMC iterations: 1000.
## [Note] Parameter Estimates:
## 0.810166
## 1.11657
## [Note] Convergence Criterion: 0.655578.
## [Note] Iteration Number: 2.
## [Note] Current number of MCMC iterations: 1000.
## [Note] Parameter Estimates:
## 0.807775
## 1.19848
## [Note] Convergence Criterion: 0.0288968.
## [Note] Iteration Number: 3.
## [Note] Current number of MCMC iterations: 1000.
## [Note] Parameter Estimates:
## 0.801477
## 1.20974
## [Note] Convergence Criterion: 0.00314139.
## [Note] Iteration Number: 4.
## [Note] Current number of MCMC iterations: 1000.
## [Note] Parameter Estimates:
## 0.814074
## 1.25737
## [Note] Convergence Criterion: 0.00612404.
## [Note] Iteration Number: 5.
## [Note] Current number of MCMC iterations: 1000.
## [Note] Parameter Estimates:
## 0.823345
## 1.26626
## [Note] Convergence Criterion: 0.00132764.
## [Note] Iteration Number: 6.
## [Note] Current number of MCMC iterations: 1000.
## [Note] Parameter Estimates:
## 0.833545
## 1.29405
## [Note] Convergence Criterion: 0.00251191.
## [Note] Iteration Number: 7.
## [Note] Current number of MCMC iterations: 1000.
## [Note] Parameter Estimates:
## 0.81979
## 1.25567

```

```
## [Note] Convergence Criterion: 0.00414933.
## Monte Carlo error too big: increasing MCMC sample size.
## Monte Carlo error too big: increasing MCMC sample size.
## Monte Carlo error too big: increasing MCMC sample size.
## Monte Carlo error too big: increasing MCMC sample size.
## Monte Carlo error too big: increasing MCMC sample size.
## Monte Carlo error too big: increasing MCMC sample size.
## Monte Carlo error too big: increasing MCMC sample size.
## Monte Carlo error too big: increasing MCMC sample size.
## Monte Carlo error too big: increasing MCMC sample size.
## [Note] Iteration Number: 8.
## [Note] Current number of MCMC iterations: 9400.
## [Note] Parameter Estimates:
## 0.823025
## 1.26278
## [Note] Convergence Criterion: 0.000165076.
```

```
pumpMLE$par
```

```
## [1] 0.8230248 1.2627842
```

Both estimates are within 0.01 of the values reported by [George et al. \(1993\)](#)³. Some discrepancy is to be expected since it is a Monte Carlo algorithm.

2.8 Creating your own functions

Now let's see an example of writing our own algorithm and using it on the model. We'll do something simple: simulating multiple values for a designated set of nodes and calculating every part of the model that depends on them. More details on programming in NIMBLE are in Part IV.

Here is our *nimbleFunction*:

```
simNodesMany <- nimbleFunction(
  setup = function(model, nodes) {
    mv <- modelValues(model)
    deps <- model$getDependencies(nodes)
    allNodes <- model$getNodeNames()
  },
  run = function(n = integer()) {
    resize(mv, n)
    for(i in 1:n) {
      model$simulate(nodes)
      model$calculate(deps)
      copy(from = model, nodes = allNodes,
           to = mv, rowTo = i, logProb = TRUE)
    }
  }
)
```

³Table 2 of the paper accidentally swapped the two estimates.

```

})

simNodesTheta1to5 <- simNodesMany(pump, "theta[1:5]")
simNodesTheta6to10 <- simNodesMany(pump, "theta[6:10]")

```

Here are a few things to notice about the `nimbleFunction`.

1. The `setup` function is written in R. It creates relevant information specific to our model for use in the run-time code.
2. The `setup` code creates a *modelValues* object to hold multiple sets of values for variables in the model provided.
3. The `run` function is written in NIMBLE. It carries out the calculations using the information determined once for each set of `model` and `nodes` arguments by the setup code. The run-time code is what will be compiled.
4. The `run` code requires type information about the argument `n`. In this case it is a scalar integer.
5. The for-loop looks just like R, but only sequential integer iteration is allowed.
6. The functions `calculate` and `simulate`, which were introduced above in R, can be used in NIMBLE.
7. The special function `copy` is used here to record values from the model into the *modelValues* object.
8. Multiple instances, or ‘specializations’, can be made by calling `simNodesMany` with different arguments. Above, `simNodesTheta1to5` has been made by calling `simNodesMany` with the `pump` model and nodes `"theta[1:5]"` as inputs to the `setup` function, while `simNodesTheta6to10` differs by providing `"theta[6:10]"` as an argument. The returned objects are objects of a uniquely generated R reference class with fields (member data) for the results of the `setup` code and a `run` method (member function).

By the way, `simNodesMany` is very similar to a standard `nimbleFunction` provided with NIMBLE, `simNodesMV`.

Now let’s execute this `nimbleFunction` in R, before compiling it.

```

set.seed(1) # make the calculation repeatable
pump$alpha <- pumpMLE$par[1]
pump$beta <- pumpMLE$par[2]
# make sure to update deterministic dependencies of the altered nodes
pump$calculate(pump$getDependencies(c("alpha","beta"), determOnly = TRUE))

## [1] 0

saveTheta <- pump$theta
simNodesTheta1to5$run(10)
simNodesTheta1to5$mv[["theta"]][1:2]

## [[1]]
## [1] 0.21801104 1.92849314 0.62185677 0.34145435 3.45061289 1.15835525

```



```
## [7] 0.99001994 0.30737332 0.09461909 0.15720154
##
## [[2]]
## [1] 0.82620401 0.08776004 0.34362701 0.29479110 0.14167506 1.15835525
## [7] 0.99001994 0.30737332 0.09461909 0.15720154
```

```
simNodesTheta1to5$mv[["logProb_x"]][1:2]
```

```
## [[1]]
## [1] -10.229575 -26.866943 -25.569779 -15.550101 -11.188351 -7.429868
## [7] -1.000761 -1.453644 -9.840589 -39.096527
##
## [[2]]
## [1] -60.920692 -1.057321 -11.034892 -11.727789 -3.427830 -7.429868
## [7] -1.000761 -1.453644 -9.840589 -39.096527
```

In this code we have initialized the values of `alpha` and `beta` to their MLE and then recorded the `theta` values to use below. Then we have requested 10 simulations from `simNodesTheta1to5`. Shown are the first two simulation results for `theta` and the log probabilities of `x`. Notice that `theta[6:10]` and the corresponding log probabilities for `x[6:10]` are unchanged because the nodes being simulated are only `theta[1:5]`. In R, this function runs slowly.

Finally, let's compile the function and run that version.

```
CsimNodesTheta1to5 <- compileNimble(simNodesTheta1to5,
                                   project = pump, resetFunctions = TRUE)
Cpump$alpha <- pumpMLE$par[1]
Cpump$beta <- pumpMLE$par[2]
Cpump$calculate(Cpump$getDependencies(c("alpha","beta"), determOnly = TRUE))
```

```
## [1] 0
```

```
Cpump$theta <- saveTheta
```

```
set.seed(1)
```

```
CsimNodesTheta1to5$run(10)
```

```
CsimNodesTheta1to5$mv[["theta"]][1:2]
```

```
## [[1]]
## [1] 0.2387677 2.0746311 0.6709612 0.3715794 3.7047288 0.4910946 0.7019477
## [8] 0.8661760 1.2961006 1.6388032
##
## [[2]]
## [1] 0.89414647 0.09737063 0.37390994 0.32147204 0.15612022 0.49109463
## [7] 0.70194774 0.86617604 1.29610065 1.63880322
```

```
CsimNodesTheta1to5$mv[["logProb_x"]][1:2]
```

```
## [[1]]
## [1] -11.732199 -29.088264 -28.278440 -18.162177 -12.306744 -2.782156
## [7] -1.042151 -1.004362 -1.894675 -3.081102
##
```

```
## [[2]]  
## [1] -66.932527 -1.104289 -12.517398 -13.876578 -3.212252 -2.782156  
## [7] -1.042151 -1.004362 -1.894675 -3.081102
```

Given the same initial values and the same random number generator seed, we got identical results for `theta[1:5]` and their dependencies, but it happened much faster.

Chapter 3

More introduction

Now that we have shown a brief example, we will introduce more about the concepts and design of NIMBLE.

One of the most important concepts behind NIMBLE is to allow a combination of high-level processing in R and low-level processing in C++. For example, when we write a Metropolis-Hastings MCMC sampler in the NIMBLE language, the inspection of the model structure related to one node is done in R, and the actual sampler calculations are done in C++. This separation between *setup* and *run* steps will become clearer as we go.

3.1 NIMBLE adopts and extends the BUGS language for specifying models

We adopted the BUGS language, and we have extended it to make it more flexible. The BUGS language became widely used in WinBUGS, then in OpenBUGS and JAGS. These systems all provide automatically-generated MCMC algorithms, but we have adopted only the language for describing models, not their systems for generating MCMCs.

NIMBLE extends BUGS by:

1. allowing you to write new functions and distributions and use them in BUGS models;
2. allowing you to define multiple models in the same code using conditionals evaluated when the BUGS code is processed;
3. supporting a variety of more flexible syntax such as R-like named parameters and more general algebraic expressions.

By supporting new functions and distributions, NIMBLE makes BUGS an extensible language, which is a major departure from previous packages that implement BUGS.

We adopted BUGS because it has been so successful, with over 30,000 users by the time they stopped counting (Lunn et al., 2009). Many papers and books provide BUGS code as a way to document their statistical models. We describe NIMBLE's version of BUGS later. The web sites for WinBUGS, OpenBUGS and JAGS provide other useful documentation on writing models in BUGS. For the most part, if you have BUGS code, you can try NIMBLE.

NIMBLE does several things with BUGS code:

1. NIMBLE creates a *model definition* object that knows everything about the variables and their relationships written in the BUGS code. Usually you'll ignore the *model definition* and let NIMBLE's default options take you directly to the next step.
2. NIMBLE creates a model object¹. This can be used to manipulate variables and operate the model from R. Operating the model includes calculating, simulating, or querying the log probability value of model nodes. These basic capabilities, along with the tools to query model structure, allow one to write programs that use the model and adapt to its structure.
3. When you're ready, NIMBLE can generate customized C++ code representing the model, compile the C++, load it back into R, and provide a new model object that uses the compiled model internally. We use the word 'compile' to refer to all of these steps together.

As an example of how radical a departure NIMBLE is from previous BUGS implementations, consider a situation where you want to simulate new data from a model written in BUGS code. Since NIMBLE creates model objects that you can control from R, simulating new data is trivial. With previous BUGS-based packages, this isn't possible.

More information about specifying and manipulating models is in Chapters 6 and 14.

3.2 nimbleFunctions for writing algorithms

NIMBLE provides *nimbleFunctions* for writing functions that can (but don't have to) use BUGS models. The main ways that *nimbleFunctions* can use BUGS models are:

1. inspecting the structure of a model, such as determining the dependencies between variables, in order to do the right calculations with each model;
2. accessing values of the model's variables;
3. controlling execution of the model's probability calculations or corresponding simulations;
4. managing *modelValues* data structures for multiple sets of model values and probabilities.

In fact, the calculations of the model are themselves constructed as *nimbleFunctions*, as are the algorithms provided in NIMBLE's algorithm library².

Programming with *nimbleFunctions* involves a fundamental distinction between two stages of processing:

1. A *setup* function within a *nimbleFunction* gives the steps that need to happen only once for each new situation (e.g., for each new model). Typically such steps include inspecting the model's variables and their relationships, such as determining which parts of a model will need to be calculated for a MCMC sampler. Setup functions are executed in R and never compiled.
2. One or more *run* functions within a *nimbleFunction* give steps that need to happen multiple times using the results of the setup function, such as the iterations of a MCMC sampler. Formally, run code is written in the NIMBLE language, which you can think of as a small subset of R along with features for operating models and related data structures. The NIMBLE language is what the NIMBLE compiler can automatically turn into C++ as part of a compiled *nimbleFunction*.

What NIMBLE does with a *nimbleFunction* is similar to what it does with a BUGS model:

¹or multiple model objects

²That's why it's easy to use new functions and distributions written as *nimbleFunctions* in BUGS code.

1. NIMBLE creates a working R version of the `nimbleFunction`. This is most useful for debugging (Section 16.7).
2. When you are ready, NIMBLE can generate C++ code, compile it, load it back into R and give you new objects that use the compiled C++ internally. Again, we refer to these steps all together as ‘compilation’. The behavior of compiled `nimbleFunctions` is usually very similar, but not identical, to their uncompiled counterparts.

If you are familiar with object-oriented programming, you can think of a `nimbleFunction` as a class definition. The `setup` function initializes a new object and `run` functions are class methods. Member data are determined automatically as the objects from a `setup` function needed in `run` functions. If no `setup` function is provided, the `nimbleFunction` corresponds to a simple (compilable) function rather than a class.

More about writing algorithms is in Chapter 16.

3.3 The NIMBLE algorithm library

In Version 1.4.0, the NIMBLE algorithm library includes:

1. MCMC with samplers including conjugate (Gibbs), slice, adaptive random walk (with options for reflection or sampling on a log scale), adaptive block random walk, and elliptical slice, among others. You can modify sampler choices and configurations from R before compiling the MCMC. You can also write new samplers as `nimbleFunctions`.
2. Reversible jump MCMC for variable selection.
3. WAIC calculation for model comparison after an MCMC algorithm has been run.
4. A set of particle filter (sequential Monte Carlo) methods including a basic bootstrap filter, auxiliary particle filter, ensemble Kalman Filter, iterated filtering 2 filter (IF2), and Liu-West filter.
5. An ascent-based Monte Carlo Expectation Maximization (MCEM) algorithm.
6. A variety of basic functions that can be used as programming tools for larger algorithms. These include:
 - a. A likelihood function for arbitrary parts of any model.
 - b. Functions to simulate one or many sets of values for arbitrary parts of any model.
 - c. Functions to calculate the summed log probability (density) for one or many sets of values for arbitrary parts of any model along with stochastic dependencies in the model structure.

More about the NIMBLE algorithm library is in Chapter 8.

Chapter 4

Installing NIMBLE

4.1 Requirements to run NIMBLE

You can run NIMBLE on any of the three common operating systems: Linux, MacOS, or Windows. The following are required to run NIMBLE.

1. [R](#), of course.
2. The [igraph](#), [coda](#), [R6](#), [pracma](#), and [numDeriv](#) R packages.
3. A working C++ compiler that NIMBLE can use from R on your system. There are standard open-source C++ compilers that the R community has already made easy to install. See [Section 4.2](#) for instructions. You don't need to know anything about C++ to use NIMBLE. This must be done before installing NIMBLE.

NIMBLE also uses a couple of C++ libraries that you don't need to install, as they will already be on your system or are provided by NIMBLE.

1. The [Eigen](#) C++ library for linear algebra. This comes with NIMBLE, or you can use your own copy.
2. The BLAS and LAPACK numerical libraries. These come with R, but see [Section 4.5.3](#) for how to use a faster version of the BLAS.

Most fairly recent versions of these requirements should work.

4.2 Installing a C++ compiler for NIMBLE to use

NIMBLE needs a C++ compiler and the standard utility *make* in order to generate and compile C++ for models and algorithms.¹

4.2.1 MacOS

On MacOS, you should install the *Xcode* command line tools. [Installing just the command line tools](#), which are available as a smaller installation than the full *XCode* development environment,

¹This differs from most packages, which might need a C++ compiler only when the package is built. If you normally install R packages using `install.packages` on Windows or MacOS, the package arrives already built to your system.

should be sufficient. Alternatively, *XCode* is freely available from the [Apple developer site](#) and the [App Store](#).

In the somewhat unlikely event you want to install from the source package rather than the CRAN binary package, the easiest approach is to use the source package provided at [R-nimble.org](#). If you do want to install from the source package provided by CRAN, you'll need to install the GNU Fortran compiler package following [these instructions from CRAN](#).

4.2.2 Linux

On Linux, you can install the GNU compiler suite (*gcc/g++*). You can use the package manager to install pre-built binaries. On Ubuntu, the following command will install or update *make*, *gcc* and *libc*.

```
sudo apt-get install build-essential
```

Older versions of *gcc* (less than version 6.3.0) may not work with NIMBLE versions 0.13.2 and newer.

4.2.3 Windows

On Windows, you should download and install *Rtools.exe* available from <https://cran.r-project.org/bin/windows/Rtools/>. Select the appropriate executable corresponding to your version of R (and follow the urge to update your version of R if you notice it is not the most recent).

Important: You must set the path so that the installer will add the location of the C++ compiler and related tools to your system's PATH variable, ensuring that R can find them. For R version 4.0 or greater (*Rtools42* or *Rtools40*) be sure to follow the instructions in the section [Putting Rtools on the PATH](#). For R version 3.6.3 or lesser (i.e., using *Rtools35.exe*) make sure to check the box labelled "Add rtools to system PATH" (page 5 of the installation pages) (it should be checked by default). After you click 'Next', you will get a page with a window for customizing the new PATH variable. You shouldn't need to do anything there, so you can simply click 'Next' again.

4.3 Installing the NIMBLE package

Since NIMBLE is an R package, you can install it in the usual way, via `install.packages("nimble")` in R or using the R CMD INSTALL method if you download the package source directly.

NIMBLE can also be obtained from the [NIMBLE website](#). To install from our website, please see our [Download page](#) for the specific invocation of `install.packages`.

To test that the installation worked and can use NIMBLE's compilation system, you can run this small test code, which should run without error and produce a value for *y* in the model.

```
code <- nimbleCode({ y ~ dnorm(0,1) })
model <- nimbleModel(code)
comp_model <- compileNimble(model)
comp_model$simulate('y')
comp_model$y
```


4.4 Troubleshooting installation problems

We have tested the installation on the three commonly used platforms – MacOS, Linux, Windows². We don't anticipate problems with installation, but we want to hear about any and help resolve them.

The following are some troubleshooting tips that have helped users in some situations.

For Windows:

- Be sure you have set `PATH` when installing Rtools (see instructions above in the Windows installation section). Alternatively, one can set the `PATH` manually using syntax similar to this (after changing `C:\\Rtools\\bin`; `C:\\Rtools\\mingw_64\\bin` to be appropriate for your system):

```
path <- Sys.getenv('PATH')
newPath <- paste("C:\\Rtools\\bin;C:\\Rtools\\mingw_64\\bin;",
                path, sep = "")
Sys.setenv(PATH = newPath)
```

- Be sure the `Rtools.exe` version matches the R version.
- Try re-installing Rtools followed by re-installing NIMBLE.
- If you're using `Rtools42` or `Rtools40`, make sure you are using a newer version of RStudio (at least 1.2.5042).
- If there are filesystem or permissions issues, it is possible to install NIMBLE in a local directory using the `lib` argument to `install.packages`.
- In the past we've heard reports from Windows users of problems when their filesystem involved a space in a directory name in the path to `RHOME`. We think this problem has been resolved.
- We've also heard reports from Windows users of problems when R is installed on a network drive. A work-around is to install locally in a directory on a drive physically on the machine.

For MacOS:

- Newly installed Xcode/command line tools may need to be started once manually to provide a one-time permission before they will work from NIMBLE.
- Upgrading your MacOS version may result in an error while installing or using NIMBLE. Please try running the following in the Terminal to reinstall Xcode/command line tools:

```
xcode-select --install
```

- If multiple C++ compilers are present on a system, be sure the `PATH` will find the right one.

All operating systems:

- To determine if the problem is with the availability of the C++ compiler or with NIMBLE itself, you can try to install the `Rcpp` package. If you can install `Rcpp` and successfully run this command in R: `Rcpp::evalCpp("2 + 2")` and get 4, that suggests the problem is with NIMBLE. If not, then the problem is probably with the C++ compiler or its use from R and not with NIMBLE itself.

²We've tested NIMBLE on Windows 7, 8 and 10.

- If problems arise from generating and compiling C++ files from the default location in R's `tempdir()`, one can use the `dirName` argument to `compileNimble` to put such files elsewhere, such as in a local working directory.

If those suggestions don't help, please post about installation problems to the [nimble-users Google group](#) or email nimble.stats@gmail.com.

4.5 Customizing your installation

For most installations, you can ignore low-level details. However, there are some options that some users may want to utilize.

4.5.1 Using your own copy of Eigen

NIMBLE uses the Eigen C++ template library for linear algebra. Version 3.4.0 of Eigen is included in the NIMBLE package and that version will be used unless the package's configuration script finds another version on the machine. This works well, and the following is only relevant if you want to use a different (e.g., newer) version.

The configuration script looks in the standard include directories, e.g. `/usr/include` and `/usr/local/include` for the header file `Eigen/Dense`. You can specify a particular location in either of two ways:

1. Set the environment variable `EIGEN_DIR` before installing the R package, for example: `export EIGEN_DIR=/usr/include/eigen3` in the bash shell.
2. Use

```
R CMD INSTALL --configure-args='--with-eigen=/path/to/eigen' nimble_VERSION.tar.gz
```

or

```
install.packages("nimble", configure.args = "--with-eigen=/path/to/eigen")
```

In these cases, the directory should be the full path to the directory that contains the Eigen directory, e.g., `/usr/include/eigen3`. It is not the full path to the Eigen directory itself, i.e., NOT `/usr/include/eigen3/Eigen`.

4.5.2 Using libnimble

NIMBLE generates specialized C++ code for user-specified models and `nimbleFunctions`. This code uses some NIMBLE C++ library classes and functions. By default, on Linux the library code is compiled once as a linkable library - `libnimble.so`. This single instance of the library is then linked with the code for each generated model. In contrast, the default for Windows and MacOS is to compile the library code as a static library - `libnimble.a` - that is compiled into each model's and each algorithm's own dynamically loadable library (DLL). This does repeat the same code across models and so occupies more memory. There may be a marginal speed advantage. If one would like to enable the linkable library in place of the static library (do this only on MacOS and other UNIX variants and not on Windows), one can install the source package with the configuration argument `--enable-dylib` set to true. First obtain the NIMBLE source package (which will have the extension `.tar.gz` from [our website](#) and then install as follows, replacing `VERSION` with the appropriate version number:

```
R CMD INSTALL --configure-args='--enable-dylib=true' nimble_VERSION.tar.gz
```

4.5.3 BLAS and LAPACK

NIMBLE also uses BLAS and LAPACK for some of its linear algebra (in particular calculating density values and generating random samples from multivariate distributions). NIMBLE will use the same BLAS and LAPACK installed on your system that R uses. Note that a fast (and where appropriate, threaded) BLAS can greatly increase the speed of linear algebra calculations. See Section A.3.1 of the [R Installation and Administration manual](#) available on CRAN for more details on providing a fast BLAS for your R installation.

4.5.4 Customizing compilation of the NIMBLE-generated C++

For each model or `nimbleFunction`, NIMBLE can generate and compile C++. To compile generated C++, NIMBLE makes system calls starting with `R CMD SHLIB` and therefore uses the regular R configuration in `${R_HOME}/etc/${R_ARCH}/Makeconf`. NIMBLE places a `Makevars` file in the directory in which the code is generated, and `R CMD SHLIB` uses this file as usual.

In all but specialized cases, the general compilation mechanism will suffice. However, one can customize this. One can specify the location of an alternative `Makevars` (or `Makevars.win`) file to use. Such an alternative file should define the variables `PKG_CPPFLAGS` and `PKG_LIBS`. These should contain, respectively, the pre-processor flag to locate the NIMBLE include directory, and the necessary libraries to link against (and their location as necessary), e.g., *Rlapack* and *Rblas* on Windows, and *libnimble*. Advanced users can also change their default compilers by editing the `Makevars` file, see Section 1.2.1 of the [Writing R Extensions manual](#) available on CRAN.

Use of this file allows users to specify additional compilation and linking flags. See the Writing R Extensions manual for more details of how this can be used and what it can contain.

Part II

Models in NIMBLE

Chapter 5

Writing models in NIMBLE’s dialect of BUGS

Models in NIMBLE are written using a variation on the BUGS language. From BUGS code, NIMBLE creates a model object. This chapter describes NIMBLE’s version of BUGS. The next chapter explains how to build and manipulate model objects.

5.1 Comparison to BUGS dialects supported by WinBUGS, OpenBUGS and JAGS

Many users will come to NIMBLE with some familiarity with WinBUGS, OpenBUGS, or JAGS, so we start by summarizing how NIMBLE is similar to and different from those before documenting NIMBLE’s version of BUGS more completely. In general, NIMBLE aims to be compatible with the original BUGS language and also JAGS’ version. However, at this point, there are some features not supported by NIMBLE, and there are some extensions that are planned but not implemented.

5.1.1 Supported features of BUGS and JAGS

1. Stochastic and deterministic¹ node declarations.
2. Most univariate and multivariate distributions.
3. Link functions.
4. Most mathematical functions.
5. ‘for’ loops for iterative declarations.
6. Arrays of nodes up to 4 dimensions.
7. Truncation and censoring as in JAGS using the `T()` notation and `dinterval`.

5.1.2 NIMBLE’s Extensions to BUGS and JAGS

NIMBLE extends the BUGS language in the following ways:

1. User-defined functions and distributions – written as `nimbleFunctions` – can be used in model code. See Chapter 13.

¹NIMBLE calls non-stochastic nodes ‘deterministic’, whereas BUGS calls them ‘logical’. NIMBLE uses ‘logical’ in the way R does, to refer to boolean (TRUE/FALSE) variables.

2. Multiple parameterizations for distributions, similar to those in R, can be used.
3. Named parameters for distributions and functions, similar to R function calls, can be used.
4. Linear algebra, including for vectorized calculations of simple algebra, can be used in deterministic declarations.
5. Distribution parameters can be expressions, as in JAGS but not in WinBUGS. Caveat: parameters to *multivariate* distributions (e.g., `dmnorm`) cannot be expressions (but an expression can be defined in a separate deterministic expression and the resulting variable then used).
6. Alternative models can be defined from the same model code by using if-then-else statements that are evaluated when the model is defined.
7. More flexible indexing of vector nodes within larger variables is allowed. For example one can place a multivariate normal vector arbitrarily within a higher-dimensional object, not just in the last index.
8. More general constraints can be declared using `dconstraint`, which extends the concept of JAGS' `dinterval`.
9. Link functions can be used in stochastic, as well as deterministic, declarations.²
10. Data values can be reset, and which parts of a model are flagged as data can be changed, allowing one model to be used for different data sets without rebuilding the model each time.
11. One can use stochastic/dynamic indexes, i.e., indexes can be other nodes or functions of other nodes. For a given dimension of a node being indexed, if the index is not constant, it must be a scalar value. So expressions such as `mu[k[i], 3]` or `mu[k[i], 1:3]` or `mu[k[i], j[i]]` are allowed, but not `mu[k[i] : (k[i]+1)]`³ Nested dynamic indexes such as `mu[k[j[i]]]` are also allowed.

5.1.3 Not-supported features of BUGS and JAGS

The following are not supported.

1. The appearance of the same node on the left-hand side of both a `<-` and a `~` declaration (used in WinBUGS for data assignment for the value of a stochastic node).
2. Multivariate nodes must appear with brackets, even if they are empty. E.g., `x` cannot be multivariate but `x[]` or `x[2:5]` can be.
3. NIMBLE generally determines the dimensionality and sizes of variables from the BUGS code. However, when a variable appears with blank indices, such as in `x.sum <- sum(x[])`, and if the dimensions of the variable are not clearly defined in other declarations, NIMBLE currently requires that the dimensions of `x` be provided when the model object is created (via `nimbleModel`).
4. Use of non-sequentially-increasing indexes in the definition of a for loop. JAGS allows syntax such as `for(i in expr)` when `expr` evaluates to an integer vector. In NIMBLE one can use the work-around of defining a constant vector, say `k`, and using `k[i]` in the body of the for loop.
5. Use of non-sequentially-increasing indexes to subset variables. JAGS allows syntax such as `sum(mu[expr])` when `expr` evaluates to an integer vector. In NIMBLE one can use the work-around of defining a `nimbleFunction` that takes `mu` and the index vector as arguments and does the calculation of interest.

²But beware of the possibility of needing to set values for 'lifted' nodes created by NIMBLE.

³In some cases, one can write a `nimbleFunction` to achieve the desired result, such as replacing `sum(mu[start[i]:end[i]])` with a `nimbleFunction` that takes `mu`, `start[i]`, and `end[i]` as arguments and does the needed summation.

5.2 Writing models

Here we introduce NIMBLE's version of BUGS. The WinBUGS, OpenBUGS and JAGS manuals are also useful resources for writing BUGS models, including many examples.

5.2.1 Declaring stochastic and deterministic nodes

BUGS is a declarative language for graphical (or hierarchical) models. Most programming languages are imperative, which means a series of commands will be executed in the order they are written. A declarative language like BUGS is more like building a machine before using it. Each line declares that a component should be plugged into the machine, but it doesn't matter in what order they are declared as long as all the right components are plugged in by the end of the code.

The machine in this case is a graphical model⁴. A *node* (sometimes called a *vertex*) holds one value, which may be a scalar or a vector. *Edges* define the relationships between nodes. A huge variety of statistical models can be thought of as graphs.

Here is the code to define and create a simple linear regression model with four observations.

```
library(nimble)
mc <- nimbleCode({
  intercept ~ dnorm(0, sd = 1000)
  slope ~ dnorm(0, sd = 1000)
  sigma ~ dunif(0, 100)
  for(i in 1:4) {
    predicted.y[i] <- intercept + slope * x[i]
    y[i] ~ dnorm(predicted.y[i], sd = sigma)
  }
})

model <- nimbleModel(mc, data = list(y = rnorm(4)))

library(igraph)

layout <- matrix(ncol = 2, byrow = TRUE,
  # These seem to be rescaled to fit in the plot area,
  # so I'll just use 0-100 as the scale
  data = c(33, 100,
           66, 100,
           50, 0, # first three are parameters
           15, 50, 35, 50, 55, 50, 75, 50, # x's
           20, 75, 40, 75, 60, 75, 80, 75, # predicted.y's
           25, 25, 45, 25, 65, 25, 85, 25) # y's
  )

sizes <- c(45, 30, 30,
          rep(20, 4),
          rep(50, 4),
```

⁴Technically, a *directed acyclic graph*

```

        rep(20, 4))

edge.color <- "black"
# c(
# rep("green", 8),
# rep("red", 4),
# rep("blue", 4),
# rep("purple", 4))
stoch.color <- "deepskyblue2"
det.color <- "orchid3"
rhs.color <- "gray73"
fill.color <- c(
  rep(stoch.color, 3),
  rep(rhs.color, 4),
  rep(det.color, 4),
  rep(stoch.color, 4)
)

plot(model$graph, vertex.shape = "crectangle",
      vertex.size = sizes,
      vertex.size2 = 20,
      layout = layout,
      vertex.label.cex = 3.0,
      vertex.color = fill.color,
      edge.width = 3,
      asp = 0.5,
      edge.color = edge.color)

```

The graph representing the model is shown in Figure 5.1. Each observation, `y[i]`, is a node whose edges say that it follows a normal distribution depending on a predicted value, `predicted.y[i]`, and standard deviation, `sigma`, which are each nodes. Each predicted value is a node whose edges say how it is calculated from `slope`, `intercept`, and one value of an explanatory variable, `x[i]`, which are each nodes.

This graph is created from the following BUGS code:

```

{
  intercept ~ dnorm(0, sd = 1000)
  slope ~ dnorm(0, sd = 1000)
  sigma ~ dunif(0, 100)
  for(i in 1:4) {
    predicted.y[i] <- intercept + slope * x[i]
    y[i] ~ dnorm(predicted.y[i], sd = sigma)
  }
}

```

In this code, stochastic relationships are declared with ‘`~`’ and deterministic relationships are declared with ‘`<-`’. For example, each `y[i]` follows a normal distribution with mean `predicted.y[i]`

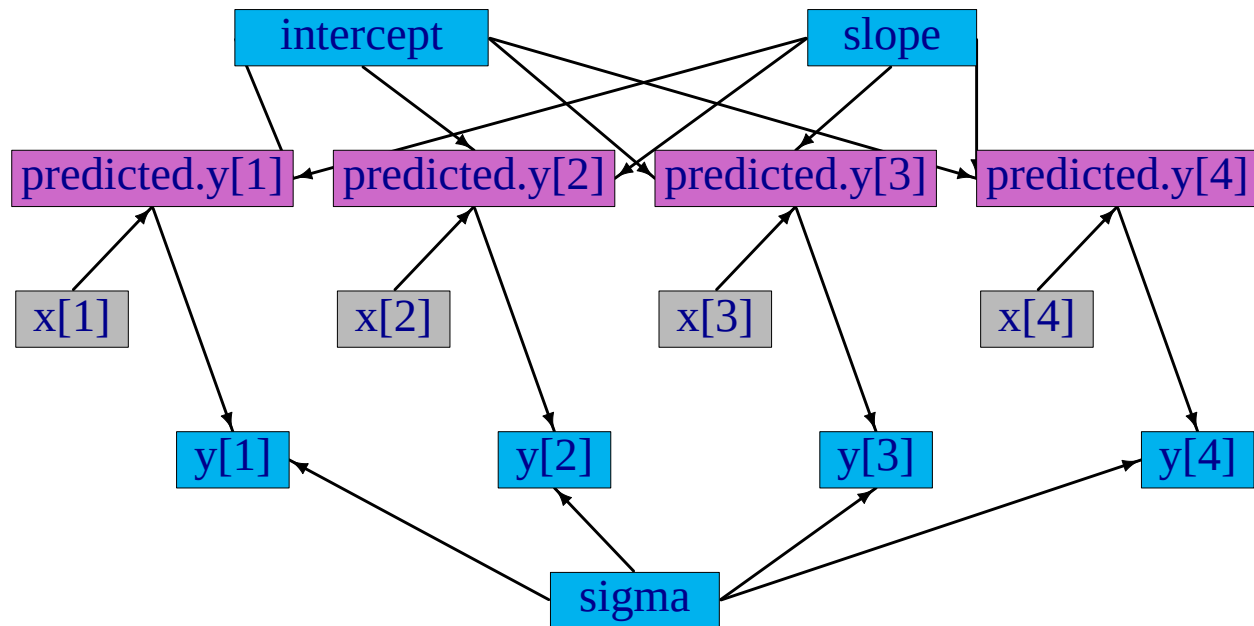


Figure 5.1: Graph of a linear regression model

and standard deviation `sigma`. Each `predicted.y[i]` is the result of `intercept + slope * x[i]`. The for-loop yields the equivalent of writing four lines of code, each with a different value of `i`. It does not matter in what order the nodes are declared. Imagine that each line of code draws part of Figure 5.1, and all that matters is that the everything gets drawn in the end. Available distributions, default and alternative parameterizations, and functions are listed in Section 5.2.4.

An equivalent graph can be created by this BUGS code:

```

{
  intercept ~ dnorm(0, sd = 1000)
  slope ~ dnorm(0, sd = 1000)
  sigma ~ dunif(0, 100)
  for(i in 1:4) {
    y[i] ~ dnorm(intercept + slope * x[i], sd = sigma)
  }
}

```

In this case, the `predicted.y[i]` nodes in Figure 5.1 will be created automatically by NIMBLE and will have a different name, generated by NIMBLE.

5.2.2 More kinds of BUGS declarations

Here are some examples of valid lines of BUGS code. This code does not describe a sensible or complete model, and it includes some arbitrary indices (e.g. `mvx[8:10, i]`) to illustrate flexibility. Instead the purpose of each line is to illustrate a feature of NIMBLE's version of BUGS.

```

{
  # 1. normal distribution with BUGS parameter order
  x ~ dnorm(a + b * c, tau)
}

```

```

# 2. normal distribution with a named parameter
y ~ dnorm(a + b * c, sd = sigma)
# 3. For-loop and nested indexing
for(i in 1:N) {
  for(j in 1:M[i]) {
    z[i,j] ~ dexp(r[ blockID[i] ])
  }
}
# 4. multivariate distribution with arbitrary indexing
for(i in 1:3)
  mvx[8:10, i] ~ dmnorm(mvMean[3:5], cov = mvCov[1:3, 1:3, i])
# 5. User-provided distribution
w ~ dMyDistribution(hello = x, world = y)
# 6. Simple deterministic node
d1 <- a + b
# 7. Vector deterministic node with matrix multiplication
d2[] <- A[ , ] %*% mvMean[1:5]
# 8. Deterministic node with user-provided function
d3 <- foo(x, hooray = y)
}

```

When a variable appears only on the right-hand side, it can be provided via **constants** (in which case it can never be changed) or via **data** or **inits**, as discussed in Chapter 6.

Notes on the comment-numbered lines are:

1. **x** follows a normal distribution with mean **a + b*c** and precision **tau** (default BUGS second parameter for **dnorm**).
2. **y** follows a normal distribution with the same mean as **x** but a named standard deviation parameter instead of a precision parameter (**sd = 1/sqrt(precision)**).
3. **z[i, j]** follows an exponential distribution with parameter **r[blockID[i]]**. This shows how for-loops can be used for indexing of variables containing multiple nodes. Variables that define for-loop indices (**N** and **M**) must also be provided as constants.
4. The arbitrary block **mvx[8:10, i]** follows a multivariate normal distribution, with a named covariance matrix instead of BUGS' default of a precision matrix. As in R, curly braces for for-loop contents are only needed if there is more than one line.
5. **w** follows a user-defined distribution. See Chapter 13.
6. **d1** is a scalar deterministic node that, when calculated, will be set to **a + b**.
7. **d2** is a vector deterministic node using matrix multiplication in R's syntax.
8. **d3** is a deterministic node using a user-provided function. See Chapter 13.

5.2.2.1 More about indexing

Examples of allowed indexing include:

- **x[i]** # a single index
- **x[i:j]** # a range of indices
- **x[i:j,k:1]** # multiple single indices or ranges for higher-dimensional arrays

- `x[i:j, ]` # blank indices indicating the full range
- `x[3*i+7]` # computed indices
- `x[(3*i):(5*i+1)]` # computed lower and upper ends of an index range
- `x[k[i]+1]` # a dynamic (and computed) index
- `x[k[j[i]]]` # nested dynamic indexes
- `x[k[i], 1:3]` # nested indexing of rows or columns

NIMBLE does not allow multivariate nodes to be used without square brackets, which is an incompatibility with JAGS. Therefore a statement like `xbar <- mean(x)` in JAGS must be converted to `xbar <- mean(x[])` (if `x` is a vector) or `xbar <- mean(x[,])` (if `x` is a matrix) for NIMBLE⁵. Section 6.1.1.5 discusses how to provide NIMBLE with dimensions of `x` when needed.

One cannot provide a vector of indices that are not constant. For example, `x[start[i]:end[i]]` is allowed only if `start` and `end` are provided as constants. Also, one cannot provide a vector, or expression evaluating to a vector (apart from use of `:`) as an index. For example, `x[inds]` is not allowed. Often one can write a `nimbleFunction` to achieve the desired result, such as defining a `nimbleFunction` that takes `x`, `start[i]`, and `end[i]` or takes `x` and `inds` as arguments and does the subsetting (possibly in combination with some other calculation). Note that `x[L:U]` and `x[inds]` are allowed in `nimbleFunction` code.

Generally NIMBLE supports R-like linear algebra expressions and attempts to follow the same rules as R about dimensions (although in some cases this is not possible). For example, `x[1:3] %*% y[1:3]` converts `x[1:3]` into a row vector and thus computes the inner product, which is returned as a 1×1 matrix (use `inprod` to get it as a scalar, which is typically easier). Like in R, a scalar index will result in dropping a dimension unless the argument `drop=FALSE` is provided. For example, `mymatrix[i, 1:3]` will be a vector of length 3, but `mymatrix[i, 1:3, drop=FALSE]` will be a 1×3 matrix. More about indexing and dimensions is discussed in Section 12.3.2.7.

5.2.3 Vectorized versus scalar declarations

Suppose you need nodes `logY[i]` that should be the log of the corresponding `Y[i]`, say for `i` from 1 to 10. Conventionally this would be created with a for loop:

```
{
  for(i in 1:10) {
    logY[i] <- log(Y[i])
  }
}
```

Since NIMBLE supports R-like algebraic expressions, an alternative in NIMBLE's dialect of BUGS is to use a vectorized declaration like this:

```
{
  logY[1:10] <- log(Y[1:10])
}
```

There is an important difference between the models that are created by the above two methods. The first creates 10 scalar nodes, `logY[1]`, ..., `logY[10]`. The second creates one vector node,

⁵In `nimbleFunctions`, as explained in later chapters, square brackets with blank indices are not necessary for multivariate objects.

`logY[1:10]`. If each `logY[i]` is used separately by an algorithm, it may be more efficient computationally if they are declared as scalars. If they are all used together, it will often make sense to declare them as a vector.

5.2.4 Available distributions

5.2.4.1 Distributions

NIMBLE supports most of the distributions allowed in BUGS and JAGS. Table 5.1 lists the distributions that are currently supported, with their default parameterizations, which match those of BUGS⁶. NIMBLE also allows one to use alternative parameterizations for a variety of distributions as described next. See Section 13.2 to learn how to write new distributions using `nimbleFunctions`.

Table 5.1: Distributions with their default order of parameters. The value of the random variable is given by x .

Name	Usage	Density	Lower	Upper
Bernoulli	<code>dbern(prob = p)</code> $0 < p < 1$	$p^x(1-p)^{1-x}$	0	1
Beta	<code>dbeta(shape1 = a,</code> <code>shape2 = b)</code> , $a > 0, b > 0$	$\frac{x^{a-1}(1-x)^{b-1}}{\beta(a,b)}$	0	1
Binomial	<code>dbin(prob = p, size = n)</code> $0 < p < 1, n \in \mathbb{N}^*$	$\binom{n}{x} p^x (1-p)^{n-x}$	0	n
CAR (intrinsic)	<code>dcar_normal(adj, weights,</code> <code>num, tau, c, zero_mean)</code>	see chapter 10 for details		
CAR (proper)	<code>dcar_proper(mu, C, adj,</code> <code>num, M, tau, gamma)</code>	see chapter 10 for details		
Categorical	<code>dcat(prob = p)</code> $p \in (\mathbb{R}^+)^N$	$\frac{p_x}{\sum_i p_i}$	1	N
Chi-square	<code>dchisq(df = k)</code> , $k > 0$	$\frac{x^{\frac{k}{2}-1} \exp(-x/2)}{2^{\frac{k}{2}} \Gamma(\frac{k}{2})}$	0	
Double exponential	<code>ddexp(location = μ,</code> <code>rate = τ)</code> , $\tau > 0$	$\frac{\tau}{2} \exp(-\tau x - \mu)$		
Dirichlet	<code>ddirch(alpha = α)</code> $\alpha_j \geq 0$	$\Gamma(\sum_j \alpha_j) \prod_j \frac{x_j^{\alpha_j-1}}{\Gamma(\alpha_j)}$	0	
Exponential	<code>dexp(rate = λ)</code> , $\lambda > 0$	$\lambda \exp(-\lambda x)$	0	
Flat	<code>dflat()</code>	$\propto 1$ (improper)		
Gamma	<code>dgamma(shape = r, rate = λ)</code> $\lambda > 0, r > 0$	$\frac{\lambda^r x^{r-1} \exp(-\lambda x)}{\Gamma(r)}$	0	
Half flat	<code>dhalfflat()</code>	$\propto 1$ (improper)	0	
Inverse gamma	<code>dinvgamma(shape = r, scale = λ)</code> $\lambda > 0, r > 0$	$\frac{\lambda^r x^{-(r+1)} \exp(-\lambda/x)}{\Gamma(r)}$	0	
LKJ Correl’n	<code>dlkj_corr_cholesky(eta = η,</code>	$\prod_{i=2}^p x_{ii}^{p-i+2\eta-2}$		

⁶Note that the same distributions are available for writing `nimbleFunctions`, but in that case the default parameterizations and function names match R’s when possible. Please see Section 12.2.4 for how to use distributions in `nimbleFunctions`.

Name	Usage	Density	Lower	Upper
Cholesky	<code>size = p</code>), $\eta > 0$			
Logistic	<code>dlogis(location = μ, rate = τ)</code> , $\tau > 0$	$\frac{\tau \exp\{(x-\mu)\tau\}}{[1+\exp\{(x-\mu)\tau\}]^2}$		
Log-normal	<code>dlnorm(meanlog = μ, taulog = τ)</code> , $\tau > 0$	$(\frac{\tau}{2\pi})^{\frac{1}{2}} x^{-1} \exp\{-\tau(\log(x) - \mu)^2/2\}$	0	
Multinomial	<code>dmulti(prob = p, size = n)</code> $\sum_j x_j = n$	$n! \prod_j \frac{p_j^{x_j}}{x_j!}$		
Multivariate normal	<code>dmnorm(mean = μ, prec = Λ)</code> Λ positive definite	$(2\pi)^{-\frac{d}{2}} \Lambda ^{\frac{1}{2}} \exp\{-\frac{(x-\mu)^T \Lambda (x-\mu)}{2}\}$		
Multivariate Student t	<code>dmvt(mu = μ, prec = Λ, df = ν)</code> , Λ positive def.	$\frac{\Gamma(\frac{\nu+d}{2})}{\Gamma(\frac{\nu}{2})(\nu\pi)^{d/2}} \Lambda ^{1/2} (1 + \frac{(x-\mu)^T \Lambda (x-\mu)}{\nu})^{-\frac{\nu+d}{2}}$		
Negative binomial	<code>dnegbin(prob = p, size = r)</code> $0 < p \leq 1, r \geq 0$	$\binom{x+r-1}{x} p^r (1-p)^x$	0	
Normal	<code>dnorm(mean = μ, tau = τ)</code> $\tau > 0$	$(\frac{\tau}{2\pi})^{\frac{1}{2}} \exp\{-\tau(x - \mu)^2/2\}$		
Poisson	<code>dpois(lambda = λ)</code> $\lambda > 0$	$\frac{\exp(-\lambda) \lambda^x}{x!}$	0	
Student t	<code>dt(mu = μ, tau = τ, df = k)</code> $\tau > 0, k > 0$	$\frac{\Gamma(\frac{k+1}{2})}{\Gamma(\frac{k}{2})} (\frac{\tau}{k\pi})^{\frac{1}{2}} \left\{1 + \frac{\tau(x-\mu)^2}{k}\right\}^{-\frac{(k+1)}{2}}$		
Uniform	<code>dunif(min = a, max = b)</code> $a < b$	$\frac{1}{b-a}$	a	b
Weibull	<code>dweib(shape = v, lambda = λ)</code> $v > 0, \lambda > 0$	$v \lambda x^{v-1} \exp(-\lambda x^v)$	0	
Wishart	<code>dwish(R = R, df = k)</code> R $p \times p$ pos. def., $k \geq p$	$\frac{ x ^{(k-p-1)/2} R ^{k/2} \exp\{-\text{tr}(Rx)/2\}}{2^{pk/2} \pi^{p(p-1)/4} \prod_{i=1}^p \Gamma((k+1-i)/2)}$		
Inverse Wishart	<code>dinvwish(S = S, df = k)</code> S $p \times p$ pos. def., $k \geq p$	$\frac{ x ^{-(k+p+1)/2} S ^{k/2} \exp\{-\text{tr}(Sx^{-1})/2\}}{2^{pk/2} \pi^{p(p-1)/4} \prod_{i=1}^p \Gamma((k+1-i)/2)}$		

5.2.4.1.1 Improper distributions Note that `dcar_normal`, `dflat` and `dhalfflat` specify improper prior distributions and should only be used when the posterior distribution of the model is known to be proper. Also for these distributions, the density function returns the unnormalized density and the simulation function returns NaN so these distributions are not appropriate for algorithms that need to simulate from the prior or require proper (normalized) densities.

5.2.4.1.2 LKJ distribution for correlation matrices NIMBLE provides the LKJ distribution via `dlkj_corr_cholesky` for correlation matrices, discussed in Section 24 of [Stan Development Team \(2021a\)](#) and based on [Lewandowski et al. \(2009\)](#). This distribution has various advantages (both from a modeling perspective and an MCMC fitting perspective) over other prior distributions for correlation or covariance matrices such as the inverse Wishart distribution.

For computational efficiency, the distribution is on the Cholesky factor of the correlation matrix

rather than the correlation matrix itself. As a result using this distribution in a model involves a bit of care, including efficiently creating a covariance matrix from the correlation matrix and a set of standard deviations.

Here is some example code illustrating how to use the distribution in a model.

```
code <- nimbleCode({
  Ustar[1:p,1:p] ~ dlkj_corr_cholesky(1.3, p)
  U[1:p,1:p] <- uppertri_mult_diag(Ustar[1:p, 1:p], sds[1:p])
  for(i in 1:n)
    y[i, 1:p] ~ dmnorm(mu[1:p], cholesky = U[1:p, 1:p], prec_param = 0)
})
```

Note that we need to take the upper triangular Cholesky factor (`Ustar`) of the correlation matrix and convert it to the upper triangular Cholesky factor (`U`) of a covariance matrix, based on a vector of standard deviations (`sds`), and use that in the parameterization of `dmnorm` that directly (and therefore efficiently) uses the Cholesky of the covariance.

Other approaches are possible, but unless care is taken they are likely to be less computationally efficient than the template above.

The template above uses a user-defined nimbleFunction to efficiently combine the Cholesky of the correlation matrix with a vector of standard deviations to produce the Cholesky of the covariance matrix. That function is given here:

```
uppertri_mult_diag <- nimbleFunction(
  run = function(mat = double(2), vec = double(1)) {
    returnType(double(2))
    p <- length(vec)
    out <- matrix(nrow = p, ncol = p, init = FALSE)
    for(i in 1:p)
      out[, i] <- mat[, i] * vec[i]
    return(out)
  })
```

Note that NIMBLE ignores the lower triangle of the Cholesky factor of the correlation matrix; the values are taken to be zero. It also ignores the `[1,1]` element, which is taken to be one. Any initial values for these elements will be ignored and simply repeated in the MCMC samples.

Assuming the `[1,1]` element is one and the lower triangle is filled with zeroes in the MCMC samples, one can reconstruct a sample, here for the `sth` sample, of the correlation matrix as follows using the matrix of MCMC samples:

```
cols <- grep("^Ustar", colnames(samples))
Ustar_sample <- matrix(samples[s, cols], p, p)
corrMatrix_sample <- crossprod(Ustar_sample) ## i.e., t(Ustar_sample) %*% Ustar_sample
```

5.2.4.2 Alternative parameterizations for distributions

NIMBLE allows one to specify distributions in model code using a variety of parameterizations, including the BUGS parameterizations. Available parameterizations are listed in Table 5.2. To

understand how NIMBLE handles alternative parameterizations, it is useful to distinguish three cases, using the `gamma` distribution as an example:

1. A *canonical* parameterization is used directly for computations⁷. For `gamma`, this is (shape, scale).
2. The BUGS parameterization is the one defined in the original BUGS language. In general this is the parameterization for which conjugate MCMC samplers can be executed most efficiently. For `dgamma`, this is (shape, rate).
3. An *alternative* parameterization is one that must be converted into the *canonical* parameterization. For `dgamma`, NIMBLE provides both (shape, rate) and (mean, sd) parameterization and creates nodes to calculate (shape, scale) from either (shape, rate) or (mean, sd). In the case of `dgamma`, the BUGS parameterization is also an *alternative* parameterization.

Since NIMBLE provides compatibility with existing BUGS and JAGS code, the order of parameters places the BUGS parameterization first. For example, the order of parameters for `dgamma` is `dgamma(shape, rate, scale, mean, sd)`. Like R, if parameter names are not given, they are taken in order, so that (shape, rate) is the default. This happens to match R's order of parameters, but it need not. If names are given, they can be given in any order. NIMBLE knows that rate is an alternative to scale and that (mean, sd) are an alternative to (shape, scale or rate).

Table 5.2: Distribution parameterizations allowed in NIMBLE. The first column indicates the supported parameterizations for distributions given in Table 5.1. The second column indicates the relationship to the *canonical* parameterization used in NIMBLE.

Parameterization	NIMBLE re-parameterization
<code>dbern(prob)</code>	<code>dbin(size = 1, prob)</code>
<code>dbeta(shape1, shape2)</code>	canonical
<code>dbeta(mean, sd)</code>	<code>dbeta(shape1 = mean^2 * (1-mean) / sd^2 - mean, shape2 = mean * (1 - mean)^2 / sd^2 + mean - 1)</code>
<code>dbin(prob, size)</code>	canonical
<code>dcat(prob)</code>	canonical
<code>dchisq(df)</code>	canonical
<code>ddexp(location, scale)</code>	canonical
<code>ddexp(location, rate)</code>	<code>ddexp(location, scale = 1 / rate)</code>
<code>ddexp(location, var)</code>	<code>ddexp(location, scale = sqrt(var / 2))</code>
<code>ddirch(alpha)</code>	canonical
<code>dexp(rate)</code>	canonical
<code>dexp(scale)</code>	<code>dexp(rate = 1/scale)</code>
<code>dgamma(shape, scale)</code>	canonical
<code>dgamma(shape, rate)</code>	<code>dgamma(shape, scale = 1 / rate)</code>
<code>dgamma(mean, sd)</code>	<code>dgamma(shape = mean^2/sd^2, scale = sd^2/mean)</code>
<code>dinvgamma(shape, rate)</code>	canonical
<code>dinvgamma(shape, scale)</code>	<code>dgamma(shape, rate = 1 / scale)</code>
<code>dlkj_corr_cholesky(eta)</code>	canonical

⁷Usually this is the parameterization in the `Rmath` header of R's C implementation of distributions.

Parameterization	NIMBLE re-parameterization
dlogis(location, scale)	canonical
dlogis(location, rate)	dlogis(location, scale = 1 / rate)
dlnorm(meanlog, sdlog)	canonical
dlnorm(meanlog, tauolog)	dlnorm(meanlog, sdlog = 1 / sqrt(tauolog))
dlnorm(meanlog, varlog)	dlnorm(meanlog, sdlog = sqrt(varlog))
dmulti(prob, size)	canonical
dmnorm(mean, cholesky, ...prec_param=1)	canonical (precision)
dmnorm(mean, cholesky, ...prec_param=0)	canonical (covariance)
dmnorm(mean, prec)	dmnorm(mean, cholesky = chol(prec), prec_param=1)
dmnorm(mean, cov)	dmnorm(mean, cholesky = chol(cov), prec_param=0)
dmvt(mu, cholesky, df, ...prec_param=1)	canonical (precision/inverse scale)
dmvt(mu, cholesky, df, ...prec_param=0)	canonical (scale)
dmvt(mu, prec, df)	dmvt(mu, cholesky = chol(prec), df, prec_param=1)
dmvt(mu, scale, df)	dmvt(mu, cholesky = chol(scale), df, prec_param=0)
dlkj_corr_cholesky(...shape, size)	canonical
dnegbin(prob, size)	canonical
dnorm(mean, sd)	canonical
dnorm(mean, tau)	dnorm(mean, sd = 1 / sqrt(tau))
dnorm(mean, var)	dnorm(mean, sd = sqrt(var))
dpois(lambda)	canonical
dt(mu, sigma, df)	canonical
dt(mu, tau, df)	dt(mu, sigma = 1 / sqrt(tau), df)
dt(mu, sigma2, df)	dt(mu, sigma = sqrt(sigma2), df)
dunif(min, max)	canonical
dweib(shape, scale)	canonical
dweib(shape, rate)	dweib(shape, scale = 1 / rate)
dweib(shape, lambda)	dweib(shape, scale = lambda^(- 1 / shape))
dwish(cholesky, df, ...scale_param=1)	canonical (scale)
dwish(cholesky, df, ...scale_param=0)	canonical (inverse scale)
dwish(R, df)	dwish(cholesky = chol(R), df, scale_param = 0)
dwish(S, df)	dwish(cholesky = chol(S), df, scale_param = 1)
dinvwish(cholesky, df, ...scale_param=1)	canonical (scale)
dinvwish(cholesky, df, ...scale_param=0)	canonical (inverse scale)
dinvwish(R, df)	dinvwish(cholesky = chol(R), df, scale_param = 0)
dinvwish(S, df)	dinvwish(cholesky = chol(S), df, scale_param = 1)

Note that for multivariate normal, multivariate t, Wishart, and Inverse Wishart, the canonical parameterization uses the Cholesky decomposition of one of the precision/inverse scale or covariance/scale matrix. For example, for the multivariate normal, if `prec_param=TRUE`, the `cholesky` argument is treated as the Cholesky decomposition of a precision matrix. Otherwise it is treated as the Cholesky decomposition of a covariance matrix.

In addition, NIMBLE supports alternative distribution names, known as aliases, as in JAGS, as specified in Table 5.3.

Table 5.3: Distributions with alternative names (aliases)

Distribution	Canonical name	Alias
Binomial	dbin	dbinom
Chi-square	dchisq	dchisqr
Double exponential	ddexp	dlaplace
Dirichlet	ddirch	ddirich
Multinomial	dmulti	dmultinom
Negative binomial	dnegbin	dnbinom
Weibull	dweib	dweibull
Wishart	dwish	dwishart

We plan to, but do not currently, include the following distributions as part of core NIMBLE: beta-binomial, Dirichlet-multinomial, F, Pareto, or forms of the multivariate t other than the standard one provided.

5.2.5 Available BUGS language functions

Tables 5.4-5.5 show the available operators and functions. Support for more general R expressions is covered in Chapter 12 about programming with `nimbleFunctions`.

For the most part NIMBLE supports the functions used in BUGS and JAGS, with exceptions indicated in the table. Additional functions provided by NIMBLE are also listed. Note that we provide distribution functions for use in calculations, namely the ‘p’, ‘q’, and ‘d’ functions. See Section 12.2.4 for details on the syntax for using distribution functions as functions in deterministic calculations, as only some parameterizations are allowed and the names of some distributions differ from those used to define stochastic nodes in a model.

Table 5.4: Functions operating on scalars, many of which can operate on each element (component-wise) of vectors and matrices. Status column indicates if the function is currently provided in NIMBLE. Vector input column indicates if the function can take a vector as an argument (i.e., if the function is vectorized).

Usage	Description	Comments	Status	Vector input
<code>x y</code> , <code>x & y</code>	logical OR () and AND(&)		yes	yes
<code>!x</code>	logical not		yes	yes

Usage	Description	Comments	Status	Vector input
$x > y$, $x \geq y$	greater than (and or equal to)		yes	yes
$x < y$, $x \leq y$	less than (and or equal to)		yes	yes
$x \neq y$, $x == y$	(not) equals		yes	yes
$x + y$, $x - y$, $x * y$	component-wise operators	mix of scalar and vector	yes	yes
x / y	component-wise division	vector x and scalar y	yes	yes
x^y , $\text{pow}(x, y)$	power	x^y ; vector x , scalar y	yes	yes
$x \% y$	modulo (remainder)		yes	no
$\text{min}(x1, x2)$, $\text{max}(x1, x2)$	min. (max.) of two scalars		yes	See <code>pmin</code> , <code>pmax</code>
$\text{exp}(x)$	exponential		yes	yes
$\text{log}(x)$	natural logarithm		yes	yes
$\text{sqrt}(x)$	square root		yes	yes
$\text{abs}(x)$	absolute value		yes	yes
$\text{step}(x)$	step function at 0	0 if $x < 0$, 1 if $x \geq 0$	yes	yes
$\text{equals}(x, y)$	equality of two scalars	1 if $x == y$, 0 if $x \neq y$	yes	yes
$\text{cube}(x)$	third power	x^3	yes	yes
$\text{sin}(x)$, $\text{cos}(x)$, $\text{tan}(x)$	trigonometric functions		yes	yes
$\text{asin}(x)$, $\text{acos}(x)$, $\text{atan}(x)$	inverse trigonometric functions		yes	yes
$\text{asinh}(x)$, $\text{acosh}(x)$, $\text{atanh}(x)$	inv. hyperbolic trig. functions		yes	yes
$\text{logit}(x)$	logit	$\log(x/(1-x))$	yes	yes
$\text{ilogit}(x)$, $\text{expit}(x)$	inverse logit	$\exp(x)/(1+\exp(x))$	yes	yes
$\text{probit}(x)$	probit (Gaussian quantile)	$\Phi^{-1}(x)$	yes	yes
$\text{iprobit}(x)$, $\text{phi}(x)$	inverse probit (Gaussian CDF)	$\Phi(x)$	yes	yes
$\text{cloglog}(x)$	complementary log log	$\log(-\log(1-x))$	yes	yes
$\text{icloglog}(x)$	inverse complementary log log	$1 - \exp(-\exp(x))$	yes	yes
$\text{ceiling}(x)$	ceiling function	$\lceil x \rceil$	yes	yes
$\text{floor}(x)$	floor function	$\lfloor x \rfloor$	yes	yes
$\text{round}(x)$	round to integer		yes	yes
$\text{trunc}(x)$	truncation to integer		yes	yes
$\text{lgamma}(x)$, $\text{loggam}(x)$	log gamma function	$\log \Gamma(x)$	yes	yes
$\text{besselK}(x, \text{nu},$ $\dots \text{expon.scaled})$	modified bessel function of the second kind	returns vector even if x a matrix/array	yes	yes
$\text{log1p}(x)$	log of $1 + x$	$\log(1 + x)$	yes	yes
$\text{lfactorial}(x)$, $\text{logfact}(x)$	log factorial	$\log x!$	yes	yes
$\text{qDIST}(x, \text{PARAMS})$	“q” distribution functions	canonical parameterization	yes	yes
$\text{pDIST}(x, \text{PARAMS})$	“p” distribution functions	canonical parameterization	yes	yes
$\text{rDIST}(x, \text{PARAMS})$	“r” distribution functions	canonical parameterization	yes	yes
$\text{dDIST}(x, \text{PARAMS})$	“d” distribution functions	canonical parameterization	yes	yes
$\text{sort}(x)$			no	
$\text{rank}(x, s)$			no	
$\text{ranked}(x, s)$			no	

order(x)

no

Table 5.5: Functions operating on vectors and matrices. Status column indicates if the function is currently provided in NIMBLE.

Usage	Description	Comments	Status
<code>inverse(x)</code>	matrix inverse	x symmetric, positive def.	yes
<code>chol(x)</code>	matrix Cholesky factorization	x symmetric, positive def. returns upper triang. matrix	yes
<code>t(x)</code>	matrix transpose	$x^\top$	yes
<code>x%*%y</code>	matrix multiply	xy ; x, y conformant	yes
<code>inprod(x, y)</code>	dot product	$x^\top y$; x and y vectors	yes
<code>solve(x, y)</code>	solve system of equations	$x^{-1}y$; y matrix or vector	yes
<code>forwardsolve(x, y)</code>	solve lower-triangular system of equations	$x^{-1}y$; x lower-triangular	yes
<code>backsolve(x, y)</code>	solve upper-triangular system of equations	$x^{-1}y$; x upper-triangular	yes
<code>logdet(x)</code>	log matrix determinant	$\log x $	yes
<code>asRow(x)</code>	convert vector to 1-row matrix	sometimes automatic	yes
<code>asCol(x)</code>	convert vector to 1-column matrix	sometimes automatic	yes
<code>sum(x)</code>	sum of elements of x		yes
<code>mean(x)</code>	mean of elements of x		yes
<code>sd(x)</code>	standard deviation of elements of x		yes
<code>prod(x)</code>	product of elements of x		yes
<code>min(x), max(x)</code>	min. (max.) of elements of x		yes
<code>pmin(x, y), pmax(x, y)</code>	vector of mins (maxs) of elements of x and y		yes
<code>interp.lin(x, v1, v2)</code>	linear interpolation		no
<code>eigen(x)\$values</code>	matrix eigenvalues	x symmetric	yes
<code>eigen(x)\$vectors</code>	matrix eigenvectors	x symmetric	yes
<code>svd(x)\$d</code>	matrix singular values		yes
<code>svd(x)\$u</code>	matrix left singular vectors		yes
<code>svd(x)\$v</code>	matrix right singular vectors		yes

See Section 13.1 to learn how to use `nimbleFunctions` to write new functions for use in BUGS code.

5.2.6 Available link functions

NIMBLE allows the link functions listed in Table 5.6.

Table 5.6: Link functions.

Link function	Description	Range	Inverse
<code>cloglog(y) <- x</code>	Complementary log log	$0 < y < 1$	<code>y <- icloglog(x)</code>
<code>log(y) <- x</code>	Log	$0 < y$	<code>y <- exp(x)</code>
<code>logit(y) <- x</code>	Logit	$0 < y < 1$	<code>y <- expit(x)</code>
<code>probit(y) <- x</code>	Probit	$0 < y < 1$	<code>y <- iprobit(x)</code>

Link function	Description	Range	Inverse
---------------	-------------	-------	---------

Link functions are specified as functions applied to a node on the left hand side of a BUGS expression. To handle link functions in deterministic declarations, NIMBLE converts the declaration into an equivalent inverse declaration. For example, `log(y) <- x` is converted into `y <- exp(x)`. In other words, the link function is just a simple variant for conceptual clarity.

To handle link functions in a stochastic declaration, NIMBLE does some processing that inserts an additional node into the model. For example, the declaration `logit(p[i]) ~ dnorm(mu[i],1)`, is equivalent to the following two declarations:

- `logit_p[i] ~ dnorm(mu[i], 1),`
- `p[i] <- expit(logit_p[i])`

where `expit` is the inverse of `logit`.

Note that NIMBLE does not provide an automatic way of initializing the additional node (`logit_p[i]` in this case), which is a parent node of the explicit node (`p[i]`), without explicitly referring to the additional node by the name that NIMBLE generates.

5.2.7 Truncation, censoring, and constraints

NIMBLE provides three ways to declare boundaries on the value of a variable, each for different situations. We introduce these and comment on their relationships to related features of JAGS and BUGS. The three methods are:

5.2.7.1 Truncation

Either of the following forms,

```
x ~ dnorm(0, sd = 10) T(0, a)
x ~ T(dnorm(0, sd = 10), 0, a)
```

declares that `x` follows a normal distribution between 0 and `a` (inclusive of 0 and `a`). Either boundary may be omitted or may be another node, such as `a` in this example. The first form is compatible with JAGS, but in NIMBLE it can only be used when reading code from a text file. When writing model code in R, the second version must be used.

Truncation means the possible values of `x` are limited a priori, hence the probability density of `x` must be normalized⁸. In this example it would be the normal probability density divided by its integral from 0 to `a`. Like JAGS, NIMBLE also provides `I` as a synonym for `T` to accommodate older BUGS code, but `T` is preferred because it disambiguates multiple usages of `I` in BUGS.

5.2.7.2 Censoring

Censoring refers to the situation where one datum gives the lower or upper bound on an unobserved random variable. This is common in survival analysis, when for an individual still surviving at the end of a study, the age of death is not known and hence is ‘censored’ (right-censoring). NIMBLE adopts JAGS syntax for censoring, as follows:

⁸NIMBLE uses the CDF and inverse CDF (quantile) functions of a distribution to do this; in some cases if one uses truncation to include only the extreme tail of a distribution, numerical difficulties can arise.

```
censored[i] ~ dinterval(t[i], c[i])
t[i] ~ dweib(r, mu[i])
```

In the case of right-censoring, `censored[i]` should be given as `data` with a value of 1 if `t[i]` is right-censored (`t[i] > c[i]`) and 0 if it is observed. The data vector for `t` should have NA (indicating missing data) for any censored `t[i]` entries. (As a result, these nodes will be sampled in an MCMC.) The vector for `c` should give the censoring times corresponding to censored entries and a value above the observed times for uncensored entries (e.g., `Inf`). The values for `c` can be provided via `constants`, `inits`, or `data`. We recommend providing via `constants` for values that are fixed and via `inits` for values that are to be estimated or might be changed for some reason.

Left-censored observations would be specified by setting `censored[i]` to 0 and `t[i]` to NA.

Important: The value of `t[i]` corresponding to each censored data point should be given a valid initial value (via `inits`) that is consistent with `censored[i]`. Otherwise, NIMBLE will initialize from the prior, which may not produce a valid initial value.

The `dinterval` is not really a distribution but rather a trick: in the above example when `censored[i] = 1` it gives a ‘probability’ of 1 if `t[i] > c[i]` and 0 otherwise. This means that `t[i] <= c[i]` is treated as impossible. More generally than simple right- or left-censoring, `censored[i] ~ dinterval(t[i], c[i, ])` is defined such that for a vector of increasing cut-points, `c[i, ]`, `t[i]` is enforced to fall within the `censored[i]`-th cutpoint interval. This is done by setting data `censored[i]` as follows:

```
censored[i] = 0 if t[i] <= c[i, 1]
censored[i] = m if c[i, m] < t[i] <= c[i, m+1] for 1 ≤ m ≤ M
censored[i] = M if c[i, M] < t[i]
```

(The `i` index is provided only for consistency with the previous example.) The most common uses of `dinterval` will be for left- and right-censored data, in which case `c[i, ]` will be a single value (and typically given as simply `c[i]`), and for interval-censored data, in which case `c[i, ]` will be a vector of two values.

Nodes following a `dinterval` distribution should normally be set as `data` with known values. Otherwise, the node may be simulated during initialization in some algorithms (e.g., MCMC) and thereby establish a permanent, perhaps unintended, constraint.

Censoring differs from truncation because censoring an observation involves bounds on a random variable that could have taken any value, while in truncation we know a priori that a datum could not have occurred outside the truncation range.

5.2.7.3 Constraints and ordering

NIMBLE provides a more general way to enforce constraints using `dconstraint(cond)`. For example, we could specify that the sum of `mu1` and `mu2` must be positive like this:

```
mu1 ~ dnorm(0, 1)
mu2 ~ dnorm(0, 1)
constraint_data ~ dconstraint( mu1 + mu2 > 0 )
```

with `constraint_data` set (as `data`) to 1. This is equivalent to a half-normal distribution on the half-plane $\mu_1 + \mu_2 > 0$. Nodes following `dconstraint` should be provided as data for the same reason of avoiding unintended initialization described above for `dinterval`.

Important: the model should be initialized so that the constraints are satisfied. Otherwise, NIMBLE will initialize from the prior, which may not produce a valid initial value and may cause algorithms (particularly) MCMC to fail.

Formally, `dconstraint(cond)` is a probability distribution on $\{0, 1\}$ such that $P(1) = 1$ if `cond` is TRUE and $P(0) = 1$ if `cond` is FALSE.

Of course, in many cases, parameterizing the model so that the constraints are automatically respected may be a better strategy than using `dconstraint`. One should be cautious about constraints that would make it hard for an MCMC or optimization to move through the parameter space (such as equality constraints that involve two or more parameters). For such restrictive constraints, general purpose algorithms that are not tailored to the constraints may fail or be inefficient. If constraints are used, it will generally be wise to ensure the model is initialized with values that satisfy them.

5.2.7.3.1 Ordering To specify an ordering of parameters, such as $\alpha_1 \leq \alpha_2 \leq \alpha_3$ one can use `dconstraint` as follows:

```
constraint_data ~ dconstraint( alpha1 <= alpha2 & alpha2 <= alpha3 )
```

Note that unlike in BUGS, one cannot specify prior ordering using syntax such as

```
alpha[1] ~ dnorm(0, 1) I(, alpha[2])
alpha[2] ~ dnorm(0, 1) I(alpha[1], alpha[3])
alpha[3] ~ dnorm(0, 1) I(alpha[2], )
```

as this does not represent a directed acyclic graph.

Also note that specifying prior ordering using `T(,)` can result in possibly unexpected results. For example:

```
alpha1 ~ dnorm(0, 1)
alpha2 ~ dnorm(0, 1) T(alpha1, )
alpha3 ~ dnorm(0, 1) T(alpha2, )
```

will enforce $\alpha_1 \leq \alpha_2 \leq \alpha_3$, but it does not treat the three parameters symmetrically. Instead it puts a marginal prior on `alpha1` that is standard normal and then constrains `alpha2` and `alpha3` to follow truncated normal distributions. This is not equivalent to a symmetric prior on the three `alphas` that assigns zero probability density when values are not in order.

NIMBLE does not support the JAGS `sort` syntax.

5.2.8 Model macros

A NIMBLE *macro* is a succinct syntax that expands to create the NIMBLE model code for part or all of a model.

The `nimbleMacros` package provides an initial set of macros, currently focused on linear modeling and include:

- LM: specify a complete generalized linear model (GLM) or generalized linear mixed model (GLMM), using standard R and `lme4` model formula syntax,
- LINPRED: specify a linear predictor using a standard R formula,
- FORLOOP: specify a for loop by providing code with index ranges.

These macros are discussed in detail in the [vignette for the nimbleMacros package](#) and in the documentation for each macro, e.g., `help(LINPRED)`.

Here is a brief example of using the LM macro to set up a GLMM.

We'll consider the Chick example from the `lme4` package. We'll set up a NIMBLE model analogous to this mixed effects model specified with `lme4::lmer`:

```
lmer(weight ~ Time + (1|Chick), data = ChickWeight)
```

with a linear effect of time and random intercepts for each chick.

Here's the specification of the model code in nimble, using the LM macro:

```
library(nimbleMacros)

code <- nimbleCode({
  LM(weight[1:N] ~ Time + (1|Chick))
})

chickConst <- list(Time = ChickWeight$Time, Chick = ChickWeight$Chick,
  N = nrow(ChickWeight))
chickData <- list(weight = ChickWeight$weight)

mod <- nimbleModel(code, constants = chickConst, data = chickData)

mod$getCode()

## {
##   "# LM"
##   "   ## FORLOOP"
##   for (i_1 in 1:N) {
##     weight[i_1] ~ dnorm(mu_[i_1], sd = sd_residual)
##   }
##   "   ## ----"
##   "   ## LINPRED"
##   "     ### nimbleMacros::FORLOOP"
##   for (i_2 in 1:N) {
##     mu_[i_2] <- beta_Intercept + beta_Time * Time[i_2] +
##       beta_Chick[Chick[i_2]]
##   }
##   "     ### ----"
##   "     ### nimbleMacros::LINPRED_PRIORS"
##   beta_Intercept ~ dnorm(0, sd = 1000)
##   beta_Time ~ dnorm(0, sd = 1000)
##   sd_Chick ~ dunif(0, 100)
```

```
##      "      #### nimbleMacros::FORLOOP"
##      for (i_3 in 1:50) {
##          beta_Chick[i_3] ~ dnorm(0, sd = sd_Chick)
##      }
##      "      #### ----"
##      "      ### ----"
##      "      ## ----"
##      sd_residual ~ dunif(0, 100)
##      "# ----"
## }
```

We can see list of the names of the model variables created by macros in the model using the `getMacroParameters()` function:

```
mod$getMacroParameters()
```

```
## $LM
## $LM[[1]]
## [1] "mu_"          "sd_residual"
##
##
## $FORLOOP
## $FORLOOP[[1]]
## NULL
##
##
## $LINPRED
## $LINPRED[[1]]
## [1] "beta_Intercept" "beta_Time"      "beta_Chick"
##
##
## $`nimbleMacros::FORLOOP`
## $`nimbleMacros::FORLOOP`[[1]]
## NULL
##
## $`nimbleMacros::FORLOOP`[[2]]
## NULL
##
##
## $`nimbleMacros::LINPRED_PRIORS`
## $`nimbleMacros::LINPRED_PRIORS`[[1]]
## [1] "beta_Intercept" "beta_Time"      "sd_Chick"        "beta_Chick"
```

That specified the entire model with a macro. One can also use macros to create the code for components of a model. Here's a brief example.

Here's a complete linear regression model (in this example omitting the random effects) of chick weight as a function of time making use of `LINPRED` and `FORLOOP` (similar to the code generated by `LM`):

```
code <- nimbleCode({
  mu[1:N] <- LINPRED(~Time[1:N])

  weight[1:N] ~ FORLOOP(dnorm(mu[1:N], sd = sigma))
  sigma ~ dunif(0, 100)
})
mod <- nimbleModel(code, constants = chickConst, data = chickData)
mod$getCode()
```

```
## {
##   "# LINPRED"
##   "  ## nimbleMacros::FORLOOP"
##   for (i_1 in 1:N) {
##     mu[i_1] <- beta_Intercept + beta_Time * Time[i_1]
##   }
##   "  ## ----"
##   "  ## nimbleMacros::LINPRED_PRIORS"
##   beta_Intercept ~ dnorm(0, sd = 1000)
##   beta_Time ~ dnorm(0, sd = 1000)
##   "  ## ----"
##   "# ----"
##   "# FORLOOP"
##   for (i_2 in 1:N) {
##     weight[i_2] ~ dnorm(mu[i_2], sd = sigma)
##   }
##   "# ----"
##   sigma ~ dunif(0, 100)
## }
```

As discussed in Section 13.4, NIMBLE users can also create new macros to use in models.

Chapter 6

Building and using models

This chapter explains how to build and manipulate model objects starting from BUGS code.

6.1 Creating model objects

NIMBLE provides two functions for creating model objects: `nimbleModel` and `readBUGSmodel`. The first, `nimbleModel`, is more general and was illustrated in Chapter 2. The second, `readBUGSmodel` provides compatibility with BUGS file formats for models, variables, data, and initial values for MCMC.

In addition one can create new model objects from existing model objects.

6.1.1 Using *nimbleModel* to create a model

`nimbleModel` processes BUGS code to determine all the nodes, variables, and their relationships in a model. Any constants must be provided at this step. Data and initial values can optionally be provided. BUGS code passed to `nimbleModel` must go through `nimbleCode`.

We look again at the pump example from the introduction:

```
pumpCode <- nimbleCode({
  for (i in 1:N){
    theta[i] ~ dgamma(alpha,beta);
    lambda[i] <- theta[i]*t[i];
    x[i] ~ dpois(lambda[i])
  }
  alpha ~ dexp(1.0);
  beta ~ dgamma(0.1,1.0);
})

pumpConsts <- list(N = 10,
  t = c(94.3, 15.7, 62.9, 126, 5.24,
        31.4, 1.05, 1.05, 2.1, 10.5))

pumpData <- list(x = c(5, 1, 5, 14, 3, 19, 1, 1, 4, 22))
```

```

pumpInits <- list(alpha = 1, beta = 1,
                 theta = rep(0.1, pumpConsts$N))

pump <- nimbleModel(code = pumpCode, name = "pump", constants = pumpConsts,
                  data = pumpData, inits = pumpInits)

```

6.1.1.1 Data and constants

NIMBLE makes a distinction between data and constants:

- *Constants* can never be changed and must be provided when a model is defined. For example, a vector of known index values, such as for block indices, helps define the model graph itself and must be provided as constants. Variables used in the index ranges of for-loops must also be provided as constants.
- *Data* is a label for the role a node plays in the model. Nodes marked as data will by default be protected from any functions that would simulate over their values (see `simulate` in Chapter 14), but it is possible to over-ride that default or to change their values by direct assignment. This allows an algorithm to be applied to many data sets in the same model without re-creating the model each time. It also allows simulation of data in a model.

WinBUGS, OpenBUGS and JAGS do not allow data values to be changed or different nodes to be labeled as data without starting from the beginning again. Hence they do not distinguish between constants and data.

For compatibility with BUGS and JAGS, NIMBLE allows both to be provided in the `constants` argument to `nimbleModel`, in which case NIMBLE handles values for stochastic nodes as data and everything else as constants.

Values for nodes that appear only on the right-hand side of BUGS declarations (e.g., covariates/predictors) can be provided as constants or as data or initial values. There is no real difference between providing as data or initial values and the values can be added after building a model via `setInits` or `setData`. However if provided as data, calls to `setInits` will not overwrite those values (though direct assignment of values will overwrite those values).

6.1.1.2 Providing (or changing) data and initial values for an existing model

Whereas constants must be provided during the call to `nimbleModel`, data and initial values can be provided later via the model member functions `setData` and `setInits`. For example, if `pumpData` is a named list of data values (as above), then `pump$setData(pumpData)` sets the named variables to the values in the list.

`setData` does two things: it sets the values of the (stochastic) data nodes, and it flags those nodes as containing data. `nimbleFunction` programmers can then use that information to control whether an algorithm should over-write data or not. For example, NIMBLE's `simulate` functions by default do not overwrite data values but can be told to do so. Values of data variables can be replaced, and the indication of which nodes should be treated as data can be reset by using the `resetData` method, e.g. `pump$resetData()`.

Data nodes cannot be deterministic, and using `setData` on deterministic nodes (or passing values for

deterministic nodes via the `data` argument to `nimbleModel`) will not flag those nodes as containing data. It will set the values of those nodes, but that will presumably be overwritten as soon as the nodes are deterministically calculated.

To change data values without any modification of which nodes are flagged as containing data, simply use R's usual assignment syntax to assign values in a compiled (or, more rarely, an uncompiled) model, e.g.,

```
cModel$y <- c(1.5, 2.5, 1.7)
```

This can be useful for running an MCMC with a different dataset of the same size (and of course the same pattern of missingness, if any) without having to rebuild and recompile the MCMC, such as in a simulation study. It is possible to change the data values in a compiled model using `setData`, but we don't recommend doing this because `setData` won't modify which nodes are flagged as containing data in the already-constructed MCMC, thereby potentially introducing confusion.

6.1.1.3 Missing data values

Sometimes one needs a model variable to have a mix of data and non-data, often due to missing data values. In NIMBLE, when data values are provided, any nodes with NA values will *not* be labeled as data. A node following a multivariate distribution must be either entirely observed or entirely missing.

Here's an example of running an MCMC on the *pump* model, with two of the observations taken to be missing. Some of the steps in this example are documented more below. NIMBLE's default MCMC configuration will treat the missing values as unknowns to be sampled, as can be seen in the MCMC output here.

```
pumpMiss <- pump$newModel()
pumpMiss$resetData()
pumpDataNew <- pumpData
pumpDataNew$x[c(1, 3)] <- NA
pumpMiss$setData(pumpDataNew)

pumpMissConf <- configureMCMC(pumpMiss)

## ===== Monitors =====
## thin = 1: alpha, beta
## ===== Samplers =====
## RW sampler (1)
##   - alpha
## posterior_predictive sampler (2)
##   - theta[] (2 elements)
## conjugate sampler (9)
##   - beta
##   - theta[] (8 elements)
## ===== Comments =====

pumpMissConf$addMonitors('x', 'alpha', 'beta', 'theta')

## thin = 1: alpha, beta, theta, x
```

```

pumpMissMCMC <- buildMCMC(pumpMissConf)
Cobj <- compileNimble(pumpMiss, pumpMissMCMC)

niter <- 10
set.seed(0)
Cobj$pumpMissMCMC$run(niter)
samples <- as.matrix(Cobj$pumpMissMCMC$mvSamples)

samples[1:5, 13:17]

```

```

##      x[1] x[2] x[3] x[4] x[5]
## [1,]  12   1   0  14   3
## [2,]   7   1  35  14   3
## [3,]  19   1   4  14   3
## [4,]  61   1  36  14   3
## [5,]  70   1  61  14   3

```

Missing values may also occur in explanatory/predictor variables. Values for such variables should be passed in via the `data` argument to `nimbleModel`, with `NA` for the missing values. In some contexts, one would want to specify distributions for such explanatory variables, for example so that an MCMC would impute the missing values.

6.1.1.4 Defining alternative models with the same code

Avoiding code duplication is a basic principle of good programming. In NIMBLE, one can use definition-time if-then-else statements to create different models from the same code. As a simple example, say we have a linear regression model and want to consider including or omitting `x[2]` as an explanatory variable:

```

regressionCode <- nimbleCode({
  intercept ~ dnorm(0, sd = 1000)
  slope1 ~ dnorm(0, sd = 1000)
  if(includeX2) {
    slope2 ~ dnorm(0, sd = 1000)
    for(i in 1:N)
      predictedY[i] <- intercept + slope1 * x1[i] + slope2 * x2[i]
  } else {
    for(i in 1:N) predictedY[i] <- intercept + slope1 * x1[i]
  }
  sigmaY ~ dunif(0, 100)
  for(i in 1:N) Y[i] ~ dnorm(predictedY[i], sigmaY)
})

includeX2 <- FALSE
modelWithoutX2 <- nimbleModel(regressionCode, constants = list(N = 30),
                             check=FALSE)
modelWithoutX2$getVarNames()

## [1] "intercept"                "slope1"

```



```
## [3] "predictedY"                "sigmaY"
## [5] "lifted_d1_over_sqrt_oPsigmaY_cP" "Y"
## [7] "x1"

includeX2 <- TRUE
modelWithX2 <- nimbleModel(regressionCode, constants = list(N = 30),
                           check = FALSE)
modelWithX2$getVarNames()

## [1] "intercept"                "slope1"
## [3] "slope2"                  "predictedY"
## [5] "sigmaY"                  "lifted_d1_over_sqrt_oPsigmaY_cP"
## [7] "Y"                       "x1"
## [9] "x2"
```

Whereas the *constants* are a property of the *model definition* – since they may help determine the model structure itself – *data* nodes can be different in different copies of the model generated from the same *model definition*. The `setData` and `setInits` described above can be used for each copy of the model.

6.1.1.5 Providing dimensions via *nimbleModel*

`nimbleModel` can usually determine the dimensions of every variable from the declarations in the BUGS code. However, it is possible to use a multivariate object only with empty indices (e.g. `x[,]`), in which case the dimensions must be provided as an argument to `nimbleModel`.

Here's an example with multivariate nodes. The first provides indices, so no `dimensions` argument is needed, while the second omits the indices and provides a `dimensions` argument instead.

```
code <- nimbleCode({
  y[1:K] ~ dmulti(p[1:K], n)
  p[1:K] ~ ddirch(alpha[1:K])
  log(alpha[1:K]) ~ dmnorm(alpha0[1:K], R[1:K, 1:K])
})

K <- 5
model <- nimbleModel(code, constants = list(n = 3, K = K,
                                           alpha0 = rep(0, K), R = diag(K)),
                   check = FALSE)

codeAlt <- nimbleCode({
  y[] ~ dmulti(p[], n)
  p[] ~ ddirch(alpha[])
  log(alpha[]) ~ dmnorm(alpha0[], R[ , ])
})

model <- nimbleModel(codeAlt, constants = list(n = 3, K = K, alpha0 = rep(0, K),
                                           R = diag(K)),
                   dimensions = list(y = K, p = K, alpha = K),
                   check = FALSE)
```

In that example, since `alpha0` and `R` are provided as constants, we don't need to specify their dimensions.

6.1.2 Creating a model from standard BUGS and JAGS input files

Users with BUGS and JAGS experience may have files set up in standard formats for use in BUGS and JAGS. `readBUGSmodel` can read in the model, data/constant values and initial values in those formats. It can also take information directly from R objects somewhat more flexibly than `nimbleModel`, specifically allowing inputs set up similarly to those for BUGS and JAGS. In either case, after processing the inputs, it calls `nimbleModel`. Note that unlike BUGS and JAGS, only a single set of initial values can be specified in creating a model. Please see `help(readBUGSmodel)` for argument details.

As an example of using `readBUGSmodel`, let's create a model for the *pump* example from BUGS.

```
pumpDir <- system.file('classic-bugs', 'vol1', 'pump', package = 'nimble')
pumpModel <- readBUGSmodel('pump.bug', data = 'pump-data.R',
                           inits = 'pump-init.R', dir = pumpDir)
```

Note that `readBUGSmodel` allows one to include `var` and `data` blocks in the model file as in some of the BUGS examples (such as *inhaler*). The `data` block pre-computes constant and data values. Also note that if `data` and `inits` are provided as files, the files should contain R code that creates objects analogous to what would populate the list if a list were provided instead. Please see the JAGS manual examples or the `classic_bugs` directory in the NIMBLE package for example syntax. NIMBLE by and large does not need the information given in a `var` block but occasionally this is used to determine dimensionality, such as in the case of syntax like `xbar <- mean(x[])` where `x` is a variable that appears only on the right-hand side of BUGS expressions.

Note that NIMBLE does not handle formatting such as in some of the original BUGS examples in which data was indicated with syntax such as `data x in 'x.txt'`.

6.1.3 Making multiple instances from the same model definition

Sometimes it is useful to have more than one copy of the same model. For example, an algorithm (i.e., `nimbleFunction`) such as an MCMC will be bound to a particular model before it is run. A user could build multiple algorithms to use the same model instance, or they may want each algorithm to have its own instance of the model.

There are two ways to create new instances of a model, shown in this example:

```
simpleCode <- nimbleCode({
  for(i in 1:N) x[i] ~ dnorm(0, 1)
})

# Return the model definition only, not a built model
simpleModelDefinition <- nimbleModel(simpleCode, constants = list(N = 10),
                                   returnDef = TRUE, check = FALSE)

# Make one instance of the model
simpleModelCopy1 <- simpleModelDefinition$newModel(check = FALSE)
# Make another instance from the same definition
simpleModelCopy2 <- simpleModelDefinition$newModel(check = FALSE)
```

```
# Ask simpleModelCopy2 for another copy of itself
simpleModelCopy3 <- simpleModelCopy2$newModel(check = FALSE)
```

Each copy of the model can have different nodes flagged as data and different values in any nodes. They cannot have different values of N because that is a constant; it must be a constant because it helps define the model.

6.2 NIMBLE models are objects you can query and manipulate

NIMBLE models are objects that can be modified and manipulated from R. In this section we introduce some basic ways to use a model object. Chapter 14 covers more topics for writing algorithms that use models.

6.2.1 What are variables and nodes?

This section discusses some basic concepts and terminology to be able to speak about NIMBLE models clearly.

Suppose we have created a model from the following BUGS code.

```
mc <- nimbleCode({
  a ~ dnorm(0, 0.001)
  for(i in 1:5) {
    y[i] ~ dnorm(a, sd = 0.1)
    for(j in 1:3)
      z[i,j] ~ dnorm(y[i], tau)
  }
  tau ~ dunif(0, 20)
  y.squared[1:5] <- y[1:5]^2
})

model <- nimbleModel(mc, data = list(z = matrix(rnorm(15), nrow = 5)))
```

In NIMBLE terminology:

- The *variables* of this model are `a`, `y`, `z`, and `y.squared`.
- The *nodes* of this model are `a`, `y[1]`, ..., `y[5]`, `z[1,1]`, ..., `z[5, 3]`, and `y.squared[1:5]`. In graph terminology, nodes are vertices in the model graph.
- The *node functions* of this model are `a ~ dnorm(0, 0.001)`, `y[i] ~ dnorm(a, 0.1)`, `z[i,j] ~ dnorm(y[i], sd = 0.1)`, and `y.squared[1:5] <- y[1:5]^2`. Each node's calculations are handled by a node function. Sometimes the distinction between nodes and node functions is important, but when it is not important we may refer to both simply as *nodes*.
- The *scalar elements* of this model include all the scalar nodes as well as the scalar elements `y.squared[1]`, ..., `y.squared[5]` of the multivariate node `y.squared[1:5]`.

6.2.2 Determining the nodes and variables in a model

One can determine the variables in a model using `getVarNames` and the nodes in a model using `getNodeNames`. Optional arguments to `getNodeNames` allow you to select only certain types of nodes, as discussed in Section 14.1.1 and in the R help for `getNodeNames`.

```
model$getVarNames()
```

```
## [1] "a"                "y"
## [3] "lifted_d1_over_sqrt_oPtau_cP" "z"
## [5] "tau"              "y.squared"
```

```
model$getNodeNames()
```

```
## [1] "a"                "tau"
## [3] "y[1]"             "y[2]"
## [5] "y[3]"             "y[4]"
## [7] "y[5]"             "lifted_d1_over_sqrt_oPtau_cP"
## [9] "y.squared[1:5]"   "z[1, 1]"
## [11] "z[1, 2]"          "z[1, 3]"
## [13] "z[2, 1]"          "z[2, 2]"
## [15] "z[2, 3]"          "z[3, 1]"
## [17] "z[3, 2]"          "z[3, 3]"
## [19] "z[4, 1]"          "z[4, 2]"
## [21] "z[4, 3]"          "z[5, 1]"
## [23] "z[5, 2]"          "z[5, 3]"
```

Note that some of the nodes may be ‘lifted’ nodes introduced by NIMBLE (Section 14.1.2). In this case `lifted_d1_over_sqrt_oPtau_cP` (this is a node for the standard deviation of the `z` nodes using NIMBLE’s canonical parameterization of the normal distribution) is the only lifted node in the model.

To determine the dependencies of one or more nodes in the model, you can use `getDependencies` as discussed in Section 14.1.3.

6.2.3 Accessing nodes

Model variables can be accessed and set just as in R using `$` and `[[ ]]`. For example

```
model$a <- 5
model$a
```

```
## [1] 5
```

```
model[["a"]]
```

```
## [1] 5
```

```
model$y[2:4] <- rnorm(3)
model$y
```

```
## [1] NA -0.7317482 0.8303732 -1.2080828 NA
```

```
model[["y"]][c(1, 5)] <- rnorm(2)
model$y

## [1] -1.0479844 -0.7317482  0.8303732 -1.2080828  1.4411577

model$z[1,]

## [1]  2.0752450  0.2199248 -0.7660820
```

While nodes that are part of a variable can be accessed as above, each node also has its own name that can be used to access it directly. For example, `y[2]` has the name ‘`y[2]`’ and can be accessed by that name as follows:

```
model[["y[2]"]]

## [1] -0.7317482

model[["y[2]"]] <- -5
model$y

## [1] -1.0479844 -5.0000000  0.8303732 -1.2080828  1.4411577

model[["z[2, 3]"]]

## [1] -0.4302118

model[["z[2:4, 1:2]"]][1, 2]

## [1] -1.46725

model$z[2, 2]

## [1] -1.46725
```

Notice that node names can include index blocks, such as `model[["z[2:4, 1:2]"]]`, and these are not strictly required to correspond to actual nodes. Such blocks can be subsequently sub-indexed in the regular R manner, such as `model[["z[2:4, 1:2]"]][1, 2]`.

6.2.4 How nodes are named

Every node has a name that is a character string including its indices, with a space after every comma. For example, `X[1, 2, 3]` has the name ‘`X[1, 2, 3]`’. Nodes following multivariate distributions have names that include their index blocks. For example, a multivariate node for `X[6:10, 3]` has the name ‘`X[6:10, 3]`’.

The definitive source for node names in a model is `getNodeNames`, described previously.

In the event you need to ensure that a name is formatted correctly, you can use the `expandNodeNames` method. For example, to get the spaces correctly inserted into ‘`X[1,1:5]`’:

```
multiVarCode <- nimbleCode({
  X[1, 1:5] ~ dmnorm(mu[], cov[,])
  X[6:10, 3] ~ dmnorm(mu[], cov[,])
})
```

```
multiVarModel <- nimbleModel(multiVarCode, dimensions =
  list(mu = 5, cov = c(5,5)), calculate = FALSE)

multiVarModel$expandNodeNames("X[1,1:5]")
```

```
## [1] "X[1, 1:5]"
```

Alternatively, for those inclined to R's less commonly used features, a nice trick is to use its `parse` and `deparse` functions.

```
deparse(parse(text = "X[1,1:5]", keep.source = FALSE)[[1]])
```

```
## [1] "X[1, 1:5]"
```

The `keep.source = FALSE` makes `parse` more efficient.

6.2.5 Why use node names?

Syntax like `model[["z[2, 3]"]]` may seem strange at first, because the natural habit of an R user would be `model[["z"]][2,3]`. To see its utility, consider the example of writing the `nimbleFunction` given in Section 2.8. By giving every scalar node a name, even if it is part of a multivariate variable, one can write functions in R or NIMBLE that access any single node by a name, regardless of the dimensionality of the variable in which it is embedded. This is particularly useful for NIMBLE, which resolves how to access a particular node during the compilation process.

6.2.6 Checking if a node holds data

Finally, you can query whether a node is flagged as data using the `isData` method applied to one or more nodes or nodes within variables:

```
model$isData('z[1]')
```

```
## [1] TRUE
```

```
model$isData(c('z[1]', 'z[2]', 'a'))
```

```
## [1] TRUE TRUE FALSE
```

```
model$isData('z')
```

```
## [1] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
```

```
model$isData('z[1:3, 1]')
```

```
## [1] TRUE TRUE TRUE
```

6.3 Using models in parallel

NIMBLE uses Reference Classes and R6 classes for models and algorithms. Objects of these classes are passed by reference and copies of such objects are simply new variable names that reference the same underlying object.

Thus to run an algorithm in parallel on a given model, one must create multiple copies of the model and algorithm, and compiled versions of these, by calling `nimbleModel`, `buildMCMC`, `compileNimble`, etc. once for each copy. In other words all such calls should be within the parallelized block of code.

For a worked example in the context of MCMC, please see [the parallelization example on our webpage](#).

Part III

Algorithms in NIMBLE

Chapter 7

MCMC

NIMBLE provides a variety of paths to creating and executing an MCMC algorithm, which differ greatly in their simplicity of use, and also in the options available and customizability.

The most direct approach to invoking the MCMC engine is using the `nimbleMCMC` function (Section 7.1). This one-line call creates and executes an MCMC, and provides a wide range of options for controlling the MCMC: specifying monitors, burn-in, and thinning, running multiple MCMC chains with different initial values, and returning posterior samples, summary statistics, and/or a WAIC value. However, this approach is restricted to using NIMBLE’s default MCMC algorithm; further customization of, for example, the specific samplers employed, is not possible.

The lengthier and more customizable approach to invoking the MCMC engine on a particular NIMBLE model object involves the following steps:

1. (Optional) Create and customize an MCMC configuration for a particular model:
 - a. Use `configureMCMC` to create an MCMC configuration (see Section 7.2). The configuration contains a list of samplers with the node(s) they will sample.
 - b. (Optional) Customize the MCMC configuration:
 - i. Add, remove, or re-order the list of samplers (Section 7.11 and `help(samplers)` in R for details), including adding your own samplers (Section 16.5);
 - ii. Change the tuning parameters or adaptive properties of individual samplers;
 - iii. Change the variables to monitor (record for output) and thinning intervals for MCMC samples.
2. Use `buildMCMC` to build the MCMC object and its samplers either from the model (using default MCMC configuration) or from a customized MCMC configuration (Section 7.3).
3. Compile the MCMC object (and the model), unless one is debugging and wishes to run the uncompiled MCMC.
4. Run the MCMC and extract the samples (Sections 7.5, 7.6 and 7.7).
5. Optionally, calculate the WAIC (Section 7.8).

End-to-end examples of MCMC in NIMBLE can be found in Sections 2.5-2.6 and Section 7.12.

7.1 One-line invocation of MCMC: *nimbleMCMC*

The most direct approach to executing an MCMC algorithm in NIMBLE is using `nimbleMCMC`. This single function can be used to create an underlying model and associated MCMC algorithm, compile both of these, execute the MCMC, and return samples, summary statistics, and a WAIC value. This approach circumvents the longer (and more flexible) approach using `nimbleModel`, `configureMCMC`, `buildMCMC`, `compileNimble`, and `runMCMC`, which is described subsequently.

The `nimbleMCMC` function provides control over the:

- number of MCMC iterations in each chain;
- number of MCMC chains to execute;
- number of burn-in samples to discard from each chain;
- thinning interval on which samples should be recorded;
- model variables to monitor and return posterior samples;
- initial values, or a function for generating initial values for each chain;
- setting the random number seed;
- returning posterior samples as a matrix or a `coda mcmc` object;
- returning posterior summary statistics; and
- returning a WAIC value calculated using post-burn-in samples from all chains.

This entry point for using `nimbleMCMC` is the `code`, `constants`, `data`, and `inits` arguments that are used for building a NIMBLE model (see Chapters 5 and 6). However, when using `nimbleMCMC`, the `inits` argument can also specify a list of lists of initial values that will be used for each MCMC chain, or a function that generates a list of initial values, which will be generated at the onset of each chain. As an alternative entry point, a NIMBLE `model` object can also be supplied to `nimbleMCMC`, in which case this model will be used to build the MCMC algorithm.

Based on its arguments, `nimbleMCMC` optionally returns any combination of

- Posterior samples,
- Posterior summary statistics, and
- WAIC value.

The above are calculated and returned for each MCMC chain, using the post-burn-in and thinned samples. Additionally, posterior summary statistics are calculated for all chains combined when multiple chains are run.

Several example usages of `nimbleMCMC` are shown below:

```
code <- nimbleCode({
  mu ~ dnorm(0, sd = 1000)
  sigma ~ dunif(0, 1000)
  for(i in 1:10)
    x[i] ~ dnorm(mu, sd = sigma)
})
data <- list(x = c(2, 5, 3, 4, 1, 0, 1, 3, 5, 3))
initsFunction <- function() list(mu = rnorm(1,0,1), sigma = runif(1,0,10))

# execute one MCMC chain, monitoring the "mu" and "sigma" variables,
# with thinning interval 10. fix the random number seed for reproducible
# results. by default, only returns posterior samples.
```

```
mcmc.out <- nimbleMCMC(code = code, data = data, inits = initsFunction,
                      monitors = c("mu", "sigma"), thin = 10,
                      niter = 20000, nchains = 1, setSeed = TRUE)

# note that the inits argument to nimbleModel must be a list of
# initial values, whereas nimbleMCMC can accept inits as a function
# for generating new initial values for each chain.
initsList <- initsFunction()
Rmodel <- nimbleModel(code, data = data, inits = initsList)

# using the existing Rmodel object, execute three MCMC chains with
# specified burn-in. return samples, summary statistics, and WAIC.
mcmc.out <- nimbleMCMC(model = Rmodel,
                      niter = 20000, nchains = 3, nburnin = 2000,
                      summary = TRUE, WAIC = TRUE)

# run ten chains, generating random initial values for each
# chain using the inits function specified above.
# only return summary statistics from each chain; not all the samples.
mcmc.out <- nimbleMCMC(model = Rmodel, nchains = 10, inits = initsFunction,
                      samples = FALSE, summary = TRUE)
```

See `help(nimbleMCMC)` for further details.

7.2 The MCMC configuration

The MCMC configuration contains information needed for building an MCMC. When no customization is needed, one can jump directly to the `buildMCMC` step below. An MCMC configuration is an object of class `MCMCconf`, which includes:

- The model on which the MCMC will operate
- The model nodes which will be sampled (updated) by the MCMC
- The samplers and their internal configurations, called control parameters
- Two sets of variables that will be monitored (recorded) during execution of the MCMC and thinning intervals for how often each set will be recorded. Two sets are allowed because it can be useful to monitor different variables at different intervals

7.2.1 Default MCMC configuration

Assuming we have a model named `Rmodel`, the following will generate a default MCMC configuration:

```
mcmcConf <- configureMCMC(Rmodel)
```

The default configuration will contain a single sampler for each node in the model, and the default ordering follows the topological ordering of the model.

7.2.1.1 Default assignment of sampler algorithms

The default sampler assigned to a stochastic node is determined by the following, in order of precedence:

1. If the node has no data nodes in its entire downstream dependency network, a `posterior_predictive` sampler is assigned. This sampler updates the node in question and all downstream stochastic nodes by simulating new values from each node's conditional distribution. As of version 0.13.0, the operation of the `posterior_predictive` sampler has changed in order to improve MCMC mixing. See Section 7.2.1.2.
2. If the node has a conjugate relationship between its prior distribution and the distributions of its stochastic dependents, a `conjugate` ('Gibbs') sampler is assigned.
3. If the node follows a multinomial distribution, then a `RW_multinomial` sampler is assigned. This is a discrete random-walk sampler in the space of multinomial outcomes.
4. If a node follows a Dirichlet distribution, then a `RW_dirichlet` sampler is assigned. This is a random walk sampler in the space of the simplex defined by the Dirichlet.
5. If a node follows an LKJ correlation distribution, then a `RW_block_lkj_corr_cholesky` sampler is assigned. This is a block random walk sampler in a transformed space where the transformation uses the signed stickbreaking approach described in Section 10.12 of [Stan Development Team \(2021b\)](#).
6. If the node follows any other multivariate distribution, then a `RW_block` sampler is assigned for all elements. This is a Metropolis-Hastings adaptive random-walk sampler with a multivariate normal proposal ([Roberts and Sahu, 1997](#)). As of NIMBLE version 1.3.0, one can instead use the `barker` sampler if one sets `nimbleOptions(MCMCuseBarkerAsDefaultMV = TRUE)`.
7. If the node is binary-valued (strictly taking values 0 or 1), then a `binary` sampler is assigned. This sampler calculates the conditional probability for both possible node values and draws the new node value from the conditional distribution, in effect making a Gibbs sampler.
8. If the node is otherwise discrete-valued, then a `slice` sampler is assigned ([Neal, 2003](#)).
9. If none of the above criteria are satisfied, then a `RW` sampler is assigned. This is a Metropolis-Hastings adaptive random-walk sampler with a univariate normal proposal distribution.

Details of each sampler and its control parameters can be found by invoking `help(samplers)`.

7.2.1.2 Sampling posterior predictive nodes

A posterior predictive node is a node that is not itself data and has no data nodes in its entire downstream (descendant) dependency network. Such nodes play no role in inference for model parameters but have often been included in BUGS models to accomplish posterior predictive checks and similar calculations.

As of version 0.13.0, NIMBLE's handling of posterior predictive nodes in MCMC sampling has changed in order to improve MCMC mixing. Samplers for nodes that are not posterior predictive nodes no longer condition on the values of the posterior predictive nodes. This produces a valid MCMC over the posterior distribution marginalizing over the posterior predictive nodes. This MCMC will generally mix better than an MCMC that conditions on the values of posterior predictive nodes, by reducing the dimensionality of the parameter space and removing the dependence between the sampled nodes and the posterior predictive nodes. At the end of each MCMC iteration, the posterior predictive nodes are sampled by `posterior_predictive` sampler(s) based on their conditional distribution(s).

7.2.1.3 Options to control default sampler assignments

Very basic control of default sampler assignments is provided via two arguments to `configureMCMC`. The `useConjugacy` argument controls whether conjugate samplers are assigned when possible, and the `multivariateNodesAsScalars` argument controls whether scalar elements of multivariate nodes are sampled individually. See `help(configureMCMC)` for usage details.

7.2.1.4 Default monitors

The default MCMC configuration includes monitors on all top-level stochastic nodes of the model. Only variables that are monitored will have their samples saved for use outside of the MCMC. MCMC configurations include two sets of monitors, each with different thinning intervals. By default, the second set of monitors (`monitors2`) is empty.

7.2.1.5 Automated parameter blocking

The automated parameter blocking algorithm is no longer actively maintained. In some cases, it may not operate correctly with more recent system features and/or distributions.

The default configuration may be replaced by one generated from an automated parameter blocking algorithm. This algorithm determines groupings of model nodes that, when jointly sampled with a `RW_block` sampler, increase overall MCMC efficiency. Overall efficiency is defined as the effective sample size of the slowest-mixing node divided by computation time. This is done by:

```
autoBlockConf <- configureMCMC(Rmodel, autoBlock = TRUE)
```

Note that this using `autoBlock = TRUE` compiles and runs MCMCs, progressively exploring different sampler assignments, so it takes some time and generates some output. It is most useful for determining effective blocking strategies that can be re-used for later runs. The additional control argument `autoIt` may also be provided to indicate the number of MCMC samples to be used in each trial of the automated blocking procedure (default 20,000).

7.2.2 Customizing the MCMC configuration

The MCMC configuration may be customized in a variety of ways, either through additional named arguments to `configureMCMC` or by calling methods of an existing `MCMCconf` object.

7.2.2.1 Controlling which nodes to sample

One can create an MCMC configuration with default samplers on just a particular set of nodes using the `nodes` argument to `configureMCMC`. The value for the `nodes` argument may be a character vector containing node and/or variable names. In the case of a variable name, a default sampler will be added for all stochastic nodes in the variable. The order of samplers will match the order of `nodes`. Any deterministic nodes will be ignored.

If a data node is included in `nodes`, *it will be assigned a sampler*. This is the only way in which a default sampler may be placed on a data node and will result in overwriting data values in the node.

7.2.2.2 Creating an empty configuration

If you plan to customize the choice of all samplers, it can be useful to obtain a configuration with no sampler assignments at all. This can be done by any of `nodes = NULL`, `nodes = character()`, or `nodes = list()`.

7.2.2.3 Overriding the default sampler control list values

The default values of control list elements for all sampling algorithms may be overridden through use of the `control` argument to `configureMCMC`, which should be a named list. Named elements in the `control` argument will be used for all default samplers and any subsequent sampler added via `addSampler` (see below). For example, the following will create the default MCMC configuration, except all RW samplers will have their initial `scale` set to 3, and none of the samplers (RW, or otherwise) will be adaptive.

```
mcmcConf <- configureMCMC(Rmodel, control = list(scale = 3, adaptive = FALSE))
```

When adding samplers to a configuration using `addSampler`, the default control list can also be over-ridden.

7.2.2.4 Adding samplers to the configuration: *addSampler*

Additional samplers may be added to a configuration using the `addSampler` method of the `MCMCconf` object. The `addSampler` method has two modes of operation, determined by the `default` argument.

When `default = TRUE`, `addSampler` will assign NIMBLE's default sampling algorithm for each node specified following the same protocol as `configureMCMC`. This may include conjugate samplers, multivariate samplers, or otherwise, also using additional arguments to guide the selection process (for example, `useConjugacy` and `multivariateNodesAsScalars`). In this mode of operation, the `type` argument is not used.

When `default = FALSE`, or when this argument is omitted, `addSampler` uses the `type` argument to specify the precise sampler to assign. Instances of this particular sampler are assigned to all nodes specified. The `type` argument may be provided as a character string or a `nimbleFunction` object. Valid character strings are indicated in `help(samplers)` (do not include "sampler_"). Added samplers can be labeled with a `name` argument, which is used in output of `printSamplers`. Writing a new sampler as a `nimbleFunction` is covered in Section 16.5.

Regardless of the mode of operation, nodes are specified using either the `target` or the `nodes` argument. The `target` argument does not undergo expansion to constituent nodes (unless `default = TRUE`), and thus only a single sampler is added. The `nodes` argument is always expanded to the underlying nodes, and separate samplers are added for each node. Either argument is provided as a character vector. Newly added samplers will be appended to the end of current sampler list. Adding a sampler for a node will *not* remove existing samplers operating on that node.

The hierarchy of precedence for control list elements for added samplers is:

1. The `control` list argument provided to `addSampler`;
2. The original `control` list argument provided to `configureMCMC`;
3. The default values, as defined in the sampling algorithm `setup` function.

See `help(addSampler)` for more details.

7.2.2.5 Printing, re-ordering, modifying and removing samplers: *printSamplers*, *removeSamplers*, *setSamplers*, and *getSamplerDefinition*

The current, ordered, list of all samplers in the MCMC configuration may be printed by calling the `printSamplers` method. When you want to see only samplers acting on specific model nodes or variables, provide those names as an argument to `printSamplers`. The `printSamplers` method accepts arguments controlling the level of detail displayed as discussed in its R help information.

```
# Print all samplers
mcmcConf$printSamplers()

# Print all samplers operating on node "a[1]",
# or any of the "beta[]" variables
mcmcConf$printSamplers(c("a[1]", "beta"))

# Print all conjugate and slice samplers
mcmcConf$printSamplers(type = c("conjugate", "slice"))

# Print all RW samplers operating on "x"
mcmcConf$printSamplers("x", type = "RW")

# Print the first 100 samplers
mcmcConf$printSamplers(1:100)

# Print all samplers in their order of execution
mcmcConf$printSamplers(executionOrder = TRUE)
```

Samplers may be removed from the configuration object using `removeSamplers`, which accepts a character vector of node or variable names, or a numeric vector of indices.

```
# Remove all samplers acting on "x" or any component of it
mcmcConf$removeSamplers("x")

# Remove all samplers acting on "alpha[1]" and "beta[1]"
mcmcConf$removeSamplers(c("alpha[1]", "beta[1]"))

# Remove the first five samplers
mcmcConf$removeSamplers(1:5)

# Providing no argument removes all samplers
mcmcConf$removeSamplers()
```

Samplers to retain may be specified reordered using `setSamplers`, which also accepts a character vector of node or variable names, or a numeric vector of indices.

```
# Set the list of samplers to those acting on any components of the
# model variables "x", "y", or "z".
mcmcConf$setSamplers(c("x", "y", "z"))

# Set the list of samplers to only those acting on model nodes
```

```
# "alpha[1]", "alpha[2]", ..., "alpha[10]"
mcmcConf$setSamplers("alpha[1:10]")

# Truncate the current list of samplers to the first 10 and the 100th
mcmcConf$setSamplers(ind = c(1:10, 100))
```

The `nimbleFunction` definition underlying a particular sampler may be viewed using the `getSamplerDefinition` method, using the sampler index as an argument. A node name argument may also be supplied, in which case the definition of the first sampler acting on that node is returned. In all cases, `getSamplerDefinition` only returns the definition of the *first* sampler specified either by index or node name.

```
# Return the definition of the third sampler in the mcmcConf object
mcmcConf$getSamplerDefinition(3)

# Return the definition of the first sampler acting on node "x",
# or the first of any indexed nodes comprising the variable "x"
mcmcConf$getSamplerDefinition("x")
```

7.2.2.6 Customizing individual sampler configurations: *getSamplers*, *setSamplers*, *setName*, *setSamplerFunction*, *setTarget*, and *setControl*

Each sampler in an `MCMCconf` object is represented by a sampler configuration as a `samplerConf` object. Each `samplerConf` is a reference class object containing the following (required) fields: `name` (a character string), `samplerFunction` (a valid `nimbleFunction` sampler), `target` (the model node to be sampled), and `control` (list of control arguments). The `MCMCconf` method `getSamplers` allows access to the `samplerConf` objects. These can be modified and then passed as an argument to `setSamplers` to over-write the current list of samplers in the MCMC configuration object. However, no checking of the validity of this modified list is performed; if the list of `samplerConf` objects is corrupted to be invalid, incorrect behavior will result at the time of calling `buildMCMC`. The fields of a `samplerConf` object can be modified using the access functions `setName(name)`, `setSamplerFunction(fun)`, `setTarget(target, model)`, and `setControl(control)`.

Here are some examples:

```
# retrieve samplerConf list
samplerConfList <- mcmcConf$getSamplers()

# change the name of the first sampler
samplerConfList[[1]]$setName("newNameForThisSampler")

# change the sampler function of the second sampler,
# assuming existence of a nimbleFunction 'anotherSamplerNF',
# which represents a valid nimbleFunction sampler.
samplerConfList[[2]]$setSamplerFunction(anotherSamplerNF)

# change the 'adaptive' element of the control list of the third sampler
control <- samplerConfList[[3]]$control
control$adaptive <- FALSE
```

```
samplerConfList[[3]]$setControl(control)

# change the target node of the fourth sampler
samplerConfList[[4]]$setTarget("y", model) # model argument required

# use this modified list of samplerConf objects in the MCMC configuration
mcmcConf$setSamplers(samplerConfList)
```

7.2.2.7 Customizing the sampler execution order

The ordering of sampler execution can be controlled as well. This allows for sampler functions to execute multiple times within a single MCMC iteration, or the execution of different sampler functions to be interleaved with one another.

The sampler execution order is set using the function `setSamplerExecutionOrder`, and the current ordering of execution is retrieved using `getSamplerExecutionOrder`. For example, assuming the MCMC configuration object `mcmcConf` contains five samplers:

```
# first sampler to execute twice, in succession:
mcmcConf$setSamplerExecutionOrder(c(1, 1, 2, 3, 4, 5))

# first sampler to execute multiple times, interleaved:
mcmcConf$setSamplerExecutionOrder(c(1, 2, 1, 3, 1, 4, 1, 5))

# fourth sampler to execute 10 times, only
mcmcConf$setSamplerExecutionOrder(rep(4, 10))

# omitting the argument to setSamplerExecutionOrder()
# resets the ordering to each sampler executing once, sequentially
mcmcConf$setSamplerExecutionOrder()

# retrieve the current ordering of sampler execution
ordering <- mcmcConf$getSamplerExecutionOrder()

# print the sampler functions in the order of execution
mcmcConf$printSamplers(executionOrder = TRUE)
```

7.2.2.8 Monitors and thinning intervals: *printMonitors*, *getMonitors*, *setMonitors*, *addMonitors*, *resetMonitors* and *setThin*

An MCMC configuration object contains two independent sets of variables to monitor, each with their own thinning interval: `thin` corresponding to `monitors`, and `thin2` corresponding to `monitors2`. Monitors operate at the *variable* level. Only entire model variables may be monitored. Specifying a monitor on a *node*, e.g., `x[1]`, will result in the entire variable `x` being monitored.

The variables specified in `monitors` and `monitors2` will be recorded (with thinning interval `thin`) into objects called `mvSamples` and `mvSamples2`, contained within the MCMC object. These are both *modelValues* objects; *modelValues* are NIMBLE data structures used to store multiple sets of

values of model variables¹. These can be accessed as the member data `mvSamples` and `mvSamples2` of the MCMC object, and they can be converted to matrices using `as.matrix` or lists using `as.list` (see Section 7.7).

Monitors may be added to the MCMC configuration either in the original call to `configureMCMC` or using the `addMonitors` method:

```
# Using an argument to configureMCMC
mcmcConf <- configureMCMC(Rmodel, monitors = c("alpha", "beta"),
                          monitors2 = "x")

# Calling a member method of the mcmcConf object
# This results in the same monitors as above
mcmcConf$addMonitors("alpha", "beta")
mcmcConf$addMonitors2("x")
```

A new set of monitor variables can be added to the MCMC configuration, overwriting the current monitors, using the `setMonitors` method:

```
# Replace old monitors, now monitor "delta" and "gamma" only
mcmcConf$setMonitors("gamma", "delta")
```

Similarly, either thinning interval may be set at either step:

```
# Using an argument to configureMCMC
mcmcConf <- configureMCMC(Rmodel, thin = 1, thin2 = 100)

# Calling a member method of the mcmcConf object
# This results in the same thinning intervals as above
mcmcConf$setThin(1)
mcmcConf$setThin2(100)
```

The current lists of monitors and thinning intervals may be displayed using the `printMonitors` method. Both sets of monitors (`monitors` and `monitors2`) may be reset to empty character vectors by calling the `resetMonitors` method. The methods `getMonitors` and `getMonitors2` return the currently specified `monitors` and `monitors2` as character vectors.

7.2.2.9 Monitoring model log-probabilities

To record model log-probabilities from an MCMC, one can add monitors for *logProb* variables (which begin with the prefix `logProb_`) that correspond to variables with (any) stochastic nodes. For example, to record and extract log-probabilities for the variables `alpha`, `sigma_mu`, and `Y`:

```
mcmcConf <- configureMCMC(Rmodel, enableWAIC = TRUE)
mcmcConf$addMonitors("logProb_alpha", "logProb_sigma_mu", "logProb_Y")
Rmcmc <- buildMCMC(mcmcConf)
Cmodel <- compileNimble(Rmodel)
Cmcmc <- compileNimble(Rmcmc, project = Rmodel)
Cmcmc$run(10000)
samples <- as.matrix(Cmcmc$mvSamples)
```

¹See Section 15.1 for general information on `modelValues`.

The `samples` matrix will contain both MCMC samples and model log-probabilities.

7.2.2.10 Using numeric samples to define a prior distribution

A set of numeric samples, perhaps generated from another MCMC algorithm, can be used to define the prior distribution of model nodes. This is accomplished using the `prior_samples` MCMC sampler. When assigning the `prior_samples` sampler to a `target` node, you must also provide a vector of numeric values (when `target` is a scalar node), or a matrix of values in the multidimensional case. A new value (or rows of values) is selected either sequentially (or randomly) from this numeric vector/matrix, and assigned into the `target` node on each MCMC iteration. See `help(prior_samples)` for more details:

```
mcmcConf <- configureMCMC(Rmodel)
mcmcConf$addSampler(target = 'theta', type = 'prior_samples', samples = rnorm(100))
```

7.3 Building and compiling the MCMC

Once the MCMC configuration object has been created, and customized to one's liking, it may be used to build an MCMC function:

```
Rmcmc <- buildMCMC(mcmcConf)
```

`buildMCMC` is a `nimbleFunction`. The returned object `Rmcmc` is an instance of the `nimbleFunction` specific to configuration `mcmcConf` (and of course its associated model).

Note that if you would like to be able to calculate the WAIC of the model, you should usually set `enableWAIC = TRUE` as an argument to `configureMCMC` (or to `buildMCMC` if not using `configureMCMC`), or set `nimbleOptions(MCMCenableWAIC = TRUE)`, which will enable WAIC calculations for all subsequently built MCMC functions. For more information on WAIC calculations, including situations in which you can calculate WAIC without having set `enableWAIC = TRUE` see Section 7.8 or `help(waic)` in R.

When no customization is needed, one can skip `configureMCMC` and simply provide a model object to `buildMCMC`. The following two MCMC functions will be identical:

```
mcmcConf <- configureMCMC(Rmodel)    # default MCMC configuration
Rmcmc1 <- buildMCMC(mcmcConf)

Rmcmc2 <- buildMCMC(Rmodel)          # uses the default configuration for Rmodel
```

For speed of execution, we usually want to compile the MCMC function to C++ (as is the case for other NIMBLE functions). To do so, we use `compileNimble`. If the model has already been compiled, it should be provided as the `project` argument so the MCMC will be part of the same compiled project. A typical compilation call looks like:

```
Cmcmc <- compileNimble(Rmcmc, project = Rmodel)
```

Alternatively, if the model has not already been compiled, they can be compiled together in one line:

```
Cmcmc <- compileNimble(Rmodel, Rmcmc)
```

Note that if you compile the MCMC with another object (the model in this case), you'll need to explicitly refer to the MCMC component of the resulting object to be able to run the MCMC:

```
Cmcmc$Rmcmc$run(niter = 1000)
```

7.4 Initializing MCMC

To see how to provide initial values, see `help(runMCMC)`, `help(nimbleMCMC)`, or `help(nimbleModel)`.

It is often important to provide valid and reasonable initial values to an MCMC, either as fixed values or via an initialization function.

Not doing so can cause slow convergence or even failure of an MCMC algorithm (e.g., if the values the model is initialized with are not valid). When starting an MCMC, when NIMBLE encounters a missing parameter value, it simulates from the prior distribution. NIMBLE can be more sensitive to missing or bad starting values than some MCMC packages.

The following cases are particularly important to consider when initializing:

- In a model with flat priors (i.e., using `x~dflat()` or `x~dhalfflat()`), NIMBLE cannot generate initial values from those priors.
- In a model with diffuse priors, initializing from the prior can give unreasonable/extreme initial values.
- In a model with stochastic indices (e.g., `x[idx[i]]` with `idx[i]` unknown), those indices should have (valid) initial values.
- In a model with constraints (via `dconstraint`), the model should be initialized such that the constraints are satisfied.
- In a model with censoring (via `dinterval`), the initial value(s) of `t` in `dinterval` for the censored node(s) should be consistent with the censoring that is specified.

That said, some MCMC algorithms (such a conjugate samplers) don't use the initial values, so flat or diffuse priors on nodes assigned such samplers will not be a problem with regard to initialization.

Many user questions and problems end up being related to missing or bad initial values. Some suggestions for diagnosing initialization problems include:

- Run `model$initializeInfo()` to find uninitialized nodes.
- Run `model$calculate()` to see if the full model log probability density can be calculated (values of `-Inf`, `NA`, `NaN` indicate there are missing or invalid initial values).
- To debug the specific node(s) whose initial value(s) are causing problems, run `model$calculate(nodes)` on specific `nodes` to find their log probability density values, looking look for values of `-Inf`, `NA`, or `NaN`.
- For the node(s) with invalid log probability density, inspect the values of the node(s) and parameter(s) of the distribution(s) assigned to those node(s). You can inspect values of a variable in the model using natural R syntax, `model$variable_name`.
- With a compiled or uncompiled model, you can run the model initialization (but no MCMC iterations) with `mcmc$run(niter = 0)`. Then inspect the values of model variables to see the results of initialization.

7.5 User-friendly execution of MCMC algorithms: *runMCMC*

Once an MCMC algorithm has been created using `buildMCMC`, the function `runMCMC` can be used to run multiple chains and extract posterior samples, summary statistics and/or a WAIC value. This is a simpler approach to executing an MCMC algorithm, than the process of executing and extracting samples as described in Sections 7.6 and 7.7.

`runMCMC` also provides several user-friendly options such as burn-in, thinning, running multiple chains, and different initial values for each chain. However, using `runMCMC` does not support several lower-level options, such as timing the individual samplers internal to the MCMC, continuing an existing MCMC run (picking up where it left off), or modifying the sampler execution ordering.

`runMCMC` takes arguments that will control the following aspects of the MCMC:

- Number of iterations in each chain;
- Number of chains;
- Number of burn-in samples to discard from each chain;
- Thinning interval for recording samples;
- Initial values, or a function for generating initial values for each chain;
- Setting the random number seed;
- Returning the posterior samples as a `coda mcmc` object;
- Returning summary statistics calculated from each chains; and
- Returning a WAIC value calculated using (post-burn-in) samples from all chains.

The following examples demonstrate some uses of `runMCMC`, and assume the existence of `Cmcmc`, a compiled MCMC algorithm.

```
# run a single chain, and return a matrix of samples
mcmc.out <- runMCMC(Cmcmc)

# run three chains of 10000 samples, discard initial burn-in of 1000,
# record samples thereafter using a thinning interval of 10,
# and return of list of sample matrices
mcmc.out <- runMCMC(Cmcmc, niter=10000, nburnin=1000, thin=10, nchains=3)

# run three chains, returning posterior samples, summary statistics,
# and the WAIC value
mcmc.out <- runMCMC(Cmcmc, nchains = 3, summary = TRUE, WAIC = TRUE)

# run two chains, and specify the initial values for each
initsList <- list(list(mu = 1, sigma = 1),
                 list(mu = 2, sigma = 10))
mcmc.out <- runMCMC(Cmcmc, nchains = 2, inits = initsList)

# run ten chains of 100,000 iterations each, using a function to
# generate initial values and a fixed random number seed for each chain.
# only return summary statistics from each chain; not all the samples.
initsFunction <- function()
  list(mu = rnorm(1,0,1), sigma = runif(1,0,100))
mcmc.out <- runMCMC(Cmcmc, niter = 100000, nchains = 10,
```



```
inits = initsFunction, setSeed = TRUE,
samples = FALSE, summary = TRUE)
```

See `help(runMCMC)` for further details.

7.6 Running the MCMC

The MCMC algorithm (either the compiled or uncompiled version) can be executed using the member method `mcmc$run` (see `help(buildMCMC)` in R). The `run` method has one required argument, `niter`, the number of iterations to be run.

The `run` method has optional arguments `nburnin`, `thin` and `thin2`. These can be used to specify the number of pre-thinning burn-in samples to discard, and the post-burnin thinning intervals for recording samples (corresponding to `monitors` and `monitors2`). If either `thin` and `thin2` are provided, they will override the thinning intervals that were specified in the original MCMC configuration object.

7.6.1 Rerunning versus restarting an MCMC

The `run` method has an optional `reset` argument. When `reset = TRUE` (the default value), the following occurs prior to running the MCMC:

1. All model nodes are checked and filled or updated as needed, in valid (topological) order. If a stochastic node is missing a value, it is populated using a call to `simulate` and its log probability value is calculated. The values of deterministic nodes are calculated from their parent nodes. If any right-hand-side-only nodes (e.g., explanatory variables) are missing a value, an error results.
2. All MCMC sampler functions are reset to their initial state: the initial values of any sampler control parameters (e.g., `scale`, `sliceWidth`, or `propCov`) are reset to their initial values, as were specified by the original MCMC configuration.
3. The internal modelValues objects `mvSamples` and `mvSamples2` are each resized to the appropriate length for holding the requested number of samples (`niter/thin`, and `niter/thin2`, respectively).

This means that one can begin a new run of an existing MCMC without having to rebuild or recompile the model or the MCMC. This can be helpful if one wants to use the same model and MCMC configuration, but with different initial values, different values of data nodes (but which nodes are data nodes must be the same), changes to covariate values (or other non-data, non-parameter values) in the model, or a different number of MCMC iterations, thinning interval or burn-in.

In contrast, when `mcmc$run(niter, reset = FALSE)` is called, the MCMC picks up from where it left off, continuing the previous chain and expanding the output as needed. No values in the model are checked or altered, and sampler functions are not reset to their initial states. (Similarly, if using WAIC, one can use `resetWAIC = FALSE` so that the WAIC calculation is based on all the samples from the expanded set of samples.)

The `run` method also has an optional `resetMV` argument. This argument is only considered when `reset` is set to `FALSE`. When `mcmc$run(niter, reset = FALSE, resetMV = TRUE)` is called the internal modelValues objects `mvSamples` and `mvSamples2` are each resized to the appropriate

length for holding the requested number of samples (`niter/thin`, and `niter/thin2`, respectively) and the MCMC carries on from where it left off. In other words, the previously obtained samples are deleted (e.g. to reduce memory usage) prior to continuing the MCMC. The default value of `resetMV` is `FALSE`.

7.6.2 Measuring sampler computation times: *getTimes*

If you want to obtain the computation time spent in each sampler, you can set `time=TRUE` as a run-time argument and then use the method `getTimes()` obtain the times. For example,

```
Cmcmc$run(niter, time = TRUE)
Cmcmc$getTimes()
```

will return a vector of the total time spent in each sampler, measured in seconds.

7.6.3 Assessing the adaption process of *RW* and *RW_block* samplers

If you'd like to see the evolution (over the iterations) of the acceptance proportion and proposal scale information, you can use some internal methods provided by NIMBLE, after setting two options to make the history accessible. Here we assume that `cMCMC` is the compiled MCMC object and `idx` is the numeric index of the sampler function you want to access from amongst the list of sampler functions that are part of the MCMC.

```
## set options to make history accessible
nimbleOptions(buildInterfacesForCompiledNestedNimbleFunctions = TRUE)
nimbleOptions(MCMCsaveHistory = TRUE)
## Next, set up and run your MCMC.
## Now access the history information:
Cmcmc$samplerFunctions[[idx]]$getScaleHistory()
Cmcmc$samplerFunctions[[idx]]$getAcceptanceHistory()
Cmcmc$samplerFunctions[[idx]]$getPropCovHistory() ## only for RW_block
```

Note that modifying elements of the control list may greatly affect the performance of the `RW_block` sampler. In particular, the sampler can take a long time to find a good proposal covariance when the elements being sampled are not on the same scale. We recommend providing an informed value for 'propCov' in this case (possibly simply a diagonal matrix that approximates the relative scales), as well as possibly providing a value of 'scale' that errs on the side of being too small. You may also consider decreasing 'adaptFactorExponent' and/or 'adaptInterval', as doing so has greatly improved performance in some cases.

7.7 Extracting MCMC samples

After executing the MCMC, the output samples can be extracted as follows:

```
mvSamples <- cmcmc$mvSamples
mvSamples2 <- cmcmc$mvSamples2
```

These *modelValues* objects can be converted into matrices using `as.matrix` or lists using `as.list`:

```
samplesMatrix <- as.matrix(mvSamples)
samplesList <- as.list(mvSamples)
```

```
samplesMatrix2 <- as.matrix(mvSamples2)
samplesList2 <- as.list(mvSamples2)
```

The column names of the matrices will be the node names of nodes in the monitored variables. Then, for example, the mean of the samples for node `x[2]` could be calculated as:

```
mean(samplesMatrix[, "x[2]"])
```

The list version will contain an element for each variable that will be the size and shape of the variable with an additional index for MCMC iteration. By default MCMC iteration will be the first index, but including `iterationAsLastIndex = TRUE` will make it the last index.

Obtaining samples as matrices or lists is most common, but see Section 15.1 for more about programming with `modelValues` objects, especially if you want to write `nimbleFunctions` to use the samples.

7.8 Calculating WAIC

The WAIC (Watanabe, 2010) can be calculated from the posterior samples produced during the MCMC algorithm. Users have two options to calculate WAIC. The main approach requires enabling WAIC when setting up the MCMC and allows access to a variety of ways to calculate WAIC. This approach does not require that any specific monitors be set² because the WAIC is calculated in an online manner, accumulating the necessary quantities to calculate WAIC as the MCMC is run.

The second approach allows one to calculate the WAIC after an MCMC has been run using an MCMC object or matrix (or dataframe) containing the posterior samples, but it requires the user to have correctly specified the monitored variables and only provides the default way to calculate WAIC. An advantage of the second approach is that one can specify additional burnin beyond that specified for the MCMC.

Here we first discuss the main approach and then close the section by showing the second approach. Specific details of the syntax are provided in `help(waic)`.

To enable WAIC for the first approach, the argument `enableWAIC = TRUE` must be supplied to `configureMCMC` (or to `buildMCMC` if not using `configureWAIC`), or the `MCMCenableWAIC` NIMBLE option must have been set to `TRUE`.

The WAIC (as well as the `pWAIC` and `lppd` values) is extracted by the member method `mcmc$getWAIC` (see `help(waic)` in R for more details) or is available as the `WAIC` element of the `runMCMC` or `nimbleMCMC` output lists. One can use the member method `mcmc$getWAICdetails` for additional quantities related to WAIC that are discussed below.

Note that there is not a unique value of WAIC for a model. WAIC is calculated from Equations 5, 12, and 13 in Gelman et al. (2014) (i.e., using `pWAIC2`), as discussed in detail in Hug and Paciorek (2021). Therefore, NIMBLE provides user control over how WAIC is calculated in two ways.

First, by default NIMBLE provides the conditional WAIC, namely the version of WAIC where all parameters directly involved in the likelihood are treated as θ for the purposes of Equation 5 from Gelman et al. (2014). Users can request the marginal WAIC (see Ariyo et al. (2020)), namely the

²Prior to version 0.12.0, NIMBLE required specific monitors, because WAIC was calculated at the end of the MCMC using the entire set of MCMC samples.

version of WAIC where latent variables are integrated over (i.e., using a marginal likelihood). This is done by providing the `waicControl` list with a `marginalizeNodes` element to `configureMCMC` or `buildMCMC` (when providing a model as the argument to `buildMCMC`). See `help(waic)` for more details.

Second, WAIC relies on a partition of the observations, i.e., ‘pointwise’ prediction. By default, in NIMBLE the sum over log pointwise predictive density values treats each data node as contributing a single value to the sum. When a data node is multivariate, that data node contributes a single value to the sum based on the joint density of the elements in the node. If one wants to group data nodes such that the joint density within each group is used, one can provide the `waicControl` list with a `dataGroups` element to `configureMCMC` or `buildMCMC` (when providing a model as the argument to `buildMCMC`). See `help(waic)` for more details.

Note that based on a limited set of simulation experiments in [Hug and Paciorek \(2021\)](#), our tentative recommendation is that users only use marginal WAIC if also using grouping.

Marginal WAIC requires using Monte Carlo simulation at each iteration of the MCMC to average over draws for the latent variables. To assess the stability of the marginal WAIC to the number of Monte Carlo iterations, one can examine how the WAIC changes with increasing iterations (up to the full number of iterations specified via the `nItsMarginal` element of the `waicControl` list) based on the `WAIC_partialMC`, `lppd_partialMC`, and `pWAIC_partialMC` elements of the detailed WAIC output.

For comparing WAIC between two models, [Vehtari et al. \(2017\)](#) discuss using the per-observation (more generally, per-data group) contributions to the overall WAIC values to get an approximate standard error for the difference in WAIC between the models. These element-wise values are available as the `WAIC_elements`, `lppd_elements`, and `pWAIC_elements` components of the detailed WAIC output.

The second overall approach to WAIC available in NIMBLE allows one to calculate WAIC post hoc after MCMC sampling using an MCMC object or matrix (or dataframe) containing posterior samples. Simply set up the MCMC and run it without enabling WAIC (but making sure to include in the monitors all stochastic parent nodes of the data nodes) and then use the function `calculateWAIC`, as shown in this example:

```
samples <- runMCMC(Cmcmc, niter = 10000, nburnin = 1000)
## Using calculateWAIC with an MCMC object
calculateWAIC(Cmcmc)
## Using calculateWAIC with a matrix of samples
calculateWAIC(samples, model)
## Specifying additional burnin, so only last 5000 (1000+4000) iterations used
calculateWAIC(Cmcmc, burnin = 4000)
```

This approach only provides conditional WAIC without any grouping of data nodes (though one can achieve grouping by grouping data nodes into multivariate nodes).

7.9 k-fold cross-validation

The `runCrossValidate` function in NIMBLE performs k-fold cross-validation on a `nimbleModel` fit via MCMC. More information can be found by calling `help(runCrossValidate)`.

7.10 Variable selection using Reversible Jump MCMC

A common method for Bayesian variable selection in regression-style problems is to define a space of different models and have MCMC sample over the models as well as the parameters for each model. The models differ in which explanatory variables are included. Often this idea is implemented in the BUGS language by writing the largest possible model and including indicator variables to turn regression parameters off and on (see model code below for an example of how indicator variables can be used). Foramally, that approach doesn't sample between different models, instead embedding the models in one large model. However, that approach can result in poor mixing and require compromises in choices of prior distributions because a given regression parameter will sample from its prior when the corresponding indicator is 0. It is also computationally wasteful, because sampling effort will be spent on a coefficient even if it has no effect because the corresponding indicator is 0.

A different view of the problem is that the different combinations of coefficients represent models of different dimensions. Reversible Jump MCMC ([Green, 1995](#)) (RJMCMC) is a general framework for MCMC simulation in which the dimension of the parameter space can vary between iterates of the Markov chain. The reversible jump sampler can be viewed as an extension of the Metropolis-Hastings algorithm onto more general state spaces. NIMBLE provides an implementation of RJMCMC for variable selection that requires the user to write the largest model of interest and supports use of indicator variables but formally uses RJMCMC sampling for better mixing and efficiency. When a coefficient is not in the model (or its indicator is 0), it will not be sampled, and it will therefore not follow its prior in that case.

In technical detail, given two models M_1 and M_2 of possibly different dimensions, the core idea of RJMCMC is to remove the difference in the dimensions of models M_1 and M_2 by supplementing the corresponding parameters θ_1 and θ_2 with auxiliary variables $u_{1 \rightarrow 2}$ and $u_{2 \rightarrow 1}$ such that $(\theta_1, u_{1 \rightarrow 2})$ and $(\theta_2, u_{2 \rightarrow 1})$ are in bijection, $(\theta_2, u_{2 \rightarrow 1}) = \Psi(\theta_1, u_{1 \rightarrow 2})$. The corresponding Metropolis-Hastings acceptance probability is generalized accounting for the proposal density for the auxiliary variables.

NIMBLE implements RJMCMC for variable selection using a univariate normal distribution centered on 0 (or some fixed value) as the proposal density for parameters being added to the model. Two ways to set up models for RJMCMC are supported, which differ by whether the inclusion probabilities for each parameter are assumed known or must be written in the model:

- If the inclusion probabilities are assumed known, then RJMCMC may be used with regular model code, i.e. model code written without heed to variable selection.
- If the inclusion probability is a parameter in the model, perhaps with its own prior, then RJMCMC requires that indicator variables be written in the model code. The indicator variables will be sampled using RJMCMC, but otherwise they are like any other nodes in the model.

The steps to set up variable selection using RJMCMC are:

1. Write the model with indicator variables if needed.
2. Configure the MCMC as usual with `configureMCMC`.
3. Modify the resulting MCMC configuration object with `configureRJ`.

The `configureRJ` function modifies the MCMC configuration to (1) assign samplers that turn on and off variable in the model and (2) modify the existing samplers for the regression coefficients to use 'toggled' versions. The toggled versions invoke the samplers only when corresponding variable

is currently in the model. In the case where indicator variables are included in the model, sampling to turn on and off variables is done using `RJ_indicator` samplers. In the case where indicator variables are not included, sampling is done using `RJ_fixed_prior` samplers.

In the following we provide an example for the two different model specifications.

7.10.1 Using indicator variables

Here we consider a normal linear regression in which two covariates `x1` and `x2` are candidates to be included in the model. The two corresponding coefficients are `beta1` and `beta2`. The indicator variables are `z1` and `z2`, and their inclusion probability is `psi`. As described below, one can also use vectors for a set of coefficients and corresponding indicator variables.

```
## Linear regression with intercept and two covariates
code <- nimbleCode({
  beta0 ~ dnorm(0, sd = 100)
  beta1 ~ dnorm(0, sd = 100)
  beta2 ~ dnorm(0, sd = 100)
  sigma ~ dunif(0, 100)
  z1 ~ dbern(psi)  ## indicator variable associated with beta1
  z2 ~ dbern(psi)  ## indicator variable associated with beta2
  psi ~ dbeta(1, 1) ## hyperprior on inclusion probability
  for(i in 1:N) {
    Ypred[i] <- beta0 + beta1 * z1 * x1[i] + beta2 * z2 * x2[i]
    Y[i] ~ dnorm(Ypred[i], sd = sigma)
  }
})

## simulate some data
set.seed(1)
N <- 100
x1 <- runif(N, -1, 1)
x2 <- runif(N, -1, 1) ## this covariate is not included
Y <- rnorm(N, 1.5 + 2 * x1, sd = 1)

## build the model and configure default MCMC

RJexampleModel <- nimbleModel(code, constants = list(N = N),
                              data = list(Y = Y, x1 = x1, x2 = x2),
                              inits = list(beta0 = 0, beta1 = 0, beta2 = 0,
                                             sigma = sd(Y), z2 = 1, z1 = 1, psi = 0.5))

RJexampleConf <- configureMCMC(RJexampleModel)

## For illustration, look at the default sampler assignments
RJexampleConf$printSamplers()

## [1] conjugate_dnorm_dnorm_additive sampler: beta0
## [2] conjugate_dnorm_dnorm_linear sampler: beta1
```

```
## [3] conjugate_dnorm_dnorm_linear sampler: beta2
## [4] RW sampler: sigma
## [5] conjugate_dbeta_dbern_identity sampler: psi
## [6] binary sampler: z1
## [7] binary sampler: z2
```

At this point we can modify the MCMC configuration object, `RJexampleConf`, to use reversible jump samplers for selection on `beta1` and `beta2`.

```
configureRJ(conf = RJexampleConf,
            targetNodes = c("beta1", "beta2"),
            indicatorNodes = c('z1', 'z2'),
            control = list(mean = c(0, 0), scale = 2))
```

The `targetNodes` argument gives the coefficients (nodes) for which we want to do variable selection. The `indicatorNodes` gives the corresponding indicator nodes, ordered to match `targetNodes`. The `control` list gives the means and scale (standard deviation) for normal reversible jump proposals for `targetNodes`. The means will typically be 0 (by default `mean = 0` and `scale = 1`), but they could be anything. An optional `control` element called `fixedValue` can be provided in the non-indicator setting; this gives the value taken by nodes in `targetNodes` when they are out of the model (by default, `fixedValue` is 0). All `control` elements can be scalars or vectors. A scalar values will be used for all `targetNodes`. A vector value must be of equal length as `targetNodes` and will be used in order.

To use RJMCMC on a vector of coefficients with a corresponding vector of indicator variables, simply provide the variable names for `targetNodes` and `indicatorNodes`. For example, `targetNodes = "beta"` is equivalent to `targetNodes = c("beta[1]", "beta[2]")` and `indicatorNodes = "z"` is equivalent to `indicatorNodes = c("z[1]", "z[2]")`. Expansion of variable names into a vector of node names will occur as described in Section 14.3.1.1. When using this method, *both* arguments must be provided as variable names to be expanded.

Next we can see the result of `configureRJ` by printing the modified list of samplers:

```
RJexampleConf$printSamplers()

## [1] conjugate_dnorm_dnorm_additive sampler: beta0
## [2] RW sampler: sigma
## [3] conjugate_dbeta_dbern_identity sampler: psi
## [4] RJ_indicator sampler: z1, mean: 0, scale: 2, targetNode: beta1
## [5] RJ_toggled sampler: beta1, samplerType: conjugate_dnorm_dnorm_linear
## [6] RJ_indicator sampler: z2, mean: 0, scale: 2, targetNode: beta2
## [7] RJ_toggled sampler: beta2, samplerType: conjugate_dnorm_dnorm_linear
```

An `RJ_indicator` sampler was assigned to each of `z[1]` and `z[2]` in place of the `binary` sampler, while the samplers for `beta[1]` and `beta[2]` have been changed to `RJ_toggled` samplers. The latter samplers contain the original samplers, in this case `conjugate_dnorm_dnorm` samplers, and use them only when the corresponding indicator variable is equal to 1 (i.e., when the coefficient is in the model).

Notice that the order of the samplers has changed, since `configureRJ` calls `removeSampler` for nodes in `targetNodes` and `indicatorNodes`, and subsequently `addSampler`, which appends the sampler to the end of current sampler list. Order can be modified by using `setSamplers`.

Also note that `configureRJ` modifies the MCMC configuration (first argument) and returns `NULL`.

7.10.2 Without indicator variables

We consider the same regression setting, but without the use of indicator variables and with fixed probabilities of including each coefficient in the model.

```
## Linear regression with intercept and two covariates
code <- nimbleCode({
  beta0 ~ dnorm(0, sd = 100)
  beta1 ~ dnorm(0, sd = 100)
  beta2 ~ dnorm(0, sd = 100)
  sigma ~ dunif(0, 100)
  for(i in 1:N) {
    Ypred[i] <- beta0 + beta1 * x1[i] + beta2 * x2[i]
    Y[i] ~ dnorm(Ypred[i], sd = sigma)
  }
})

## build the model
RJexampleModel2 <- nimbleModel(code, constants = list(N = N),
                               data = list(Y = Y, x1 = x1, x2 = x2),
                               inits = list(beta0 = 0, beta1 = 0, beta2 = 0,
                                             sigma = sd(Y)))

RJexampleConf2 <- configureMCMC(RJexampleModel2)

## print NIMBLE default samplers
RJexampleConf2$printSamplers()

## [1] conjugate_dnorm_dnorm_additive sampler: beta0
## [2] conjugate_dnorm_dnorm_linear sampler: beta1
## [3] conjugate_dnorm_dnorm_linear sampler: beta2
## [4] RW sampler: sigma

In this case, since there are no indicator variables, we need to pass to configureRJ the prior inclusion probabilities for each node in targetNodes, by specifying either one common value or a vector of values for the argument priorProb. This case does not allow for a stochastic prior.

configureRJ(conf = RJexampleConf2,
            targetNodes = c("beta1", "beta2"),
            priorProb = 0.5,
            control = list(mean = 0, scale = 2, fixedValue = c(1.5, 0)))

## print samplers after configureRJ
RJexampleConf2$printSamplers()

## [1] conjugate_dnorm_dnorm_additive sampler: beta0
## [2] RW sampler: sigma
## [3] RJ_fixed_prior sampler: beta1, priorProb: 0.5, mean: 0, scale: 2, fixedValue: 1.5
## [4] RJ_toggled sampler: beta1, samplerType: conjugate_dnorm_dnorm_linear, fixedValue: 1.5
```



```
## [5] RJ_fixed_prior sampler: beta2, priorProb: 0.5, mean: 0, scale: 2, fixedValue: 0
## [6] RJ_toggled sampler: beta2, samplerType: conjugate_dnorm_dnorm_linear, fixedValue: 0
```

Since there are no indicator variables, the `RJ_fixed_prior` sampler is assigned directly to each of `beta[1]` and `beta[2]` along with the `RJ_toggled` sampler. The former sets a coefficient to its `fixedValue` when it is out of the model. The latter invokes the regular sampler for the coefficient only if it is in the model at a given iteration.

If `fixedValue` is given when using `indicatorNodes` the values provided in `fixedValue` are ignored. However the same behavior can be obtained in this situation, using a different model specification. For example, the model in 7.10.1 can be modified to have `beta1` equal to 1.5 when not in the model as follows:

```
for(i in 1:N) {
  Ypred[i] <- beta0 + (1 - z1) * 1.5 * beta1 * x1[i] +
               z1 * beta1 * x1[i] + beta2 * z2 * x2[i]
  Y[i] ~ dnorm(Ypred[i], sd = sigma)
}
```

7.11 Samplers provided with NIMBLE

NIMBLE provides a variety of other commonly-used samplers, some for general-purpose use and some for specialized use with particular distributions or situations. Some that are worth particular consideration as they may perform better than the samplers that NIMBLE assigns by default are:

1. the Hamiltonian Monte Carlo (HMC) sampler in the `nimbleHMC` package,
2. the slice sampler for sampling scalar parameters in place of the default RW (Metropolis) sampler,
3. the Barker sampler for sampling multiple parameters in place of the default `RW_block` (block Metropolis) sampler,
4. the noncentered sampler (`noncentered`) for jointly sampling random effects with their hyperparameters,
5. the Pólya-gamma sampler (`polyagamma`) for sampling regression coefficients in logistic regression specifications, and
6. the reversible-jump sampler for selection of coefficients in regression specifications.

Most documentation of MCMC samplers provided with NIMBLE can be found by invoking `help(samplers)` in R.

In the remainder of this section, we provide additional explanation of conjugate samplers, the HMC sampler in the `nimbleHMC` package, and how complete customization can be achieved by making a sampler use an arbitrary log-likelihood function, such as to build a particle MCMC algorithm.

7.11.1 Conjugate (‘Gibbs’) samplers

By default, `configureMCMC()` and `buildMCMC()` will assign conjugate samplers to all nodes satisfying a conjugate relationship, unless the option `useConjugacy = FALSE` is specified.

The current release of NIMBLE supports conjugate sampling of the relationships listed in Table 7.1³.

³NIMBLE’s internal definitions of these relationships can be viewed with

Table 7.1: Conjugate relationships supported by NIMBLE's MCMC engine.

Prior Distribution	Sampling (Dependent Node) Distribution	Parameter
Beta	Bernoulli	prob
	Binomial	prob
	Negative Binomial	prob
Dirichlet	Multinomial	prob
	Categorical	prob
Flat	Normal	mean
	Lognormal	meanlog
Gamma	Poisson	lambda
	Normal	tau
	Lognormal	taulog
	Gamma	rate
	Inverse Gamma	scale
	Exponential	rate
	Double Exponential	rate
	Weibull	lambda
Halfflat	Normal	sd
	Lognormal	sdlog
Inverse Gamma	Normal	var
	Lognormal	varlog
	Gamma	scale
	Inverse Gamma	rate
	Exponential	scale
	Double Exponential	scale
Normal	Normal	mean
	Lognormal	meanlog
Multivariate Normal	Multivariate Normal	mean
Wishart	Multivariate Normal	prec
Inverse Wishart	Multivariate Normal	cov

Conjugate sampler functions may (optionally) dynamically check that their own posterior likelihood calculations are correct. If incorrect, a warning is issued. However, this functionality will roughly double the run-time required for conjugate sampling. By default, this option is disabled in NIMBLE. This option may be enabled with `nimbleOptions(verifyConjugatePosteriors = TRUE)`.

If one wants information about conjugate node relationships for other purposes, they can be obtained using the `checkConjugacy` method on a model. This returns a named list describing all conjugate relationships. The `checkConjugacy` method can also accept a character vector argument specifying a subset of node names to check for conjugacy.

`nimble:::conjugacyRelationshipsInputList.`

7.11.2 Hamiltonian Monte Carlo (HMC)

As of version 1.0.0, NIMBLE provides support for automatic differentiation (AD), as described in Chapter 17. AD is needed for Hamiltonian Monte Carlo (HMC) sampling. The HMC algorithm is provided through the `nimbleHMC` package, which requires version 1.0.0 (or higher) of the `nimble` package.

HMC is an MCMC sampling algorithm that works by a physical analogy of frictionless motion on a surface that is the negative log posterior density. HMC often achieves good mixing, although at high computational cost. HMC sampling can operate on any continuous-valued model dimensions, including multiple dimensions simultaneously. HMC sampling is more flexibly applicable using `nimbleHMC` than in other implementations, since HMC can be applied to some parts of a model while other samplers are used on other (potentially overlapping) parts of the same model. Indeed, Neal (2011) discusses this mixed application of HMC with other sampling methods.

The `nimbleHMC` package provides two distinct HMC samplers, which implement two versions of No-U-Turn (NUTS) HMC sampling. HMC-NUTS sampling is a self-tuning version of standard HMC, which self-tunes the step-size and the number of integration steps used in the numeric integrator’s exploration of parameter space. The `NUTS_classic` sampler in `nimbleHMC` implements the original (“classic”) HMC-NUTS algorithm developed in Hoffman and Gelman (2014), which was the seminal version of HMC-NUTS. The `NUTS` sampler in `nimbleHMC` is a modern version of HMC-NUTS sampling with improved adaptation routines and convergence criteria, which matches the HMC sampler available in version 2.32.2 of Stan (Stan Development Team, 2023).

The samplers provided in `nimbleHMC` can be used in `nimble`’s general MCMC system, and may be used in combination with other samplers provided with `nimble` operating on different (or overlapping) parts of the model. For convenience, `nimbleHMC` provides several different ways to set up and run HMC which allow different degrees of control:

- `nimbleHMC` is analogous to `nimbleMCMC`: It accepts model `code`, `data`, `inits` and `constants` arguments, and performs all steps from building the model to running one or more MCMC chains. Use the `nimbleHMC` function if you want one call to apply NUTS sampling to all continuous-valued model dimensions, and `nimble`’s default samplers for all discrete-valued dimensions.
- `buildHMC` is analogous to `buildMCMC`: It accepts a model object as an argument, and builds an MCMC algorithm which applies NUTS sampling to all continuous-valued model dimensions, and `nimble`’s default samplers for all discrete-valued dimensions. Use `buildHMC` if you have created a model object, and want to automatically build the default HMC algorithm for your model.
- `configureHMC` is analogous to `configureMCMC`: It accepts a model object as an argument, and sets up a default MCMC configuration containing one NUTS sampler operating on all continuous-valued model dimensions, and `nimble`’s default samplers for all discrete-valued dimensions. You can also specify which `nodes` should have samplers assigned. You can then modify the MCMC configuration (use `addSampler` and other methods) to further customize it before building and running the MCMC. Use `configureHMC` if you are familiar with MCMC configuration objects, and may want to further customize your MCMC using HMC or other samplers or monitors before building the MCMC.
- `addHMC` manages use of `addSampler` to specifically add an HMC sampler for a chosen set of nodes to an existing MCMC configuration. It optionally removes other samplers operating on the specified nodes. Use `addHMC` to add an HMC sampler operating one or more target

nodes to an existing MCMC configuration object.

- For the most fine-grained control, you can use the `addSampler` method specifying the the argument `type="NUTS"` or `type="NUTS_classic"` to add HMC sampler(s) to an MCMC configuration object.

As of version 1.1.0, we now support the following new functionality for AD in models:

- Dynamic indexing is supported, although of course one cannot take derivatives with respect to the actual index values, since they are discrete. This means the index values cannot be sampled by HMC.
- One can take derivatives with respect to nodes assigned CAR priors, so HMC sampling can be performed on variables assigned a CAR prior(i.e., `dcar_normal` and `dcar_proper`).
- One can take derivatives with respect to the parameters of `dcat`, `dconstraint`, and `dinterval`. Hence HMC sampling is supported for *parameters* of nodes that have `dcat`, `dconstraint`, and `dinterval` dependencies. (However, the nodes that follow one of these distributions are discrete, so derivatives with respect to those nodes themselves are not supported, and HMC cannot sample those nodes.)
- Truncated distributions can be used, but one cannot take derivatives with respect to their parameters or the nodes themselves. This means that HMC sampling **cannot** be performed on nodes that have truncated nodes as dependencies.

More information is available from the help pages, e.g. `help(nimbleHMC)`, `help(buildHMC)`, `help(configureHMC)`, or `help(addHMC)`.

7.11.2.1 HMC example

Here we will configure and build an MCMC to use the NUTS HMC sampler on a GLMM example. In this case we'll use HMC sampling for the whole model, i.e., all parameters and latent states.

We'll use a Poisson Generalized Linear Mixed Model (GLMM) as a simple example model for HMC (and also for Laplace approximation in 9.1). There will be 10 groups (indexed by `i`) of 5 observations (indexed by `j`) each. Each observation has a covariate, `X`, and each group has a random effect `ran_eff`. Here is the model code:

```
model_code <- nimbleCode({
  # priors
  intercept ~ dnorm(0, sd = 100)
  beta ~ dnorm(0, sd = 100)
  sigma ~ dunif(0, 10)
  # random effects and data
  for(i in 1:10) {
    # random effects
    ran_eff[i] ~ dnorm(0, sd = sigma)
    for(j in 1:5) {
      # data
      y[i,j] ~ dpois(exp(intercept + beta*X[i,j] + ran_eff[i]))
    }
  }
})
```

We'll simulate some values for `X`.

```
set.seed(123)
X <- matrix(rnorm(50), nrow = 10)
```

In order to allow an algorithm to use AD for a specific model, that model must be created with `buildDerivs = TRUE`. (Note that if you use the one-step method `nimbleHMC` described above, then you do not need to build the model yourself and so you do not need to provide this `buildDerivs` option.)

```
inits <- list(intercept = 0, beta = 0.2, sigma = 0.5)
model <- nimbleModel(model_code, constants = list(X = X), inits = inits,
  calculate = FALSE, buildDerivs = TRUE) # Here is the argument needed for AD.
```

To finish setting up the example we need to put data values into the model. We could have provided data in the call to `nimbleModel`, but instead we will simulate them using the model itself. Specifically, we will set parameter values, simulate data values, and then set those as the data to use.

```
model$calculate() # This will return NA because the model is not fully initialized.

## [1] NA

model$simulate(model$getDependencies('ran_eff'))
model$calculate() # Now the model is fully initialized: all nodes have valid values.

## [1] -80.74344

model$setData('y') # Now the model has y marked as data, with values from simulation.
```

Finally, we will make a compiled version of the model.

```
Cmodel <- compileNimble(model)
```

Now we will build an HMC algorithm for the parameters and random effects in the model.

```
library(nimbleHMC)
HMC <- buildHMC(model)

## ===== Monitors =====
## thin = 1: beta, intercept, sigma
## ===== Samplers =====
## NUTS sampler (1)
## - intercept, beta, sigma, ran_eff[1], ran_eff[2], ran_eff[3], ran_eff[4], ran_eff[5], ran_eff[6]

CHMC <- compileNimble(HMC, project = model)
# The compiled HMC will use the compiled model.
samples <- runMCMC(CHMC, niter = 1000, nburnin = 500) # short run for illustration

## [Note] NUTS sampler (nodes: intercept, beta, sigma, ran_eff[1], ran_eff[2], ran_eff[3], ran_eff[4], ran_eff[5], ran_eff[6])
## Since `warmupMode` is 'default' and `nburnin` > 0,
## the number of warmup iterations is equal to `nburnin`.
## The burnin samples will be discarded, and all samples returned will be post-warmup
```

We will not investigate results in detail, but here is a summary to see that reasonable results were

generated.

```
summary(coda::as.mcmc(samples))

##
## Iterations = 1:500
## Thinning interval = 1
## Number of chains = 1
## Sample size per chain = 500
##
## 1. Empirical mean and standard deviation for each variable,
##    plus standard error of the mean:
##
##           Mean      SD Naive SE Time-series SE
## beta      0.1790 0.1438 0.006432      0.008174
## intercept -0.2524 0.4463 0.019960      0.056677
## sigma     0.7912 0.4091 0.018296      0.051351
##
## 2. Quantiles for each variable:
##
##           2.5%      25%      50%      75%  97.5%
## beta      -0.1299 0.08311 0.1873 0.280133 0.4415
## intercept -1.6013 -0.42914 -0.1681 -0.002989 0.3520
## sigma     0.2935 0.53669 0.6886 0.943581 2.0846
```

7.11.2.2 Automatic parameter transformation

In the example above, `sigma` is a parameter that must be non-negative. HMC works only on unconstrained parameter spaces. NIMBLE's HMC functionality automatically transforms any constrained parameters to be in an unconstrained parameter space, runs the MCMC, and back-transforms to the original space before providing the MCMC samples. The details of the parameter transformation system can be found in Section 17.7.

7.11.3 Particle filter samplers

The `nimbleSMC` package provides samplers that perform particle MCMC, primarily intended for sampling top-level parameters in state-space or hidden Markov models of time-series data. This sampler is described in Section 8.1.2.

7.11.4 Customized log-likelihood evaluations: *RW_llFunction* sampler

Sometimes it is useful to control the log-likelihood calculations used for an MCMC updater instead of simply using the model. For example, one could use a sampler with a log-likelihood that analytically (or numerically) integrates over latent model nodes. Or one could use a sampler with a log-likelihood that comes from a stochastic approximation such as a particle filter (see below), allowing composition of a particle MCMC (PMCMC) algorithm (Andrieu et al., 2010). The `RW_llFunction` sampler handles this by using a Metropolis-Hastings algorithm with a normal proposal distribution and a user-provided log-likelihood function. To allow compiled execution, the log-likelihood function must be provided as a specialized instance of a `nimbleFunction`. The log-likelihood function

may use the same model as the MCMC as a setup argument (as does the example below), but if so the state of the model should be unchanged during execution of the function (or you must understand the implications otherwise).

The `RW_llFunction` sampler can be customized using the `control` list argument to set the initial proposal distribution scale and the adaptive properties for the Metropolis-Hastings sampling. In addition, the `control` list argument must contain a named `llFunction` element. This is the specialized `nimbleFunction` that calculates the log-likelihood; it must accept no arguments and return a scalar double number. The return value must be the total log-likelihood of all stochastic dependents of the `target` nodes – and, if `includesTarget = TRUE`, of the target node(s) themselves – or whatever surrogate is being used for the total log-likelihood. This is a required `control` list element with no default. See `help(samplers)` for details.

Here is a complete example:

```
code <- nimbleCode({
  p ~ dunif(0, 1)
  y ~ dbin(p, n)
})

Rmodel <- nimbleModel(code, data = list(y=3), inits = list(p=0.5, n=10))

llFun <- nimbleFunction(
  setup = function(model) { },
  run = function() {
    y <- model$y
    p <- model$p
    n <- model$n
    ll <- lfactorial(n) - lfactorial(y) - lfactorial(n-y) +
      y * log(p) + (n-y) * log(1-p)
    returnType(double())
    return(ll[1])
  }
)

RllFun <- llFun(Rmodel)

mcmcConf <- configureMCMC(Rmodel, nodes = NULL)

mcmcConf$addSampler(target = "p", type = "RW_llFunction",
  control = list(llFunction = RllFun, includesTarget = FALSE))

Rmcmc <- buildMCMC(mcmcConf)
```

Note that we need to return `ll[1]` and not just `ll` because there are no scalar variables in compiled models. Hence `y` and other variables, and therefore `ll`, are of dimension 1 (i.e., vectors / one-dimensional arrays), so we need to specify the first element in order to have the return type be a scalar.

7.12 Detailed MCMC example: *litters*

Here is a detailed example of specifying, building, compiling, and running two MCMC algorithms. We use the `litters` example from the BUGS examples.

```
#####
#### model configuration ####
#####

# define our model using BUGS syntax
litters_code <- nimbleCode({
  for (i in 1:G) {
    a[i] ~ dgamma(1, .001)
    b[i] ~ dgamma(1, .001)
    for (j in 1:N) {
      r[i,j] ~ dbin(p[i,j], n[i,j])
      p[i,j] ~ dbeta(a[i], b[i])
    }
    mu[i] <- a[i] / (a[i] + b[i])
    theta[i] <- 1 / (a[i] + b[i])
  }
})

# list of fixed constants
constants <- list(G = 2,
                  N = 16,
                  n = matrix(c(13, 12, 12, 11, 9, 10, 9, 9, 8, 11, 8, 10, 13,
                               10, 12, 9, 10, 9, 10, 5, 9, 9, 13, 7, 5, 10, 7, 6,
                               10, 10, 10, 7), nrow = 2))

# list specifying model data
data <- list(r = matrix(c(13, 12, 12, 11, 9, 10, 9, 9, 8, 10, 8, 9, 12, 9,
                          11, 8, 9, 8, 9, 4, 8, 7, 11, 4, 4, 5, 5, 3, 7, 3,
                          7, 0), nrow = 2))

# list specifying initial values
inits <- list(a = c(1, 1),
              b = c(1, 1),
              p = matrix(0.5, nrow = 2, ncol = 16),
              mu = c(.5, .5),
              theta = c(.5, .5))

# build the R model object
Rmodel <- nimbleModel(litters_code,
                      constants = constants,
                      data = data,
                      inits = inits)
```

```
#####
#### MCMC configuration and building ####
#####

# generate the default MCMC configuration;
# only wish to monitor the derived quantity "mu"
mcmcConf <- configureMCMC(Rmodel, monitors = "mu")

# check the samplers assigned by default MCMC configuration
mcmcConf$printSamplers()

# double-check our monitors, and thinning interval
mcmcConf$printMonitors()

# build the executable R MCMC function
mcmc <- buildMCMC(mcmcConf)

# let's try another MCMC, as well,
# this time using the crossLevel sampler for top-level nodes

# generate an empty MCMC configuration
# we need a new copy of the model to avoid compilation errors
Rmodel2 <- Rmodel$newModel()
mcmcConf_CL <- configureMCMC(Rmodel2, nodes = NULL, monitors = "mu")

# add two crossLevel samplers
mcmcConf_CL$addSampler(target = c("a[1]", "b[1]"), type = "crossLevel")
mcmcConf_CL$addSampler(target = c("a[2]", "b[2]"), type = "crossLevel")

# let's check the samplers
mcmcConf_CL$printSamplers()

# build this second executable R MCMC function
mcmc_CL <- buildMCMC(mcmcConf_CL)

#####
#### compile to C++, and run ####
#####

# compile the two copies of the model
Cmodel <- compileNimble(Rmodel)
Cmodel2 <- compileNimble(Rmodel2)

# compile both MCMC algorithms, in the same
# project as the R model object
# NOTE: at this time, we recommend compiling ALL
```



```

# executable MCMC functions together
Cmcmc <- compileNimble(mcmc, project = Rmodel)
Cmcmc_CL <- compileNimble(mcmc_CL, project = Rmodel2)

# run the default MCMC function,
# and example the mean of mu[1]
Cmcmc$run(1000)
cSamplesMatrix <- as.matrix(Cmcmc$mvSamples) # alternative: as.list
mean(cSamplesMatrix[, "mu[1]"])

# run the crossLevel MCMC function,
# and examine the mean of mu[1]
Cmcmc_CL$run(1000)
cSamplesMatrix_CL <- as.matrix(Cmcmc_CL$mvSamples)
mean(cSamplesMatrix_CL[, "mu[1]"])

#####
#### run multiple MCMC chains ####
#####

# run 3 chains of the crossLevel MCMC
samplesList <- runMCMC(Cmcmc_CL, niter=1000, nchains=3)

lapply(samplesList, dim)

```

7.13 Comparing different MCMCs with *compareMCMCs*

Please see the `compareMCMCs` package for the features previously provided by the `compareMCMCs` function (and `MCMCsuite` function) in NIMBLE (until version 0.8.0). The `compareMCMCs` package provides tools to automatically run MCMC in nimble (including multiple sampler configurations), WinBUGS, OpenBUGS, JAGS, Stan, or any other engine for which you provide a simple common interface. The package makes it easy to manage comparison metrics and generate html pages with comparison figures.

7.14 Running MCMC chains in parallel

It is possible to run multiple chains in parallel using standard R parallelization packages such as `parallel`, `foreach`, and `future`. However, you must create separate copies of all model and MCMC objects using `nimbleModel`, `buildMCMC`, `compileNimble`, etc. This is because NIMBLE uses Reference Classes and R6 classes, so copying such objects simply creates a new variable name that refers to the original object.

Thus, in your parallel loop or `lapply`-style statement, you should run `nimbleModel` and all subsequent calls to create and compile the model and MCMC algorithm within the parallelized block of code, once for each MCMC chain being run in parallel.

For a worked example, please see [the parallelization example on our webpage](#).

Chapter 8

Particle Filters, PMCMC, and MCEM

The NIMBLE algorithm library includes a growing set of non-MCMC algorithms, including likelihood-based algorithms. These include:

- A suite of sequential Monte Carlo (particle filtering) algorithms (particle filters, particle MCMC, iterated particle filter, and ensemble Kalman filter);
- Monte Carlo expectation maximization (MCEM) for maximum likelihood estimation;
- Laplace approximation and adaptive Gauss-Hermite quadrature (AGHQ) for maximum likelihood estimation of Laplace/AGHQ-approximated marginal likelihood; and
- nested approximation of Bayesian models using INLA-like methods.

In this chapter we discuss sequential Monte Carlo methods and MCEM and in the next chapter we discuss methods related to Laplace approximation.

8.1 Particle filters / sequential Monte Carlo and iterated filtering

As of Version 0.10.0 of NIMBLE, all of NIMBLE’s sequential Monte Carlo/particle filtering functionality lives in the `nimbleSMC` package, described in [Michaud et al. \(2021\)](#). Please load this package before trying to use these algorithms.

8.1.1 Filtering algorithms

NIMBLE includes algorithms for four different types of sequential Monte Carlo (also known as particle filters), which can be used to sample from the latent states and approximate the log likelihood of a state-space model. These include the bootstrap filter, the auxiliary particle filter, the Liu-West filter, and the ensemble Kalman filter. The iterated filtering version 2 (IF2) is a related method for maximum-likelihood estimation. Each of these is built with the eponymous functions `buildBootstrapFilter`, `buildAuxiliaryFilter`, `buildLiuWestFilter`, `buildEnsembleKF`, and `buildIteratedFilter2`. Each method requires setup arguments `model` and `nodes`; the latter should be a character vector specifying latent model nodes. In addition, each method can be customized using a `control` list argument. Details on the control options and specifics of the algorithms can be found in the help pages for the functions.

Once built, each filter can be run by specifying the number of particles. Each filter has a `modelValues` object named `mvEWSamples` that is populated with equally-weighted samples from the posterior distribution of the latent states (and in the case of the Liu-West filter, the posterior distribution of the top level parameters as well) as the filter is run. The bootstrap, auxiliary, and Liu-West filters, as well as the IF2 method, also have another `modelValues` object, `mvWSamples`. This has unequally-weighted samples from the posterior distribution of the latent states, along with weights for each particle. In addition, the bootstrap and auxiliary particle filters return estimates of the log-likelihood of the given state-space model.

We first create a linear state-space model to use as an example for our particle filter algorithms.

```
# Building a simple linear state-space model.
# x is latent space, y is observed data
timeModelCode <- nimbleCode({
  x[1] ~ dnorm(mu_0, 1)
  y[1] ~ dnorm(x[1], 1)
  for(i in 2:t){
    x[i] ~ dnorm(x[i-1] * a + b, 1)
    y[i] ~ dnorm(x[i] * c, 1)
  }

  a ~ dunif(0, 1)
  b ~ dnorm(0, 1)
  c ~ dnorm(1,1)
  mu_0 ~ dnorm(0, 1)
})

# simulate some data
t <- 25; mu_0 <- 1
x <- rnorm(1, mu_0, 1)
y <- rnorm(1, x, 1)
a <- 0.5; b <- 1; c <- 1
for(i in 2:t){
  x[i] <- rnorm(1, x[i-1] * a + b, 1)
  y[i] <- rnorm(1, x[i] * c, 1)
}

# build the model
rTimeModel <- nimbleModel(timeModelCode, constants = list(t = t),
                          data <- list(y = y), check = FALSE )

# Set parameter values and compile the model
rTimeModel$a <- 0.5
rTimeModel$b <- 1
rTimeModel$c <- 1
rTimeModel$mu_0 <- 1

cTimeModel <- compileNimble(rTimeModel)
```

8.1.1.1 Bootstrap filter

Here is an example of building and running the bootstrap filter.

```
# Build bootstrap filter
rBootF <- buildBootstrapFilter(rTimeModel, "x",
                              control = list(thresh = 0.8, saveAll = TRUE,
                                              smoothing = FALSE))

# Compile filter
cBootF <- compileNimble(rBootF, project = rTimeModel)
# Set number of particles
parNum <- 5000
# Run bootstrap filter, which returns estimate of model log-likelihood
bootLLEst <- cBootF$run(parNum)
# The bootstrap filter can also return an estimate of the effective
# sample size (ESS) at each time point
bootESS <- cBootF$returnESS()
```

8.1.1.2 Auxiliary particle filter

Next, we provide an example of building and running the auxiliary particle filter. Note that a filter cannot be built on a model that already has a filter specialized to it, so we create a new copy of our state space model first.

```
# Copy our state-space model for use with the auxiliary filter
auxTimeModel <- rTimeModel$newModel(replicate = TRUE)
compileNimble(auxTimeModel)
# Build auxiliary filter
rAuxF <- buildAuxiliaryFilter(auxTimeModel, "x",
                              control = list(thresh = 0.5, saveAll = TRUE))

# Compile filter
cAuxF <- compileNimble(rAuxF, project = auxTimeModel)
# Run auxiliary filter, which returns estimate of model log-likelihood
auxLLEst <- cAuxF$run(parNum)
# The auxiliary filter can also return an estimate of the effective
# sample size (ESS) at each time point
auxESS <- cAuxF$returnESS()
```

8.1.1.3 Liu and West filter

Now we give an example of building and running the Liu and West filter, which can sample from the posterior distribution of top-level parameters as well as latent states. Note that the Liu-West filter often performs poorly and is provided primarily for didactic purposes. The Liu and West filter accepts an additional `params` argument, specifying the top-level parameters to be sampled.

```
# Copy model
LWTimeModel <- rTimeModel$newModel(replicate = TRUE)
compileNimble(LWTimeModel)
# Build Liu-West filter, also
# specifying which top level parameters to estimate
```

```

rLWF <- buildLiuWestFilter(LWTimeModel, "x", params = c("a", "b", "c"),
                           control = list(saveAll = FALSE))

## Warning in buildLiuWestFilter(LWTimeModel, "x", params = c("a", "b", "c"), :
## The Liu-West filter often performs poorly and is provided primarily for didactic
## purposes.

# Compile filter
cLWF <- compileNimble(rLWF, project = LWTimeModel)
# Run Liu-West filter
cLWF$run(parNum)

```

8.1.1.4 Ensemble Kalman filter

Next we give an example of building and running the ensemble Kalman filter, which can sample from the posterior distribution of latent states.

```

# Copy model
ENKFTimeModel <- rTimeModel$newModel(replicate = TRUE)
compileNimble(ENKFTimeModel)
# Build and compile ensemble Kalman filter
rENKF <- buildEnsembleKF(ENKFTimeModel, "x",
                          control = list(saveAll = FALSE))
cENKF <- compileNimble(rENKF, project = ENKFTimeModel)
# Run ensemble Kalman filter
cENKF$run(parNum)

```

Once each filter has been run, we can extract samples from the posterior distribution of our latent states as follows:

```

# Equally-weighted samples (available from all filters)
bootEWSamp <- as.matrix(cBootF$mvEWSamples) # alternative: as.list
auxEWSamp <- as.matrix(cAuxF$mvEWSamples)
LWFEWSamp <- as.matrix(cLWF$mvEWSamples)
ENKFEWSamp <- as.matrix(cENKF$mvEWSamples)

# Unequally-weighted samples, along with weights (available
# from bootstrap, auxiliary, and Liu and West filters)
bootWSamp <- as.matrix(cBootF$mvWSamples, "x")
bootWts <- as.matrix(cBootF$mvWSamples, "wts")
auxWSamp <- as.matrix(xAuxF$mvWSamples, "x")
auxWts <- as.matrix(cAuxF$mvWSamples, "wts")

# Liu and West filter also returns samples
# from posterior distribution of top-level parameters:
aEWSamp <- as.matrix(cLWF$mvEWSamples, "a")

```

8.1.1.5 Iterated filtering 2 (IF2)

The IF2 method (Ionides et al., 2015) accomplishes maximum likelihood estimation using a scheme wherein both latent states and parameters are represented by particles that are weighted and re-sampled during the iterations. Iterations include perturbations to the parameter particles following a schedule of decreasing magnitude to yield convergence to the MLE.

Here we apply IF2 to Nile River flow data, specifying a changepoint in the year the Aswan Dam was constructed, as the dam altered river flows.

```
library(FKF)

flowCode <- nimbleCode({
  for(t in 1:n)
    y[t] ~ dnorm(x[t], sd = sigmaMeasurements)
  x[1] ~ dnorm(x0, sd = sigmaInnovations)
  for(t in 2:n)
    x[t] ~ dnorm((t-1==28)*meanShift1899 + x[t-1], sd = sigmaInnovations)
  logSigmaInnovations ~ dnorm(0, sd = 100)
  logSigmaMeasurements ~ dnorm(0, sd = 100)
  sigmaInnovations <- exp(logSigmaInnovations)
  sigmaMeasurements <- exp(logSigmaMeasurements)
  x0 ~ dnorm(1120, var = 100)
  meanShift1899 ~ dnorm(0, sd = 100)
})

flowModel <- nimbleModel(flowCode, data = list(y = Nile),
  constants = list(n = length(Nile)),
  inits = list(logSigmaInnovations = log(sd(Nile)),
    logSigmaMeasurements = log(sd(Nile)),
    meanShift1899 = -100))
```

Note that the prior distributions for the parameters are not used by IF2, except possibly to obtain boundaries of valid parameter values (not the case here).

Now we build the filter, specifying user-controlled standard deviations (in this case the same as the perturbation `sigma` values) for use in generating the initial particles for the parameters via the control list.

```
filter <- buildIteratedFilter2(model = flowModel,
  nodes = 'x',
  params = c('logSigmaInnovations',
    'logSigmaMeasurements',
    'meanShift1899'),
  baselineNode = 'x0',
  control = list(sigma = c(0.1, 0.1, 5),
    initParamSigma = c(0.1, 0.1, 5)))

cFlowModel <- compileNimble(flowModel)
cFilter <- compileNimble(filter, project = flowModel)
```

We now run the algorithm with 1000 particles for 100 iterations with the schedule parameter equal

to 0.2.

In addition to the estimates, we can extract the values of the log-likelihood, the estimates and the standard deviation of the parameter particles as they evolve over the iterations, in order to assess convergence.

```
set.seed(1)
est <- cFilter$run(m = 1000, niter = 100, alpha = 0.2)

cFilter$estimates[95:100,] ## Last 5 iterations of parameter values

##           [,1]      [,2]      [,3]
## [1,]  0.013306153  4.822613 -269.7837
## [2,] -0.013961938  4.857470 -271.4965
## [3,]  0.002183997  4.850475 -271.1919
## [4,]  0.015598044  4.851961 -272.3889
## [5,] -0.005571944  4.842719 -272.6390
## [6,]  0.042982317  4.837347 -271.1800

cFilter$logLik[90:100] ## Last 5 iterations of log likelihood values

## [1] -627.0497 -626.9570 -626.9856 -626.9954 -626.7448 -626.7521 -626.8472
## [8] -626.7398 -626.9869 -627.1354 -626.6648
```

Comparing to use of the the Kalman Filter from the FKF package, we see the log-likelihood is fairly similar:

```
dtpred <- matrix(0, ncol = length(Nile))
dtpred[28] <- est[3]
ct <- matrix(0)
Zt <- Tt <- matrix(1)

fkfResult <- fkf(HHt = matrix(exp(2*est[1])),
                 GGt = matrix(exp(2*est[2])),
                 yt = rbind(Nile),
                 a0 = 1120,
                 P0 = matrix(100),
                 dt = dtpred, ct = ct, Zt = Zt, Tt = Tt)

fkfResult$logLik

## [1] -626.5521
```

8.1.2 Particle MCMC (PMCMC)

Particle MCMC (PMCMC) is a method that uses MCMC for top-level model parameters and uses a particle filter to approximate the time-series likelihood for use in determining MCMC acceptance probabilities (Andrieu et al., 2010). NIMBLE implements PMCMC by providing random-walk Metropolis-Hastings samplers for model parameters that make use of particle filters in this way. These samplers can use NIMBLE's bootstrap filter or auxiliary particle filter, or they can use a user-defined filter. Whichever filter is specified will be used to obtain estimates of the likelihood of the state-space model (marginalizing over the latent states), which is used for calculation of the

Metropolis-Hastings acceptance probability. The `RW_PF` sampler uses a univariate normal proposal distribution, and can be used to sample scalar top-level parameters. The `RW_PF_block` sampler uses a multivariate normal proposal distribution and can be used to jointly sample vectors of top-level parameters. The PMCMC samplers can be specified with a call to `addSampler` with `type = "RW_PF"` or `type = "RW_PF_block"`, a syntax similar to the other MCMC samplers listed in Section 7.11.

The `RW_PF` sampler and `RW_PF_block` sampler can be customized using the `control` list argument to set the adaptive properties of the sampler and options for the particle filter algorithm to be used. In addition, providing `pfOptimizeNparticles=TRUE` in the `control` list will use an experimental algorithm to estimate the optimal number of particles to use in the particle filter. See `help(samplers)` for details. The MCMC configuration for the `timeModel` in the previous section will serve as an example for the use of our PMCMC sampler. Here we use the identity matrix as our proposal covariance matrix.

```
timeConf <- configureMCMC(rTimeModel, nodes = NULL) # empty MCMC configuration

# Add random walk PMCMC sampler with particle number optimization.
timeConf$addSampler(target = c("a", "b", "c", "mu_0"), type = "RW_PF_block",
                    control = list(propCov= diag(4), adaptScaleOnly = FALSE,
                                   latents = "x", pfOptimizeNparticles = TRUE))
```

The `type = "RW_PF"` and `type = "RW_PF*block"` samplers default to using a bootstrap filter. The adaptation control parameters `adaptive`, `adaptInterval`, and `adaptScaleOnly` work in the same way as for an RW and RW*block samplers. However, it is not clear if the same approach to adaptation works well for PMCMC, so one should consider turning off adaptation and using a well-chosen proposal covariance.

It is also possible that more efficient results can be obtained by using a custom filtering algorithm. Choice of filtering algorithm can be controlled by the `pfType` control list entry. The `pfType` entry can be set either to `'bootstrap'` (the default), `'auxiliary'`, or the name of a user-defined nimbleFunction that returns a likelihood approximation.

Any user-defined filtering nimbleFunction named in the `pfType` control list entry must satisfy the following:

1. The `nimbleFunction` must be the result of a call to `nimbleFunction()`.
2. The `nimbleFunction` must have setup code that accepts the following (and only the following) arguments:
 - `model`, the NIMBLE model object that the MCMC algorithm is defined on.
 - `latents`, a character vector specifying the latent model nodes over which the particle filter will stochastically integrate over to estimate the log-likelihood function.
 - `control`, an R list object. Note that the `control` list can be used to pass in any additional information or arguments that the custom filter may require.
3. The `nimbleFunction` must have a `run` function that accepts a single integer argument (the number of particles to use), and returns a scalar double (the log-likelihood estimate).
4. The `nimbleFunction` must define, in setup code, a `modelValues` object named `mvEWSamples` that is used to contain equally weighted samples of the latent states (that is, the `latents`

argument to the setup function). Each time the `run()` method of the `nimbleFunction` is called with number of particles `m`, the `mvEWSamples` `modelValues` object should be resized to be of size `m` via a call to `resize(mvEWSamples, m)`.

8.2 Monte Carlo Expectation Maximization (MCEM)

Suppose we have a model with missing data – or latent variables or random effects, which can be viewed as missing data – and we would like to maximize the marginal likelihood of the model, integrating over the missing data. A brute-force method for doing this is MCEM. This is an EM algorithm in which the missing data are simulated via Monte Carlo (often MCMC, when the full conditional distributions cannot be directly sampled from) at each iteration. MCEM can be slow but is also a general workhorse method that is applicable to a wide range of problems.

`buildMCEM` provides an (optionally) ascent-based MCEM algorithm based on Caffo et al. (2005). `buildMCEM` was re-written for nimble version 1.2.0 and is not compatible with previous versions. For the E (expectation) step, `buildMCEM` uses an MCMC for the latent states given the data and current parameters. For the M (maximization) step, `buildMCEM` uses `optim`.

The ascent-based feature uses an estimate of the standard error of the move in one iteration to determine if that move is clearly uphill or is swamped by noise due to using a Monte Carlo (MCMC) sample. In the latter case, the MCMC sample size is increased until there is a clear uphill move. In a similar manner, convergence is determined based on whether the uphill move is confidently less than a tolerance level taken to indicate that the algorithm is at the MLE. The associated tuning parameters, along with details on control over each step, are described at `help(buildMCEM)`.

The maximization part of each MCEM iteration will use gradients from automatic differentiation if they are supported for the model and `buildDerivs=TRUE` when `nimbleModel` was called to create the model. If any parameters have constraints on valid values (e.g. greater than 0), by default a parameter transformation will be set up (see `help(parameterTransform)`) and optimization will be done in the transformed (unconstrained) space.

Additionally, the MCEM algorithm can provide an estimate of the asymptotic covariance matrix of the parameters. See 8.2.1.

We will revisit the *pump* example to illustrate the use of NIMBLE's MCEM algorithm.

```
pump <- nimbleModel(code = pumpCode, name = "pump", constants = pumpConsts,
                   data = pumpData, inits = pumpInits, check = FALSE,
                   buildDerivs=TRUE)

Cpump <- compileNimble(pump)

# build an MCEM algorithm with ascent-based convergence criterion
pumpMCEM <- buildMCEM(model = pump,
                      latentNodes = "theta")
# The latentNodes="theta" would be determined by default but is shown for clarity.
CpumpMCEM <- compileNimble(pumpMCEM, project=pump)
```

The first argument to `buildMCEM`, `model`, is a NIMBLE model. At the moment, the model provided cannot be part of another MCMC sampler.

```
pumpMLE <- CpumpMCEM$findMLE(initM = 1000)
```

```
## [Note] Iteration Number: 1.
## [Note] Current number of MCMC iterations: 1000.
## [Note] Parameter Estimates:
## 0.818983
## 1.14746
## [Note] Convergence Criterion: 0.684673.
## [Note] Iteration Number: 2.
## [Note] Current number of MCMC iterations: 1000.
## [Note] Parameter Estimates:
## 0.818793
## 1.21849
## [Note] Convergence Criterion: 0.0209188.
## [Note] Iteration Number: 3.
## [Note] Current number of MCMC iterations: 1000.
## [Note] Parameter Estimates:
## 0.819104
## 1.24505
## [Note] Convergence Criterion: 0.00392522.
## [Note] Iteration Number: 4.
## [Note] Current number of MCMC iterations: 1000.
## [Note] Parameter Estimates:
## 0.838402
## 1.29963
## [Note] Convergence Criterion: 0.00657192.
## [Note] Iteration Number: 5.
## [Note] Current number of MCMC iterations: 1000.
## [Note] Parameter Estimates:
## 0.821287
## 1.26525
## [Note] Convergence Criterion: 0.00361366.
## [Note] Iteration Number: 6.
## [Note] Current number of MCMC iterations: 1000.
## [Note] Parameter Estimates:
## 0.835258
## 1.29028
## [Note] Convergence Criterion: 0.00249731.
## Monte Carlo error too big: increasing MCMC sample size.
## [Note] Iteration Number: 7.
## [Note] Current number of MCMC iterations: 1167.
## [Note] Parameter Estimates:
## 0.829192
## 1.28844
## [Note] Convergence Criterion: 0.000914589.
```

```
pumpMLE$par
```

```
## [1] 0.8291919 1.2884366
```

For this model, we can check the result by analytically integrating over the latent nodes (possible for this model but often not feasible), which gives maximum likelihood estimates of $\hat{\alpha} = 0.823$ and $\hat{\beta} = 1.261$. Thus the MCEM seems to do pretty well, though tightening the convergence criteria may be warranted in actual usage.

For more complicated models, it is worth exploring the tuning parameters (see `help(buildMCEM)`). MCEM inherits from the EM algorithm the difficulty that the convergence path can be slow, depending on the model. This can depend on the parameterization, such as whether random effects are centered or uncentered, so it can also be worth exploring different ways to write a model.

8.2.1 Estimating the asymptotic covariance From MCEM

The `vcov` method of an MCEM `nimbleFunction` calculates a Monte Carlo estimate of the asymptotic covariance of the parameters based on the Hessian matrix at the MLE, using the method of [Louis \(1982\)](#). Arguments to this method allow control over whether to generate a new MCMC sample for this purpose and other details.

Continuing the above example, here is the covariance matrix:

```
pumpCov <- CpumpMCEM$vcov(pumpMLE$par)
pumpCov
```

```
##           [,1]      [,2]
## [1,] 0.1295558 0.2263726
## [2,] 0.2263726 0.6758945
```

Chapter 9

Laplace, AGHQ, and nested approximations

The NIMBLE algorithm library includes a growing set of non-MCMC algorithms including likelihood-based algorithms.

In this chapter we discuss Laplace approximation and the closely related adaptive Gauss-Hermite quadrature (AGHQ) approximation for maximum likelihood estimation, as well as INLA-like nested approximations that use Laplace/AGHQ methods for approximate marginalization for deterministic Bayesian estimation.

9.1 Laplace approximation and adaptive Gauss-Hermite quadrature

As of NIMBLE version 1.4.0, NIMBLE's Laplace and AGHQ approximation live in the `nimbleQuad` package. Please load that package before trying to use these algorithms.

Many hierarchical models include continuous random effects that must be integrated over to obtain the (marginal) likelihood of the parameters given the data. Laplace approximation and adaptive Gauss-Hermite quadrature (AGHQ) are often accurate and fast approximations for doing so. Laplace is simply AGHQ with a single quadrature point (the conditional mode of the random effects).

NIMBLE provides these algorithms via `buildLaplace` and `buildAGHQuad` (the former simply calls the latter), which take advantage of the automatic differentiation features introduced in version 1.0.0.

Next we will show how to use nimble's Laplace approximation, which uses derivatives internally, to get maximum (approximate) likelihood estimates for a GLMM model. Additional details on Laplace and on AGHQ can be found by running `help(buildLaplace)`.

Laplace approximation is equivalent to first-order adaptive Gauss-Hermite quadrature, which is also available (via `buildAGHQ` and `runAGHQ`), although here we will focus on Laplace approximation only. In this context, Laplace approximation approximates the integral over continuous random effects needed to calculate the likelihood. Hence, it gives an approximate likelihood (often quite accurate) that can be used for maximum likelihood estimation. Note that the Laplace approximation uses

second derivatives, and the gradient of the Laplace approximation (used for finding the MLE efficiently) uses third derivatives. These are described in detail by [Skaug and Fournier \(2006\)](#) and [Fournier et al. \(2012\)](#).

9.1.1 GLMM example

We'll re-introduce the simple Poisson Generalized Linear Mixed Model (GLMM) example model from 7.11.2.1 and use Laplace approximation on it. There will be 10 groups (i) of 5 observations (j) each. Each observation has a covariate, X , and each group has a random effect `ran_eff`. Here is the model code:

```
model_code <- nimbleCode({
  # priors
  intercept ~ dnorm(0, sd = 100)
  beta ~ dnorm(0, sd = 100)
  sigma ~ dhalfflat()
  # random effects and data
  for(i in 1:10) {
    # random effects
    ran_eff[i] ~ dnorm(0, sd = sigma)
    for(j in 1:5) {
      # data
      y[i,j] ~ dpois(exp(intercept + beta*X[i,j] + ran_eff[i]))
    }
  }
})
```

Note that we changed the prior on `sigma` to avoid having an upper bound. Prior distributions are not included in maximum likelihood using the Laplace approximation but do indicate the range of valid values. We recommend caution in using priors for variance component parameters (standard deviations, variances, precisions) that have a finite upper bound (e.g., `sigma ~ dunif(0, 100)`), because the probit transformation applied in that case may result in poor optimization performance.

We'll simulate some values for X .

```
set.seed(123)
X <- matrix(rnorm(50), nrow = 10)
```

Next, we build the model, including `buildDerivs=TRUE`, which is needed to use derivatives with a model. Internally Laplace/AGHQ approximation use derivatives from NIMBLE's automatic differentiation (AD) system to build the approximation to the marginal likelihood.

```
model <- nimbleModel(model_code, constants = list(X = X), calculate = FALSE,
  buildDerivs = TRUE) # Here is the argument needed for AD.
```

As preparation for the Laplace examples below, we need to finish setting up the GLMM. We could have provided data in the call to `nimbleModel`, but instead we will simulate it using the model itself. Specifically, we will set parameter values, simulate data values, and then set those as the data to use.

```

model$intercept <- 0
model$beta <- 0.2
model$sigma <- 0.5
model$calculate() # This will return NA because the model is not fully initialized.

## [1] NA

model$simulate(model$getDependencies('ran_eff'))
model$calculate() # Now the model is fully initialized: all nodes have valid values.

## [1] -78.44085

model$setData('y') # Now the model has y marked as data, with values from simulation.

```

Finally, we will make a compiled version of the model.

```
Cmodel <- compileNimble(model)
```

9.1.2 Using Laplace approximation

To create a Laplace approximation specialized to the parameters of interest for this model, we use the nimbleFunction `buildLaplace`. For many models, the setup code in `buildLaplace` will automatically determine the random effects to be integrated over and the associated nodes to calculate. In fact, if you omit the parameter nodes, it will assume that all top-level nodes in the model should be treated as parameters. If fine-grained control is needed, these various sets of nodes can be input directly into `buildLaplace`. To see what default handling of nodes is being done for your model, use `setupMargNodes` with the same node inputs as `buildLaplace`.

```

library(nimbleQuad)

glmm_laplace <- buildLaplace(model, paramNodes = c('intercept', 'beta', 'sigma'))
Cglmm_laplace <- compileNimble(glmm_laplace, project = model)

```

With the compiled Laplace approximation, we can now find the MLE and related information such as standard errors.

```

results <- runLaplace(Cglmm_laplace)
results$summary

## $params
##           estimate stdError
## intercept -0.1491944 0.2464880
## beta       0.1935212 0.1467230
## sigma      0.5703362 0.2066515
##
## $randomEffects
##           estimate stdError
## ran_eff[1] -0.33711373 0.4305830
## ran_eff[2] -0.02964535 0.3987838
## ran_eff[3]  0.40575212 0.3858675
## ran_eff[4]  1.04768889 0.3779772

```

```
## ran_eff[5] -0.36731650 0.4290567
## ran_eff[6] 0.26907207 0.3863272
## ran_eff[7] -0.54950702 0.4654195
## ran_eff[8] -0.11864461 0.4175452
## ran_eff[9] 0.10006643 0.3926128
## ran_eff[10] -0.04411292 0.3971147
##
## $vcov
##          intercept          beta          sigma
## intercept 0.060756334 -0.002691102 -0.014082732
## beta      -0.002691102 0.021527634 -0.005098509
## sigma     -0.014082732 -0.005098509 0.042704860
##
## $logLik
## [1] -63.44875
##
## $df
## [1] 3
##
## $originalScale
## [1] TRUE
```

One of output elements is the maximized log likelihood (`logLik`), which is useful for model comparison.

`buildLaplace` actually offers several choices in how computations are done, differing in how they use *double tapering* for derivatives or not. In some cases one or another choice might be more efficient. See `help(buildLaplace)` if you want to explore it further.

Finally, let's confirm that it worked by comparing to results from package `glmmTMB`. In this case, `nimble`'s Laplace approximation is faster than `glmmTMB` (on the machine used here), but that is not the point of this example. Here our interest is in checking that `nimble`'s Laplace approximation worked correctly in a case where we have an established tool such as `glmmTMB`.

```
library(glmmTMB)
y <- as.numeric(model$y) # Re-arrange inputs for call to glmmTMB
X <- as.numeric(X)
group <- rep(1:10, 5)
data <- as.data.frame(cbind(X,y,group))
tmb_fit <- glmmTMB(y ~ X + (1 | group), family = poisson, data = data)
summary(tmb_fit)
```

```
## Family: poisson ( log )
## Formula:          y ~ X + (1 | group)
## Data: data
##
##      AIC      BIC   logLik deviance df.resid
##   132.9   138.6    -63.4   126.9      47
##
## Random effects:
```



```
##
## Conditional model:
##   Groups Name      Variance Std.Dev.
##   group  (Intercept) 0.3253   0.5703
## Number of obs: 50, groups:  group, 10
##
## Conditional model:
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.1492     0.2465  -0.605   0.545
## X           0.1935     0.1467   1.319   0.187
```

```
logLik(tmb_fit)
```

```
## 'log Lik.' -63.44875 (df=3)
```

The results match within numerical tolerance typical of optimization problems. Specifically, the coefficients for `(Intercept)` and `X` match nimble's `Intercept` and `beta`, the random effects standard deviation for `group` matches nimble's `sigma`, and the standard errors match.

9.1.3 Using the Laplace approximation methods directly

If one wants finer grain control over using the approximation, one can use the methods provided by `buildLaplace`. These include calculating the Laplace approximation for some input parameter values, calculating its gradient, and maximizing the Laplace-approximated likelihood. Here we'll show some of these steps.

```
# Get the Laplace approximation for one set of parameter values.
Cglmm_laplace$calcLaplace(c(0, 0, 1))
```

```
## [1] -65.57246
```

```
Cglmm_laplace$gr_Laplace(c(0, 0, 1)) # Get the corresponding gradient.
```

```
## [1] -1.866840  8.001648 -4.059555
```

```
MLE <- Cglmm_laplace$findMLE(c(0, 0, 1)) # Find the (approximate) MLE.
MLE$par      # MLE parameter values
```

```
## [1] -0.1491982  0.1935269  0.5703413
```

```
MLE$value     # MLE log likelihood value
```

```
## [1] -63.44875
```

The final outputs show the MLE for `intercept`, `beta`, and `sigma`, followed by the maximum (approximate) likelihood.

More information about the MLE can be obtained in two ways. The `summary` method can give estimated random effects and standard errors as well as the variance-covariance matrix for the parameters and/or the random effects. The `summaryLaplace` function returns similar information but with names included in a more useful way. Here is some example code (results not shown):

```
Cglmm_laplace$summary(MLE)$randomEffects$estimate
```

```
## [1] -0.33711487 -0.02964301 0.40575572 1.04769223 -0.36731868 0.26907375
## [7] -0.54951161 -0.11864177 0.10006955 -0.04411124
```

```
summaryLaplace(Cglmm_laplace, MLE)$params
```

```
##           estimate stdError
## intercept -0.1491982 0.2464895
## beta      0.1935269 0.1467230
## sigma     0.5703413 0.2066531
```

9.1.4 Changing the optimization methods

When finding the MLE via Laplace approximation or adaptive Gauss-Hermite quadrature (AGHQ), there are two numerical optimizations: (1) maximizing the joint log-likelihood of random effects and data given a set of parameter values to construct the approximation to the marginal log-likelihood at the given parameter values, and (2) maximizing the approximation to the marginal log-likelihood over the parameter values. Optimization (1) is the “inner” optimization and optimization (2) is the “outer” optimization.

Finding the MLE via Laplace approximation may be sensitive to the optimization methods used, in particular the choice of optimizer for the inner optimization, and the “BFGS” optimizer available through `optim()` may not perform well for inner optimization.

As of version 1.3.0, the default choices for both the inner and outer optimization use R’s `nlminb` optimizer. Users can choose a different optimizer for both of the optimizations.

To change the inner or outer optimizers, one can use the `innerOptimMethod` and `outerOptimMethod` elements of the `control` list argument to `buildLaplace`. One can modify various settings that control the behavior of the inner and outer optimizers via `control` as well. See `help(buildLaplace)` for more details.

Once a Laplace approximation is built, one can use `updateSettings` to modify the choices of optimizers and various settings that control the behavior of the inner and outer optimizers (see `help(buildLaplace)` for details).

By default, NIMBLE provides various optimization methods available through R’s `optim()` as well as R’s `nlminb` method and the BOBYQA method from the `nloptr` package (by specifying `bobyqa`). Users can also provide their own optimization function in R that they can then use with Laplace approximation. User optimization functions must have a particular set and order of arguments and must first be registered with NIMBLE via `nimOptimMethod`. See `help(nimOptim)` for more details.

Here’s an example of setting up the Newton method optimizer from the TMB package as the inner optimizer for use with NIMBLE’s Laplace approximation. (Note that NIMBLE and TMB have distinct AD systems and Laplace approximation implementations; here we simply use the `TMB::newton` optimization function.)

```
library(TMB)
library(Matrix)
```

```
## Create an R wrapper function that has the interface needed for NIMBLE
## and wraps the optimizer of interest.
nimbleTMBnewton <- function(par, fn, gr, he, lower, upper, control, hessian) {
```

```

## Wrap `he` as return value needs to be of class `dsCMatrix`.
he_matrix <- function(p) Matrix(he(p), doDiag = FALSE, sparse = TRUE)
invalid <- function(x) is.null(x) || is.na(x) || is.infinite(x)
if(invalid(control$trace)) control$trace <- 1
if(invalid(control$maxit)) control$maxit <- 100
if(invalid(control$reltol)) control$reltol <- 1e-8
res <- newton(par, fn, gr, he_matrix,
              trace = control$trace, maxit = control$maxit, tol = control$reltol)
## Additional arguments (e.g., `alpha` and `tol10`) can be hard-coded in `newton()` call.
ans <- list(
  ## What is handled in the return is fairly particular, so often needs conversion
  ## from a given method such as TMB::newton.
  par = res$par,
  value = res$value,
  counts = c(0, 0, 0), # Counts of fn/gr/he calls, but these are not counted by TMB::newton.
  evaluations = res$iterations,
  hessian = NULL, # TMB::newton gives a `dsCMatrix` but we need a base R matrix.
  message = "ran by TMB::newton",
  convergence = 0 # TMB::newton does not return a convergence code so give 0 (converged).
)
return(ans)
}

## Register the optimizer with NIMBLE.
nimOptimMethod("nimbleTMBnewton", nimbleTMBnewton)

## Use the optimizer for the inner optimization when finding the Laplace MLE.
glmm_laplace <- buildLaplace(model, c('intercept','beta','sigma'),
                             control = list(innerOptimMethod = "nimbleTMBnewton"))

```

9.2 Nested approximation (INLA-like) methods

NIMBLE version 1.4.0 introduces a nested approximation method that provides approximate posterior inference using methodology similar to the well-known INLA approach (Rue et al., 2009, Martins et al. (2013)), implemented in the R-INLA package and to the related methods for extended Gaussian latent models (EGLMs) of Stringer et al. (2023), implemented in the `aghq` R package.

In general, such nested approximations build on Laplace approximation, which provides an approximate marginal posterior for model (hyper)parameters, integrating (marginalizing) over latent nodes. Then instead of maximizing the approximation, one approximates the marginal posterior of the (hyper)parameters on a carefully-chosen set of points. Inference for individual (hyper)parameters is done by numerical approximation, numerical integration, or sampling from an approximation to the marginal posterior. Inference for the latent nodes is done via numerical integration or via sampling from a mixture (over the hyperparameter points) of multivariate normal distributions.

Our implementation in NIMBLE borrows heavily from the INLA and EGLM approaches.

Here we list some of the similarities and differences from INLA and the EGLM approach (`aghq`

package):

- Like EGLM, we use automatic differentiation to calculate the Laplace-approximated marginal likelihood.
- Like EGLM, we take the latent nodes to be only the latent stochastic parameters in the model, without including the additive predictor values as done in INLA.
- For marginal inference on a chosen univariate (hyper)parameter we provide the univariate asymmetric Gaussian approximation used by INLA and also (for increased accuracy at additional computational expense) numerical integration via AGHQ as in EGLM.
- For joint inference on the (hyper)parameters we provide simulation from the joint asymmetric Gaussian approximation as done in INLA.
- For inference on the latent nodes, we provide joint simulation from a multivariate normal mixture over the (hyper)parameter grid points as done in EGLM and also available in INLA. Unlike in INLA, we do not provide univariate latent inference using deterministic nested Laplace approximation. The simulation-based approach may not be as accurate, but it allows for joint inference, including inference on quantities that depend on more than one latent node.
- Unlike either EGLM or INLA, latent nodes are not required to have a joint normal distribution, though accuracy may be less when the latent nodes have other distributions.
- For latent nodes whose conditional distributions factor into univariate conditionally independent sets, the Laplace approximation is a product of univariate approximations, and one can instead use NIMBLE’s AGHQ approximation for higher accuracy.
- We allow the user to choose the grid used for the (hyper)parameters. By default for $d > 2$ parameters, we use the CCD grid used by INLA, but one can choose to use the AGHQ grid as used in EGLM or provide one’s own grid. In the future we may provide additional grids, such as sparse grids.

9.2.1 Overview of the methodology

NIMBLE’s nested approximation consists of several pieces, with different options that allow a user to choose how the approximation is done. We briefly describe the pieces here. For simplicity, we refer to the (hyper)parameters simply as “parameters”.

9.2.1.1 Marginalization over the latent nodes

We approximately marginalize (integrate) over the latent nodes to approximate the marginal joint distribution of the parameters. This uses Laplace approximation and is sometimes referred to as the “inner” marginalization. The Laplace approximation is computed using the gradient and Hessian with respect to the latent nodes for a given set of parameter values.

In the case that the conditional distributions of the latent nodes factor into univariate conditionally-independent sets of nodes (conditional on the parameters), this approximation by default uses the product of univariate Laplace approximations and in NIMBLE can be made more accurate by using AGHQ with more than one quadrature point.

9.2.1.2 Approximating the marginal parameter density on a grid

In order to simulate from the posterior distribution for the latent nodes (or to estimate the marginal likelihood), one needs to evaluate the joint parameter posterior density on a grid of points.

NIMBLE primarily offers the options of using a CCD grid (as used by INLA) or an AGHQ grid (as

used in the EGLM approach). While the AGHQ grid is expected to be more accurate, the CCD grid uses fewer points and is therefore less computationally intensive.

For $d \leq 2$ NIMBLE defaults to the AGHQ grid and otherwise uses the CCD grid.

NIMBLE also allows users to provide their own grid.

9.2.1.3 Joint inference for latent nodes

NIMBLE provides the ability to simulate the latent nodes from a mixture of multivariate Gaussian distributions, mixing over the parameter values at the grid points discussed above. The multivariate Gaussian distribution is based on the Laplace approximation for the latent nodes, which provides the mean and variance conditional on the parameter value at the grid point. The weights in the mixture are based on the Laplace-approximated marginal density (for the parameters and data, jointly), with stratified sampling to reduce variance.

INLA adjusts the latent node marginals via a Gaussian copula to correspond with a skew-normal approximation [TODO: check on terminology here]. We have not implemented such an approximation so no adjustment is done in NIMBLE.

9.2.1.4 Univariate inference for parameters

NIMBLE offers two approaches for univariate marginal inference for individual parameters. The first is a computationally-efficient integration-free method that mimics that used by INLA (Martins et al., 2013). This uses a joint asymmetric Gaussian distribution approximation as the marginal joint posterior of the parameters and allows one to calculate the density at individual evaluation points for the univariate marginals. NIMBLE calculates the univariate marginal density on a fine grid of evaluation points, from which one can build a spline-based approximation to the marginal density that can be used to compute moments and quantiles for univariate inference.

The second, more accurate approach, is to use $d - 1$ -dimensional AGHQ to integrate over the Laplace-approximated joint marginal distribution for the parameters. This can greatly improve accuracy, but can be computationally expensive unless d is quite small (e.g., 2-4), or even if the number of latent node elements is quite large (which makes the Laplace approximation expensive). Because of the expense, we only take this approach if requested by the user, and we allow the user to choose the specific parameter(s) for which they want to estimate marginals with this approach.

9.2.1.5 Joint inference for parameters

If one is interested in joint inference for the parameters (or inference on functions of more than one parameter), NIMBLE provides the ability to simulate from the asymmetric multivariate Gaussian distribution approximation to the marginal joint distribution of the parameters that is used by INLA (Martins et al., 2013). The approximation is done in the transformed space after transforming the parameters to be unconstrained (this can sometimes reduce the dimension of the parameter vector such as with the Dirichlet, Wishart, and LKJ distributions). In the transformed space, a two-piece split normal distribution (which allows for skew in both directions) is used for each univariate component after rotating the transformed parameters based on the Hessian at the maximum to account for the correlation structure.

As also done in INLA, we use a Gaussian copula to transform the simulated parameter values, in a univariate fashion, to match the univariate marginal distributions estimated either from the

integration-free or AGHQ approaches.

[Paul, I think we want to mention the use of the skew terms in the asymmetric Gaussian. Can you work that in wherever you think that appropriate, perhaps in some additional text giving a bit more detail on the asymmetric Gaussian?]

9.2.1.6 Approximate marginal likelihood

The marginal likelihood for a model integrates over all parameters (including both ‘parameters’ discussed above and latent nodes). This can be useful for model selection, but is not well-defined if any prior distributions are improper and is not reliable for diffuse prior distributions.

In NIMBLE’s nested approximation, the marginal likelihood is approximated either based on the approximate Gaussian distribution for the parameters used by INLA or via AGHQ using the AGHQ grid weights and associated Laplace-approximated parameter density values.

9.2.1.7 Determining latent nodes and parameters

By default, NIMBLE’s nested approximation will try to automatically determine the node sets, selecting random effects and regression fixed effects (coefficients) as the “latent nodes” and other unknown quantities (e.g., variance components and parameters of random effects distributions) as “parameters” (also often referred to as “hyperparameters”).

Given the computations involved, it is best to have the number of parameters be relatively small, such as fewer than 10.

Note that the treatment of fixed effects differs between the nested approximation and NIMBLE’s Laplace approximation. In Laplace approximation, fixed effects are treated as parameters of interest and maximized with respect to, rather than being marginalized over. In the nested approximation, they are marginalized over and inference on them can be done based on simulated values. One advantage of grouping fixed effects with random effects is that one can make simulation-based inference on quantities that depend on both sets of effects, such a regression-style linear predictors.

It is also possible to choose which nodes go in which set (Section 9.2.2.2). One might choose to include fixed effects in the parameter set, akin to how Laplace approximation works. Another situation where one might configure this manually is if random effects in a model are specified in a “non-centered” parameterization such as:

$$\eta_i = \mu + \sigma b_i b_i \sim \mathcal{N}(0, 1)$$

In this parameterization, the random effects, b_i , do not depend on any parameters because μ and σ are involved directly in the linear predictor, η_i . This stands in contrast to the common centered parameterization, with $b_i \sim \mathcal{N}(\mu, \sigma)$. Because each b_i does not depend on any parameters, NIMBLE’s nested approximation node determination algorithm would by default put the b_i nodes into the parameter set, which in general would result in very slow computation (because there would be a large number of parameters). (And in reality this would generally lead to a failure to build the algorithm because there would be no latent nodes found.)

One further note is that it can be advantageous computationally to have fixed effects in the parameter set. This can sometimes allow NIMBLE to set up a product of lower-dimensional Laplace approximations instead of one higher-dimensional Laplace approximation. However, the benefit of

having a product of Laplace approximations trades off with the higher-dimensionality of the parameter vector, which increases computation. How these trade off in a particular modeling context can be worth experimentation.

9.2.1.8 Computational bottlenecks

To sample the latent nodes, the algorithm needs to find the weights and the parameters of the multivariate normal approximation to the latent nodes at each point in the grid of parameters. For a large number of parameters this can be expensive, because Laplace approximation is done at many grid points. For a large number of latent node elements, even a single Laplace approximation at a single set of parameter values can be expensive. Computing the parameters of this approximation involves optimization over a space whose dimension is the number of latent node elements, as well as computation of the Hessian at the maximum. Furthermore, for inference, one then needs to simulate from the high-dimensional multivariate normal.

To approximate the univariate marginals for the parameters via AGHQ, this requires $d - 1$ -dimensional AGHQ, which can be expensive, because the number of quadrature points grows as $(d - 1)^k$ where k is the number of Gauss-Hermite quadrature grid points in one dimension. Often this would be chosen to be small, such as 3 or 5, but even then the computations increase rapidly in d .

9.2.2 Using NIMBLE's nested approximation

Next we'll give example usage of NIMBLE's nested approximation. Further details are available in the usual R help content for the various functions.

The core functions are `buildNestedApprox` and `runNestedApprox`. These are analagous to the “build” and “run” functions for MCMC and for Laplace.

- `buildNestedApprox` sets up the approximation for a model of interest and allows the user to control various aspects of how the approximation is done. It returns an uncompiled nested approximation algorithm.
- `runNestedApprox` runs the basic steps of the approximation, giving initial univariate marginal parameter inference (and optionally allowing the user to request latent node or parameter samples). It returns a nested approximation (`nestedApprox`) object.
- Various additional functions can be applied to the nested approximation object to do further components of the approximation, including:
 - `improveParamMarginals`: use AGHQ to improve marginal parameter inference relative to the default inference based on the asymmetric Gaussian approximation.
 - `sampleLatents`: sample from the approximate joint distribution of the latent nodes (the mixture over parameter grid points of multivariate normal distributions).
 - `sampleParams`: sample from the asymmetric Gaussian approximation for the parameters.
 - `calcMarginalLogLikImproved`: use AGHQ to improve estimation of the marginal likelihood relative to the default estimate based on the asymmetric Gaussian approximation.
 - `{d,e,q,r}marginal` and `plotMarginal`: estimate the density (`d`), expectations (`e`), and quantiles (`q`) for a univariate marginal and sample from the marginal (`r`) and plot the marginal density.

9.2.2.1 Example use

[This is a normal-normal case so one can marginalize exactly and do inference (MCMC or MLE or otherwise) on the analytically-marginalized model. We could look for a different example, such as the E. cervi example from our trainings.]

[I think we probably want to shift to parameterizing in terms of std. dev. for interpretability of output and better choice of priors]

We'll use the penicillin example from the `faraway` package, which has data on penicillin production as a function of treatment (four levels) and blend (five levels). The treatment is considered as a fixed effect (note the constant, large variance/small precision for `b[i]`) while the blend/block is considered as a random effect. Note that because of the normal likelihood and normal priors for the latent nodes, the conditional distribution for the latent nodes given the data and hyperparameters is a multivariate normal, so the Laplace approximation in this case is exact. In many uses of nested approximation, the likelihood is not normal, but the latent node distribution is.

```
data(penicillin, package="faraway")

code <- nimbleCode({
  for(i in 1:n) {
    mu[i] <- inprod(b[1:nTreat], x[i, 1:nTreat]) + re[blend[i]]
    y[i] ~ dnorm(mu[i], tau = Tau)
  }
  Tau ~ dgamma(1, 5e-05)
  Tau_re ~ dgamma(1, 5e-05)
  for( i in 1:nTreat ){ b[i] ~ dnorm(0, tau = 0.001) }
  for( i in 1:nBlend ){ re[i] ~ dnorm(0, tau = Tau_re) }
})
X <- model.matrix(~treat, data = penicillin)
data = list(y = penicillin$yield)
constants = list(nTreat = 4, nBlend = 5, n = nrow(penicillin),
  x = X, blend = as.numeric(penicillin$blend))
inits <- list(Tau = 1, Tau_re = 1, b = c(mean(data$y), rep(0,3)), re = rep(0,5))

model <- nimbleModel(code, data = data, constants = constants,
  inits = inits, buildDerivs = TRUE)
comp_model <- compileNimble(model)
```

The prior distributions for the precision parameters, `Tau` and `Tau_re`, are chosen to match INLA's default priors, and for a real analysis, more careful consideration of these would be worthwhile.

Given a NIMBLE model, we can build (and compile) our nested approximation.

```
library(nimbleQuad)
approx <- buildNestedApprox(model = model)
```

```
## Building nested posterior approximation for the following node sets:
##   - parameter nodes: Tau, Tau_re
##   - latent nodes: b (4 elements), re (5 elements)
##   with AGHQ grid for the parameters and Laplace approximation for the latent nodes.
```



```
## Building Laplace approximation.
```

```
comp_approx <- compileNimble(approx, project = model)
```

```
## Compiling
```

```
## [Note] This may take a minute.
```

```
## [Note] Use 'showCompilerOutput = TRUE' to see C++ compilation details.
```

Note that NIMBLE has automatically put the two variance components into the parameters and the fixed and random effects into the latent nodes. As described below, one can change the node sets if desired by passing `paramNodes` and/or `latentNodes`.

Because there are only two parameter nodes, an AGQH grid for the parameters is used by default.

Next we run our nested approximation, getting back initial inference on the parameters.

```
# To prefer fixed rather than scientific notation for easier viewing.
```

```
options(scipen = 2)
```

```
result <- runNestedApprox(comp_approx)
```

```
## method AGHQ quadRules[i] CCD
```

```
##
```

```
## method AGHQ quadRules[i] AGHQ
```

```
##
```

```
## Finding posterior mode for parameter(s).
```

For small models this initial step generally won't take too long, but even for large models or a large number of parameters, this step will generally be faster than the steps shown below because the asymmetric Gaussian distribution approximation for the parameters can be computed quickly, only requiring maximization and computation of the Hessian for the Laplace approximation (that said that could take some time when there is a large number of latent nodes) followed by some simple calculations.

Note that given the vague priors for the parameters used here, the marginal likelihood is probably not useful.

Now, let's refine our estimation of the parameters by using AGHQ in place of the asymmetric Gaussian approximation. That does $d - 1$ dimensional AGHQ (simply 1-d AGHQ) for each parameter requested (by default all of the parameters).

```
result$improveParamMarginals(nodes = 'Tau_re')
```

```
## Calculating inner AGHQ/Laplace approximation at (5) marginal points with 3 quadrature grid p
```

```
## Model (hyper)parameters:
```

```
##           mean           sd           2.5%           25%           50%
## Tau       0.03723785    0.01250226  0.01796493  0.02824207  0.0355641
## Tau_re 21390.01121712 19227.71139513 616.12651240 6156.68361271 15964.3802887
##           75%           97.5%
## Tau       0.04430652    0.06636311
## Tau_re 31747.36659283 69218.96635695
##
```

```
## Marginal log-likelihood (asymmetric Gaussian approximation): -87.86014(*)
```

```
## (*) Marginal log-likelihood is invalid for improper priors and may not be useful
## for non-informative priors.
```

We see that the inference for `Tau_re` has changed somewhat, in particular the more extreme quantiles. Looking at the values on the standard deviation rather than the precision scale would probably be more interpretable.

There are two quantities that control the accuracy of the AGHQ marginal estimate. The first is the number of grid points in the AGHQ numerical integration where k (`nQuad`) is the number of points in one dimension. More is better for accuracy but the computational cost grows as $(d-1)^k$. The marginal density is estimated at each of a set of evaluation points of values of the parameter (`nMarginalGrid` points), using AGHQ for each point, after which a spline is fit to estimate the full marginal density.

By default we use three quadrature points in each dimension and five evaluation points.

If we increase the number of evaluation points, we see it has some effect on the inference.

```
result$improveParamMarginals(nodes = 'Tau_re', nMarginalGrid = 9)
```

```
## Calculating inner AGHQ/Laplace approximation at (9) marginal points with 3 quadrature grid p
```

```
## Model (hyper)parameters:
```

```
##              mean              sd              2.5%              25%              50%
## Tau          0.03723785      0.01250226    0.01796493    0.02824207    0.0355641
## Tau_re 19769.06345669 19213.06376564 509.02664088 5772.43820045 14049.6526705
##              75%              97.5%
## Tau          0.04430652      0.06636311
## Tau_re 28069.93694960 69207.20084103
##
```

```
## Marginal log-likelihood (asymmetric Gaussian approximation): -87.86014(*)
```

```
## (*) Marginal log-likelihood is invalid for improper priors and may not be useful
## for non-informative priors.
```

If we increase the accuracy of the AGHQ integration, there is little effect.

```
result$improveParamMarginals(nodes = 'Tau_re', nMarginalGrid = 9, nQuad = 5)
```

```
## Calculating inner AGHQ/Laplace approximation at (9) marginal points with 5 quadrature grid p
```

```
## Model (hyper)parameters:
```

```
##              mean              sd              2.5%              25%              50%
## Tau          0.03723785      0.01250226    0.01796493    0.02824207    0.0355641
## Tau_re 19769.08046879 19213.06498944 509.03625965 5772.45467883 14049.6714696
##              75%              97.5%
## Tau          0.04430652      0.06636311
## Tau_re 28069.95527479 69207.21901592
##
```

```
## Marginal log-likelihood (asymmetric Gaussian approximation): -87.86014(*)
```

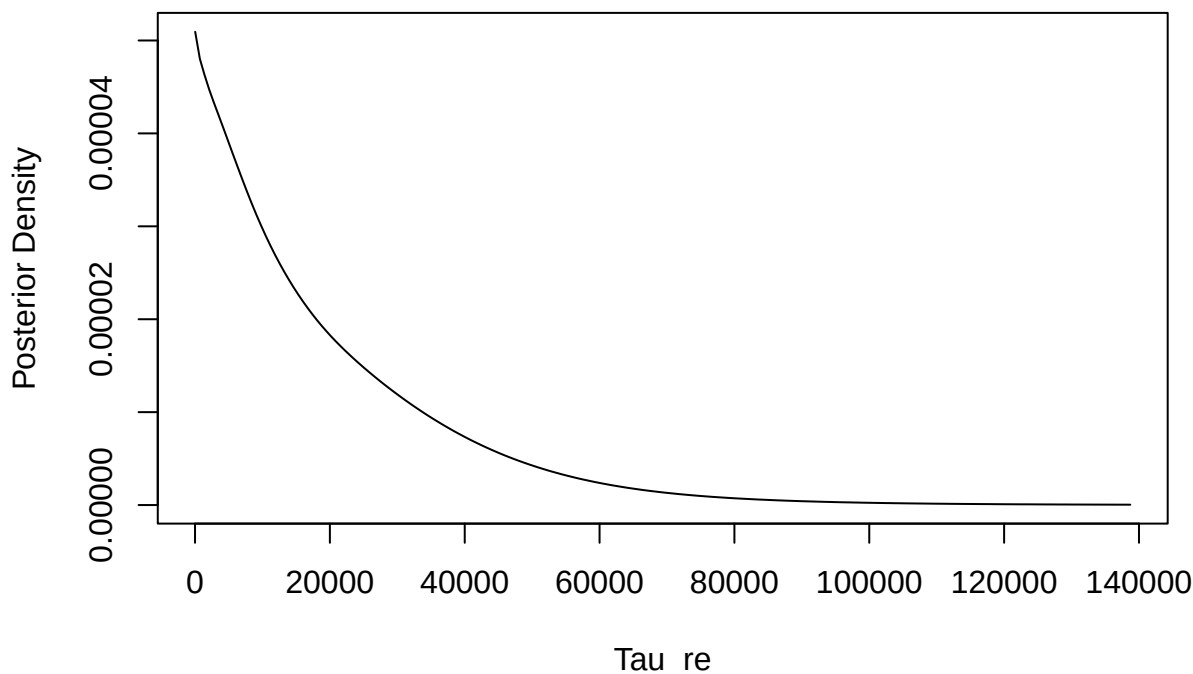
```
## (*) Marginal log-likelihood is invalid for improper priors and may not be useful
## for non-informative priors.
```

We can estimate other expectations or quantiles than those reported and plot a density estimate:

```
result$qmarginal('Tau_re', quantiles = c(.05, .95))
```

```
##      0.05      0.95
## 1031.073 56887.131
```

```
result$plotMarginal('Tau_re')
```



```
## Compute the posterior mean of the random effects standard deviation instead of precision.
prec_to_sd <- function(x) 1/sqrt(x)
result$emarginal('Tau_re', prec_to_sd)
```

```
## [1] 0.01236354
```

Next, we turn to inference on the latent nodes. This is straightforward in principle but can be computationally costly because we sample from a mixture of multivariate normal distributions. This requires Laplace approximation at each of the parameter grid points, where the Laplace approximation can be costly if there are many latent nodes or many grid points. And it requires sampling from the multivariate normal distributions, which can be costly if there are many latent nodes.

```
latent_sample <- result$sampleLatents(n = 1000)
```

```
## method AGHQ quadRules[i] CCD
```

```
##
```

```
## method AGHQ quadRules[i] AGHQ
```

```
##
```

```
## Calculating inner AGHQ/Laplace approximation at 9 outer (parameter) grid points (one dot per
```

```
apply(latent_sample, 2, quantile, c(.025, .25, .5, .75, .975))
```

```
##      b[1]      b[2]      b[3]      b[4]      re[1]      re[2]
```

```
## 2.5% 79.16658 -5.4108622 -1.899548 -4.1033667 -0.0363411644 -0.0351552192
## 25% 82.10253 -0.9164199 2.948704 0.3491649 -0.0077175956 -0.0076002049
## 50% 83.62043 1.2990240 5.184559 2.3957180 0.0000184762 0.0001110363
## 75% 85.19945 3.5704082 7.551397 4.5236792 0.0078309245 0.0085215982
## 97.5% 88.35162 8.2557027 11.964100 8.9400303 0.0355062614 0.0389319919
##          re[3]          re[4]          re[5]
## 2.5% -0.0353717900 -0.0329743269 -0.03310659935
## 25% -0.0080197945 -0.0069093706 -0.00712277642
## 50% -0.0007879027 -0.0001664026 -0.00007946043
## 75% 0.0066856277 0.0083026308 0.00764220204
## 97.5% 0.0370267681 0.0346506873 0.03374579369
```

As a side effect, in the `result` object, we now include a second estimate of the marginal likelihood based on the evaluation of the Laplace approximation on the parameter grid. As note above, with vague priors, the marginal likelihood is probably not useful.

Note that one can optionally return the parameter values corresponding with each latent sample to see the explicit mixture being used, by specifying `includeParams=TRUE`. In this case, with a parameter grid of nine points (three quadrature points in each of two dimensions), there are nine unique sets of parameter values in the sample from the mixture of multivariate normal distributions.

Finally, if one wants joint inference for multiple parameters (or a function of more than one parameter), one can sample from the approximate Gaussian approximation, with a copula-based adjustment so the univariate marginals of the sample match the (current) marginal estimates. This adjustment will use the initial estimate, if `improveParamMarginals` has not been run, or the improved marginal estimates for whichever parameters `improveParamMarginals` has been run for (only `Tau_re` so far in this example).

We'll use the sample to compute a simple functional of multiple parameters, which is a main use case for having a sample rather than simply using the univariate marginals shown above. In this case, we look to see if the random effects variance exceeds the observation variance.

```
param_sample <- result$sampleParams(n = 1000)

mean(1/param_sample[, 'Tau_re'] > 1/param_sample[, 'Tau'])

## [1] 0
```

9.2.2.2 Setting latent nodes and parameters

As discussed above, NIMBLE tries to automatically determine sets of latent nodes and parameters, with random and fixed effects generally included in the latent nodes. In some cases this automatic determination gives sets that are not natural choices because of the model structure and in other cases, a user might want to choose the sets themselves.

Here's the NIMBLE code for a toy GLMM model for illustration.

```
code <- nimbleCode({
  for(i in 1:n) {
    y[i] ~ dbern(p[i])
    logit(p[i]) <- beta1 * x[i] + b[i]
    b[i] ~ dnorm(mu, sd = sigma)
```

```

    }
    mu ~ dflat()
    sigma ~ dunif(0, 100)
    beta1 ~ dflat()
  })
n <- 30
x <- rnorm(n)
y <- rbinom(n, 1, 0.5)

model <- nimbleModel(code, constants = list(n = n), data = list(y = y, x = x))

## Defining model

## Building model

## Setting data and initial values

## Running calculate on model
##   [Note] Any error reports that follow may simply reflect missing values in model variables

## Checking model sizes and dimensions

##   [Note] This model is not fully initialized. This is not an error.
##           To see which variables are not initialized, use model$initializeInfo().
##           For more information on model initialization, see help(modelInitialization).

approx <- buildNestedApprox(model)

## Building nested posterior approximation for the following node sets:
##   - parameter nodes: mu, sigma
##   - latent nodes: beta1, b (30 elements)
##   with AGHQ grid for the parameters and Laplace approximation for the latent nodes.

## Building Laplace approximation.

```

We see that NIMBLE has put the fixed effect (`beta1`) and random effects (`b[i]`) in the latent nodes and the random effects parameters, *mu* and *sigma*, into the parameter set.

In the model above, the role of the overall intercept is played by the random effects mean. We could instead have centered the random effects on zero and introduced an explicit intercept.

```

code <- nimbleCode({
  for(i in 1:n) {
    y[i] ~ dbern(p[i])
    logit(p[i]) <- beta0 + beta1 * x[i] + b[i]
    b[i] ~ dnorm(0, sd = sigma)
  }
  mu ~ dflat()
  sigma ~ dunif(0, 100)
  beta0 ~ dflat()
  beta1 ~ dflat()
})

```

```

n <- 30
x <- rnorm(n)
y <- rbinom(n, 1, 0.5)

model <- nimbleModel(code, constants = list(n = n), data = list(y = y, x = x))

## Defining model
## Building model
## Setting data and initial values
## Running calculate on model
## [Note] Any error reports that follow may simply reflect missing values in model variables
## Checking model sizes and dimensions
## [Note] This model is not fully initialized. This is not an error.
##         To see which variables are not initialized, use model$initializeInfo().
##         For more information on model initialization, see help(modelInitialization).

approx <- buildNestedApprox(model)

## Building nested posterior approximation for the following node sets:
## - parameter nodes: sigma
## - latent nodes: beta0, beta1, b (30 elements)
## with AGHQ grid for the parameters and Laplace approximation for the latent nodes.

## Building Laplace approximation.

```

The split into latent nodes and parameters is equivalent.

In contrast, if we use a non-centered parameterization (this arises often when using HMC as a better parameterization), then NIMBLE tries to put everything into the parameters, and an error occurs.

```

code <- nimbleCode({
  for(i in 1:n) {
    y[i] ~ dbern(p[i])
    logit(p[i]) <- beta0 + beta1 * x[i] + sigma*b[i]
    b[i] ~ dnorm(0, 1)
  }
  mu ~ dflat()
  sigma ~ dunif(0, 100)
  beta0 ~ dflat()
  beta1 ~ dflat()
})
n <- 30
x <- rnorm(n)
y <- rbinom(n, 1, 0.5)

model <- nimbleModel(code, constants = list(n = n), data = list(y = y, x = x))
approx <- buildNestedApprox(model)

```

```
## Error in buildNestedApprox(model): No latent nodes detected in model. Check the model structure
```

Instead, we could manually specify the parameters. [possible condIndepSets bug causing 2 laplace approximations here...]

```
code <- nimbleCode({
  for(i in 1:n) {
    y[i] ~ dbern(p[i])
    logit(p[i]) <- beta0 + beta1 * x[i] + sigma*b[i]
    b[i] ~ dnorm(0, 1)
  }
  mu ~ dflat()
  sigma ~ dunif(0, 100)
  beta0 ~ dflat()
  beta1 ~ dflat()
})
n <- 30
x <- rnorm(n)
y <- rbinom(n, 1, 0.5)

model <- nimbleModel(code, constants = list(n = n), data = list(y = y, x = x))
approx <- buildNestedApprox(model, paramNodes = "sigma")
```

In some cases it can be computationally advantageous to move fixed effects into the parameter set so that the Laplace approximation can be done as a product of lower-dimensional Laplace approximations. Here's an example of doing that manually.

```
code <- nimbleCode({
  for(i in 1:n) {
    y[i] ~ dbern(p[i])
    logit(p[i]) <- beta0 + beta1 * x[i] + sigma*b[i]
    b[i] ~ dnorm(0, 1)
  }
  mu ~ dflat()
  sigma ~ dunif(0, 100)
  beta0 ~ dflat()
  beta1 ~ dflat()
})
n <- 30
x <- rnorm(n)
y <- rbinom(n, 1, 0.5)

model <- nimbleModel(code, constants = list(n = n), data = list(y = y, x = x))

## Defining model
## Building model
## Setting data and initial values
```

```

## Running calculate on model
##   [Note] Any error reports that follow may simply reflect missing values in model variables
## Checking model sizes and dimensions
##   [Note] This model is not fully initialized. This is not an error.
##           To see which variables are not initialized, use model$initializeInfo().
##           For more information on model initialization, see help(modelInitialization).
approx <- buildNestedApprox(model, paramNodes = c("beta0", "beta1", "sigma"))

## Building nested posterior approximation for the following node sets:
##   - parameter nodes: beta0, beta1, sigma
##   - latent nodes: mu, b (30 elements)
##   with CCD grid for the parameters and Laplace approximation for the latent nodes.
## Building 31 individual Laplace approximations (one dot for each): .....

```

Note the messaging indicating that 30 Laplace approximations (one for each conditionally-independent set of nodes containing a y_i, b_i pair) were built. (In other, perhaps more common situations, one might have grouping structure such that a single random effect is associated with multiple observations such that the random effect and associated observations are conditionally independent of other random effects and observations given the parameters.)

Chapter 10

Spatial models

NIMBLE supports two variations of conditional autoregressive (CAR) model structures: the improper intrinsic Gaussian CAR (ICAR) model, and a proper Gaussian CAR model. This includes distributions to represent these spatially-dependent model structures in a BUGS model, as well as specialized MCMC samplers for these distributions.

10.1 Intrinsic Gaussian CAR model: *dcar_normal*

The intrinsic Gaussian conditional autoregressive (ICAR) model used to model dependence of block-level values (e.g., spatial areas or temporal blocks) is implemented in NIMBLE as the `dcar_normal` distribution. Additional details for using this distribution are available using `help('CAR-Normal')`.

ICAR models are improper priors for random fields (e.g., temporal or spatial processes). The prior is a joint prior across a collection of latent process values. For more technical details on CAR models, including higher-order CAR models, please see [Rue and Held \(2005\)](#), [Banerjee et al. \(2015\)](#), and [Paciorek \(2009\)](#). Since the distribution is improper it should not be used as the distribution for data values, but rather to specify a prior for an unknown process. As discussed in the references above, the distribution can be seen to be a proper density in a reduced dimension subspace; thus the impropriety only holds on one or more linear combinations of the latent process values.

In addition to our focus here on CAR modeling for spatial data, the ICAR model can also be used in other contexts, such as for temporal data in a discrete time context.

10.1.1 Specification and density

NIMBLE uses the same parameterization as WinBUGS / GeoBUGS for the `dcar_normal` distribution, providing compatibility with existing WinBUGS code. NIMBLE also provides the WinBUGS name `car.normal` as an alias.

10.1.1.1 Specification

The `dcar_normal` distribution is specified for a set of N spatially dependent regions as:

```
x[1:N] ~ dcar_normal(adj, weights, num, tau, c, zero_mean)
```

The `adj`, `weights` and `num` parameters define the adjacency structure and associated weights of the spatially-dependent field. See `help('CAR-Normal')` for details of these parameters. When specifying a CAR distribution, these parameters must have constant values. They do not necessarily have to be specified as `constants` when creating a model object using `nimbleModel`, but they should be defined in a static way: as right-hand-side only variables with initial values provided as `constants`, `data` or `inits`, or using fixed numerical deterministic declarations. Each of these two approaches for specifying values are shown in the example.

The adjacency structure defined by `adj` and the associated `weights` must be symmetric. That is, if region i is neighbor of region j , then region j must also be a neighbor of region i . Further, the weights associated with these reciprocating relationships must be equal. NIMBLE performs a check of these symmetries and will issue an error message if asymmetry is detected.

The scalar precision `tau` may be treated as an unknown model parameter and itself assigned a prior distribution. Care should be taken in selecting a prior distribution for `tau`, and WinBUGS suggests that users be prepared to carry out a sensitivity analysis for this choice.

When specifying a higher-order CAR process, the number of constraints `c` can be explicitly provided in the model specification. This would be the case, for example, when specifying a thin-plate spline (second-order) CAR model, for which `c` should be 2 for a one-dimensional process and 3 for a two-dimensional (e.g., spatial) process, as discussed in [Rue and Held \(2005\)](#) and [Paciorek \(2009\)](#). If `c` is omitted, NIMBLE will calculate `c` as the number of disjoint groups of regions in the adjacency structure, which implicitly assumes a first-order CAR process for each group.

By default there is no zero-mean constraint imposed on the CAR process, and thus the mean is implicit within the CAR process values, with an implicit improper flat prior on the mean. To avoid non-identifiability, one should not include an additional parameter for the mean (e.g., do not include an intercept term in a simple CAR model with first-order neighborhood structure). When there are disjoint groups of regions and the constraint is not imposed, there is an implicit distinct improper flat prior on the mean for each group, and it would not make sense to impose the constraint since the constraint holds across all regions. Similarly, if one sets up a neighborhood structure for higher-order CAR models, it would not make sense to impose the zero-mean constraint as that would account for only one of the eigenvalues that are zero. Imposing this constraint (by specifying the parameter `zero_mean = 1`) allows users to model the process mean separately, and hence a separate intercept term should be included in the model.

NIMBLE provides a convenience function `as.carAdjacency` for converting other representations of the adjacency information into the required `adj`, `weights`, `num` format. This function can convert:

- A symmetric adjacency matrix of weights (with diagonal elements equal to zero), using `as.carAdjacency(weightMatrix)`
- Two length- N lists with numeric vector elements giving the neighboring indices and associated weights for each region, using `as.carAdjacency(neighborList, weightList)`

These conversions should be done in R, and the resulting `adj`, `weights`, `num` vectors can be passed as `constants` into `nimbleModel`.

10.1.1.2 Density

For process values $x = (x_1, \dots, x_N)$ and precision τ , the improper CAR density is given as:

$$p(x|\tau) \propto \tau^{(N-c)/2} e^{-\frac{\tau}{2} \sum_{i \neq j} w_{ij} (x_i - x_j)^2}$$

where the summation over all (i, j) pairs, with the weight between regions i and j given by w_{ij} , is equivalent to summing over all pairs for which region i is a neighbor of region j . Note that the value of c modifies the power to which the precision is raised, accounting for the impropriety of the density based on the number of zero eigenvalues in the implicit precision matrix for x .

For the purposes of MCMC sampling the individual CAR process values, the resulting conditional prior of region i is:

$$p(x_i|x_{-i}, \tau) \sim N\left(\frac{1}{w_{i+}} \sum_{j \in \mathcal{N}_i} w_{ij} x_j, w_{i+} \tau\right)$$

where x_{-i} represents all elements of x except x_i , the neighborhood $\mathcal{N}_i$ of region i is the set of all j for which region j is a neighbor of region i , $w_{i+} = \sum_{j \in \mathcal{N}_i} w_{ij}$, and the Normal distribution is parameterized in terms of precision.

10.1.2 Example

Here we provide an example model using the intrinsic Gaussian `dcar_normal` distribution. The CAR process values are used in a spatially-dependent Poisson regression.

To mimic the behavior of WinBUGS, we specify `zero_mean = 1` to enforce a zero-mean constraint on the CAR process, and therefore include a separate intercept term `alpha` in the model. Note that we do not necessarily recommend imposing this constraint, per the discussion earlier in this chapter.

```
code <- nimbleCode({
  alpha ~ dflat()
  beta ~ dnorm(0, 0.0001)
  tau ~ dgamma(0.001, 0.001)
  s[1:N] ~ dcar_normal(adj[1:L], weights[1:L], num[1:N], tau, zero_mean = 1)
  for(i in 1:N) {
    log(lambda[i]) <- alpha + beta*x[i] + s[i]
    y[i] ~ dpois(lambda[i])
  }
})

L <- 8
constants <- list(N = 4, L = L, num = c(3, 2, 2, 1), weights = rep(1, L),
  adj = c(2,3,4,1,3,1,2,1), x = c(0, 2, 2, 8))
data <- list(y = c(6, 9, 7, 12))
inits <- list(alpha = 0, beta = 0, tau = 1, s = c(0, 0, 0, 0))
Rmodel <- nimbleModel(code, constants, data, inits)
```

The resulting model may be carried through to MCMC sampling. NIMBLE will assign a specialized sampler to the update the elements of the CAR process. See Chapter 7 for information about NIMBLE's MCMC engine, and Section 10.3 for details on MCMC sampling of the CAR processes.

10.2 Proper Gaussian CAR model: *dcar_proper*

The proper Gaussian conditional autoregressive model used to model dependence of block-level values (e.g., spatial areas or temporal blocks) is implemented in NIMBLE as the `dcar_proper` distribution. Additional details of using this distribution are available using `help('CAR-Proper')`.

Proper CAR models are proper priors for random fields (e.g., temporal or spatial processes). The prior is a joint prior across a collection of latent process values. For more technical details on proper CAR models please see [Banerjee et al. \(2015\)](#), including considerations of why the improper CAR model may be preferred.

In addition to our focus here on CAR modeling for spatial data, the proper CAR model can also be used in other contexts, such as for temporal data in a discrete time context.

10.2.1 Specification and density

NIMBLE uses the same parameterization as WinBUGS / GeoBUGS for the `dcar_proper` distribution, providing compatibility with existing WinBUGS code. NIMBLE also provides the WinBUGS name `car.proper` as an alias.

10.2.1.1 Specification

The `dcar_proper` distribution is specified for a set of N spatially dependent regions as:

```
x[1:N] ~ dcar_proper(mu, C, adj, num, M, tau, gamma)
```

There is no option of a zero-mean constraint for proper CAR process, and instead the mean for each region is specified by the `mu` parameter. The elements of `mu` can be assigned fixed values or may be specified using one common, or multiple, prior distributions.

The `C`, `adj`, `num` and `M` parameters define the adjacency structure, normalized weights, and conditional variances of the spatially-dependent field. See `help('CAR-Proper')` for details of these parameters. When specifying a CAR distribution, these parameters must have constant values. They do not necessarily have to be specified as `constants` when creating a model object using `nimbleModel`, but they should be defined in a static way: as right-hand-side only variables with initial values provided as `constants`, `data` or `inits`, or using fixed numerical deterministic declarations.

The adjacency structure defined by `adj` must be symmetric. That is, if region i is neighbor of region j , then region j must also be a neighbor of region i . In addition, the normalized weights specified in `C` must satisfy a symmetry constraint jointly with the conditional variances given in `M`. This constraint requires that $M^{-1}C$ is symmetric, where M is a diagonal matrix of conditional variances and C is the normalized (each row sums to one) weight matrix. Equivalently, this implies that $C_{ij}M_{jj} = C_{ji}M_{ii}$ for all pairs of neighboring regions i and j . NIMBLE performs a check of these symmetries and will issue an error message if asymmetry is detected.

Two options are available to simplify the process of constructing the `C` and `M` arguments; both options are demonstrated in the example. First, these arguments may be omitted from the `dcar_proper` specification. In this case, values of `C` and `M` will be generated that correspond to all weights being equal to one, or equivalently, a symmetric weight matrix containing only zeros and ones. Note that `C` and `M` should either *both* be provided, or *both* be omitted from the specification.

Second, a convenience function `as.carCM` is provided to generate the `C` and `M` arguments corresponding to a specified set of symmetric unnormalized weights. If `weights` contains the non-zero weights corresponding to an unnormalized weight matrix (`weights` is precisely the argument that can be used in the `dcar_normal` specification), then a list containing `C` and `M` can be generated using `as.carCM(adj, weights, num)`. In this case, the resulting `C` contains the row-normalized weights, and the resulting `M` is a vector of the inverse row-sums of the unnormalized weight matrix.

The scalar precision `tau` may be treated as an unknown model parameter and itself assigned a prior distribution. Care should be taken in selecting a prior distribution for `tau`, and WinBUGS suggests that users be prepared to carry out a sensitivity analysis for this choice.

An appropriate value of the `gamma` parameter ensures the propriety of the `dcar_proper` distribution. The value of `gamma` must lie between fixed bounds, which are given by the reciprocals of the largest and smallest eigenvalues of $M^{-1/2}CM^{1/2}$. These bounds can be calculated using the function `carBounds` or separately using the functions `carMinBound` and `carMaxBound`. For compatibility with WinBUGS, NIMBLE provides `min.bound` and `max.bound` as aliases for `carMinBound` and `carMaxBound`. Rather than selecting a fixed value of `gamma` within these bounds, it is recommended that `gamma` be assigned a uniform prior distribution over the region of permissible values.

Note that when `C` and `M` are omitted from the `dcar_proper` specification (and hence all weights are taken as one), or `C` and `M` are calculated from a symmetric weight matrix using the utility function `as.carCM`, then the bounds on `gamma` are necessarily $(-1, 1)$. In this case, `gamma` can simply be assigned a prior over that region. This approach is shown in both examples.

10.2.1.2 Density

The proper CAR density is given as:

$$p(x|\mu, C, M, \tau, \gamma) \sim \text{MVN}\left(\mu, \frac{1}{\tau}(I - \gamma C)^{-1}M\right)$$

where the multivariate normal distribution is parameterized in terms of covariance.

For the purposes of MCMC sampling the individual CAR process values, the resulting conditional prior of region i is:

$$p(x_i|x_{-i}, \mu, C, M, \tau, \gamma) \sim \text{N}\left(\mu_i + \sum_{j \in \mathcal{N}_i} \gamma C_{ij}(x_j - \mu_j), \frac{M_{ii}}{\tau}\right)$$

where x_{-i} represents all elements of x except x_i , the neighborhood $\mathcal{N}_i$ of region i is the set of all j for which region j is a neighbor of region i , and the Normal distribution is parameterized in terms of variance.

10.2.2 Example

We provide two example models using the proper Gaussian `dcar_proper` distribution. In both, the CAR process values are used in a spatially-dependent logistic regression to model binary presence/absence data. In the first example, the `C` and `M` parameters are omitted, which uses weights equal to one for all neighbor relationships. In the second example, symmetric unnormalized weights are specified, and `as.carCM` is used to construct the `C` and `M` parameters to the `dcar_proper` distribution.

```

# omitting C and M sets all non-zero weights to one
code <- nimbleCode({
  mu0 ~ dnorm(0, 0.0001)
  tau ~ dgamma(0.001, 0.001)
  gamma ~ dunif(-1, 1)
  s[1:N] ~ dcar_proper(mu[1:N], adj=adj[1:L], num=num[1:N], tau=tau,
                      gamma=gamma)

  for(i in 1:N) {
    mu[i] <- mu0
    logit(p[i]) <- s[i]
    y[i] ~ dbern(p[i])
  }
})

adj <- c(2, 1, 3, 2, 4, 3)
num <- c(1, 2, 2, 1)
constants <- list(adj = adj, num = num, N = 4, L = 6)
data <- list(y = c(1, 0, 1, 1))
inits <- list(mu0 = 0, tau = 1, gamma = 0, s = rep(0, 4))
Rmodel <- nimbleModel(code, constants, data, inits)

# specify symmetric unnormalized weights, use as.carCM to generate C and M
code <- nimbleCode({
  mu0 ~ dnorm(0, 0.0001)
  tau ~ dgamma(0.001, 0.001)
  gamma ~ dunif(-1, 1)
  s[1:N] ~ dcar_proper(mu[1:N], C[1:L], adj[1:L], num[1:N], M[1:N], tau,
                      gamma)

  for(i in 1:N) {
    mu[i] <- mu0
    logit(p[i]) <- s[i]
    y[i] ~ dbern(p[i])
  }
})

weights <- c(2, 2, 3, 3, 4, 4)
CM <- as.carCM(adj, weights, num)
constants <- list(C = CM$C, adj = adj, num = num, M = CM$M, N = 4, L = 6)
Rmodel <- nimbleModel(code, constants, data, inits)

```

Each of the resulting models may be carried through to MCMC sampling. NIMBLE will assign a specialized sampler to update the elements of the CAR process. See Chapter 7 for information about NIMBLE's MCMC engine, and Section 10.3 for details on MCMC sampling of the CAR processes.

10.3 MCMC Sampling of CAR models

NIMBLE's MCMC engine provides specialized samplers for the `dcar_normal` and `dcar_proper` distributions. These samplers perform sequential univariate updates on the components of the CAR process. Internally, each sampler assigns one of three specialized univariate samplers to each component, based on inspection of the model structure:

1. A conjugate sampler in the case of conjugate Normal dependencies.
2. A random walk Metropolis-Hastings sampler in the case of non-conjugate dependencies.
3. A posterior predictive sampler in the case of no dependencies.

Note that these univariate CAR samplers are not the same as NIMBLE's standard `conjugate`, `RW`, and `posterior_predictive` samplers, but rather specialized versions for operating on a CAR distribution. Details of these assignments are strictly internal to the CAR samplers.

In future versions of NIMBLE we expect to provide block samplers that update the entire CAR process as a single sample. This may provide improved MCMC performance by accounting for dependence between elements, particularly when conjugacy is available.

10.3.1 Initial values

Valid initial values should be provided for all elements of the process specified by a CAR structure before running an MCMC. This ensures that the conditional prior distribution is well-defined for each region. A simple and safe choice of initial values is setting all components of the process equal to zero, as is done in the preceding CAR examples.

For compatibility with WinBUGS, NIMBLE also allows an initial value of `NA` to be provided for regions with zero neighbors. This particular initialization is required in WinBUGS, so this allows users to make use of existing WinBUGS code.

10.3.2 Zero-neighbor regions

Regions with zero neighbors (defined by a 0 appearing in the `num` parameter) are a special case for both the `dcar_normal` and `dcar_proper` distribution. The corresponding neighborhood $\mathcal{N}$ of such a region contains no elements, and hence the conditional prior is improper and uninformative, tantamount to a `dflat` prior distribution. Thus, the conditional posterior distribution of those regions is entirely determined by the dependent nodes, if any. Sampling of these zero-neighbor regions proceeds as:

1. In the conjugate case, sampling proceeds according to the conjugate posterior.
2. In the non-conjugate case, sampling proceeds using random walk Metropolis-Hastings, where the posterior is determined entirely by the dependencies.
3. In the case of no dependents, the posterior is entirely undefined. Here, no changes will be made to the process value, and it will remain equal to its initial value throughout. By virtue of having no neighbors, this region does not contribute to the density evaluation of the subsuming `dcar_normal` node nor to the conditional prior of any other regions, hence its value (even `NA`) is of no consequence.

This behavior is different from that of WinBUGS, where the value of zero-neighbor regions of `car.normal` nodes is set to and fixed at zero.

Also note that for the `dcar_proper` distribution if any regions have zero neighbors the joint density of the process cannot be calculated. As a result one cannot do MCMC sampling of unknown parameters affecting the mean (μ) of the process, though one can instead use an uncentered parameterization in which the mean is added to the process rather than the process being centered on the mean. Or one could remove such regions from the process and model them separately.

10.3.3 Zero-mean constraint

A zero-mean constraint is available for the intrinsic Gaussian `dcar_normal` distribution. This constraint on the ICAR process values is imposed during MCMC sampling, if the argument `zero_mean = 1`, mimicking the behavior of WinBUGS. Following the univariate updates on each component, the mean is subtracted away from all process values, resulting in a zero-mean process.

Note that this is *not* equivalent to sampling under the constraint that the mean is zero (see p. 36 of [Rue and Held \(2005\)](#)) so should be treated as an ad hoc approach and employed with caution.

Chapter 11

Bayesian nonparametric models

11.1 Bayesian nonparametric mixture models

NIMBLE provides support for Bayesian nonparametric (BNP) mixture modeling. The current implementation provides support for hierarchical specifications involving Dirichlet process (DP) mixtures (Ferguson, 1973, 1974; Lo, 1984; Escobar, 1994; Escobar and West, 1995). These allow one to avoid specifying a particular parametric distribution for a given node (parameter) in a model. Instead one can use a DP mixture as a much more general, nonparametric distribution. For example, a normal distribution for a random effect could be replaced by a DP mixture of normal distributions, with the number of components of the mixture being determined from the data and not fixed in advance.

We'll first introduce the general, technical definition of a DP mixture model before describing the Chinese Restaurant Process representation, which may be more interpretable for many readers. More specifically, a DP mixture model for a random variable y_i takes the form

$$y_i \mid G \stackrel{iid}{\sim} \int h(y_i \mid \theta) G(d\theta),$$
$$G \mid \alpha, G_0 \sim DP(\alpha, G_0),$$

where $h(\cdot \mid \theta)$ is a suitable kernel (i.e., probability density/mass function) with parameter θ , and α and G_0 are the concentration and baseline distribution parameters of the DP, respectively. DP mixture models can be written with different levels of hierarchy, all being equivalent to the model above. While “y” would often be used as notation for a data value, it is used generically here, noting that often DP mixtures are used for random effects rather than directly for observations.

When the random distribution (also referred to as a random ‘measure’) G is integrated out from the model, the DP mixture model can be written using latent or membership variables, z_i , following a Chinese Restaurant Process (CRP) distribution (Blackwell and MacQueen, 1973), discussed in Section 11.2. The model takes the form

$$y_i \mid \tilde{\theta}, z_i \stackrel{ind}{\sim} h(\cdot \mid \tilde{\theta}_{z_i}),$$
$$z \mid \alpha \sim CRP(\alpha), \quad \tilde{\theta}_j \stackrel{iid}{\sim} G_0,$$

where $\text{CRP}(\alpha)$ denotes the CRP distribution with concentration parameter α . Put in perhaps more intuitive terms, z_i says which cluster/group the i th unit is in, and the parameter $\tilde{\theta}_j$ for group j is distributed according to the G_0 baseline distribution. The parameter α controls how dispersed the clustering is, described more in the next section.

If a stick-breaking representation (Sethuraman, 1994), discussed in section 11.3, is assumed for the random distribution (measure) G , then the model takes the form

$$y_i \mid \theta^*, v \stackrel{\text{ind}}{\sim} \sum_{l=1}^{\infty} \left\{ v_l \prod_{m < l} (1 - v_m) \right\} h(\cdot \mid \theta_l^*),$$

$$v_l \mid \alpha \stackrel{\text{iid}}{\sim} \text{Beta}(1, \alpha), \quad \theta_l^* \stackrel{\text{iid}}{\sim} G_0.$$

More general representations of the random distribution can be specify by considering $v_l \mid \nu_l, \alpha_l \stackrel{\text{ind}}{\sim} \text{Beta}(\nu_l, \alpha_l)$. Finite dimensional approximations can be obtained by truncating the infinite sum to have L components.

Different representations of DP mixtures lead to different computational algorithms. NIMBLE supports sampling algorithms based on the CRP representation, as well as on the stick-breaking representation. NIMBLE includes definitions of structures required to implement the CRP and stick-breaking distributions, and the associated MCMC algorithms.

11.2 Chinese Restaurant Process model

The CRP is a distribution over the space of partitions of positive integers and is implemented in NIMBLE as the `dCRP` distribution. More details for using this distribution are available using `help(CRP)`.

The CRP can be described as a stochastic process in which customers arrive at a restaurant, potentially with an infinite number of tables. Each customer sits at an empty or occupied table according to probabilities that depend on the number of customers in the occupied tables. Thus, the CRP partitions the set of customers, through their assignment to tables in the restaurant.

11.2.1 Specification and density

NIMBLE parametrizes the `dCRP` distribution by a concentration parameter and a size parameter.

11.2.1.1 Specification

The `dCRP` distribution is specified in NIMBLE for a membership vector $\mathbf{z}$ as

`z[1:N] ~ dCRP(conc, size)`

The `conc` parameter is the concentration parameter of the CRP, controlling the probability of a customer sitting on a new table, i.e., creating a new cluster. The `size` parameter defines the size of the set of integers to be partitioned.

The `conc` parameter is a positive real value that can be treated as known or unknown. When a gamma prior is assumed for the `conc` parameter, a specialized sampler is assigned. See more on this in section 11.4.1.

The **size** parameter is a positive integer that has to be fixed and equal to the length of vector $\mathbf{z}$. It defines the set of consecutive integers from 1 to N to be partitioned. Each element in $\mathbf{z}$ can be an integer from 1 to N , and repetitions are allowed.

11.2.1.2 Density

The CRP distribution partitions the set of positive integers $1, \dots, N$, into $N^* \leq N$ disjoint subsets, indicating to which subset each element belongs. For instance, if $N = 6$, the set $\{1, 2, 3, 4, 5, 6\}$ can be partitioned into the subsets $S_1 = \{1, 2, 6\}$, $S_2 = \{4, 5\}$, and $S_3 = \{3\}$. Note that $N^* = 3$, and this is one partition from out of 203 possibilities. The CRP-distributed vector z encodes this partition and its observed values would be $(1, 1, 3, 2, 2, 1)$, for this example. In mixture modeling, this indicates that observations 1, 2, and 6 belong to cluster 1, observations 4 and 5 to cluster 2, and observation 3 to cluster 3. Note that this representation is not unique, vector $(2, 2, 1, 3, 3, 2)$ encodes the same partition.

The joint probability function of $z = (z_1, \dots, z_N)$, with concentration parameter α , is given by

$$p(z \mid \alpha) \propto \frac{\Gamma(\alpha)}{\Gamma(\alpha + n)} \alpha^{N^*(z)} \prod_{k=1}^{N^*(z)} \Gamma(m_k(z)),$$

where $m_k(z)$ denotes the number of elements in z that are equal to k , $N^*(z)$ denotes the number of unique elements in z , and $\Gamma(\cdot)$ denotes the gamma function. The full conditional distribution for z_i given z_{-i} is

$$p(z_i = m \mid z_{-i}, \alpha) = \frac{1}{n - 1 + \alpha} \sum_{j \neq i} 1_{\{z_j\}}(m) + \frac{\alpha}{n - 1 + \alpha} 1_{\{z^{new}\}}(m),$$

where z_{-i} denotes vector z after removing its i -th component, z^{new} is a value not in z_{-i} , and 1_A denotes the indicator function at set A .

Note that the probability of creating a new cluster is proportional to α : the larger the concentration parameter, the more clusters are created.

11.2.2 Example

The following example illustrates how to use NIMBLE to perform single density estimation for real-valued data, under a BNP approach, using the dCRP distribution. (Note that the BNP approach is also often used to perform density estimation on random effects.) The model is given by

$$\begin{aligned} y_i \mid \tilde{\theta}, \tilde{\sigma}^2, z_i &\stackrel{iid}{\sim} N(\tilde{\theta}_{z_i}, \tilde{\sigma}_{z_i}^2), \quad i = 1, \dots, N, \\ z &\sim \text{CRP}(\alpha), \quad \alpha \sim \text{Gamma}(1, 1), \\ \tilde{\theta}_j &\stackrel{iid}{\sim} N(0, 100), \quad \tilde{\sigma}_j^2 \stackrel{iid}{\sim} \text{InvGamma}(1, 1), \quad j = 1, \dots, M. \end{aligned}$$

```
code <- nimbleCode({
  z[1:N] ~ dCRP(alpha, size = N)
  alpha ~ dgamma(1, 1)
  for(i in 1:M) {
```

```

    thetatilde[i] ~ dnorm(0, var = 100)
    s2tilde[i] ~ dinvgamma(1, 1)
  }
  for(i in 1:N)
    y[i] ~ dnorm(thetatilde[z[i]], var = s2tilde[z[i]])
})

set.seed(1)
constants <- list(N = 100, M = 50)
data <- list(y = c(rnorm(50, -5, sqrt(3)), rnorm(50, 5, sqrt(4))))
inits <- list(thetatilde = rnorm(constants$M, 0, 10),
             s2tilde = rinvgamma(constants$M, 1, 1),
             z = sample(1:10, size = constants$N, replace = TRUE),
             alpha = 1)
model <- nimbleModel(code, constants, data, inits)

```

The model can be fitted through MCMC sampling. NIMBLE will assign a specialized sampler to update z and α . See Chapter 7 for information about NIMBLE’s MCMC engine, and Section 11.4.1 for details on MCMC sampling of the CRP.

One of the advantages of BNP mixture models is that the number of clusters is treated as random. Therefore, in MCMC sampling, the number of cluster parameters varies with the iteration. Since NIMBLE does not currently allow dynamic length allocation, the number of unique cluster parameters, N^* , has to be fixed. One safe option is to set this number to N , but this is inefficient, both in terms of computation and in terms of storage, because in practice it is often that $N^* < N$. In addition, configuring and building the MCMC can be slow (and use a lot of memory) for large N . In an effort to mitigate these inefficiencies, we allow the user to set $N^* = M$, with $M < N$, as seen in the example above. However, if this number is too small and is exceeded in any iteration a warning is issued.

11.2.3 Extensions

The BNP functionality in NIMBLE was extended in version 0.10.0 to more general models. These extensions enable users, for instance, to use a DP or DPM prior for the distribution of the random effects in a generalized linear mixed effects model with mutiple measurements over time or multiple trials per participant.

The following example illustrates how to use NIMBLE in a random effects model with repeated measurements per subject using a DP prior for the distribution of the subject’s random effects. The model is given by

$$y_{i,j} \mid \tilde{\theta}, z_i, \sigma^2 \sim N(\tilde{\theta}_{z_i}, \sigma^2), \quad i = 1, \dots, N, \quad j = 1, \dots, J,$$

$$z \sim \text{CRP}(\alpha), \quad \alpha \sim \text{Gamma}(1, 1),$$

$$\tilde{\theta}_m \stackrel{iid}{\sim} N(0, 100), \quad m = 1, \dots, M, \quad \sigma^2 \sim \text{InvGamma}(1, 1).$$

The corresponding NIMBLE code is

```
code <- nimbleCode({
  z[1:N] ~ dCRP(alpha, size = N)
  alpha ~ dgamma(1, 1)
  sigma2 ~ dinvgamma(1, 1)
  for(i in 1:M) {
    thetatilde[i] ~ dnorm(0, var = 100)
  }
  for(i in 1:N) {
    for(j in 1:J) {
      y[i, j] ~ dnorm(thetatilde[z[i]], var = sigma2)
    }
  }
})

set.seed(1)
constants <- list(N = 10, J = 5, M = 5)
data <- list(y = matrix(c(rnorm(25, -25, 1), rnorm(25, 25, 1)), ncol=constants$J,
                        nrow=constants$N, byrow=TRUE))
inits <- list(thetatilde = rnorm(constants$M, 0, 10),
              z = sample(1:5, size = constants$N, replace = TRUE),
              alpha = 1,
              sigma2 = 1)
modelRandEff <- nimbleModel(code, constants, data, inits)
```

Alternatively, each subject could have a vector of parameters being clustered. For example in the model above one could instead specify a vector of means, such as `thetaTilde[z[i], j]`, instead of a single mean. This allows group-specific parameters to also vary across the repeated measurements.

As before, the model can be fitted through MCMC sampling. NIMBLE will assign a specialized sampler to update `z` and `alpha`. See Chapter 7 for information about NIMBLE's MCMC engine, and Section 11.4.1 for details on MCMC sampling of the CRP.

11.3 Stick-breaking model

In NIMBLE, weights defined by sequentially breaking a stick, as in the stick-breaking process, are implemented as the `stick_breaking` link function. More details for using this function are available using `help(stick_breaking)`.

11.3.1 Specification and function

NIMBLE parametrizes the `stick_breaking` function by vector of values in $(0, 1)$.

11.3.1.1 Function

The weights $(w_1, \dots, w_L)$ follow a finite stick-breaking construction if

$$\begin{aligned}
w_1 &= v_1, \\
w_l &= v_l \prod_{m < l} (1 - v_m), \quad l = 2, \dots, L-1 \\
w_L &= \prod_{m < L} (1 - v_m).
\end{aligned}$$

for $v_l \in [0, 1], l = 1, \dots, L-1$.

11.3.1.2 Specification

The `stick_breaking` function is specified in NIMBLE for a vector `w` of probabilities as

```
w[1:L] <- stick_breaking(v[1:(L-1)])
```

The argument `v` is a vector of values between 0 and 1 defining the sequential breaking points of the stick after removing the previous portions already broken off. It is of length $L-1$, implicitly assuming that its last component is equal to 1.

In order to complete the definition of the weights in the stick-breaking representation of G , a prior distribution on $(0, 1)$ should to be assumed for $v_l, l = 1, \dots, L-1$, for instance a beta prior.

11.3.2 Example

Here we illustrate how to use NIMBLE for the example described in section 11.2.2, but considering a stick-breaking representation for G . The model is given by

$$\begin{aligned}
y_i &| \theta^*, \sigma^{*2}, z_i \stackrel{iid}{\sim} N(\theta^*_{z_i}, \sigma^{*2}_{z_i}), \quad i = 1, \dots, N, \\
z_i &\sim \text{Categorical}(w), \quad i = 1, \dots, N, \\
v_l &\stackrel{iid}{\sim} \text{Beta}(1, \alpha), \quad l = 1, \dots, L-1, \\
\alpha &\sim \text{Gamma}(1, 1), \\
\theta_l^* &\stackrel{iid}{\sim} N(0, 100), \quad \sigma_l^{*2} \stackrel{iid}{\sim} \text{InvGamma}(1, 1), \quad l = 1, \dots, L.
\end{aligned}$$

where $w_1 = v_1, w_l = v_l \prod_{m < l} (1 - v_m)$, for $l = 1, \dots, L-1$, and $w_L = \prod_{m < L} (1 - v_m)$.

```
code <- nimbleCode({
  for(i in 1:(L-1)){
    v[i] ~ dbeta(1, alpha)
  }
  alpha ~ dgamma(1, 1)
  w[1:L] <- stick_breaking(v[1:(L-1)])
  for(i in 1:L) {
    thetastar[i] ~ dnorm(0, var = 100)
    s2star[i] ~ dinvgamma(1, 1)
  }
  for(i in 1:N) {
    z[i] ~ dcat(w[1:L])
  }
})
```

```

    y[i] ~ dnorm(thetastar[z[i]], var = s2star[z[i]])
  }
})

set.seed(1)
constants <- list(N = 100, L=50)
data <- list(y = c(rnorm(50, -5, sqrt(3)), rnorm(50, 5, sqrt(4))))
inits <- list(thetastar = rnorm(constants$L, 0, 100),
             s2star = rinvgamma(constants$L, 1, 1),
             z = sample(1:10, size = constants$N, replace = TRUE),
             v = rbeta(constants$L, 1, 1),
             alpha = 1)
modelSB <- nimbleModel(code, constants, data, inits)

```

The resulting model may be carried through to MCMC sampling. NIMBLE will assign a specialized sampler to update v . See Chapter 7 for information about NIMBLE’s MCMC engine, and Section 11.4.2 for details on MCMC sampling of the stick-breaking weights.

11.4 MCMC sampling of BNP models

BNP models can be specified in different, yet equivalent, manners. Examples 11.2.2 and 11.3.2 are examples of density estimation for real-valued data, and are specified through the CRP and the stick-breaking process, respectively. Different specifications lead NIMBLE to assign different sampling algorithms for the model. When the model is specified through a CRP, a collapsed sampler (Neal, 2000) is assigned. Under this specification, the random measure G is integrated out from the model. When a stick-breaking representation is used, a blocked Gibbs sampler is assigned, see Ishwaran and James (2001) and Ishwaran and James (2002).

11.4.1 Sampling CRP models

NIMBLE’s MCMC engine provides specialized samplers for the dCRP distribution, updating each component of the membership vector sequentially. Internally, the sampler is assigned based on inspection of the model structure, evaluating conjugacy between the mixture kernel and the baseline distribution, as follows:

1. A conjugate sampler in the case of the baseline distribution being conjugate for the mixture kernel.
2. A non-conjugate sampler in the case of the baseline distribution not being conjugate for the mixture kernel.

Note that both samplers are specialized versions that operate on a vector having a CRP distribution. Details of these assignments are strictly internal to the CRP samplers. The current release of NIMBLE supports conjugate sampling for the dCRP distribution for the relationships listed in Table 11.1. Additional relationships are provided in Table 11.2 for normal mixture kernels when both mean and variance are unknown.

Table 11.1: Conjugate relationships for the dCRP distribution supported by NIMBLE’s MCMC engine.

Baseline Distribution	Mixture (Dependent Node) Distribution	Parameter
Beta	Bernoulli	prob
	Binomial	prob
	Negative Binomial	prob
Dirichlet	Multinomial	prob
Gamma	Poisson	lambda
	Normal	tau
	Gamma	rate
	Inverse Gamma	scale
	Exponential	rate
	Weibull	lambda
Inverse Gamma	Normal	var
Normal	Normal	mean
Multivariate Normal	Multivariate Normal	mean
Wishart	Multivariate Normal	prec
Inverse Wishart	Multivariate Normal	cov

Table 11.2: Additional conjugate relationships for the dCRP distribution supported by NIMBLE’s MCMC engine.

Baseline Distribution	Mixture (Dependent Node) Distribution	Parameter
Normal-Gamma	Normal	mean, tau
Normal-Inverse Gamma	Normal	mean, var
Multivariate Normal-Wishart	Multivariate Normal	mean, prec
Multivariate Normal-Inverse Wishart	Multivariate Normal	mean, cov

To reduce computation and improve mixing, we only sample the parameters of the clusters (e.g., $\tilde{\theta}_j$ in 11.2.2 when the associated cluster is occupied, using the `CRP_cluster_wrapper` sampler, which wraps around an underlying actual sampler. In addition, this approach requires that any sampler assigned to parameters of the base measure, G_0 , (i.e., unknown parameters in the prior for $\tilde{\theta}_j$) ignore cluster parameters associated with clusters that are not occupied, since their current values are meaningless. We assign a special slice sampler that determines the occupied clusters in any given iteration, called `slice_CRP_base_param`. Note that if you choose to use a different sampler for the base measure parameters, you should also avoid using the `CRP_cluster_wrapper` sampler.

Finally, a specialized sampler is assigned to the `conc` hyper parameter when a gamma hyper prior is assigned, see section 6 in Escobar and West (1995) for more details. Otherwise, a random walk Metropolis-Hastings sampler is assigned.

11.4.1.1 Initial values

Valid initial values should be provided for all elements of the process specified by a CRP structure before running the MCMC. A simple and safe choice for $\mathbf{z}$ is to provide a sample of size N , the same

as its length, of values between 1 and some reasonable number of clusters (less than or equal to the length of $\mathbf{z}$), with replacement, as done in the preceding CRP example. For the concentration parameter, a safe initial value is 1.

11.4.1.2 Sampling the random measure

In BNP models, it is often of interest to make inference about the unknown measure G . NIMBLE provides the `getSamplesDPmeasure` sampler to generate posterior samples of the random measure G when a CRP structure is involved in the model.

The `getSamplesDPmeasure` function has two arguments: `MCMC` and `epsilon`.

The `MCMC` argument is a compiled or uncompiled MCMC object. The MCMC object should monitor the membership (or clustering) variable, the cluster parameters, all stochastic nodes of the cluster parameters, and the concentration parameter, if it is random. Use the `monitors` argument when configuring the MCMC to ensure these variables are monitored.

The `epsilon` argument is used to determine the truncation level of the representation of the random measure. Its default value is $1e - 04$.

The sampler is used only after the MCMC for the model has been run; more details are available from `help(getSamplesDPmeasure)`.

Posterior samples of the random measure G are (almost surely) of the form $\sum_{l=1}^{\infty} w_l \delta_{\theta_l}$, where δ_{θ} is the Dirac measure at θ , w_l are stick-breaking weights, and θ_l are atoms (or point masses). The variables that define the stick-breaking weights are iid $Beta(1, \alpha + N)$ distributed, where α is the concentration parameter of the CRP distribution and N is the sample size. Independently, the atoms, θ_l , are iid and follow a distribution of the form

$$(\alpha + N)^{-1} \sum_{k=1}^{N^*(z)} m_k(z) \delta_{\{\tilde{\theta}_k\}} + \alpha(\alpha + N)^{-1} G_0,$$

where G_0 is the prior baseline distribution, and z and $\tilde{\theta}_k$ are posterior samples of the labeling vector and cluster parameters, respectively. Their values, together with the values of α (if random) are obtained from the MCMC's output. Expressions $m_k(z)$ and $N^*(z)$ are defined as in section 11.2.

The `getSamplesDPmeasure` function provides samples of a truncated version of the infinite mixture to a level L . The truncation level L is such that the tail probability left from the approximation is at most `epsilon`, denoted ϵ . The following relationship determines the truncation level: $L = \lceil \log(\epsilon) / \log[(\alpha + N) / (\alpha + N + 1)] \rceil$. The value of L varies at each iteration of the MCMC's output when α is random, while it is the same at each iteration when α is fixed. For more details about sampling the random measure and its truncation level see [Gelfand and Kottas \(2002\)](#).

Because of the discrete nature of the atom's distribution, the stick-breaking representation truncated to a level L will generally have repeated values. Therefore, in order to reduce the output's dimensionality, the stick-breaking weights of identical atoms are added up. This results in samples of G that have only the 'unique' atoms in $\{\theta_l\}_{l=1}^L$.

The following code exemplifies how to generate samples from G after defining the model as in Section 11.2.2.

```

cmodel <- compileNimble(model)

monitors <- c('z', 'thetatile', 's2tilde' , 'alpha')
modelConf <- configureMCMC(model, monitors = monitors)
modelMCMC <- buildMCMC(modelConf)

cmodelMCMC <- compileNimble(modelMCMC, project = model)
cmodelMCMC$run(1000)

samplesG <- getSamplesDPmeasure(cmodelMCMC)

```

11.4.2 Sampling stick-breaking models

NIMBLE's MCMC engine provides specialized samplers for the beta-distributed random variables that are the arguments to the stick-breaking function, updating each component of the weight vector sequentially. The sampler is assigned based on inspection of the model structure. Specifically, the specialized sampler is assigned when the membership vector has a categorical distribution, its weights are defined by a stick-breaking function, and the vector defining the weights follows a beta distribution.

11.4.2.1 Initial values

Valid initial values should be provided for all elements of the stick-breaking function and membership variable before running the MCMC. A simple and safe choice for z is to provide a sample of size N , of values between 1 and some value less than L , with replacement, as done in the preceding stick-breaking example. For the stick variables, safe initial values can be simulated from a beta distribution.

Part IV

Programming with NIMBLE

Overview

Part IV is the programmer's guide to NIMBLE. At the heart of programming in NIMBLE are `nimbleFunctions`. These support two principal features: (1) a *setup* function that is run once for each model, nodes, or other setup arguments, and (2) *run* functions that will be compiled to C++ and are written in a subset of R enhanced with features to operate models. Formally, what can be compiled comprises the NIMBLE language, which is designed to be R-like.

This part of the manual is organized as follows:

- Chapter 12 describes how to write simple `nimbleFunctions`, which have no setup code and hence don't interact with models, to compile parts of R for fast calculations. This covers the subset of R that is compilable, how to declare argument types and return types, and other information.
- Chapter 13 explains how to write `nimbleFunctions` that can be included in BUGS code as user-defined distributions or user-defined functions.
- Chapter 14 introduces more features of NIMBLE models that are useful for writing `nimbleFunctions` to use models, focusing on how to query model structure and carry out model calculations.
- Chapter 15 introduces two kinds of data structures: `modelValues` are used for holding multiple sets of values of model variables; `nimbleList` data structures are similar to R lists but require fixed element names and types, allowing the NIMBLE compiler to use them.
- Chapter 16 draws on the previous chapters to show how to write `nimbleFunctions` that work with models, or more generally that have a setup function for any purpose. Typically a setup function queries model structure (Chapter 14) and may establish some `modelValues` or `nimbleList` data structures or configurations (Chapter 15). Then *run* functions written in the same way as simple `nimbleFunctions` (Chapter 12) along with model operations (Chapter 14) define algorithm computations that can be compiled via C++.

Chapter 12

Writing simple nimbleFunctions

12.1 Introduction to simple nimbleFunctions

nimbleFunctions are the heart of programming in NIMBLE. In this chapter, we introduce simple *nimbleFunctions* that contain only one function to be executed, in either compiled or uncompiled form, but no setup function or additional methods.

Defining a simple *nimbleFunction* is like defining an R function: `nimbleFunction` returns a function that can be executed, and it can also be compiled. Simple *nimbleFunctions* are useful for doing math or the other kinds of processing available in NIMBLE when no model or `modelValues` is needed. These can be used for any purpose in R programming. They can also be used as new functions and distributions in NIMBLE's extension of BUGS (Chapter 13).

Here's a basic example implementing the textbook calculation of least squares estimation of linear regression parameters¹:

```
solveLeastSquares <- nimbleFunction(  
  run = function(X = double(2), y = double(1)) { # type declarations  
    ans <- inverse(t(X) %*% X) %*% (t(X) %*% y)  
    return(ans)  
    returnType(double(2)) # return type declaration  
  } )
```

```
X <- matrix(rnorm(400), nrow = 100)  
y <- rnorm(100)  
solveLeastSquares(X, y)
```

```
##           [,1]  
## [1,]  0.088547575  
## [2,]  0.002025635  
## [3,] -0.090387283  
## [4,] -0.072660391
```

¹Of course, in general, explicitly calculating the inverse is not the recommended numerical recipe for least squares.

```
CsolveLeastSquares <- compileNimble(solveLeastSquares)
CsolveLeastSquares(X, y)
```

```
##           [,1]
## [1,]  0.088547575
## [2,]  0.002025635
## [3,] -0.090387283
## [4,] -0.072660391
```

In this example, we fit a linear model for 100 random response values (y) to four columns of randomly generated explanatory variables (X). We ran the nimbleFunction `solveLeastSquares` uncompiled, natively in R, allowing testing and debugging (Section 16.7). Then we compiled it and showed that the compiled version does the same thing, but faster². NIMBLE’s compiler creates C++ that uses the Eigen (<http://eigen.tuxfamily.org>) library for linear algebra.

Notice that the actual NIMBLE code is written as an R function definition that is passed to `nimbleFunction` as the `run` argument. Hence we call it the *run* code. *run* code is written in the NIMBLE language. This is similar to a narrow subset of R with some additional features. Formally, we view it as a distinct language that encompasses what can be compiled from a nimbleFunction.

To write nimbleFunctions, you will need to learn:

- what R functions are supported for NIMBLE compilation and any ways they differ from their regular R counterparts;
- how NIMBLE handles types of variables;
- how to declare types of `nimbleFunction` arguments and return values;
- that compiled nimbleFunctions always pass arguments to each other by reference.

The next sections cover each of these topics in turn.

12.2 R functions (or variants) implemented in NIMBLE

12.2.1 Finding help for NIMBLE’s versions of R functions

Often, R help pages are available for NIMBLE’s versions of R functions using the prefix ‘nim’ and capitalizing the next letter. For example, help on NIMBLE’s version of `numeric` can be found by `help(nimNumeric)`. In some cases help is found directly using the name of the function as it appears in R.

12.2.2 Basic operations

Basic R operations supported for NIMBLE compilation are listed in Table 12.1.

²On the machine this is being written on, the compiled version runs a few times faster than the uncompiled version. However we refrain from formal speed tests.

Table 12.1: Basic R manipulation functions in NIMBLE. For some of these functions, in addition to the usual use of `help()`, you can find help in R for NIMBLE’s version of a function by using the “nim” prefix and capitalizing the next letter. E.g. `help(nimC)` for help with `c()`.

Function	Comments (differences from R)
<code>c()</code>	No <code>recursive</code> argument.
<code>rep()</code>	No <code>rep.int</code> or <code>rep.len</code> arguments.
<code>seq()</code> and <code>:</code>	Negative integer sequences from <code>:</code> , e.g. <code>2:1</code> do not work.
<code>which()</code>	No <code>arr.ind</code> or <code>useNames</code> arguments.
<code>diag()</code>	Works like R in three ways: <code>diag(vector)</code> returns a matrix with <code>vector</code> on the diagonal; <code>diag(matrix)</code> returns the diagonal vector of <code>matrix</code> ; <code>diag(n)</code> returns an $n \times n$ identity matrix. No <code>nrow</code> or <code>ncol</code> arguments.
<code>diag()<-</code>	Works for assigning the diagonal vector of a matrix.
<code>dim()</code>	Works on a vector as well as higher-dimensional arguments.
<code>length()</code>	
<code>is.na()</code>	Does not correctly handle NAs (or NaNs) from R that are type 'logical' or 'integer', so convert these using <code>as.numeric()</code> before passing from R to NIMBLE.
<code>is.nan()</code>	
<code>any()</code>	One argument only; NAs treated as FALSE.
<code>all()</code>	One argument only; NAs treated as FALSE.
<code>numeric()</code>	Allows additional arguments to control initialization.
<code>logical()</code>	Allows additional arguments to control initialization.
<code>integer()</code>	Allows additional arguments to control initialization.
<code>matrix()</code>	Allows additional arguments to control initialization.
<code>array()</code>	Allows additional arguments to control initialization.
indexing	Arbitrary integer and logical indexing is supported for objects of one or two dimensions. For higher-dimensional objects, only <code>:</code> indexing works, and then only to create an object of at most two dimensions.

In addition to the above functions, we provide functions `any_na()` and `any_nan()` for finding if there are any NA or NaN values in a vector. These are equivalent to using `any(is.na())` and `any(is.nan())` in R, with the exception noted in the table above regarding `is.na()`.

Other R functions with numeric arguments and return value can be called during compiled execution by wrapping them as a `nimbleRcall` (see Section 12.7).

Next we cover some of these functions in more detail.

12.2.2.1 *numeric, integer, logical, matrix and array*

`numeric`, `integer`, or `logical` will create a 1-dimensional vector of floating-point (or ‘double’ [precision]), integer, or logical values, respectively. The `length` argument specifies the vector length (default 0), and the `value` argument specifies the initial value either as a scalar (used for all vector elements, with default 0) or a vector. If a vector and its length is not equal to `length`, the remaining values will be filled by R-style recycling if the `recycle` argument is `TRUE` (which is the default). The

`init` argument specifies whether or not to initialize the elements in compiled code (default `TRUE`). If first use of the variable does not rely on initial values, using `init = FALSE` will yield slightly more efficient performance.

`matrix` creates a 2-dimensional matrix object of either floating-point (if `type = "double"`, the default), integer (if `type = "integer"`), or logical (if `type = "logical"`) values. As in R, `nrow` and `ncol` arguments specify the number of rows and columns, respectively. The `value` and `init` arguments are used in the same way as for `numeric`, `integer`, and `logical`.

`array` creates a vector or higher-dimensional object, depending on the `dim` argument, which takes a vector of sizes for each dimension. The `type`, `value` and `init` argument behave the same as for `matrix`.

The best way to create an identity matrix is with `diag(n)`, which returns an $n \times n$ identity matrix. NIMBLE also provides a deprecated nimbleFunction `identityMatrix` that does the same thing.

Examples of these functions, and the related function `'setSize'` for changing the size of a numeric object, are given in Section 12.3.2.4.

12.2.2.2 *length* and *dim*

`length` behaves like R's `length` function. It returns the *entire* length of `X`. That means if `X` is multivariate, `length(X)` returns the product of the sizes of the dimensions. NIMBLE's version of `dim`, which has synonym `nimDim`, behaves like R's `dim` function for matrices or arrays and like R's `length` function for vectors. In other words, regardless of whether the number of dimensions is 1 or more, it returns a vector of the sizes.

12.2.2.3 Depreciated creation of non-scalar objects using *declare*

Previous versions of NIMBLE provided a function `declare` for declaring variables. The more R-like functions `numeric`, `integer`, `logical`, `matrix` and `array` are intended to replace `declare`, but `declare` is still supported for backward compatibility. In a future version of NIMBLE, `declare` may be removed.

12.2.3 Math and linear algebra

Numeric scalar and matrix mathematical operations are listed in Tables 12.2-12.3.

As in R, many scalar operations in NIMBLE will work component-wise on vectors or higher dimensional objects. For example if `B` and `C` are vectors, `A = B + C` will add them and create vector `C` by component-wise addition of `B` and `C`. In the current version of NIMBLE, component-wise operations generally only work for vectors and matrices, not arrays with more than two dimensions. The only exception is assignment: `A = B` will work up to NIMBLE's current limit of four dimensions.

Table 12.2: Functions operating on scalars, many of which can operate on each element (component-wise) of vectors and matrices. Status column indicates if the function is currently provided in NIMBLE. Vector input column indicates if the function can take a vector as an argument (i.e., if the function is vectorized).

Usage	Description	Comments	Status	Vector input
<code>x y, x & y</code>	logical OR () and AND(&)		yes	yes
<code>!x</code>	logical not		yes	yes
<code>x > y, x >= y</code>	greater than (and or equal to)		yes	yes
<code>x < y, x <= y</code>	less than (and or equal to)		yes	yes
<code>x != y, x == y</code>	(not) equals		yes	yes
<code>x + y, x - y, x * y</code>	component-wise operators	mix of scalar and vector	yes	yes
<code>x / y</code>	component-wise division	vector x and scalar y	yes	yes
<code>x^y, pow(x, y)</code>	power	x^y ; vector x , scalar y	yes	yes
<code>x %% y</code>	modulo (remainder)		yes	no
<code>min(x1, x2), max(x1, x2)</code>	min. (max.) of two scalars		yes	See <code>pmin</code> , <code>pmax</code>
<code>exp(x)</code>	exponential		yes	yes
<code>log(x)</code>	natural logarithm		yes	yes
<code>sqrt(x)</code>	square root		yes	yes
<code>abs(x)</code>	absolute value		yes	yes
<code>step(x)</code>	step function at 0	0 if $x < 0$, 1 if $x \geq 0$	yes	yes
<code>equals(x, y)</code>	equality of two scalars	1 if $x == y$, 0 if $x != y$	yes	yes
<code>cube(x)</code>	third power	x^3	yes	yes
<code>sin(x), cos(x), tan(x)</code>	trigonometric functions		yes	yes
<code>asin(x), acos(x), atan(x)</code>	inverse trigonometric functions		yes	yes
<code>asinh(x), acosh(x), atanh(x)</code>	inv. hyperbolic trig. functions		yes	yes
<code>logit(x)</code>	logit	$\log(x/(1-x))$	yes	yes
<code>ilogit(x), expit(x)</code>	inverse logit	$\exp(x)/(1+\exp(x))$	yes	yes
<code>probit(x)</code>	probit (Gaussian quantile)	$\Phi^{-1}(x)$	yes	yes
<code>iprobit(x), phi(x)</code>	inverse probit (Gaussian CDF)	$\Phi(x)$	yes	yes
<code>cloglog(x)</code>	complementary log log	$\log(-\log(1-x))$	yes	yes
<code>icloglog(x)</code>	inverse complementary log log	$1 - \exp(-\exp(x))$	yes	yes
<code>ceiling(x)</code>	ceiling function	$\lceil x \rceil$	yes	yes
<code>floor(x)</code>	floor function	$\lfloor x \rfloor$	yes	yes
<code>round(x)</code>	round to integer		yes	yes
<code>trunc(x)</code>	truncation to integer		yes	yes
<code>lgamma(x), loggam(x)</code>	log gamma function	$\log \Gamma(x)$	yes	yes
<code>besselK(x, nu, ...expon.scaled)</code>	modified bessel function of the second kind	returns vector even if x a matrix/array	yes	yes
<code>log1p(x)</code>	log of $1 + x$	$\log(1 + x)$	yes	yes
<code>lfactorial(x),</code>	log factorial	$\log x!$	yes	yes

Usage	Description	Comments	Status	Vector input
<code>logfact(x)</code>				
<code>qDIST(x, PARAMS)</code>	“q” distribution functions	canonical parameterization	yes	yes
<code>pDIST(x, PARAMS)</code>	“p” distribution functions	canonical parameterization	yes	yes
<code>rDIST(x, PARAMS)</code>	“r” distribution functions	canonical parameterization	yes	yes
<code>dDIST(x, PARAMS)</code>	“d” distribution functions	canonical parameterization	yes	yes
<code>sort(x)</code>			no	
<code>rank(x, s)</code>			no	
<code>ranked(x, s)</code>			no	
<code>order(x)</code>			no	

Table 12.3: Functions operating on vectors and matrices. Status column indicates if the function is currently provided in NIMBLE.

Usage	Description	Comments	Status
<code>inverse(x)</code>	matrix inverse	x symmetric, positive def.	yes
<code>chol(x)</code>	matrix Cholesky factorization	x symmetric, positive def. returns upper triang. matrix	yes
<code>t(x)</code>	matrix transpose	$x^\top$	yes
<code>x%*%y</code>	matrix multiply	xy ; x, y conformant	yes
<code>inprod(x, y)</code>	dot product	$x^\top y$; x and y vectors	yes
<code>solve(x,y)</code>	solve system of equations	$x^{-1}y$; y matrix or vector	yes
<code>forwardsolve(x, y)</code>	solve lower-triangular system of equations	$x^{-1}y$; x lower-triangular	yes
<code>backsolve(x, y)</code>	solve upper-triangular system of equations	$x^{-1}y$; x upper-triangular	yes
<code>logdet(x)</code>	log matrix determinant	$\log x $	yes
<code>asRow(x)</code>	convert vector to 1-row matrix	sometimes automatic	yes
<code>asCol(x)</code>	convert vector to 1-column matrix	sometimes automatic	yes
<code>sum(x)</code>	sum of elements of x		yes
<code>mean(x)</code>	mean of elements of x		yes
<code>sd(x)</code>	standard deviation of elements of x		yes
<code>prod(x)</code>	product of elements of x		yes
<code>min(x), max(x)</code>	min. (max.) of elements of x		yes
<code>pmin(x, y), pmax(x,y)</code>	vector of mins (maxs) of elements of x and y		yes
<code>interp.lin(x, v1, v2)</code>	linear interpolation		no
<code>eigen(x)</code>	matrix eigendecomposition; returns a nimbleList of type <code>eigenNimbleList</code>	x symmetric	yes
<code>svd(x)</code>	matrix singular value decomposition; returns a nimbleList of type <code>svdNimbleList</code>		yes

More information on the nimbleLists returned by the `eigen` and `svd` functions in NIMBLE can be found in Section 15.2.2.

12.2.4 Distribution functions

Distribution ‘d’, ‘r’, ‘p’, and ‘q’ functions can all be used from `nimbleFunctions` (and in BUGS model code), but care is needed in the syntax, as follows.

- Names of the distributions generally (but not always) match those of R, which sometimes differ from BUGS. See the list below.
- Supported parameterizations are also indicated in the list below.
- For multivariate distributions (multivariate normal, Dirichlet, and Wishart), ‘r’ functions only return one random draw at a time, and the first argument must always be 1.
- R’s recycling rule (re-use of an argument as needed to accommodate longer values of other arguments) is generally followed, but the returned object is always a scalar or a vector, not a matrix or array.

As in R (and `nimbleFunctions`), arguments are matched by order or by name (if given). Standard arguments to distribution functions in R (`log`, `log.p`, `lower.tail`) can be used and have the same defaults. User-defined distributions for BUGS (Chapter 13) can also be used from `nimbleFunctions`.

For standard distributions, we rely on R’s regular help pages (e.g., `help(dgamma)`). For distributions unique to NIMBLE (e.g., `dexp_nimble`, `ddirch`), we provide help pages.

Supported distributions, listed by their ‘d’ function, include:

- `dbinom(x, size, prob, log)`
- `dcat(x, prob, log)`
- `dmulti(x, size, prob, log)`
- `dnbinom(x, size, prob, log)`
- `dpois(x, lambda, log)`
- `dbeta(x, shape1, shape2, log)`
- `dchisq(x, df, log)`
- `ddexp(x, location, rate, log)`
- `ddexp(x, location, scale, log)`
- `dexp(x, rate, log)`
- `dexp_nimble(x, rate, log)`
- `dexp_nimble(x, scale, log)`
- `dgamma(x, shape, rate, log)`
- `dgamma(x, shape, scale, log)`
- `dinvgamma(x, shape, rate, log)`
- `dinvgamma(x, shape, scale, log)`
- `dlnorm(x, meanlog, sdlog, log)`
- `dlogis(x, location, scale, log)`
- `dnorm(x, mean, sd, log)`
- `dt_nonstandard(x, df, mu, sigma, log)`
- `dt(x, df, log)`
- `dunif(x, min, max, log)`
- `dweibull(x, shape, scale, log)`
- `ddirch(x, alpha, log)`
- `dlkj_corr_cholesky(x, shape, size, log)`
- `dmnorm_chol(x, mean, cholesky, prec_param, log)`

- `dmvt_chol(x, mu, cholesky, df, prec_param, log)`
- `dwish_chol(x, cholesky, df, scale_param, log)`
- `dinvwish_chol(x, cholesky, df, scale_param, log)`

In the last three, `cholesky` stands for Cholesky decomposition of the relevant matrix; `prec_param` is a logical indicating whether the Cholesky is of a precision matrix (TRUE) or covariance matrix (FALSE)³; and `scale_param` is a logical indicating whether the Cholesky is of a scale matrix (TRUE) or an inverse scale matrix (FALSE).

12.2.5 Flow control: *if-then-else*, *for*, *while*, and *stop*

These basic flow-control structures use the same syntax as in R. However, `for`-loops are limited to sequential integer indexing. For example, `for(i in 2:5) {...}` works as it does in R. Decreasing index sequences are not allowed. Unlike in R, `if` is not itself a function that returns a value.

We plan to include more flexible `for`-loops in the future, but for now we've included just one additional useful feature: `for(i in seq_along(NFL))` will work as in R, where `NFL` is a `nimbleFunctionList`. This is described in Section 16.4.8.

`stop`, or equivalently `nimStop`, throws control to R's error-handling system and can take a character argument that will be displayed in an error message.

12.2.6 *print* and *cat*

`print`, or equivalently `nimPrint`, prints an arbitrary set of outputs in order and adds a newline character at the end. `cat` or `nimCat` is identical, except without a newline at the end.

12.2.7 Checking for user interrupts: *checkInterrupt*

When you write algorithms that will run for a long time in C++, you may want to explicitly check whether a user has tried to interrupt the execution (i.e., by pressing Control-C). Simply include `checkInterrupt` in run code at places where a check should be done. If there has been an interrupt waiting to be handled, the process will stop and return control to R.

12.2.8 Optimization: *optim* and *nimOptim*

NIMBLE provides a `nimOptim` function that partially implements R's `optim` function with some minor differences. `nimOptim` can use methods from R's `optim`, including 'Nelder-Mead', 'BFGS', 'CG', 'L-BFGS-B', but it does not support methods 'SANN' and 'Brent'. As of version 1.2.0, it also allows users to provide their own optimization methods as R functions (although this feature is subject to change in future versions). Support for R's `nlminb` is provided in this way. NIMBLE's `nimOptim` allows gradients to be supplied using user-provided functions if available or finite differences otherwise. Currently `nimOptim` does not support extra parameters to the function being optimized (via `\dots`), but a work-around is to create a new `nimbleFunction` that calls another one with the additional parameters. Finally, `nimOptim` requires a `nimbleList` datatype for the `control` parameter, whereas R's `optim` uses a simple R `list`. To define the `control` parameter, create a default value with the `nimOptimDefaultControl` function, and set any desired fields.

See `help(nimOptim)` for details, including example usage.

³For the multivariate `t`, these are more properly termed the 'inverse scale' and 'scale' matrices

12.2.9 Integration: *integrate* and *nimIntegrate*

NIMBLE provides a `nimIntegrate` function that implements R's `integrate` function, which carries out adaptive Gauss-Kronrod quadrature to integrate a function of one variable over a finite or infinite interval.

In addition to general use in a `nimbleFunction`, this can, of course, be used in a user-defined function in model code, e.g., to implement models that involve an integral, such as certain point process and survival models.

One could also consider using `nimIntegrate` to numerically integrate over a parameter in a model, such as to remove a parameter that is not mixing well in an MCMC. Note that NIMBLE provides Laplace (and the closely-related adaptive Gauss-Hermite quadrature) approximation for integrating over continuous parameters such as random effects. See section 9.1 and `help(Laplace)` for more information on those.

Here is an example of using `nimIntegrate` to implement a vectorized von Mises distribution.

```
integrand <- nimbleFunction(
  run = function(x = double(1), theta = double(1)) {
    return( exp(theta[1] * cos(x) + theta[2] * cos(2 * (x + theta[3]))) )
    returnType(double(1))
  })

dGenVonMises <- nimbleFunction(
  run = function(x = double(1), mu1 = double(), mu2 = double(),
    kappa1 = double(), kappa2 = double(),
    limits = double(1), log = integer(0, default = 0)){
    range <- limits[2] - limits[1]
    mu1R <- (mu1 - range/2)/range*2*pi
    mu2R <- (mu2 - range/2)/range*2*pi
    z <- (x - range/2)/range*2*pi
    d <- (mu1R - mu2R) %% pi
    num <- exp(kappa1 * cos(z - mu1R) + kappa2 * cos(2 * (z - mu2R)))
    tmp <- c(kappa1, kappa2, d)
    den <- nimIntegrate(integrand, lower = 0, upper = 2*pi, tmp)[1]
    dens <- num/den
    result <- dens*2*pi/range
    if(log) {
      return(log(result))
    } else return(result)
    returnType(double(1))
  }
)

cgvonmises <- compileNimble(dGenVonMises)
x <- 0:360
# plot(x, circular::dgenvonmises(circular(x, type = "angles", units = "degrees"), -1, 1, 2, 1))
# plot(x, cgvonmises(x, mu1=20, mu2=300, kappa1=2, kappa2=1, limits = c(0, 360)), type = 'l')
```

12.2.10 ‘nim’ synonyms for some functions

NIMBLE uses some keywords, such as `dim` and `print`, in ways similar but not identical to R. In addition, there are some keywords in NIMBLE that have the same names as R functions with quite different functionality. For example, `step` is part of the BUGS language, but it is also an R function for stepwise model selection. And `equals` is part of the BUGS language but is also used in the `testthat` package, which we use in testing NIMBLE.

NIMBLE tries to avoid conflicts by replacing some keywords immediately upon creating a nimble-Function. These replacements include

- `c` → `nimC`
- `copy` → `nimCopy`
- `dim` → `nimDim`
- `print` → `nimPrint`
- `cat` → `nimCat`
- `step` → `nimStep`
- `equals` → `nimEquals`
- `rep` → `nimRep`
- `round` → `nimRound`
- `seq` → `nimSeq`
- `stop` → `nimStop`
- `switch` → `nimSwitch`
- `numeric`, `integer`, `logical` → `nimNumeric`, `nimInteger`, `nimLogical`
- `matrix`, `array` → `nimMatrix`, `nimArray`
- `optim` → `nimOptim`
- `integrate` → `nimIntegrate`

This system gives programmers the choice between using the keywords like `nimPrint` directly, to avoid confusion in their own code about which ‘print’ is being used, or to use the more intuitive keywords like `print` but remember that they are not the same as R’s functions.

12.3 How NIMBLE handles types of variables

Variables in the NIMBLE language are statically typed. Once a variable is used for one type, it can’t subsequently be used for a different type. This rule facilitates NIMBLE’s compilation to C++. The NIMBLE compiler often determines types automatically, but sometimes the programmer needs to explicitly provide them.

The elemental types supported by NIMBLE include *double* (floating-point), *integer*, *logical*, and *character*. The *type* of a numeric or logical object refers to the number of dimensions and the elemental type of the elements. Hence if `x` is created as a double matrix, it can only be used subsequently for a double matrix. The size of each dimension is not part of its type and thus can be changed. Up to four dimensions are supported for double, integer, and logical. Only vectors (one dimension) are supported for character. Unlike R, NIMBLE supports true scalars, which have 0 dimensions.

12.3.1 nimbleList data structures

A `nimbleList` is a data structure that can contain arbitrary other NIMBLE objects, including other `nimbleList`s. Like other NIMBLE types, `nimbleList`s are strongly typed: each `nimbleList` is created from a configuration that declares what types of objects it will hold. `nimbleList`s are covered in Chapter 15.2.

12.3.2 How numeric types work

R's dynamic types support easy programming because one type can sometimes be transformed to another type automatically when an expression is evaluated. NIMBLE's static types makes it stricter than R.

12.3.2.1 When NIMBLE can automatically set a numeric type

When a variable is first created by assignment, its type is determined automatically by that assignment. For example, if `x` has not appeared before, then

```
x <- A %*% B # assume A and B are double matrices or vectors
```

will create `x` to be a double matrix of the correct size (determined during execution).

12.3.2.1.1 Avoid changing types of a variable within a `nimbleFunction` Because NIMBLE is statically typed, you cannot use the same variable name for two objects of different types (including objects of different dimensions).

Suppose we have (implicitly) created `x` as a double matrix. If `x` is used subsequently, it can only be used as a double matrix. This is true even if it is assigned a new value, which will again set its size automatically but cannot change its type.

```
x <- A %*% B # assume A and B are double matrices or vectors
x <- nimMatrix(0, nrow = 5, ncol = 2) # OK: 'x' is still a double matrix
x <- rnorm(10) # NOT OK: 'x' is a double vector
```

12.3.2.2 When a numeric object needs to be created before being used

If the contents of a variable are to be populated by assignment into some indices in steps, the variable must be created first. Further, it must be large enough for its eventual contents; it will not be automatically resized if assignments are made beyond its current size. For example, in the following code, `x` must be created before being filled with contents for specific indices.

```
x <- numeric(10)

for(i in 1:10)
  x[i] <- foo(y[i])
```

12.3.2.3 How NIMBLE handles NA

NIMBLE supports use of `NA` with some caveats. In the compiled version of a `nimbleFunction`:

- `NA` values in a logical scalar or vector will not work;

- Assigning NA to a new variable will make that variable have type double (in R, this would be a logical!);
- Non-trivial use of NA in integer variables may fail by having the value become a large (negative) number;
- Use of NA in doubles should generally work.

These issues arise because NIMBLE uses R's encoding of NA values in C(++) but uses native compiled math and type-casting in C++, which do not always preserve R's NA encodings. In summary, NA should generally work for doubles and should be used cautiously (i.e. after you test that what you need to work actually works) for integers. For some needs, NaN is a suitable alternative to NA.

12.3.2.4 Changing the sizes of existing objects: *setSize*

`setSize` changes the size of an object, preserving its contents in column-major order.

```
# Example of creating and resizing a floating-point vector
# myNumericVector will be of length 10, with all elements initialized to 2
myNumericVector <- numeric(10, value = 2)
# resize this numeric vector to be length 20; last 10 elements will be 0
setSize(myNumericVector, 20)
```

```
# Example of creating a 1-by-10 matrix with values 1:10 and resizing it
myMatrix <- matrix(1:10, nrow = 1, ncol = 10)
# resize this matrix to be a 10-by-10 matrix
setSize(myMatrix, c(10, 10))
# The first column will have the 1:10
```

12.3.2.5 Confusions between scalars and length-one vectors

In R, there is no such thing as a true scalar; scalars can always be treated as vectors of length one. NIMBLE allows true scalars, which can create confusions. For example, consider the following code:

```
myfun <- nimbleFunction(
  run = function(i = integer()) { # i is an integer scalar
    randomValues <- rnorm(10) # double vector
    a <- randomValues[i] # double scalar
    b <- randomValues[i:i] # double vector
    d <- a + b # double vector
    f <- c(i) # integer vector
  })
```

In the line that creates `b`, the index range `i:i` is not evaluated until run time. Even though `i:i` will always evaluate to simply `i`, the compiler does not determine that. Since there is a vector index range provided, the result of `randomValues[i:i]` is determined to be a vector. The following line then creates `d` as a vector, because a vector plus a scalar returns a vector. Another way to create a vector from a scalar is to use `c`, as illustrated in the last line.

12.3.2.6 Confusions between vectors and one-column or one-row matrices

Consider the following code:

```
myfun <- nimbleFunction(  
  run = function() {  
    A <- matrix(value = rnorm(9), nrow = 3)  
    B <- rnorm(3)  
    Cmatrix <- A %*% B           # double matrix, one column  
    Cvector <- (A %*% B)[,1]     # double vector  
    Cmatrix <- (A %*% B)[,1]     # error, vector assigned to matrix  
    Cmatrix[,1] <- (A %*% B)[,1] # ok, if Cmatrix is large enough  
  })
```

This creates a matrix A, a vector B, and matrix-multiplies them. The vector B is automatically treated as a one-column matrix in matrix algebra computations. The result of matrix multiplication is always a matrix, but a programmer may expect a vector, since they know the result will have one column. To make it a vector, simply extract the first column. More information about such handling is provided in the next section.

12.3.2.7 Understanding dimensions and sizes from linear algebra

As much as possible, NIMBLE behaves like R when determining types and sizes returned from linear algebra expressions, but in some cases this is not possible because R uses run-time information while NIMBLE must determine dimensions at compile time. For example, when matrix multiplying a matrix by a vector, R treats the vector as a one-column matrix unless treating it as a one-row matrix is the only way to make the expression valid, as determined at run time. NIMBLE usually must assume during compilation that it should be a one-column matrix, unless it can determine not just the number of dimensions but the actual sizes during compilation. When needed `asRow` and `asCol` can control how a vector will be treated as a matrix.

Here is a guide to such issues. Suppose `v1` and `v2` are vectors, and `M1` is a matrix. Then

- `v1 + M1` promotes `v1` to a 1-column matrix if `M1` is a one-column matrix. Otherwise, this results in a run-time error. This behavior occurs for all component-wise binary functions.
- `v1 %*% M1` defaults to promoting `v1` to a 1-row matrix, unless it is known at compile-time that `M1` has 1 row, in which case `v1` is promoted to a 1-column matrix.
- `M1 %*% v1` defaults to promoting `v1` to a 1-column matrix, unless it is known at compile time that `M1` has 1 column, in which case `v1` is promoted to a 1-row matrix.
- `v1 %*% v2` promotes `v1` to a 1-row matrix and `v2` to a 1-column matrix, so the returned values is a 1x1 matrix with the inner product of `v1` and `v2`. If you want the inner product as a scalar, use `inprod(v1, v2)`.
- `asRow(v1)` explicitly promotes `v1` to a 1-row matrix. Therefore `v1 %*% asRow(v2)` gives the outer product of `v1` and `v2`.
- `asCol(v1)` explicitly promotes `v1` to a 1-column matrix.
- The default promotion for a vector is to a 1-column matrix. Therefore, `v1 %*% t(v2)` is equivalent to `v1 %*% asRow(v2)`.
- When indexing, dimensions with scalar indices will be dropped. For example, `M1[1,]` and `M1[,1]` are both vectors. If you do not want this behavior, use `drop=FALSE` just as in R. For

example, `M1[1,,drop=FALSE]` is a matrix.

- The left-hand side of an assignment can use indexing, but if so it must already be correctly sized for the result. For example, `Y[5:10, 20:30] <- x` will not work – and could crash your R session with a segmentation fault – if `Y` is not already at least 10x30 in size. This can be done by `setSize(Y, c(10, 30))`. See Section 12.3.2.4 for more details. Note that non-indexed assignment to `Y`, such as `Y <- x`, will automatically set `Y` to the necessary size.

Here are some examples to illustrate the above points, assuming `M2` is a square matrix.

- `Y <- v1 + M2 %*% v2` will return a 1-column matrix. If `Y` is created by this statement, it will be a 2-dimensional variable. If `Y` already exists, it must already be 2-dimensional, and it will be automatically re-sized for the result.
- `Y <- v1 + (M2 %*% v2)[,1]` will return a vector. `Y` will either be created as a vector or must already exist as a vector and will be re-sized for the result.

12.3.2.8 Size warnings and the potential for crashes

For matrix algebra, NIMBLE cannot ensure perfect behavior because sizes are not known until run time. Therefore, it is possible for you to write code that will crash your R session. In Version 1.4.0, NIMBLE attempts to issue a warning if sizes are not compatible, but it does not halt execution. Therefore, if you execute `A <- M1 %*% M2`, and `M1` and `M2` are not compatible for matrix multiplication, NIMBLE will output a warning that the number of rows of `M1` does not match the number of columns of `M2`. After that warning the statement will be executed and may result in a crash. Another easy way to write code that will crash is to do things like `Y[5:10, 20:30] <- x` without ensuring `Y` is at least 10x30. In the future we hope to prevent crashes, but in Version 1.4.0 we limit ourselves to trying to provide useful information.

12.4 Declaring argument and return types

NIMBLE requires that types of arguments and the type of the return value be explicitly declared.

As illustrated in the example in Section 12.1, the syntax for a type declaration is:

```
type(nDim, sizes)
```

where `type` is `double`, `integer`, `logical` or `character`. (In more general `nimbleFunction` programming, a type can also be a `nimbleList` type, discussed in Section 15.2.)

For example `run = function(x = double(1)) { ... }` sets the single argument of the `run` function to be a vector of numeric values of unknown size.

For `type(nDim, sizes)`, `nDim` is the number of dimensions, with 0 indicating scalar and omission of `nDim` defaulting to a scalar. `sizes` is an optional vector of fixed, known sizes.

For example, `double(2, c(4, 5))` declares a 4×5 matrix. If sizes are omitted, they will be set either by assignment or by `setSize`.

In the case of scalar arguments only, a default value can be provided. For example, to provide 1.2 as a default:

```
myfun <- nimbleFunction(
  run = function(x = double(0, default = 1.2)) {
  })
```

Functions with return values must have their return type explicitly declared using `returnType`, which can occur anywhere in the run code. For example `returnType(integer(2))` declares the return type to be a matrix of integers. A return type of `void()` means there is no return value, which is the default if no `returnType` statement is included.

Note that because all values in models are stored as doubles and because of some limitations in NIMBLE's automatic casting, non-scalar return values of user-defined distributions must be doubles.

12.5 Compiled nimbleFunctions pass arguments by reference

Uncompiled nimbleFunctions pass arguments like R does, by copy. If `x` is passed as an argument to function `foo`, and `foo` modifies `x` internally, it is modifying its copy of `x`, not the original `x` that was passed to it.

Compiled nimbleFunctions pass arguments to other compiled nimbleFunctions by reference (or pointer). This is very different. Now if `foo` modifies `x` internally, it is modifying the same `x` that was passed to it. This allows much faster execution but is obviously a fundamentally different behavior.

Uncompiled execution of nimbleFunctions is primarily intended for debugging. However, debugging of how nimbleFunctions interact via arguments requires testing the compiled versions.

12.6 Calling external compiled code

If you have a function in your own compiled C or C++ code and an appropriate header file, you can generate a nimbleFunction that wraps access to that function, which can then be used in other nimbleFunctions. See `help(nimbleExternalCall)` for an example. This also contains an example of using an externally compiled function in the BUGS code of a model.

12.7 Calling uncompiled R functions from compiled nimbleFunctions

Sometimes one may want to combine R functions with compiled nimbleFunctions. Obviously a compiled nimbleFunction can be called from R. An R function with numeric inputs and output can be called from compiled nimbleFunctions. The call to the R function is wrapped in a nimbleFunction returned by `nimbleRcall`. See `help(nimbleRcall)` for an example, including an example of using the resulting function in the BUGS code of a model.

Chapter 13

Creating user-defined distributions and functions for models

NIMBLE allows you to define your own functions and distributions as *nimbleFunctions* for use in model code. As a result, NIMBLE frees you from being constrained to the functions and distributions discussed in Chapter 5. For example, instead of setting up a Dirichlet prior with multinomial data and needing to use MCMC, one could recognize that this results in a Dirichlet-multinomial distribution for the data and provide that as a user-defined distribution instead.

Since NIMBLE allows you to wrap calls to external compiled code or arbitrary R functions as *nimbleFunctions*, and since you can define model functions and distributions as *nimbleFunctions*, you can combine these features to build external compiled code or arbitrary R functions into a model. See Sections 12.6-12.7.

As of version 1.2.0, NIMBLE also supports using more advanced *nimbleFunctions*, those with `setup` code and possibly multiple methods (see Chapter 16), to provide functions and/or distributions for use in models. This allows the functions and/or distributions to store information internally. See section 13.3 below.

Note that NIMBLE generally expects user-defined distributions or functions to be defined in the global environment. If you define them in a function (which would generally be the case if you are using them in the context of parallelization), one approach would be to assign them to the global environment in your function:

```
## for a user-defined function
assign('myfun', myfun, envir = .GlobalEnv)
## for a user-defined distribution
assign('dfoo', dfoo, envir = .GlobalEnv)
assign('rfoo', rfoo, envir = .GlobalEnv)
## similarly for 'p' and 'q' functions if you define them
```

13.1 User-defined functions

To provide a new function for use in BUGS code, simply create a *nimbleFunction* as discussed in Chapter 12. Then use it in your BUGS code. That's it.

Writing `nimbleFunctions` requires that you declare the dimensionality of arguments and the returned object (Section 12.4). Make sure that the dimensionality specified in your `nimbleFunction` matches how you use it in BUGS code. For example, if you define scalar parameters in your BUGS code you will want to define `nimbleFunctions` that take scalar arguments. Here is an example that returns twice its input argument:

```
timesTwo <- nimbleFunction(
  run = function(x = double(0)) {
    returnType(double(0))
    return(2*x)
  })

code <- nimbleCode({
  for(i in 1:3) {
    mu[i] ~ dnorm(0, 1)
    mu_times_two[i] <- timesTwo(mu[i])
  }
})
```

The `x = double(0)` argument and `returnType(double(0))` establish that the input and output will both be zero-dimensional (scalar) numbers.

You can define `nimbleFunctions` that take inputs and outputs with more dimensions. Here is an example that takes a vector (1-dimensional) as input and returns a vector with twice the input values:

```
vectorTimesTwo <- nimbleFunction(
  run = function(x = double(1)) {
    returnType(double(1))
    return(2*x)
  }
)

code <- nimbleCode({
  for(i in 1:3) {
    mu[i] ~ dnorm(0, 1)
  }
  mu_times_two[1:3] <- vectorTimesTwo(mu[1:3])
})
```

There is a subtle difference between the `mu_times_two` variables in the two examples. In the first example, there are individual nodes for each `mu_times_two[i]`. In the second example, there is a single multivariate node, `mu_times_two[1:3]`. Each implementation could be more efficient for different needs. For example, suppose an algorithm modifies the value of `mu[2]` and then updates nodes that depend on it. In the first example, `mu_times_two[2]` would be updated. In the second example `mu_times_two[1:3]` would be updated because it is a single, vector node.

At present in compiled use of a model, you cannot provide a scalar argument where the user-defined `nimbleFunction` expects a vector; unlike in R, scalars are not simply vectors of length 1.

13.2 User-defined distributions

To provide a user-defined distribution, you need to define density ('d') and (optionally) simulation ('r') `nimbleFunctions` for your distribution. In many cases you can then simply use your distribution in BUGS code as you would any distribution already provided by NIMBLE, while in a few special cases¹ you need to explicitly register your distribution as described in Section 13.2.1.

You need to provide the simulation ('r') function if any algorithm used with a model that uses the distribution needs to simulate from the distribution. This is not the case for NIMBLE's built-in MCMC sampler functions, and therefore the simulation function is not generally required for standard MCMC in NIMBLE. However, the 'r' function is necessary for initialization of nodes that are assigned the user-defined distribution if no initial value is provided, and for sampling posterior predictive nodes (those nodes with no downstream data dependencies) that are assigned the user-defined distribution.

You can optionally provide distribution ('p') and quantile ('q') functions, which will allow truncation to be applied to a user-defined distribution. You can also provide a list of alternative parameterizations, but only if you explicitly register the distribution.

Here is an extended example of providing a univariate exponential distribution (solely for illustration as this is already provided by NIMBLE) and a multivariate Dirichlet-multinomial distribution.

```
dmyexp <- nimbleFunction(
  run = function(x = double(0), rate = double(0, default = 1),
    log = integer(0, default = 0)) {
    returnType(double(0))
    logProb <- log(rate) - x*rate
    if(log) return(logProb)
    else return(exp(logProb))
  })

rmyexp <- nimbleFunction(
  run = function(n = integer(0), rate = double(0, default = 1)) {
    returnType(double(0))
    if(n != 1) print("rmyexp only allows n = 1; using n = 1.")
    dev <- runif(1, 0, 1)
    return(-log(1-dev) / rate)
  })

pmyexp <- nimbleFunction(
  run = function(q = double(0), rate = double(0, default = 1),
    lower.tail = integer(0, default = 1),
    log.p = integer(0, default = 0)) {
    returnType(double(0))
    if(!lower.tail) {
      logp <- -rate * q
      if(log.p) return(logp)
    }
  })
```

¹These include providing alternative parameterizations, specifying the range of the distribution, or specifying that the distribution is a discrete distribution.

```

        else return(exp(logp))
      } else {
        p <- 1 - exp(-rate * q)
        if(!log.p) return(p)
        else return(log(p))
      }
    })

qmyexp <- nimbleFunction(
  run = function(p = double(0), rate = double(0, default = 1),
    lower.tail = integer(0, default = 1),
    log.p = integer(0, default = 0)) {
    returnType(double(0))
    if(log.p) p <- exp(p)
    if(!lower.tail) p <- 1 - p
    return(-log(1 - p) / rate)
  })

ddirchmulti <- nimbleFunction(
  run = function(x = double(1), alpha = double(1), size = double(0),
    log = integer(0, default = 0)) {
    returnType(double(0))
    logProb <- lgamma(size+1) - sum(lgamma(x+1)) + lgamma(sum(alpha)) -
      sum(lgamma(alpha)) + sum(lgamma(alpha + x)) -
      lgamma(sum(alpha) + size)
    if(log) return(logProb)
    else return(exp(logProb))
  })

rdirchmulti <- nimbleFunction(
  run = function(n = integer(0), alpha = double(1), size = double(0)) {
    returnType(double(1))
    if(n != 1) print("rdirchmulti only allows n = 1; using n = 1.")
    p <- rdirch(1, alpha)
    return(rmulti(1, size = size, prob = p))
  })

code <- nimbleCode({
  y[1:K] ~ ddirchmulti(alpha[1:K], n)
  for(i in 1:K) {
    alpha[i] ~ dmyexp(1/3)
  }
})

model <- nimbleModel(code, constants = list(K = 5, n = 10))

```

The distribution-related functions should take as input the parameters for a single parameterization,

which will be the canonical parameterization that NIMBLE will use.

Here are more details on the requirements for distribution-related `nimbleFunctions`, which follow R's conventions:

- Your distribution-related functions must have names that begin with 'd', 'r', 'p' and 'q'. The name of the distribution must not be identical to any of the NIMBLE-provided distributions.
- All simulation ('r') functions must take `n` as their first argument. Note that you may simply have your function only handle `n=1` and return an warning for other values of `n`.
- NIMBLE uses doubles for numerical calculations, so we suggest simply using doubles in general, even for integer-valued parameters or values of random variables. In fact, non-scalars *must* be declared as doubles.
- All density functions must have as their last argument `log` and implement return of the log probability density. NIMBLE algorithms typically use only `log = 1` (i.e., `TRUE`), but we recommend you implement the `log = 0` (i.e., `FALSE`) case for completeness.
- All distribution and quantile functions must have their last two arguments be (in order) `lower.tail` and `log.p`. These functions must work for `lower.tail = 1` (i.e., `TRUE`) and `log.p = 0` (i.e., `FALSE`), as these are the inputs we use when working with truncated distributions. It is your choice whether you implement the necessary calculations for other combinations of these inputs, but again we recommend doing so for completeness.
- Define the `nimbleFunctions` in R's global environment. Don't expect R's standard scoping to work².

13.2.1 Using *registerDistributions* for alternative parameterizations and providing other information

Behind the scenes, NIMBLE uses the function `registerDistributions` to set up new distributions for use in BUGS code. In some circumstances, you will need to call `registerDistributions` directly to provide information that NIMBLE can't obtain automatically from the `nimbleFunctions` you write.

The cases in which you'll need to explicitly call `registerDistributions` are when you want to do any of the following:

- provide alternative parameterizations,
- indicate a distribution is discrete, and
- provide the range of possible values for a distribution.

If you would like to allow for multiple parameterizations, you can do this via the `Rdist` element of the list provided to `registerDistributions` as illustrated below. If you provide CDF ('p') and inverse CDF (quantile, i.e. 'q') functions, be sure to specify `pqAvail = TRUE` when you call `registerDistributions`. Here's an example of using `registerDistributions` to provide an alternative parameterization (scale instead of rate) and to provide the range for the user-defined exponential distribution. We can then use the alternative parameterization in our BUGS code.

```
registerDistributions(list(
  dmyexp = list(
    BUGSdist = "dmyexp(rate, scale)",
    Rdist = "dmyexp(rate = 1/scale)",
```

²NIMBLE can't use R's standard scoping because it doesn't work for R reference classes, and `nimbleFunctions` are implemented as custom-generated reference classes.

```

    altParams = c("scale = 1/rate", "mean = 1/rate"),
    pqAvail = TRUE,
    range = c(0, Inf)
  )
))

code <- nimbleCode({
  y[1:K] ~ ddirchmulti(alpha[1:K], n)
  for(i in 1:K) {
    alpha[i] ~ T(dmyexp(scale = 3), 0, 100)
  }
})

model <- nimbleModel(code, constants = list(K = 5, n = 10),
  inits = list(alpha = rep(1, 5)))

```

There are a few rules for how you specify the information about a distribution that you provide to `registerDistributions`:

- The function name in the `BUGSdist` entry in the list provided to `registerDistributions` will be the name you can use in BUGS code.
- The names of your `nimbleFunctions` must match the function name in the `Rdist` entry. If missing, the `Rdist` entry defaults to be the same as the `BUGSdist` entry.
- Your distribution-related functions must take as arguments the parameters in default order, starting as the second argument and in the order used in the parameterizations in the `Rdist` argument to `registerDistributions` or the `BUGSdist` argument if there are no alternative parameterizations.
- You must specify a `types` entry in the list provided to `registerDistributions` if the distribution is multivariate or if any parameter is non-scalar.

Further details on using `registerDistributions` can be found via R help on `registerDistributions`. NIMBLE uses the same list format as `registerDistributions` to define its distributions. This list can be found in the `R/distributions_inputList.R` file in the package source code directory or as the R list `nimble::distributionsInputList`.

13.3 Advanced user-defined functions and distributions

Sometimes it is useful for a user-defined function or distribution to use external information and/or hold variables that persist between calls. One can do this by creating `nimbleFunctions` that use `setup` code and have one or more methods. If you are familiar with object-oriented programming, this variety of `nimbleFunction` defines a class. Here we will give a toy example that is not fully explained because the NIMBLE programming concepts are covered in Chapter 16. After studying that chapter, the example here should make more sense.

As a toy example, consider this basic linear regression model with one explanatory variable:

```

code <- nimbleCode({
  beta0 ~ dnorm(0, sd = 100)
  beta1 ~ dnorm(0, sd = 100)

```

```

sigma ~ dhalfflat()
for(i in 1:n) {
  y[i] ~ dnorm(beta0 + beta1*x[i], sd = sigma)
}
})
# Simulate some data:
x <- runif(10)
y <- rnorm(10, 3 + 0.5 * x, sd = 0.2)
model <- nimbleModel(code, constants = list(n = 10, x = x),
                     data = list(y = y),
                     inits = list(beta0=3.1,beta1=0.4,sigma=0.3))

```

Now suppose we want to rewrite this model in an equivalent way by providing a “distribution” that actually contains all the steps to calculate the log likelihood for the linear regression internally (for all of `y[1:10]` at once), including that it contains the explanatory variable `x`. We could do that as follows:

```

nimbleOptions(allowNFObjInModel=TRUE) # This option should be TRUE by default,
                                         # but we include this step to be sure.
linear_reg <- nimbleFunction(
  setup = function(predictor) {}, # Call x "predictor" here for clarity
  methods = list(
    dcalc = function(x = double(1), # Note that this will be y!
                     beta0=double(), beta1=double(), sigma=double(),
                     log=integer(0, default=0)) {
      llh <- sum(dnorm(x, beta0+beta1*predictor, sd=sigma, log=TRUE))
      if(log) return(llh)
      return(exp(llh))
      returnType(double())
    })
my_linear_reg <- linear_reg(x)
code2 <- nimbleCode({
  beta0 ~ dnorm(0, sd = 100)
  beta1 ~ dnorm(0, sd = 100)
  sigma ~ dhalfflat()
  y[1:n] ~ my_linear_reg$dcalc(beta0, beta1, sigma)
})
model <- nimbleModel(code2, constants = list(n = 10),
                     data = list(y = y),
                     inits = list(beta0=3.1,beta1=0.4,sigma=0.3))

```

What is going on, and why would one even think of doing it this way? `linear_reg` here is defined as a `nimbleFunction` to create objects that will hold values of `x` internally and use them when the `dcalc` function (or “method”) is called. `my_linear_reg` is one instance of `linear_reg` that holds the specific values of `x` provided. The method `dcalc` has the format (arguments, return type, and prefix “d”) of a user-defined distribution, so it can be used that way via `my_linear_reg$dcalc`. One can provide a corresponding `rcalc` method, but as above this is only required if it will actually be needed by an algorithm. The suffix “calc” could be replaced with anything, but a user-defined

distribution must have the prefix “d” (and can even be named just “d”), and a corresponding simulation must begin with “r”. Any method in an object like `my_linear_reg` can be used as a user-defined function, and the same `nimbleFunction` (`linear_reg`) can define multiple user-defined functions and/or distributions.

(Note the deliberate illustration of a potential confusion: The random variable argument to a user-defined distribution is always named “x”, which in this example will get the values of `y[1:10]`.)

Now, *why* would one want to do this? There are several reasons. First, sometimes instead of a short vector like `x`, there may be quite large data objects involved in some calculation. It can be wasteful (of time and memory when building and compiling models) to provide large objects in models for the sole purpose of passing them to user-defined functions or distributions. It can work well to access them internally as shown with `x` in this example. One can also consider caching intermediate values from costly calculations or other tricks to speed up calculations.

It is vital when using this approach to understand clearly how an algorithm will use a model, and there is potential to create problems. For example, NIMBLE’s MCMC system uses a set of samplers, each of which assumes the model is fully up-to-date when it begins its operations. “Up-to-date” means that model variables all have the correct current values and all deterministic and stochastic calculations are up-to-date with those values. Sometimes, samplers copy values around to achieve this. However, a sampler has no way to know what variables a `nimbleFunction` object (e.g. `my_linear_reg`) is managing internally, and thus has no way to keep them up to date, if necessary. Techniques such as caching or “memoization” should be careful in this regard.

One can also register a distribution in a `nimbleFunction` object manually, for the reasons given above. The name of the distribution is simply the string “`my_linear_reg$dcalc`”, to use the current example.

13.4 User-defined model macros

As discussed in Section 5.2.8, users can use macros that provide a succinct syntax for expressing components of a model. The `nimbleMacros` package provides an initial set of macros, focused on the components of generalized linear (mixed) models.

In addition, NIMBLE users can write their own macros for use in model code.

As a toy example, suppose you wanted to create a macro called `MYDIST`, which creates a normal distribution with mean 0 and provided standard deviation. The macro would look like this in the NIMBLE model code:

```
y ~ MYDIST(sdev = 2)
```

Note that by convention, we define the macro name in all capital letters.

We want the final, processed output to look like:

```
y ~ dnorm(0, sd = 2)
```

One would then need to create the `MYDIST` macro using `buildMacro`.

See the [vignette for the `nimbleMacros` package](#) and `help(buildMacro)` for more information about how to use `buildMacro` to create a new macro.

Chapter 14

Working with NIMBLE models

Here we describe how one can get information about NIMBLE models and carry out operations on a model. While all of this functionality can be used from R, its primary use occurs when writing `nimbleFunctions` (see Chapter 16). Information about node types, distributions, and dimensions can be used to determine algorithm behavior in *setup* code of `nimbleFunctions`. Information about node or variable values or the parameter and bound values of a node would generally be used for algorithm calculations in *run* code of `nimbleFunctions`. Similarly, carrying out numerical operations on a model, including setting node or variable values, would generally be done in run code.

14.1 The variables and nodes in a NIMBLE model

Section 6.2 defines what we mean by variables and nodes in a NIMBLE model and discusses how to determine and access the nodes in a model and their dependency relationships. Here we'll review and go into more detail on the topics of determining the nodes and node dependencies in a model.

14.1.1 Determining the nodes in a model

One can determine the variables in a model using `getVarNames` and the nodes in a model using `getNodeNames`, with optional arguments allowing you to select only certain types of nodes. We illustrate here with the pump model from Chapter 2.

```
pump$getVarNames()
```

```
## [1] "lifted_d1_over_beta" "theta"          "lambda"
## [4] "x"                  "alpha"         "beta"
```

```
pump$getNodeNames()
```

```
## [1] "alpha"          "beta"           "lifted_d1_over_beta"
## [4] "theta[1]"       "theta[2]"       "theta[3]"
## [7] "theta[4]"       "theta[5]"       "theta[6]"
## [10] "theta[7]"       "theta[8]"       "theta[9]"
## [13] "theta[10]"      "lambda[1]"      "lambda[2]"
## [16] "lambda[3]"      "lambda[4]"      "lambda[5]"
## [19] "lambda[6]"      "lambda[7]"      "lambda[8]"
```

```
## [22] "lambda[9]"      "lambda[10]"     "x[1]"
## [25] "x[2]"           "x[3]"           "x[4]"
## [28] "x[5]"           "x[6]"           "x[7]"
## [31] "x[8]"           "x[9]"           "x[10]"
```

```
pump$getNodeNames(determOnly = TRUE)
```

```
## [1] "lifted_d1_over_beta" "lambda[1]"      "lambda[2]"
## [4] "lambda[3]"           "lambda[4]"      "lambda[5]"
## [7] "lambda[6]"           "lambda[7]"      "lambda[8]"
## [10] "lambda[9]"           "lambda[10]"
```

```
pump$getNodeNames(stochOnly = TRUE)
```

```
## [1] "alpha"      "beta"      "theta[1]"  "theta[2]"  "theta[3]"  "theta[4]"
## [7] "theta[5]"   "theta[6]"  "theta[7]"  "theta[8]"  "theta[9]"  "theta[10]"
## [13] "x[1]"       "x[2]"      "x[3]"      "x[4]"      "x[5]"      "x[6]"
## [19] "x[7]"       "x[8]"      "x[9]"      "x[10]"
```

```
pump$getNodeNames(dataOnly = TRUE)
```

```
## [1] "x[1]" "x[2]" "x[3]" "x[4]" "x[5]" "x[6]" "x[7]" "x[8]" "x[9]"
## [10] "x[10]"
```

You can see one lifted node (see next section), `lifted_d1_over_beta`, involved in a reparameterization to NIMBLE's canonical parameterization of the gamma distribution for the `theta` nodes.

We can determine the set of nodes contained in one or more nodes or variables using `expandNodeNames`, illustrated here for an example with multivariate nodes. The `returnScalarComponents` argument also allows us to return all of the scalar elements of multivariate nodes.

```
multiVarCode2 <- nimbleCode({
  X[1, 1:5] ~ dmnorm(mu[, ], cov[, ])
  X[6:10, 3] ~ dmnorm(mu[, ], cov[, ])
  for(i in 1:4)
    Y[i] ~ dnorm(mn, 1)
})

multiVarModel2 <- nimbleModel(multiVarCode2,
  dimensions = list(mu = 5, cov = c(5,5)),
  calculate = FALSE)

multiVarModel2$expandNodeNames("Y")

## [1] "Y[1]" "Y[2]" "Y[3]" "Y[4]"

multiVarModel2$expandNodeNames(c("X", "Y"), returnScalarComponents = TRUE)

## [1] "X[1, 1]" "X[1, 2]" "X[1, 3]" "X[6, 3]" "X[7, 3]" "X[8, 3]"
## [7] "X[9, 3]" "X[10, 3]" "X[1, 4]" "X[1, 5]" "Y[1]" "Y[2]"
```



```
## [13] "Y[3]"      "Y[4]"
```

As discussed in Section 6.2.6, you can determine whether a node is flagged as data using `isData`.

14.1.2 Understanding lifted nodes

In some cases, NIMBLE introduces new nodes into the model that were not specified in the BUGS code for the model, such as the `lifted_d1_over_beta` node in the introductory example. For this reason, it is important that programs written to adapt to different model structures use NIMBLE's systems for querying the model graph. For example, a call to `pump$getDependencies("beta")` will correctly include `lifted_d1_over_beta` in the results. If one skips this step and assumes the nodes are only those that appear in the BUGS code, one may not get correct results.

It can be helpful to know the situations in which lifted nodes are generated. These include:

1. When distribution parameters are expressions, NIMBLE creates a new deterministic node that contains the expression for a given parameter. The node is then a direct descendant of the new deterministic node. This is an optional feature, but it is currently enabled in all cases.
2. As discussed in Section 5.2.6, the use of link functions causes new nodes to be introduced. This requires care if you need to initialize values in stochastic declarations with link functions.
3. Use of alternative parameterizations of distributions, described in Section 5.2.4 causes new nodes to be introduced. For example when a user provides the precision of a normal distribution as `tau`, NIMBLE creates a new node `sd <- 1/sqrt(tau)` and uses `sd` as a parameter in the normal distribution. If many nodes use the same `tau`, only one new `sd` node will be created, so the computation `1/sqrt(tau)` will not be repeated redundantly.

14.1.3 Determining dependencies in a model

Next we'll see how to determine the node dependencies (or 'descendants' or child nodes) in a model. There are a variety of arguments to `getDependencies` that allow one to specify whether to include the node itself, whether to include deterministic or stochastic or data dependents, etc. By default `getDependencies` returns descendants up to the next stochastic node on all edges emanating from the node(s) specified as input. This is what would be needed to calculate a Metropolis-Hastings acceptance probability in MCMC, for example.

```
pump$getDependencies("alpha")
```

```
## [1] "alpha"      "theta[1]"   "theta[2]"   "theta[3]"   "theta[4]"   "theta[5]"
## [7] "theta[6]"   "theta[7]"   "theta[8]"   "theta[9]"   "theta[10]"
```

```
pump$getDependencies(c("alpha", "beta"))
```

```
## [1] "alpha"      "beta"      "lifted_d1_over_beta"
## [4] "theta[1]"   "theta[2]"   "theta[3]"
## [7] "theta[4]"   "theta[5]"   "theta[6]"
## [10] "theta[7]"   "theta[8]"   "theta[9]"
## [13] "theta[10]"
```

```
pump$getDependencies("theta[1:3]", self = FALSE)
```

```
## [1] "lambda[1]" "lambda[2]" "lambda[3]" "x[1]"      "x[2]"      "x[3]"
```

```
pump$getDependencies("theta[1:3]", stochOnly = TRUE, self = FALSE)
```

```
## [1] "x[1]" "x[2]" "x[3]"
```

```
# get all dependencies, not just the direct descendants
```

```
pump$getDependencies("alpha", downstream = TRUE)
```

```
## [1] "alpha"      "theta[1]"   "theta[2]"   "theta[3]"   "theta[4]"
## [6] "theta[5]"   "theta[6]"   "theta[7]"   "theta[8]"   "theta[9]"
## [11] "theta[10]"  "lambda[1]"  "lambda[2]"  "lambda[3]"  "lambda[4]"
## [16] "lambda[5]"  "lambda[6]"  "lambda[7]"  "lambda[8]"  "lambda[9]"
## [21] "lambda[10]" "x[1]"       "x[2]"       "x[3]"       "x[4]"
## [26] "x[5]"       "x[6]"       "x[7]"       "x[8]"       "x[9]"
## [31] "x[10]"
```

```
pump$getDependencies("alpha", downstream = TRUE, dataOnly = TRUE)
```

```
## [1] "x[1]" "x[2]" "x[3]" "x[4]" "x[5]" "x[6]" "x[7]" "x[8]" "x[9]"
## [10] "x[10]"
```

In addition, one can determine parent nodes using `getParents`.

```
pump$getParents("alpha")
```

```
## character(0)
```

14.2 Accessing information about nodes and variables

14.2.1 Getting distributional information about a node

We briefly demonstrate some of the functionality for information about a node here, but refer readers to the R help on `modelBaseClass` for full details.

Here is an example model, with use of various functions to determine information about nodes or variables.

```
code <- nimbleCode({
  for(i in 1:4)
    y[i] ~ dnorm(mu, sd = sigma)
  mu ~ T(dnorm(0, 5), -20, 20)
  sigma ~ dunif(0, 10)
})
m <- nimbleModel(code, data = list(y = rnorm(4)),
                  inits = list(mu = 0, sigma = 1))
m$isEndNode('y')
```

```
## y[1] y[2] y[3] y[4]
## TRUE TRUE TRUE TRUE
```

```
m$getDistribution('sigma')
```

```
## sigma
```

```
## "dunif"
m$isDiscrete(c('y', 'mu', 'sigma'))

## y[1] y[2] y[3] y[4] mu sigma
## FALSE FALSE FALSE FALSE FALSE FALSE

m$isDeterm('mu')

## mu
## FALSE

m$getDimension('mu')

## value
## 0

m$getDimension('mu', includeParams = TRUE)

## value mean sd tau var
## 0 0 0 0 0
```

Note that any variables provided to these functions are expanded into their constituent node names, so the length of results may not be the same length as the input vector of node and variable names. However the order of the results should be preserved relative to the order of the inputs, once the expansion is accounted for.

14.2.2 Getting information about a distribution

One can also get generic information about a distribution based on the name of the distribution using the function `getDistributionInfo`. In particular, one can determine whether a distribution was provided by the user (`isUserDefined`), whether a distribution provides CDF and quantile functions (`pqDefined`), whether a distribution is a discrete distribution (`isDiscrete`), the parameter names (include alternative parameterizations) for a distribution (`getParamNames`), and the dimension of the distribution and its parameters (`getDimension`). For more extensive information, please see the R help for `getDistributionInfo`.

14.2.3 Getting distribution parameter values for a node

The function `getParam` provides access to values of the parameters of a node's distribution. The two arguments must be the name of one (stochastic) node and the name of a parameter for the distribution followed by that node. The parameter does not have to be one of the parameters used when the node was declared. Alternative parameterization values can also be obtained. See Section 5.2.4.1 for available parameterizations. (These can also be seen in `nimble::distributionsInputList`.)

Here is an example:

```
gammaModel <- nimbleModel(
  nimbleCode({
    a ~ dlnorm(0, 1)
    x ~ dgamma(shape = 2, scale = a)
  }), data = list(x = 2.4), inits = list(a = 1.2))
gammaModel$getParam('x', 'scale')
```

```
## [1] 1.2
```

```
gammaModel$getParam('x', 'rate')
```

```
## [1] 0.8333333
```

`getParam` is part of the NIMBLE language, so it can be used in run code of `nimbleFunctions`.

14.2.4 Getting distribution bounds for a node

The function `getBound` provides access to the lower and upper bounds of the distribution for a node. In most cases these bounds will be fixed based on the distribution, but for the uniform distribution the bounds are the parameters of the distribution, and when truncation is used (Section 5.2.7), the bounds will be determined by the truncation. Like the functions described in the previous section, `getBound` can be used as global function taking a model as the first argument, or it can be used as a model member function. The next two arguments must be the name of one (stochastic) node and either "lower" or "upper" indicating whether the lower or upper bound is desired. For multivariate nodes the bound is a scalar that is the bound for all elements of the node, as we do not handle truncation for multivariate nodes.

Here is an example:

```
exampleModel <- nimbleModel(
  nimbleCode({
    y ~ T(dnorm(mu, sd = sig), a, Inf)
    a ~ dunif(-1, b)
    b ~ dgamma(1, 1)
  }), inits = list(a = -0.5, mu = 1, sig = 1, b = 4),
  data = list(y = 4))
getBound(exampleModel, 'y', 'lower')
```

```
## [1] -0.5
```

```
getBound(exampleModel, 'y', 'upper')
```

```
## [1] Inf
```

```
exampleModel$b <- 3
exampleModel$calculate(exampleModel$getDependencies('b'))
```

```
## [1] -4.386294
```

```
getBound(exampleModel, 'a', 'upper')
```

```
## [1] 3
```

```
exampleModel$getBound('b', 'lower')
```

```
## [1] 0
```

`getBound` is part of the NIMBLE language, so it can be used in run code of `nimbleFunctions`. In fact, we anticipate that most use of `getBound` will be for algorithms, such as for the reflection version of the random walk MCMC sampler.

14.3 Carrying out model calculations

14.3.1 Core model operations: calculation and simulation

The four basic ways to operate a model are to calculate nodes, simulate into nodes, get the log probabilities (or probability densities) that have already been calculated, and compare the log probability of a new value to that of an old value. In more detail:

- **calculate** For a stochastic node, **calculate** determines the log probability value, stores it in the appropriate **logProb** variable, and returns it. For a deterministic node, **calculate** executes the deterministic calculation and returns 0.
- **simulate** For a stochastic node, **simulate** generates a random draw. For deterministic nodes, **simulate** is equivalent to **calculate** without returning 0. **simulate** always returns NULL (or **void** in C++).
- **getLogProb** **getLogProb** simply returns the most recently calculated log probability value, or 0 for a deterministic node.
- **calculateDiff** **calculateDiff** is identical to **calculate**, but it returns the new log probability value minus the one that was previously stored. This is useful when one wants to change the value or values of node(s) in the model (e.g., by setting a value or **simulate**) and then determine the change in the log probability, such as needed for a Metropolis-Hastings acceptance probability.

Each of these functions is accessed as a member function of a model object, taking a vector of node names as an argument¹. If there is more than one node name, **calculate** and **getLogProb** return the sum of the log probabilities from each node, while **calculateDiff** returns the sum of the new values minus the old values. Next we show an example using **simulate**.

14.3.1.1 Example: simulating arbitrary collections of nodes

```
mc <- nimbleCode({
  a ~ dnorm(0, 0.001)
  for(i in 1:5) {
    y[i] ~ dnorm(a, 0.1)
    for(j in 1:3)
      z[i,j] ~ dnorm(y[i], sd = 0.1)
  }
  y.squared[1:5] <- y[1:5]^2
})

model <- nimbleModel(mc, data = list(z = matrix(rnorm(15), nrow = 5)))

model$a <- 1
model$y

## [1] NA NA NA NA NA

model$simulate("y[1:3]")
# simulate(model, "y[1:3]")
```

¹Standard usage is `model$calculate(nodes)` but `calculate(model, nodes)` is synonymous.

```

model$y

## [1] 0.8023941 1.5781841 4.4900193      NA      NA
model$simulate("y")
model$y

## [1] 6.540423 -2.016232 6.199039 -1.740852 1.842280
model$z

##           [,1]      [,2]      [,3]
## [1,] -1.6988735 -0.6230498 0.9158482
## [2,] 0.2318525 -0.4395223 0.3201767
## [3,] -0.1190907 -0.5052968 -0.3666873
## [4,] 1.7724929 0.1860351 -0.9406113
## [5,] 0.3434222 0.1764178 0.6347029
model$simulate(c("y[1:3]", "z[1:5, 1:3]"))
model$y

## [1] 1.7031972 0.1243384 5.4090152 -1.7408521 1.8422796
model$z

##           [,1]      [,2]      [,3]
## [1,] -1.6988735 -0.6230498 0.9158482
## [2,] 0.2318525 -0.4395223 0.3201767
## [3,] -0.1190907 -0.5052968 -0.3666873
## [4,] 1.7724929 0.1860351 -0.9406113
## [5,] 0.3434222 0.1764178 0.6347029
model$simulate(c("z[1:5, 1:3]"), includeData = TRUE)
model$z

##           [,1]      [,2]      [,3]
## [1,] 1.63730605 1.7692504 1.70187168
## [2,] 0.03119038 0.2458073 -0.08449563
## [3,] 5.35640000 5.2548750 5.42844735
## [4,] -1.71440982 -1.8527256 -1.67575678
## [5,] 1.73898956 1.9081997 1.86606253

```

The example illustrates a number of features:

1. `simulate(model, nodes)` is equivalent to `model$simulate(nodes)`. You can use either, but the latter is encouraged and the former may be deprecated in the future.
2. Inputs like `"y[1:3]"` are automatically expanded into `c("y[1]", "y[2]", "y[3]")`. In fact, simply `"y"` will be expanded into all nodes within `y`.
3. An arbitrary number of nodes can be provided as a character vector.
4. Simulations will be done in the order provided, so in practice the nodes should often be obtained by functions such as `getDependencies`. These return nodes in topologically-sorted order, which means no node is manipulated before something it depends on.

5. The data nodes `z` were not simulated into until `includeData = TRUE` was used.

Use of `calculate`, `calculateDiff` and `getLogProb` are similar to `simulate`, except that they return a value (described above) and they have no `includeData` argument.

14.3.2 Pre-defined nimbleFunctions for operating on model nodes: *simNodes*, *calcNodes*, and *getLogProbNodes*

`simNodes`, `calcNodes` and `getLogProbNodes` are basic nimbleFunctions that simulate, calculate, or get the log probabilities (densities), respectively, of the same set of nodes each time they are called. Each of these takes a model and a character string of node names as inputs. If `nodes` is left blank, then all the nodes of the model are used.

For `simNodes`, the nodes provided will be topologically sorted to simulate in the correct order. For `calcNodes` and `getLogProbNodes`, the nodes will be sorted and dependent nodes will be included. Recall that the calculations must be up to date (from a `calculate` call) for `getLogProbNodes` to return the values you are probably looking for.

```
simpleModelCode <- nimbleCode({
  for(i in 1:4){
    x[i] ~ dnorm(0,1)
    y[i] ~ dnorm(x[i], 1) # y depends on x
    z[i] ~ dnorm(y[i], 1) # z depends on y
    # z conditionally independent of x
  }
})

simpleModel <- nimbleModel(simpleModelCode, check = FALSE)
cSimpleModel <- compileNimble(simpleModel)

# simulates all the x's and y's
rSimXY <- simNodes(simpleModel, nodes = c('x', 'y'))

# calls calculate on x and its dependents (y, but not z)
rCalcXDep <- calcNodes(simpleModel, nodes = 'x')

# calls getLogProb on x's and y's
rGetLogProbXDep <- getLogProbNodes(simpleModel,
                                   nodes = 'x')

# compiling the functions
cSimXY <- compileNimble(rSimXY, project = simpleModel)
cCalcXDep <- compileNimble(rCalcXDep, project = simpleModel)
cGetLogProbXDep <- compileNimble(rGetLogProbXDep, project = simpleModel)

cSimpleModel$x

## [1] NA NA NA NA
```

```

cSimpleModel$y

## [1] NA NA NA NA
# simulating x and y
cSimXY$run()
cSimpleModel$x

## [1] 0.71527595 -0.93848304 0.09535401 -0.46281942
cSimpleModel$y

## [1] -0.7536062 -0.7857965 1.8691166 -1.1108904
cCalcXDep$run()

## [1] -11.03292
# gives correct answer because logProbs
# updated by 'calculate' after simulation
cGetLogProbXDep$run()

## [1] -11.03292
cSimXY$run()

# gives old answer because logProbs
# not updated after 'simulate'
cGetLogProbXDep$run()

## [1] -11.03292
cCalcXDep$run()

## [1] -13.16811

```

14.3.3 Accessing log probabilities via *logProb* variables

For each variable that contains at least one stochastic node, NIMBLE generates a model variable with the prefix 'logProb_'. In general users will not need to access these logProb variables directly but rather will use `getLogProb`. However, knowing they exist can be useful, in part because these variables can be monitored in an MCMC.

When the stochastic node is scalar, the logProb variable will have the same size. For example:

```

model$logProb_y

## [1] NA NA NA NA NA
model$calculate("y")

## [1] -11.79727
model$logProb_y

```



```
## [1] -2.094955 -2.108570 -3.042202 -2.445845 -2.105703
```

Creation of `logProb` variables for stochastic multivariate nodes is trickier, because they can represent an arbitrary block of a larger variable. In general NIMBLE records the `logProb` values using the lowest possible indices. For example, if `x[5:10, 15:20]` follows a Wishart distribution, its log probability (density) value will be stored in `logProb_x[5, 15]`. When possible, NIMBLE will reduce the dimensions of the corresponding `logProb` variable. For example, in

```
for(i in 1:10) x[i,] ~ dmnorm(mu[], prec[,])
```

`x` may be 10×20 (dimensions must be provided), but `logProb_x` will be 10×1 . For the most part you do not need to worry about how NIMBLE is storing the log probability values, because you can always get them using `getLogProb`.

Chapter 15

Data structures in NIMBLE

NIMBLE provides several data structures useful for programming.

We'll first describe *modelValues*, which are containers designed for storing values for models. Then in Section 15.2 we'll describe *nimbleLists*, which have a similar purpose to lists in R, allowing you to store heterogeneous information in a single object.

modelValues can be created in either R or in *nimbleFunction* setup code. *nimbleLists* can be created in R code, in *nimbleFunction* setup code, and in *nimbleFunction* run code, from a *nimbleList* definition created in R or setup code. Once created, *modelValues* and *nimbleLists* can then be used either in R or in *nimbleFunction* setup or run code. If used in run code, they will be compiled along with the *nimbleFunction*.

15.1 The *modelValues* data structure

modelValues are containers designed for storing values for models. They may be used for model outputs or model inputs. A *modelValues* object will contain *rows* of variables. Each row contains one object of each variable, which may be multivariate. The simplest way to build a *modelValues* object is from a model object. This will create a *modelValues* object with the same variables as the model. Although they were motivated by models, one is free to set up a *modelValues* with any variables one wants.

As with the material in the rest of this chapter, *modelValues* objects will generally be used in *nimbleFunctions* that interact with models (see Chapter 16)¹. *modelValues* objects can be defined either in setup code or separately in R (and then passed as an argument to setup code). The *modelValues* object can then be used in run code of *nimbleFunctions*.

15.1.1 Creating *modelValues* objects

Here is a simple example of creating a *modelValues* object:

```
pumpModelValues = modelValues(pumpModel, m = 2)
pumpModel$x
```

```
## [1] 5 1 5 14 3 19 1 1 4 22
```

¹One may want to read this section after an initial reading of Chapter 16.

```
pumpModelValues$x
```

```
## [[1]]
##  [1] NA NA NA NA NA NA NA NA NA NA NA
##
## [[2]]
##  [1] NA NA NA NA NA NA NA NA NA NA NA
```

In this example, `pumpModelValues` has the same variables as `pumpModel`, and we set `pumpModelValues` to have `m = 2` rows. As you can see, the rows are stored as elements of a list.

Alternatively, one can define a `modelValues` object manually by first defining a `modelValues` *configuration* via the `modelValuesConf` function, and then creating an instance from that configuration, like this:

```
mvConf = modelValuesConf(vars = c('a', 'b', 'c'),
                          type = c('double', 'int', 'double'),
                          size = list(a = 2, b = c(2,2), c = 1) )

customMV = modelValues(mvConf, m = 2)
customMV$a
```

```
## [[1]]
##  [1] NA NA
##
## [[2]]
##  [1] NA NA
```

The arguments to `modelValuesConf` are matching lists of variable names, types, and sizes. See `help(modelValuesConf)` for more details. Note that in R execution, the types are not enforced. But they will be the types created in C++ code during compilation, so they should be specified carefully.

The object returned by `modelValues` is an uncompiled `modelValues` object. When a `nimbleFunction` is compiled, any `modelValues` objects it uses are also compiled. A NIMBLE model always contains a `modelValues` object that it uses as a default location to store the values of its variables.

Here is an example where the `customMV` created above is used as the `setup` argument for a `nimbleFunction`, which is then compiled. Its compiled `modelValues` is then accessed with `$`.

```
# simple nimbleFunction that uses a modelValues object
resizeMV <- nimbleFunction(
  setup = function(mv){},
  run = function(k = integer() ){
    resize(mv,k)} )

rResize <- resizeMV(customMV)
cResize <- compileNimble(rResize)
cResize$run(5)
cCustomMV <- cResize$mv
```

```
# cCustomMV is a compiled modelValues object
cCustomMV[['a']]
```

```
## [[1]]
## [1] NA NA
##
## [[2]]
## [1] NA NA
##
## [[3]]
## [1] 0 0
##
## [[4]]
## [1] 0 0
##
## [[5]]
## [1] 0 0
```

Compiled modelValues objects can be accessed and altered in all the same ways as uncompiled ones. However, only uncompiled modelValues can be used as arguments to setup code in nimbleFunctions.

In the example above a modelValues object is passed to setup code, but a modelValues configuration can also be passed, with creation of modelValues object(s) from the configuration done in setup code.

15.1.2 Accessing contents of modelValues

The values in a modelValues object can be accessed in several ways from R, and in fewer ways from NIMBLE.

```
# sets the first row of a to (0, 1). R only.
customMV[['a']] [[1]] <- c(0,1)

# sets the second row of a to (2, 3)
customMV['a', 2] <- c(2,3)

# can access subsets of each row
customMV['a', 2][2] <- 4

# accesses all values of 'a'. Output is a list. R only.
customMV[['a']]
```

```
## [[1]]
## [1] 0 1
##
## [[2]]
## [1] 2 4
```

```
# sets the first row of b to a matrix with values 1. R only.
customMV[['b']] [[1]] <- matrix(1, nrow = 2, ncol = 2)
```

```
# sets the second row of b. R only.
customMV[['b']][[2]] <- matrix(2, nrow = 2, ncol = 2)

# make sure the size of inputs is correct
# customMV['a', 1] <- 1:10 "
# problem: size of 'a' is 2, not 10!
# will cause problems when compiling nimbleFunction using customMV
```

Currently, only the syntax `customMV["a", 2]` works in the NIMBLE language, not `customMV[["a"]][[2]]`.

We can query and change the number of rows using `getsize` and `resize`, respectively. These work in both R and NIMBLE. Note that we don't specify the variables in this case: all variables in a `modelValues` object will have the same number of rows.

```
getsize(customMV)
```

```
## [1] 2
```

```
resize(customMV, 3)
getsize(customMV)
```

```
## [1] 3
```

```
customMV$a
```

```
## [[1]]
## [1] 0 1
##
## [[2]]
## [1] 2 4
##
## [[3]]
## [1] NA NA
```

Often it is useful to convert a `modelValues` object to a matrix for use in R. For example, we may want to convert MCMC output into a matrix, for use with the `coda` package for processing MCMC samples, or into a list. This can be done with the `as.matrix` and `as.list` methods for `modelValues` objects, respectively. `as.matrix` will generate column names from every scalar element of variables (e.g. “b[1, 1]”, “b[2, 1]”, etc.). The rows of the `modelValues` will be the rows of the matrix, with any matrices or arrays converted to a vector based on column-major ordering.

```
as.matrix(customMV, 'a') # convert 'a'
```

```
##      a[1] a[2]
## [1,]    0    1
## [2,]    2    4
## [3,]   NA   NA
```

```
as.matrix(customMV) # convert all variables
```

```
##      a[1] a[2] b[1, 1] b[2, 1] b[1, 2] b[2, 2] c[1]
## [1,]    0    1      1      1      1      1    NA
```

```
## [2,]    2    4        2        2        2        2    NA
## [3,]   NA   NA        NA        NA        NA        NA   NA
```

`as.list` will return a list with an element for each variable. Each element will be an array of the same size and shape as the variable with an additional index for the iteration (e.g., MCMC iteration when used for MCMC output). By default iteration is the first index, but it can be the last if needed.

```
as.list(customMV, 'a')    # get only 'a'
```

```
## $a
##      [,1] [,2]
## [1,]    0    1
## [2,]    2    4
## [3,]   NA   NA
```

```
as.list(customMV)        # get all variables
```

```
## $a
##      [,1] [,2]
## [1,]    0    1
## [2,]    2    4
## [3,]   NA   NA
##
```

```
## $b
## , , 1
##
##      [,1] [,2]
## [1,]    1    1
## [2,]    2    2
## [3,]   NA   NA
##
## , , 2
##
```

```
##      [,1] [,2]
## [1,]    1    1
## [2,]    2    2
## [3,]   NA   NA
##
##
```

```
## $c
##      [,1]
## [1,]   NA
## [2,]   NA
## [3,]   NA
```

```
as.list(customMV, 'a', iterationAsLastIndex = TRUE) # put iteration in last index
```

```
## $a
##      [,1] [,2] [,3]
```

```
## [1,]    0    2  NA
## [2,]    1    4  NA
```

If a variable is a scalar, using `unlist` in R to extract all rows as a vector can be useful.

```
customMV['c', 1] <- 1
customMV['c', 2] <- 2
customMV['c', 3] <- 3
unlist(customMV['c', ])
```

```
## [1] 1 2 3
```

Once we have a `modelValues` object, we can see the structure of its contents via the `varNames` and `sizes` components of the object.

```
customMV$varNames
```

```
## [1] "a" "b" "c"
```

```
customMV$sizes
```

```
## $a
## [1] 2
##
## $b
## [1] 2 2
##
## $c
## [1] 1
```

As with most NIMBLE objects, `modelValues` are passed by reference, not by value. That means any modifications of `modelValues` objects in either R functions or `nimbleFunctions` will persist outside of the function. This allows for more efficient computation, as stored values are immediately shared among `nimbleFunctions`.

```
alter_a <- function(mv){
  mv['a',1][1] <- 1
}
customMV['a', 1]
```

```
## [1] 0 1
```

```
alter_a(customMV)
customMV['a',1]
```

```
## [1] 1 1
```

However, when you retrieve a variable from a `modelValues` object, the result is a standard R list, which is subsequently passed by value, as usual in R.

15.1.2.1 Automating calculation and simulation using `modelValues`

The `nimbleFunctions` `simNodesMV`, `calcNodesMV`, and `getLogProbsMV` can be used to operate on a model based on rows in a `modelValues` object. For example, `simNodesMV` will simulate in the

model multiple times and record each simulation in a row of its `modelValues`. `calcNodesMV` and `getLogProbsMV` iterate over the rows of a `modelValues`, copy the nodes into the model, and then do their job of calculating or collecting log probabilities (densities), respectively. Each of these returns a numeric vector with the summed log probabilities of the chosen nodes from each row. `calcNodesMV` will save the log probabilities back into the `modelValues` object if `saveLP = TRUE`, a run-time argument.

Here are some examples:

```
mv <- modelValues(simpleModel)
rSimManyXY <- simNodesMV(simpleModel, nodes = c('x', 'y'), mv = mv)
rCalcManyXDepts <- calcNodesMV(simpleModel, nodes = 'x', mv = mv)
rGetLogProbMany <- getLogProbNodesMV(simpleModel, nodes = 'x', mv = mv)

cSimManyXY <- compileNimble(rSimManyXY, project = simpleModel)
cCalcManyXDepts <- compileNimble(rCalcManyXDepts, project = simpleModel)
cGetLogProbMany <- compileNimble(rGetLogProbMany, project = simpleModel)

cSimManyXY$run(m = 5) # simulating 5 times
cCalcManyXDepts$run(saveLP = TRUE) # calculating

## [1] -11.24771 -26.42169 -15.48325 -17.24588 -16.43909

cGetLogProbMany$run() #

## [1] -11.24771 -26.42169 -15.48325 -17.24588 -16.43909

result <- as.matrix(cSimManyXY$mv) # extract simulated values
```

15.2 The nimbleList data structure

`nimbleLists` provide a container for storing different types of objects in NIMBLE, similar to the list data structure in R. Before a `nimbleList` can be created and used, a *definition*² for that `nimbleList` must be created that provides the names, types, and dimensions of the elements in the `nimbleList`. `nimbleList` definitions must be created in R (either in R's global environment or in setup code), but the `nimbleList` instances can be created in run code.

Unlike lists in R, `nimbleLists` must have the names and types of all list elements provided by a definition before the list can be used. A `nimbleList` definition can be made by using the `nimbleList` function in one of two manners. The first manner is to provide the `nimbleList` function with a series of expressions of the form `name = type(nDim)`, similar to the specification of run-time arguments to `nimbleFunctions`. The types allowed for a `nimbleList` are the same as those allowed as run-time arguments to a `nimbleFunction`, described in Section 12.4. For example, the following line of code creates a `nimbleList` definition with two elements: `x`, which is a scalar integer, and `Y`, which is a matrix of doubles.

```
exampleNimListDef <- nimbleList(x = integer(0), Y = double(2))
```

²The *configuration* for a `modelValues` object is the same concept as a *definition* here; in a future release of NIMBLE we may make the usage more consistent between `modelValues` and `nimbleLists`.

The second method of creating a `nimbleList` definition is by providing an R list of *nimbleType* objects to the `nimbleList()` function. A `nimbleType` object can be created using the `nimbleType` function, which must be provided with three arguments: the **name** of the element being created, the **type** of the element being created, and the **dim** of the element being created. For example, the following code creates a list with two `nimbleType` objects and uses these objects to create a `nimbleList` definition.

```
nimbleListTypes <- list(nimbleType(name = 'x', type = 'integer', dim = 0),
                        nimbleType(name = 'Y', type = 'double', dim = 2))

# this nimbleList definition is identical to the one created above
exampleNimListDef2 <- nimbleList(nimbleListTypes)
```

Creating definitions using a list of `nimbleTypes` can be useful, as it allows for programmatic generation of `nimbleList` elements.

Once a `nimbleList` definition has been created, new instances of `nimbleLists` can be made from that definition using the **new** member function. The **new** function can optionally take initial values for the list elements as arguments. Below, we create a new `nimbleList` from our `exampleNimListDef` and specify values for the two elements of our list:

```
exampleNimList <- exampleNimListDef$new(x = 1, Y = diag(2))
```

Once created, `nimbleList` elements can be accessed using the `$` operator, just as with lists in R. For example, the value of the `x` element of our `exampleNimbleList` can be set to 7 using

```
exampleNimList$x <- 7
```

`nimbleList` definitions can be created either in R's global environment or in **setup** code of a `nimbleFunction`. Once a `nimbleList` definition has been made, new instances of `nimbleLists` can be created using the **new** function in R's global environment, in **setup** code, or in **run** code of a `nimbleFunction`.

`nimbleLists` can also be passed as arguments to **run** code of `nimbleFunctions` and returned from `nimbleFunctions`. To use a `nimbleList` as a **run** function argument, the name of the `nimbleList` definition should be provided as the argument type, with a set of parentheses following. To return a `nimbleList` from the **run** code of a `nimbleFunction`, the **returnType** of that function should be the name of the `nimbleList` definition, again using a following set of parentheses.

Below, we demonstrate a function that takes the `exampleNimList` as an argument, modifies its `Y` element, and returns the `nimbleList`.

```
mynf <- nimbleFunction(
  run = function(vals = exampleNimListDef()){
    onesMatrix <- matrix(value = 1, nrow = 2, ncol = 2)
    vals$Y <- onesMatrix
    returnType(exampleNimListDef())
    return(vals)
  })

# pass exampleNimList as argument to mynf
mynf(exampleNimList)
```

```
## nimbleList object of type nfRefClass_27
## Field "x":
## [1] 7
## Field "Y":
##      [,1] [,2]
## [1,]    1    1
## [2,]    1    1
```

`nimbleList` arguments to run functions are passed by reference – this means that if an element of a `nimbleList` argument is modified within a function, that element will remain modified when the function has finished running. To see this, we can inspect the value of the `Y` element of the `exampleNimList` object and see that it has been modified.

```
exampleNimList$Y
```

```
##      [,1] [,2]
## [1,]    1    1
## [2,]    1    1
```

In addition to storing basic data types, `nimbleLists` can also store other `nimbleLists`. To achieve this, we must create a `nimbleList` definition that declares the types of nested `nimbleLists` a `nimbleList` will store. Below, we create two types of `nimbleLists`: the first, named `innerNimList`, will be stored inside the second, named `outerNimList`:

```
# first, create definitions for both inner and outer nimbleLists
innerNimListDef <- nimbleList(someText = character(0))
outerNimListDef <- nimbleList(xList = innerNimListDef(),
                             z = double(0))

# then, create outer nimbleList
outerNimList <- outerNimListDef$new(z = 3.14)

# access element of inner nimbleList
outerNimList$xList$someText <- "hello, world"
```

Note that definitions for inner, or nested, `nimbleLists` must be created before the definition for an outer `nimbleList`.

15.2.1 Pre-defined `nimbleList` types

Several types of `nimbleLists` are predefined in NIMBLE for use with particular kinds of `nimbleFunctions`.

- `eigenNimbleList` and `svdNimbleList` are the return types of the `eigen` and `svd` functions (Section @ref(sec:eigen-nimFunctions)).
- `waicNimbleList` is the type provided for WAIC output from an MCMC.
- `waicDetailsNimbleList` is the type provided for detailed WAIC output from an MCMC.
- `optimResultNimbleList` is the return type of `nimOptim`.
- `optimControlNimbleList` is the type for the control list input to `nimOptim`.

15.2.2 Using *eigen* and *svd* in nimbleFunctions

NIMBLE has two linear algebra functions that return nimbleLists. The `eigen` function takes a symmetric matrix, `x`, as an argument and returns a nimbleList of type `eigenNimbleList`. nimbleLists of type `eigenNimbleList` have two elements: `values`, a vector with the eigenvalues of `x`, and `vectors`, a square matrix with the same dimension as `x` whose columns are the eigenvectors of `x`. The `eigen` function has two additional arguments: `symmetric` and `only.values`. The `symmetric` argument can be used to specify if `x` is a symmetric matrix or not. If `symmetric = FALSE` (the default value), `x` will be checked for symmetry. Eigendecompositions in NIMBLE for symmetric matrices are both faster and more accurate. Additionally, eigendecompositions of non-symmetric matrices can have complex entries, which are not supported by NIMBLE. If a complex entry is detected, NIMBLE will issue a warning and that entry will be set to NaN. The `only.values` argument defaults to `FALSE`. If `only.values = TRUE`, the `eigen` function will not calculate the eigenvectors of `x`, leaving the `vectors` nimbleList element empty. This can reduce calculation time if only the eigenvalues of `x` are needed.

The `svd` function takes an $n \times p$ matrix `x` as an argument, and returns a nimbleList of type `svdNimbleList`. nimbleLists of type `svdNimbleList` have three elements: `d`, a vector with the singular values of `x`, `u` a matrix with the left singular vectors of `x`, and `v`, a matrix with the right singular vectors of `x`. The `svd` function has an optional argument `vectors` which defaults to a value of "full". The `vectors` argument can be used to specify the number of singular vectors that are returned. If `vectors = "full"`, `v` will be an $n \times n$ matrix and `u` will be an $p \times p$ matrix. If `vectors = "thin"`, `v` will be an $n \times m$ matrix, where $m = \min(n, p)$, and `u` will be an $m \times p$ matrix. If `vectors = "none"`, the `u` and `v` elements of the returned nimbleList will not be populated.

nimbleLists created by either `eigen` or `svd` can be returned from a nimbleFunction, using `returnType(eigenNimbleList())` or `returnType(svdNimbleList())` respectively. nimbleLists created by `eigen` and `svd` can also be used within other nimbleLists by specifying the nimbleList element types as `eigenNimbleList()` and `svdNimbleList()`. The below example demonstrates the use of `eigen` and `svd` within a nimbleFunction.

```
eigenListFunctionGenerator <- nimbleFunction(
  setup = function(){
    demoMatrix <- diag(4) + 2
    eigenAndSvdListDef <- nimbleList(demoEigenList = eigenNimbleList(),
                                     demoSvdList = svdNimbleList())
    eigenAndSvdList <- eigenAndSvdListDef$new()
  },
  run = function(){
    # we will take the eigendecomposition and svd of a symmetric matrix
    eigenAndSvdList$demoEigenList <<- eigen(demoMatrix, symmetric = TRUE,
                                             only.values = TRUE)

    eigenAndSvdList$demoSvdList <<- svd(demoMatrix, vectors = 'none')
    returnType(eigenAndSvdListDef())
    return(eigenAndSvdList)
  })
eigenListFunction <- eigenListFunctionGenerator()

outputList <- eigenListFunction$run()
```

```
outputList$demoEigenList$values
```

```
## [1] 9 1 1 1
```

```
outputList$demoSvdList$d
```

```
## [1] 9 1 1 1
```

The eigenvalues and singular values returned from the above function are the same since the matrix being decomposed was symmetric. However, note that both eigendecompositions and singular value decompositions are numerical procedures, and computed solutions may have slight differences even for a symmetric input matrix.

Chapter 16

Writing nimbleFunctions to interact with models

16.1 Overview

When you write an R function, you say what the input arguments are, you provide the code for execution, and in that code you give the value to be returned¹. Using the `function` keyword in R triggers the operation of creating an object that is the function.

Creating `nimbleFunctions` is similar, but there are two kinds of code and two steps of execution:

1. *Setup* code is provided as a regular R function, but the programmer does not control what it returns. Typically the inputs to *setup* code are objects like a model, a vector of nodes, a `modelValues` object or a `modelValues` configuration, or another `nimbleFunction`. The setup code, as its name implies, sets up information for run-time code. It is executed in R, so it can use any aspect of R.
2. *Run* code is provided in the NIMBLE language, which was introduced in Chapter 12. This is similar to a narrow subset of R, but it is important to remember that it is different – defined by what can be compiled – and much more limited. *Run* code can use the objects created by the *setup* code. In addition, some information on variable types must be provided for input arguments, the return value, and in some circumstances for local variables. There are two kinds of *run* code:
 - a. There is always a primary function, given as the argument `run`².
 - b. There can optionally be other functions, or ‘methods’ in the language of object-oriented programming, that share the same objects created by the *setup* function.

Here is a small example to fix ideas:

```
logProbCalcPlus <- nimbleFunction(  
  setup = function(model, node) {  
    dependentNodes <- model$getDependencies(node)  
    valueToAdd <- 1
```

¹Normally this is the value of the last evaluated code, or the argument to `return`.

²This can be omitted if you don’t need it.

```

    },
    run = function(P = double(0)) {
      model[[node]] <- P + valueToAdd
      return(model$calculate(dependentNodes))
      returnType(double(0))
    })

code <- nimbleCode({
  a ~ dnorm(0, 1)
  b ~ dnorm(a, 1)
})
testModel <- nimbleModel(code, check = FALSE)
logProbCalcPlusA <- logProbCalcPlus(testModel, "a")
testModel$b <- 1.5
logProbCalcPlusA$run(0.25)

## [1] -2.650377

dnorm(1.25,0,1,TRUE)+dnorm(1.5,1.25,1,TRUE) # direct validation

## [1] -2.650377

testModel$a # "a" was set to 0.5 + valueToAdd

## [1] 1.25

```

The call to the R function called `nimbleFunction` returns a function, similarly to defining a function in R. That function, `logProbCalcPlus`, takes arguments for its `setup` function, executes it, and returns an object, `logProbCalcPlusA`, that has a `run` member function (method) accessed by `$run`. In this case, the `setup` function obtains the stochastic dependencies of the `node` using the `getDependencies` member function of the model (see Section 14.1.3) and stores them in `dependentNodes`. In this way, `logProbCalcPlus` can adapt to any model. It also creates a variable, `valueToAdd`, that can be used by the `nimbleFunction`.

The object `logProbCalcPlusA`, returned by `logProbCalcPlus`, is permanently bound to the results of the processed `setup` function. In this case, `logProbCalcPlusA$run` takes a scalar input value, `P`, assigns `P + valueToAdd` to the given node in the model, and returns the sum of the log probabilities of that node and its stochastic dependencies³. We say `logProbCalcPlusA` is an ‘instance’ of `logProbCalcPlus` that is ‘specialized’ or ‘bound’ to `a` and `testModel`. Usually, the `setup` code will be where information about the model structure is determined, and then the `run` code can use that information without repeatedly, redundantly recomputing it. A `nimbleFunction` can be called repeatedly (one can think of it as a generator), each time returning a specialized `nimbleFunction`.

Readers familiar with object-oriented programming may find it useful to think in terms of class definitions and objects. `nimbleFunction` creates a class definition. Each specialized `nimbleFunction` is one object in the class. The setup arguments are used to define member data in the object.

³Note the use of the global assignment operator to assign into the model. This is necessary for assigning into variables from the `setup` function, at least if you want to avoid warnings from R. These warnings come from R’s reference class system.

16.2 Using and compiling nimbleFunctions

To compile the nimbleFunction, together with its model, we use `compileNimble`:

```
CnfDemo <- compileNimble(testModel, logProbCalcPlusA)
CtestModel <- CnfDemo$testModel
ClogProbCalcPlusA <- CnfDemo$logProbCalcPlusA
```

These have been initialized with the values from their uncompiled versions and can be used in the same way:

```
CtestModel$a      # values were initialized from testModel
```

```
## [1] 1.25
```

```
CtestModel$b
```

```
## [1] 1.5
```

```
lpA <- ClogProbCalcPlusA$run(1.5)
lpA
```

```
## [1] -5.462877
```

```
# verify the answer:
```

```
dnorm(CtestModel$b, CtestModel$a, 1, log = TRUE) +
  dnorm(CtestModel$a, 0, 1, log = TRUE)
```

```
## [1] -5.462877
```

```
CtestModel$a      # a was modified in the compiled model
```

```
## [1] 2.5
```

```
testModel$a      # the uncompiled model was not modified
```

```
## [1] 1.25
```

16.3 Writing setup code

16.3.1 Useful tools for setup functions

The setup function is typically used to determine information on nodes in a model, set up model-Values or nimbleList objects, set up (nested) nimbleFunctions or nimbleFunctionLists, and set up any persistent numeric objects. For example, the setup code of an MCMC nimbleFunction creates the nimbleFunctionList of sampler nimbleFunctions. The values of numeric objects created in setup code can be modified by run code and will persist across calls.

Some of the useful tools and objects to create in setup functions include:

- **vectors of node names, often from a model** Often these are obtained from the `getNodeNames`, `getDependencies`, and other methods of a model, described in Sections 14.1-14.2.
- **modelValues objects** These are discussed in Sections 15.1 and 16.4.4.

- **nimbleList objects** New instances of `nimbleLists` can be created from a `nimbleList` definition in either setup or run code. See Section 15.2 for more information.
- **specializations of other nimbleFunctions** A useful NIMBLE programming technique is to have one `nimbleFunction` contain other `nimbleFunctions`, which it can use in its run-time code (Section 16.4.7).
- **lists of other nimbleFunctions** In addition to containing single other `nimbleFunctions`, a `nimbleFunction` can contain a list of other `nimbleFunctions` (Section 16.4.8).

If one wants a `nimbleFunction` that does get specialized but has empty setup code, use `setup = function() {}` or `setup = TRUE`.

16.3.2 Accessing and modifying numeric values from setup

While models and nodes created during setup cannot be modified⁴, numeric values and `modelValues` can be, as illustrated by extending the example from above.

```
logProbCalcPlusA$valueToAdd # in the uncompiled version
```

```
## [1] 1
```

```
logProbCalcPlusA$valueToAdd <- 2
```

```
ClogProbCalcPlusA$valueToAdd # or in the compiled version
```

```
## [1] 1
```

```
ClogProbCalcPlusA$valueToAdd <- 3
```

```
ClogProbCalcPlusA$run(1.5)
```

```
## [1] -16.46288
```

```
CtestModel$a # a == 1.5 + 3
```

```
## [1] 4.5
```

16.3.3 Determining numeric types in nimbleFunctions

For numeric variables from the `setup` function that appear in the `run` function or other member functions (or are declared in `setupOutputs`), the type is determined from the values created by the setup code. The types created by setup code must be consistent across all specializations of the `nimbleFunction`. For example if `X` is created as a matrix (two-dimensional double) in one specialization but as a vector (one-dimensional double) in another, there will be a problem during compilation. The sizes may differ in each specialization.

Treatment of vectors of length one presents special challenges because they could be treated as scalars or vectors. Currently they are treated as scalars. If you want a vector, ensure that the length is greater than one in the setup code and then use `setSize` in the run-time code.

16.3.4 Control of setup outputs

Sometimes setup code may create variables that are not used in run code. By default, NIMBLE inspects run code and omits variables from setup that do not appear in run code from compila-

⁴Actually, they can be, but only for uncompiled `nimbleFunctions`.

tion. However, sometimes a programmer may want to force a numeric or character variable to be included in compilation, even if it is not used directly in run code. As shown below, such variables can be directly accessed in one `nimbleFunction` from another, which provides a way of using `nimbleFunctions` as general data structures. To force NIMBLE to include variables during compilation, for example `X` and `Y`, simply include

```
setupOutputs(X, Y)
```

anywhere in the setup code.

16.4 Writing run code

In Chapter 12 we described the functionality of the NIMBLE language that could be used in run code without setup code (typically in cases where no models or `modelValues` are needed). Next we explain the additional features that allow use of models and `modelValues` in the run code.

16.4.1 Driving models: *calculate*, *calculateDiff*, *simulate*, *getLogProb*

These four functions are the primary ways to operate a model. Their syntax was explained in Section 14.3. Except for `getLogProb`, it is usually important for the `nodes` vector to be sorted in topological order. Model member functions such as `getDependencies` and `expandNodeNames` will always return topologically sorted node names.

Most R-like indexing of a node vector is allowed within the argument to `calculate`, `calculateDiff`, `simulate`, and `getLogProb`. For example, all of the following are allowed:

```
myModel$calculate(nodes)
myModel$calculate(nodes[i])
myModel$calculate(nodes[1:3])
myModel$calculate(nodes[c(1,3)])
myModel$calculate(nodes[2:i])
myModel$calculate(nodes[ values(model, nodes) + 0.1 < x ])
```

Note that in the last line of code, one must have that the length of `nodes` is equal to that of `values(model, nodes)`, which means that all the nodes in `nodes` must be scalar nodes.

Also note that one cannot create new vectors of nodes in run code. They can only be indexed within a call to `calculate`, `calculateDiff`, `simulate` or `getLogProb`.

16.4.2 Getting and setting variable and node values

16.4.2.1 Using indexing with nodes

Here is an example that illustrates getting and setting of nodes, subsets of nodes, or variables. Note the following:

- In `model[[v]]`, `v` can only be a single node or variable name, not a vector of multiple nodes nor an element of such a vector (`model[[ nodes[i] ]]` does not work). The node itself may be a vector, matrix or array node.
- In fact, `v` can be a node-name-like character string, even if it is not actually a node in the model. See example 4 in the code below.

- One can also use `model$varName`, with the caveat that `varName` must be a variable name. This usage would only make sense for a `nimbleFunction` written for models known to have a specific variable. (Note that if `a` is a scalar node in `model`, then `model[['a']]` will be a scalar but `model$a` will be a vector of length 1.
- one should use the `<<-` global assignment operator to assign into model nodes.

Note that NIMBLE does not allow variables to change dimensions. Model nodes are the same, and indeed are more restricted because they can't change sizes. In addition, NIMBLE distinguishes between scalars and vectors of length 1. These rules, and ways to handle them correctly, are illustrated in the following code as well as in Section 12.3.

```
code <- nimbleCode({
  z ~ dnorm(0, sd = sigma)
  sigma ~ dunif(0, 10)
  y[1:n] ~ dmnorm(zeroes[1:n], cov = C[1:5, 1:5])
})
n <- 5
m <- nimbleModel(code, constants = list(n = n, zeroes = rep(0, n),
                                         C = diag(n)))
cm <- compileNimble(m)

nfGen <- nimbleFunction(
  setup = function(model) {
    # node1 and node2 would typically be setup arguments, so they could
    # have different values for different models. We are assigning values
    # here so the example is clearer.
    node1 <- 'sigma'           # a scalar node
    node2 <- 'y[1:5]'          # a vector node
    notReallyANode <- 'y[2:4]' # y[2:4] allowed even though not a node!
  },
  run = function(vals = double(1)) {
    tmp0 <- model[[node1]]      # 1. tmp0 will be a scalar
    tmp1 <- model[[node2]]      # 2. tmp1 will be a vector
    tmp2 <- model[[node2]][1]    # 3. tmp2 will be a scalar
    tmp3 <- model[[notReallyANode]] # 4. tmp3 will be a vector
    tmp4 <- model$y[3:4]         # 5. hard-coded access to a model variable
    # 6. node1 is scalar so can be assigned a scalar:
    model[[node1]] <<- runif(1)
    model[[node2]][1] <<- runif(1)
    # 7. an element of node2 can be assigned a scalar
    model[[node2]] <<- runif(length(model[[node2]]))
    # 8. a vector can be assigned to the vector node2
    model[[node2]][1:3] <<- vals[1:3]
    # elements of node2 can be indexed as needed
    returnType(double(1))
    out <- model[[node2]] # we can return a vector
    return(out)
  }
}
```

```

)

Rnf <- nfGen(m)
Cnf <- compileNimble(Rnf)
Cnf$run(rnorm(10))

## [1] -1.4887622 -1.1607519  1.4577382  0.4855725  0.6932443

```

Use of `[[ ]]` allows one to programmatically access a node based on a character variable containing the node name; this character variable would generally be set in setup code. In contrast, use of `$` hard codes the variable name and would not generally be suitable for nimbleFunctions intended for use with arbitrary models.

16.4.2.2 Getting and setting more than one model node or variable at a time using values

Sometimes it is useful to set a collection of nodes or variables at one time. For example, one might want a nimbleFunction that will serve as the objective function for an optimizer. The input to the nimbleFunction would be a vector, which should be used to fill a collection of nodes in the model before calculating their log probabilities. This can be done using `values`:

```

# get values from a set of model nodes into a vector
P <- values(model, nodes)
# or put values from a vector into a set of model nodes
values(model, nodes) <- P

```

where the first line would assign the collection of values from `nodes` into `P`, and the second would do the inverse. In both cases, values from nodes with two or more dimensions are flattened into a vector in column-wise order.

`values(model, nodes)` may be used as a vector in other expressions, e.g.,

```
Y <- A %*% values(model, nodes) + b
```

One can also index elements of nodes in the argument to `values`, in the same manner as discussed for `calculate` and related functions in Section 16.4.1.

Note again the potential for confusion between scalars and vectors of length 1. `values` returns a vector and expects a vector when used on the left-hand side of an assignment. If only a single value is being assigned, it must be a vector of length 1, not a scalar. This can be achieved by wrapping a scalar in `c()` when necessary. For example:

```

# c(rnorm(1)) creates vector of length one:
values(model, nodes[1]) <- c(rnorm(1))
# won't compile because rnorm(1) is a scalar
# values(model, nodes[1]) <- rnorm(1)

out <- values(model, nodes[1]) # out is a vector
out2 <- values(model, nodes[1])[1] # out2 is a scalar

```

16.4.3 Getting parameter values and node bounds

Sections 14.2.3-14.2.4 describe how to get the parameter values for a node and the range (bounds) of possible values for the node using `getParam` and `getBound`. Both of these can be used in run code.

16.4.4 Using `modelValues` objects

The `modelValues` structure was introduced in Section 15.1. Inside `nimbleFunctions`, `modelValues` are designed to easily save values from a model object during the running of a `nimbleFunction`. A `modelValues` object used in run code must always exist in the setup code, either by passing it in as a setup argument or creating it in the setup code.

To illustrate this, we will create a `nimbleFunction` for computing importance weights for importance sampling. This function will use two `modelValues` objects. `propModelValues` will contain a set of values simulated from the importance sampling distribution and a field `propLL` for their log probabilities (densities). `savedWeights` will contain the difference in log probability (density) between the model and the `propLL` value provided for each set of values.

```
# Accepting modelValues as a setup argument
swConf <- modelValuesConf(vars = "w",
                          types = "double",
                          sizes = 1)
setup = function(propModelValues, model, savedWeightsConf){
  # Building a modelValues in the setup function
  savedWeights <- modelValues(conf = savedWeightsConf)
  # List of nodes to be used in run function
  modelNodes <- model$getNodeNames(stochOnly = TRUE,
                                   includeData = FALSE)
}
```

The simplest way to pass values back and forth between models and `modelValues` inside of a `nimbleFunction` is with `copy`, which has the synonym `nimCopy`. See `help(nimCopy)` for argument details.

Alternatively, the values may be accessed via indexing of individual rows, using the notation `mv[var, i]`, where `mv` is a `modelValues` object, `var` is a variable name (not a node name), and `i` is a row number. Likewise, the `getsize` and `resize` functions can be used as discussed in Section 15.1. However the function `as.matrix` does not work in run code.

Here is a run function to use these `modelValues`:

```
run = function(){
  # gets the number of rows of propSamples
  m <- getsize(propModelValues)

  # resized savedWeights to have the proper rows
  resize(savedWeights, m)
  for(i in 1:m){
    # Copying from propSamples to model.
    # Node names of propSamples and model must match!
```

```

    nimCopy(from = propModelValues, to = model, row = i,
            nodes = modelNodes, logProb = FALSE)
    # calculates the log likelihood of the model
    targLL <- model$calculate()
    # retrieves the saved log likelihood from the proposed model
    propLL <- propModelValues["propLL",i][1]
    # saves the importance weight for the i-th sample
    savedWeights["w", i][1] <- exp(targLL - propLL)
  }
  # does not return anything
}

```

Once the `nimbleFunction` is built, the `modelValues` object can be accessed using `$`, which is shown in more detail below. In fact, since `modelValues`, like most NIMBLE objects, are reference class objects, one can get a reference to them before the function is executed and then use that reference afterwards.

```

# simple model and modelValues for example use with code above
targetModelCode <- nimbleCode({
  x ~ dnorm(0,1)
  for(i in 1:4)
    y[i] ~ dnorm(0,1)
})

# code for proposal model
propModelCode <- nimbleCode({
  x ~ dnorm(0,2)
  for(i in 1:4)
    y[i] ~ dnorm(0,2)
})

# creating the models
targetModel = nimbleModel(targetModelCode, check = FALSE)
propModel = nimbleModel(propModelCode, check = FALSE)
cTargetModel = compileNimble(targetModel)
cPropModel = compileNimble(propModel)

sampleMVConf = modelValuesConf(vars = c("x", "y", "propLL"),
  types = c("double", "double", "double"),
  sizes = list(x = 1, y = 4, propLL = 1) )

sampleMV <- modelValues(sampleMVConf)

# nimbleFunction for generating proposal sample
PropSamp_Gen <- nimbleFunction(
  setup = function(mv, propModel){
    nodeNames <- propModel$getNodeNames()

```

```

    },
    run = function(m = integer() ){
      resize(mv, m)
      for(i in 1:m){
        propModel$simulate()
        nimCopy(from = propModel, to = mv, nodes = nodeNames, row = i)
        mv["propLL", i][1] <- propModel$calculate()
      }
    }
  )

# nimbleFunction for calculating importance weights
# uses setup and run functions as defined in previous code chunk
impWeights_Gen <- nimbleFunction(setup = setup,
                                run = run)

# making instances of nimbleFunctions
# note that both functions share the same modelValues object
RPropSamp <- PropSamp_Gen(sampleMV, propModel)
RImpWeights <- impWeights_Gen(sampleMV, targetModel, swConf)

# compiling
CPropSamp <- compileNimble(RPropSamp, project = propModel)
CImpWeights <- compileNimble(RImpWeights, project = targetModel)

# generating and saving proposal sample of size 10
CPropSamp$run(10)

# calculating the importance weights and saving to mv
CImpWeights$run()

# retrieving the modelValues objects
# extracted objects are C-based modelValues objects

savedPropSamp_1 = CImpWeights$propModelValues
savedPropSamp_2 = CPropSamp$mv

# Subtle note: savedPropSamp_1 and savedPropSamp_2
# both provide interface to the same compiled modelValues objects!
# This is because they were both built from sampleMV.

savedPropSamp_1["x",1]

## [1] -0.3919851

```



```

savedPropSamp_2["x",1]

## [1] -0.3919851
savedPropSamp_1["x",1] <- 0 # example of directly setting a value
savedPropSamp_2["x",1]

## [1] 0
# viewing the saved importance weights
savedWeights <- CImpWeights$savedWeights
unlist(savedWeights[["w"]])

## [1] 0.5024644 1.7765357 3.5155732 0.3893390 0.5530012 0.5085069 2.3890876
## [8] 0.6548453 0.5940573 0.4943950
# viewing first 3 rows -- note that savedPropSamp_1["x", 1] was altered
as.matrix(savedPropSamp_1)[1:3, ]

##      propLL[1]      x[1]      y[1]      y[2]      y[3]      y[4]
## [1,] -4.951100  0.00000000 -0.3951785 -1.0040021 -0.6420246  0.5993671
## [2,] -7.476891  0.35317374 -0.4510107  1.7080201  1.0471602 -0.5225390
## [3,] -8.841966 -0.06913039 -2.2884703  0.6767462  0.3955090 -0.3519276

```

Importance sampling could also be written using simple vectors for the weights, but we illustrated putting them in a `modelValues` object along with model variables.

16.4.5 Using model variables and modelValues in expressions

Each way of accessing a variable, node, or `modelValues` can be used amidst mathematical expressions, including with indexing, or passed to another `nimbleFunction` as an argument. For example, the following two statements would be valid:

```
model[["x[2:8, ]"]] [2:4, 1:3] %*% Z
```

if `Z` is a vector or matrix, and

```
C[6:10] <- mv[v, i] [1:5, k] + B
```

if `B` is a vector or matrix.

The NIMBLE language allows scalars, but models defined from BUGS code are never created as purely scalar nodes. Instead, a single node such as defined by `z ~ dnorm(0, 1)` is implemented as a vector of length 1, similar to R. When using `z` via `model$z` or `model[["z"]]`, NIMBLE will try to do the right thing by treating this as a scalar. In the event of problems⁵, a more explicit way to access `z` is `model$z[1]` or `model[["z"]][1]`.

16.4.6 Including other methods in a nimbleFunction

Other methods can be included with the `methods` argument to `nimbleFunction`. These methods can use the objects created in setup code in just the same ways as the run function. In fact, the run function is just a default main method name. Any method can then call another method.

⁵Please tell us!

```

methodsDemo <- nimbleFunction(
  setup = function() {sharedValue <- 1},
  run = function(x = double(1)) {
    print("sharedValues = ", sharedValue, "\n")
    increment()
    print("sharedValues = ", sharedValue, "\n")
    A <- times(5)
    return(A * x)
    returnType(double(1))
  },
  methods = list(
    increment = function() {
      sharedValue <<- sharedValue + 1
    },
    times = function(factor = double()) {
      return(factor * sharedValue)
      returnType(double())
    })
)

methodsDemo1 <- methodsDemo()
methodsDemo1$run(1:10)

```

```

## sharedValues = 1
##
## sharedValues = 2
## [1] 10 20 30 40 50 60 70 80 90 100

```

```

methodsDemo1$sharedValue <- 1
CmethodsDemo1 <- compileNimble(methodsDemo1)
CmethodsDemo1$run(1:10)

```

```

## sharedValues = 1
##
## sharedValues = 2
## [1] 10 20 30 40 50 60 70 80 90 100

```

16.4.7 Using other nimbleFunctions

One nimbleFunction can use another nimbleFunction that was passed to it as a setup argument or was created in the setup function. This can be an effective way to program. When a nimbleFunction needs to access a setup variable or method of another nimbleFunction, use \$.

```

usePreviousDemo <- nimbleFunction(
  setup = function(initialSharedValue) {
    myMethodsDemo <- methodsDemo()
  },
  run = function(x = double(1)) {
    myMethodsDemo$sharedValue <<- initialSharedValue

```

```

    print(myMethodsDemo$sharedValue)
    A <- myMethodsDemo$run(x[1:5])
    print(A)
    B <- myMethodsDemo$times(10)
    return(B)
    returnType(double())
  })

usePreviousDemo1 <- usePreviousDemo(2)
usePreviousDemo1$run(1:10)

## 2
## sharedValues = 2
##
## sharedValues = 3
##
## 15 30 45 60 75

## [1] 30
CusePreviousDemo1 <- compileNimble(usePreviousDemo1)
CusePreviousDemo1$run(1:10)

## 2
## sharedValues = 2
##
## sharedValues = 3
##
## 15
## 30
## 45
## 60
## 75

## [1] 30

```

16.4.8 Virtual nimbleFunctions and nimbleFunctionLists

Often it is useful for one `nimbleFunction` to have a list of other `nimbleFunctions`, all of whose methods have the same arguments and return types. For example, NIMBLE's MCMC engine contains a list of samplers that are each `nimbleFunctions`.

To make such a list, NIMBLE provides a way to declare the arguments and return types of methods: virtual `nimbleFunctions` created by `nimbleFunctionVirtual`. Other `nimbleFunctions` can inherit from virtual `nimbleFunctions`, which in R is called 'containing' them. Readers familiar with object oriented programming will recognize this as a simple class inheritance system. In Version 1.4.0 it is limited to simple, single-level inheritance.

Here is how it works:

```

baseClass <- nimbleFunctionVirtual(
  run = function(x = double(1)) {returnType(double())},
  methods = list(
    foo = function() { returnType(double()) }
  ))

derived1 <- nimbleFunction(
  contains = baseClass,
  setup = function() {},
  run = function(x = double(1)) {
    print("run 1")
    return(sum(x))
    returnType(double())
  },
  methods = list(
    foo = function() {
      print("foo 1")
      return(rnorm(1, 0, 1))
      returnType(double())
    }
  )
)

derived2 <- nimbleFunction(
  contains = baseClass,
  setup = function() {},
  run = function(x = double(1)) {
    print("run 2")
    return(prod(x))
    returnType(double())
  },
  methods = list(
    foo = function() {
      print("foo 2")
      return(runif(1, 100, 200))
      returnType(double())
    }
  )
)

useThem <- nimbleFunction(
  setup = function() {
    nfl <- nimbleFunctionList(baseClass)
    nfl[[1]] <- derived1()
    nfl[[2]] <- derived2()
  },
  run = function(x = double(1)) {

```

```

    for(i in seq_along(nfl)) {
      print( nfl[[i]]$run(x) )
      print( nfl[[i]]$foo() )
    }
  }
)

```

```

useThem1 <- useThem()
set.seed(1)
useThem1$run(1:5)

```

```

## run 1
## 15
## foo 1
## -0.6264538
## run 2
## 120
## foo 2
## 157.2853

```

```

CuseThem1 <- compileNimble(useThem1)
set.seed(1)
CuseThem1$run(1:5)

```

```

## run 1
## 15
## foo 1
## -0.626454
## run 2
## 120
## foo 2
## 157.285

```

One can also use `seq_along` with `nimbleFunctionLists` (and only with `nimbleFunctionLists`). As in R, `seq_along(myFunList)` is equivalent to `1:length(myFunList)` if the length of `myFunList` is greater than zero. It is an empty sequence if the length is zero.

Virtual `nimbleFunctions` cannot define setup values to be inherited.

16.4.9 Character objects

NIMBLE provides limited uses of character objects in run code. Character vectors created in setup code will be available in run code, but the only thing you can really do with them is include them in a `print` or `stop` statement.

Note that character vectors of model node and variable names are processed during compilation. For example, in `model[[node]]`, `node` may be a character object, and the NIMBLE compiler processes this differently than `print("The node name was ", node)`. In the former, the NIMBLE compiler sets up a C++ pointer directly to the `node` in the `model`, so that the character content of `node` is never needed in C++. In the latter, `node` is used as a C++ string and therefore is needed in C++.

16.4.10 User-defined data structures

Before the introduction of `nimbleLists` in Version 0.6-4, NIMBLE did not explicitly have user-defined data structures. An alternative way to create a data structure in NIMBLE is to use `nimbleFunctions` to achieve a similar effect. To do so, one can define setup code with whatever variables are wanted and ensure they are compiled using `setupOutputs`. Here is an example:

```
dataNF <- nimbleFunction(
  setup = function() {
    X <- 1
    Y <- as.numeric(c(1, 2))
    Z <- matrix(as.numeric(1:4), nrow = 2)
    setupOutputs(X, Y, Z)
  })

useDataNF <- nimbleFunction(
  setup = function(myDataNF) {},
  run = function(newX = double(), newY = double(1), newZ = double(2)) {
    myDataNF$X <- newX
    myDataNF$Y <- newY
    myDataNF$Z <- newZ
  })

myDataNF <- dataNF()
myUseDataNF <- useDataNF(myDataNF)
myUseDataNF$run(as.numeric(100), as.numeric(100:110),
  matrix(as.numeric(101:120), nrow = 2))
myDataNF$X

## [1] 100
myDataNF$Y

## [1] 100 101 102 103 104 105 106 107 108 109 110
myDataNF$Z

##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10]
## [1,]  101  103  105  107  109  111  113  115  117  119
## [2,]  102  104  106  108  110  112  114  116  118  120
myUseDataNF$myDataNF$X

## [1] 100
nimbleOptions(buildInterfacesForCompiledNestedNimbleFunctions = TRUE)
CmyUseDataNF <- compileNimble(myUseDataNF)
CmyUseDataNF$run(-100, -(100:110), matrix(-(101:120), nrow = 2))
CmyDataNF <- CmyUseDataNF$myDataNF
CmyDataNF$X

## [1] -100
```

```
CmyDataNF$Y
```

```
## [1] -100 -101 -102 -103 -104 -105 -106 -107 -108 -109 -110
```

```
CmyDataNF$Z
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10]
## [1,] -101 -103 -105 -107 -109 -111 -113 -115 -117 -119
## [2,] -102 -104 -106 -108 -110 -112 -114 -116 -118 -120
```

You'll notice that:

- After execution of the compiled function, access to the X, Y, and Z is the same as for the uncompiled case. This occurs because `CmyUseDataNF` is an interface to the compiled version of `myUseDataNF`, and it provides access to member objects and functions. In this case, one member object is `myDataNF`, which is an interface to the compiled version of `myUseDataNF$myDataNF`, which in turn provides access to X, Y, and Z. To reduce memory use, NIMBLE defaults to *not* providing full interfaces to nested nimbleFunctions like `myUseDataNF$myDataNF`. In this example we made it provide full interfaces by setting the `buildInterfacesForCompiledNestedNimbleFunctions` option via `nimbleOptions` to `TRUE`. If we had left that option `FALSE` (its default value), we could still get to the values of interest using `valueInCompiledNimbleFunction(CmyDataNF, 'X')`
- We need to take care that at the time of compilation, the X, Y and Z values contain doubles via `as.numeric` so that they are not compiled as integer objects.
- The `myDataNF` could be created in the setup code. We just provided it as a setup argument to illustrate that option.

16.5 Example: writing user-defined samplers to extend NIMBLE's MCMC engine

One important use of `nimbleFunctions` is to write additional samplers that can be used in NIMBLE's MCMC engine. This allows a user to write a custom sampler for one or more nodes in a model, as well as for programmers to provide general samplers for use in addition to the library of samplers provided with NIMBLE.

The following code illustrates how a NIMBLE developer would implement and use a Metropolis-Hastings random walk sampler with fixed proposal standard deviation.

```
my_RW <- nimbleFunction(

  contains = sampler_BASE,

  setup = function(model, mvSaved, target, control) {
    # proposal standard deviation
    scale <- if(!is.null(control$scale)) control$scale else 1
    calcNodes <- model$getDependencies(target)
  },

  run = function() {
```

```

# initial model logProb
model_lp_initial <- getLogProb(model, calcNodes)
# generate proposal
proposal <- rnorm(1, model[[target]], scale)
# store proposal into model
model[[target]] <- proposal
# proposal model logProb
model_lp_proposed <- model$calculate(calcNodes)

# log-Metropolis-Hastings ratio
log_MH_ratio <- model_lp_proposed - model_lp_initial

# Metropolis-Hastings step: determine whether or
# not to accept the newly proposed value
u <- runif(1, 0, 1)
if(u < exp(log_MH_ratio)) jump <- TRUE
else                        jump <- FALSE

# keep the model and mvSaved objects consistent
if(jump) copy(from = model, to = mvSaved, row = 1,
              nodes = calcNodes, logProb = TRUE)
else      copy(from = mvSaved, to = model, row = 1,
              nodes = calcNodes, logProb = TRUE)
},

methods = list(  reset = function () {}  )
)

```

The name of this sampler function, for the purposes of using it in an MCMC algorithm, is `my_RW`. Thus, this sampler can be added to an existing MCMC configuration object `conf` using:

```

mcmcConf$addSampler(target = 'x', type = 'my_RW',
                    control = list(scale = 0.1))

```

To be used within the MCMC engine, sampler functions definitions must adhere exactly to the following:

- The `nimbleFunction` must include the contains statement `contains = sampler_BASE`.
- The `setup` function must have the four arguments `model`, `mvSaved`, `target`, `control`, in that order.
- The `run` function must accept no arguments, and have no return value. Further, after execution it must leave the `mvSaved` `modelValues` object as an up-to-date copy of the values and `logProb` values in the `model` object.
- The `nimbleFunction` must have a member method called `reset`, which takes no arguments and has no return value.

The purpose of the `setup` function is generally two-fold. First, to extract control parameters from the `control` list; in the example, the proposal standard deviation `scale`. It is good practice to

specify default values for any control parameters that are not provided in the `control` argument, as done in the example. Second, to generate any sets of nodes needed in the `run` function. In many sampling algorithms, as here, `calcNodes` is used to represent the target node(s) and dependencies up to the first layer of stochastic nodes, as this is precisely what is required for calculating the Metropolis-Hastings acceptance probability. These probability calculations are done using `model$calculate(calcNodes)`.

The purpose of the `mvSaved` `modelValues` object is to store the state of the model, including both node values and log probability values, as it exists before any changes are made by the sampler. This allows restoration of the state when a proposal is rejected, as can be seen in the example above. When a proposal is accepted, one should copy from the model into the `mvSaved` object. NIMBLE's MCMC engine expects that `mvSaved` contains an up-to-date copy of model values and `logProb` values at the end of the run code of a sampler.

Note that NIMBLE generally expects the user-defined sampler to be defined in the global environment. If you define it in a function (which would generally be the case if you are using it in the context of parallelization), one approach would be to assign the user-defined sampler to the global environment in your function:

```
assign('sampler_yourSampler', sampler_yourSampler, envir = .GlobalEnv)
```

16.5.1 User-defined samplers and posterior predictive nodes

As of version 0.13.0, NIMBLE's handling of posterior predictive nodes in MCMC sampling has changed in order to improve MCMC mixing. Samplers for nodes that are not posterior predictive nodes no longer condition on the values of the posterior predictive nodes. This is done by two changes:

- turning off posterior predictive nodes as dependencies in the use of `getDependencies` when building an MCMC, and
- moving all posterior predictive samplers to be last in the order of samplers used in an MCMC iteration.

The first change calls for careful consideration when writing new samplers. It is done by making `buildMCMC` set a NIMBLE system option (described below) that tells `getDependencies` to ignore posterior predictive nodes (defined as nodes that are themselves not data and have no data nodes in their entire downstream (descendant) dependency network) before it builds the samplers (i.e., runs the setup code of each sampler). That way, the setup code of all samplers that use `getDependencies` will automatically comply with the new system.

User-defined samplers that determine dependencies using `getDependencies` should automatically work in the new system (although it is worth checking). However if a user-defined sampler manually specifies dependencies, and if those dependencies include posterior predictive nodes, the MCMC results could be incorrect. Whether incorrect results occur could depend on other parts of the model structure connected to node(s) sampled by the user-defined sampler (i.e., the target node(s)) and/or posterior predictive nodes. Therefore, it is strongly recommended that all user-defined samplers rely on `getDependencies` to determine which nodes depend on the target node(s).

There are several new NIMBLE system options available to take control of the behavior just described.

- `getDependenciesIncludesPredictiveNodes` determines whether posterior predictive nodes are included in results of `getDependencies`. This defaults to `TRUE`, so that outside of building samplers, the behavior of `getDependencies` should be identical to previous versions of NIMBLE.
- `MCMCusePredictiveDependenciesInCalculations` gives the value to which `getDependenciesIncludesPredictiveNodes` will be set while building samplers. This defaults to `FALSE`.
- `MCMCorderPosteriorPredictiveSamplersLast` determines whether posterior predictive samplers are moved to the end of the sampler list. This default to `TRUE`. Behavior prior to version 0.13.0 corresponds to `FALSE`.

Here are some examples of how to use these options:

- If you want to have MCMC samplers condition on posterior predictive nodes, do `nimbleOptions(MCMCusePredictiveDependenciesInCalculations = TRUE)`.
- If you want to get identical MCMC samplers, including their ordering, as in versions prior to 0.13.0, do `nimbleOptions(MCMCusePredictiveDependenciesInCalculations = TRUE)` and `nimbleOptions(MCMCorderPosteriorPredictiveSamplersLast = FALSE)`.
- If you want to experiment with the behavior of `getDependencies` when ignoring posterior predictive nodes, do `nimbleOptions(getDependenciesIncludesPredictiveNodes = FALSE)`.

16.6 Copying nimbleFunctions (and NIMBLE models)

NIMBLE relies heavily on R's reference class system. When models, `modelValues`, and `nimbleFunctions` with setup code are created, NIMBLE generates a new, customized reference class definition for each. As a result, objects of these types are passed by reference and hence modified in place by most NIMBLE operations. This is necessary to avoid a great deal of copying and returning and having to reassign large objects, both in processing models and `nimbleFunctions` and in running algorithms.

One cannot generally copy NIMBLE models or `nimbleFunctions` (specializations or generators) in a safe fashion, because of the references to other objects embedded within NIMBLE objects. However, the model member function `newModel` will create a new copy of the model from the same model definition (Section 6.1.3). This new model can then be used with newly instantiated `nimbleFunctions`.

The reliable way to create new copies of `nimbleFunctions` is to re-run the R function called `nimbleFunction` and record the result in a new object. For example, say you have a `nimbleFunction` called `foo` and 1000 instances of `foo` are compiled as part of an algorithm related to a model called `model1`. If you then need to use `foo` in an algorithm for another model, `model2`, doing so may work without any problems. However, there are cases where the NIMBLE compiler will tell you during compilation that the second set of `foo` instances cannot be built from the previous compiled version. A solution is to re-define `foo` from the beginning – i.e. call `nimbleFunction` again – and then proceed with building and compiling the algorithm for `model2`.

16.7 Debugging nimbleFunctions

One of the main reasons that NIMBLE provides an R (uncompiled) version of each `nimbleFunction` is for debugging. One can call `debug` on `nimbleFunction` methods (in particular the main `run`

method, e.g., `debug(mynf$run)` and then step through the code in R using R’s debugger. One can also insert `browser` calls into run code and then run the `nimbleFunction` from R.

In contrast, directly debugging a compiled `nimbleFunction` is difficult, although those familiar with running R through a debugger and accessing the underlying C code may be able to operate similarly with NIMBLE code. We often resort to using `print` statements for debugging compiled code. Expert users fluent in C++ may also try setting `nimbleOptions(pauseAfterWritingFiles = TRUE)` and adding debugging code into the generated C++ files.

16.8 Timing nimbleFunctions with *run.time*

If your `nimbleFunctions` are correct but slow to run, you can use benchmarking tools to look for bottlenecks and to compare different implementations. If your functions are very long-running (say 1ms or more), then standard R benchmarking tools may suffice, e.g. the `microbenchmark` package

```
library(microbenchmark)
microbenchmark(myCompiledFunVersion1(1.234),
               myCompiledFunVersion2(1.234)) # Beware R <--> C++ overhead!
```

If your `nimbleFunctions` are very fast, say under 1ms, then `microbenchmark` will be inaccurate due to R-to-C++ conversion overhead (that won’t happen in your actual functions). To get timing information in C++, NIMBLE provides a `run.time` function that avoids the R-to-C++ overhead.

```
myMicrobenchmark <- compileNimble(nimbleFunction(
  run = function(iters = integer(0)){
    time1 <- run.time({
      for (t in 1:iters) myCompiledFunVersion1(1.234)
    })
    time2 <- run.time({
      for (t in 1:iters) myCompiledFunVersion2(1.234)
    })
    return(c(time1, time2))
    returnType(double(1))
  })
print(myMicroBenchmark(100000))
```

16.9 Clearing and unloading compiled objects

Sometimes it is useful to clear all the compiled objects from a project and unload the shared library produced by your C++ compiler. To do so, you can use `clearCompiled(obj)` where `obj` is a compiled object such as a compiled model or `nimbleFunction` (e.g., a compiled MCMC algorithm). This will clear *all* compiled objects associated with your NIMBLE project. For example, if `cModel` is a compiled model, `clearCompiled(cModel)` will clear both the model and all associated `nimbleFunctions` such as compiled MCMCs that use that model. Be careful, use of `clearCompiled` can be dangerous. There is some risk that if you have copies of the R objects that interfaced to compiled C++ objects that have been removed, and you attempt to use those R objects after clearing their compiled counterparts, you will crash R. We have tried to minimize that risk, but we can’t guarantee safe behavior.

16.10 Reducing memory usage

NIMBLE can create a lot of objects in its processing, and some of them use R features such as reference classes that are heavy in memory usage. We have noticed that building large models can use lots of memory. To help alleviate this, we provide two options, which can be controlled via `nimbleOptions`.

As noted above, the option `buildInterfacesForCompiledNestedNimbleFunctions` defaults to `FALSE`, which means NIMBLE will not build full interfaces to compiled `nimbleFunctions` that only appear within other `nimbleFunctions`. If you want access to all such `nimbleFunctions`, use the option `buildInterfacesForCompiledNestedNimbleFunctions = TRUE`. This will use more memory but can be useful for debugging.

The option `clearNimbleFunctionsAfterCompiling` is more drastic, and it is experimental, so ‘buyer beware’. This will clear much of the contents of an uncompiled `nimbleFunction` object after it has been compiled in an effort to free some memory. We expect to be able to keep making NIMBLE more efficient – faster execution and lower memory use – in the future.

Part V

Automatic Derivatives in NIMBLE

Chapter 17

Automatic Derivatives

As of version 1.0.0, NIMBLE can automatically provide numerically accurate derivatives of potentially arbitrary order for most calculations in models and/or `nimbleFunctions`. This feature enables methods such as Hamiltonian Monte Carlo (HMC, see package `nimbleHMC`), Laplace (and AGHQ) approximation, and fast optimization with methods that use function gradients.

Automatic (or algorithmic) differentiation (AD) refers to the method of carrying derivative information through a set of mathematical operations. When done this way, derivatives are numerically accurate to the precision of the computer. This is distinct from finite difference methods, used by R packages such as `numDeriv` (Gilbert and Varadhan, 2019) and `pracma` (Borchers, 2022), which approximate derivatives by calculating function values at extremely nearby points. Finite difference methods are slower and less accurate than AD. It is also distinct from writing separate functions for each derivative, which can become very complicated and sometimes slower than AD. NIMBLE uses the CppAD package (Bell, 2022) as its AD engine, following TMB’s success in doing so (Kristensen et al., 2016). A general reference on AD is Griewank and Walther (2008).

Using a packaged AD algorithm should be as simple as setting `buildDerivs=TRUE` in the model. On the other hand, writing new algorithms (as `nimbleFunctions`) that use AD requires understanding how the AD system works internally, including what can go wrong. Calls to `compileNimble` that include AD features will result in slower C++ compilation.

We will introduce NIMBLE’s AD features step by step, from simply turning them on for a model to using them in your own `nimbleFunctions`. We will show you:

1. how to turn on derivatives in a model for use in algorithms such as Laplace approximation (or Hamiltonian Monte Carlo (HMC), see 7.11.2).
2. how to modify user-defined functions and distributions to support derivatives.
3. what functions are supported and not supported for derivatives.
4. basics of obtaining derivatives in your own algorithms written as `nimbleFunctions`.
5. advanced methods of obtaining derivatives in `nimbleFunctions`, including *double-taping*.
6. how to get derivatives involving model calculations.
7. automatic parameter transformations to give any model an unconstrained parameter space for algorithms to work in.
8. an example showing use of nimble’s derivatives for maximum likelihood estimation.

First, make sure to set the option to enable derivative features. This should be `TRUE` by default,

but just in case:

```
nimbleOptions(enableDerivs = TRUE)
```

17.1 How to turn on derivatives in a model

To allow algorithms to use automatic derivatives for a model, include `buildDerivs=TRUE` in the call to `nimbleModel`. If you want derivatives to be set up for all models, you can run `nimbleOptions(buildModelDerivs = TRUE)` and omit the `buildDerivs` argument.

17.2 How to support derivatives in user-defined functions and distributions

It is possible to activate derivative support for a user-defined function or distribution used in a model. Simply set `buildDerivs=TRUE` in `nimbleFunction`.

Here is an extremely toy example. Let's say we want a model with one node that follows a user-defined distribution, which will happen to be the same as `dnorm` (a normal distribution) for illustration.

The model code is:

```
toyCode <- nimbleCode({
  x ~ d_my_norm(mean = mu, sd = sigma)
  mu ~ dunif(-10, 10)
  sigma ~ dunif(0, 5)
})
```

The user-defined distribution is:

```
d_my_norm <- nimbleFunction(
  run = function(x = double(), mean = double(), sd = double(),
                 log = integer(0, default = 0)) {
    ans <- -log(sqrt(2*pi)*sd) - 0.5*((x-mean)/sd)^2
    if(log) return(ans)
    return(exp(ans))
    returnType(double())
  },
  buildDerivs=TRUE
)
```

Now we can build the model with `buildDerivs=TRUE` and compile it:

```
# Don't worry about the warnings from nimbleModel in this case.
toyModel <- nimbleModel(toyCode, inits = list(mu = 0, sigma = 1, x = 0),
                        buildDerivs = TRUE)
```

Now `toyModel` can be used in algorithms such as HMC or Laplace approximation, or new ones you might write.

Note that in models, all model variables are implemented as doubles, even when they will actually

only ever take integer values. Therefore all arguments to user-defined distributions and functions used in models should be declared as type `double`. The `ADbreak` method discussed in Section 17.4.2 should be used if you want to stop derivative tracking on a model variable.

17.3 What operations are and aren't supported for AD

Much of the math supported by NIMBLE will work for AD. Here are some details on what is and isn't supported, as well as on some options to control how linear algebra is handled.

Features that are not supported for AD include:

- cumulative (“p”) and quantile (“q”) functions for distributions;
- truncated distributions (because they use cumulative distribution functions), for any derivatives;
- discrete distributions, for derivatives with respect to the random variable;
- random number generation;
- some specific functions including `step`, `%% (mod)`, `eigen`, `svd`, and `besselK`; and
- most model operations (other than `model$calculate`, which is supported) including `model$calculateDiff`, `model$simulate`, `model$getBound`, and `model$getParam`.
- dynamic indexing that would produce a nonscalar result, such as `p[z[i], 1:3]`. Dynamic indexing that produces a scalar result, such as `p[z[i], j]`, is allowed and can be used as a work-around, e.g., defining `ptmp` as `for(j in 1:3) ptmp[i,j] <- p[z[i],j]` and using it in place of `p[z[i], 1:3]`.

Note that functions and distributions that are not supported can still be used in a model, as long as no algorithm tries to take derivatives with respect to node(s) that is (are) not supported. For example, `x ~ dcat(p[1:3])` declares `x` to be a discrete random variable following a categorical distribution. An algorithm can use derivatives with respect to `p[1:3]` (or with respect to parent nodes of `p[1:3]`) because the calculation of the log probability of `x` given `p[1:3]` is continuous with respect to `p`. However, derivatives with respect to `x` will not work. In the case of truncated distributions, derivatives with respect to neither the parameters nor to the random variable are supported.

Some details on what is supported include:

- `round`, `floor`, `ceil`, and `trunc` are all supported with derivatives defined to be zero everywhere.
- `pow(a, b)` requires positive `a` and `b`. Note that if `b` is coded as a (possibly negative) integer, `pow_int` will be used. For example `pow(tau, -2)` will be converted to `pow_int(tau, -2)`.
- A new function `pow_int(a, b)` returns `pow(a, round(b))` and thus sets all derivatives with respect to `b` to zero. This allows valid derivatives with respect to `a` even if it takes a negative value.

For the linear algebra functions `%*%`, `chol`, `forwardsolve`, `backsolve`, and `inverse`, there are special extensions provided by NIMBLE for CppAD called (in CppAD jargon) “atomics”. By default, these atomics will be used and often improve efficiency. There may be cases where they decrease efficiency, which might include when the matrix operands are small or contain many zeros. To compare results with and without use of the atomics, they can be turned off with a set of options:

```
nimbleOptions(useADmatMultAtomic = FALSE) # for %*%
nimbleOptions(useADcholAtomic = FALSE)    # for chol
nimbleOptions(useADmatInverseAtomic = FALSE) # for inverse
nimbleOptions(useADsolveAtomic = FALSE)    # for forwardsolve and backsolve
```

When a linear algebra atomic is turned off, the AD system simply uses all the scalar operations that compose the linear algebra operation.

Another important feature of AD is that sometimes values get “baked in” to AD calculations, meaning they are used *and permanently retained* from the first set of calculations and then can’t have their value changed later (unless an algorithm does a “reset”, described below). For people writing user-defined distributions and functions, a brief summary of what can get baked in includes:

- the extent (iteration values) of any for-loops.
- values of any arguments that are not of type ‘double’, including e.g. the ‘log’ argument in `d_my_norm`. (‘d’ functions called from models always have `log = TRUE`, so in that case it is not a problem.)
- the evaluation path followed by any if-then-else calls.
- values of any integer indices.

See below for more thorough explanations.

17.4 Basics of obtaining derivatives in `nimbleFunctions`

Now that we have seen a derivative-enabled algorithm work, let’s see how to obtain derivatives in methods you write. From here on, this part of the chapter is oriented towards algorithm developers. We’ll start by showing how derivatives work in `nimbleFunctions` *without* using a model. The AD system allows you to obtain derivatives of one function or method from another.

Let’s get derivatives of the function $y = \exp(-d * x)$ where x is a vector, d is a scalar, and y is a vector.

```
derivs_demo <- nimbleFunction(
  setup = function() {},
  run = function(d = double(), x = double(1)) {
    return(exp(-d*x))
    returnType(double(1))
  },
  methods = list(
    derivsRun = function(d = double(), x = double(1)) {
      wrt <- 1:(1 + length(x)) # total length of d and x
      return(derivs(run(d, x), wrt = wrt, order = 0:2))
      returnType(ADNimbleList())
    }
  ),
  buildDerivs = 'run'
)
```

Here are some things to notice:

- Having `setup` code allows the `nimbleFunction` to have multiple methods (i.e. to behave like a class definition in object-oriented programming). Some `nimbleFunctions` don't have `setup` code, but `setup` code is required when there will be a call to `derivs`. If, as here, you don't need the `setup` to do anything, you can simply use `setup=TRUE`, which is equivalent to `setup=function(){}.`
- Any arguments to `run` that are real numbers (i.e. regular double precision numbers, but not integers or logicals) will have derivatives tracked when called through `derivs`.
- The “with-respect-to” (`wrt`) argument gives indices of the arguments for which you want derivatives. In this case, we're including all elements of `d` and `x`. The indices form one sequence over all arguments.
- `order` is a vector of derivative orders to return, which can include any of 0 (the function value, not a derivative!), 1 (1st order), or 2 (2nd order). Higher-order derivatives can be obtained by double-taping, described below.
- Basics of `nimbleFunction` programming are covered in Chapters 12-16. These include type declarations (`double()` for scalar, `double(1)` for vector), the distinction between `setup` code and `run` code, and how to write more methods (of which `run` is simply a default name when there is only one method).
- `derivs` can alternatively be called `nimDerivs`. In fact the former will be converted to the latter internally.

Let's see this work.

```
my_derivs_demo <- derivs_demo()
C_my_derivs_demo <- compileNimble(my_derivs_demo)
d <- 1.2
x <- c(2.1, 2.2)
C_my_derivs_demo$derivsRun(d, x)
```

```
## nimbleList object of type NIMBLE_ADCLASS
## Field "value":
## [1] 0.08045961 0.07136127
## Field "jacobian":
##           [,1]      [,2]      [,3]
## [1,] -0.1689652 -0.09655153 0.00000000
## [2,] -0.1569948 0.00000000 -0.08563352
## Field "hessian":
## , , 1
##
##           [,1]      [,2] [,3]
## [1,] 0.3548269 0.1222986 0
## [2,] 0.1222986 0.1158618 0
## [3,] 0.0000000 0.0000000 0
##
## , , 2
##
##           [,1] [,2]      [,3]
## [1,] 0.3453885 0 0.1170325
## [2,] 0.0000000 0 0.0000000
## [3,] 0.1170325 0 0.1027602
```

We can see that using `order = 0:2` results in the the value (“0th” order result, i.e. the value returned by `run(d, x)`), the Jacobian (matrix of first order derivatives), and the Hessian (array of second order derivatives). The `run` function here has taken three inputs (in the order `d`, `x[1]`, `x[2]`) and returned two outputs (the first and second elements of `exp(-d*x)`).

The *i*-th Jacobian row contains the first derivatives of the *i*-th output with respect to `d`, `x[1]`, and `x[2]`, in that order. The first and second indices of the Hessian array follow the same ordering as the columns of the Jacobian. The third index of the Hessian array is for the output index. For example, `hessian[3,1,2]` is the second derivative of the second output value (third index = 2) with respect to `x[2]` (first index = 3) and `d` (second index = 1).

(Although it may seem inconsistent to have the output index be first for Jacobian and last for Hessian, it is consistent with some standards for how these objects are created in other packages and used mathematically.)

When a function being called for derivatives (`run` in this case) has non-scalar arguments (`x` in this case), the indexing of inputs goes in order of arguments, and in column-major order within arguments. For example, if we had arguments `x = double(1)` and `z = double(2)` (a matrix), the inputs would be ordered as `x[1]`, `x[2]`, ..., `z[1, 1]`, `z[2, 1]`, ..., `z[1, 2]`, `z[2, 2]`, ..., etc.

17.4.1 Checking derivatives with uncompiled execution

Derivatives can also be calculated in uncompiled execution, but they will be much slower and less accurate: slower because they are run in R, and less accurate because they use finite element methods (from packages `pracma` and/or `numDeriv`). Uncompiled execution is mostly useful for checking that compiled derivatives are working correctly, because although they are slower and less accurate, they are also much simpler internally and thus provide good checks on compiled results. For example:

```
my_derivs_demo$derivsRun(d, x)

## nimbleList object of type NIMBLE_ADCLASS
## Field "value":
## [1] 0.08045961 0.07136127
## Field "jacobian":
##           [,1]      [,2]      [,3]
## [1,] -0.1689652 -0.09655153 0.00000000
## [2,] -0.1569948 0.00000000 -0.08563352
## Field "hessian":
## , , 1
##
##           [,1]      [,2] [,3]
## [1,] 0.3548269 0.1222986 0
## [2,] 0.1222986 0.1158618 0
## [3,] 0.0000000 0.0000000 0
##
## , , 2
##
##           [,1] [,2]      [,3]
## [1,] 0.3453885 0 0.1170325
```

```
## [2,] 0.0000000    0 0.0000000
## [3,] 0.1170325    0 0.1027602
```

We can see that the results are very close, but typically not identical, to those from the compiled version.

17.4.2 Holding some local variables out of derivative tracking

Sometimes one wants to omit tracking of certain variables in derivative calculations. There are three ways to do this: naming variables in an `ignore` set; ensuring a variable's type is `integer` (not simply `numeric` with an integer value) or `logical`; or using the `ADbreak` function. The differences between these have to do with the fact that the variable types used for tracking derivatives in C++ are special. The `ignore` method marks a variable as a regular type rather than a derivative-tracking type. The `integer` or `logical` method results in a regular (integer or logical) variable simply because those types are never used in derivative-tracking. The `ADbreak` method assigns one derivative-tracked variable to another (derivative-tracked or not) variable while severing any actual derivative tracking between them.

17.4.2.1 Holding variables out of derivative tracking using `ignore`

Here is an example of `ignore`, while `ADbreak` is described below.

Say that we write `exp(-d*x)` using a for-loop instead of a vectorized operation as above. It wouldn't make sense to track derivatives for the for-loop index (`i`), and indeed it would cause `compileNimble` to fail. The code to use a for-loop index `i` while telling `derivs` to *ignore* derivatives for `i` is:

```
derivs_demo2 <- nimbleFunction(
  setup = function() {},
  run = function(d = double(), x = double(1)) {
    ans <- numeric(length = length(x))
    for(i in 1:length(x))
      ans[i] <- exp(-d*x[i])
    return(ans)
    returnType(double(1))
  },
  methods = list(
    derivsRun = function(d = double(), x = double(1)) {
      wrt <- 1:(1 + length(x)) # total length of d and x
      return(derivs(run(d, x), wrt = wrt, order = 0:2))
      returnType(ADNimbleList())
    }
  ),
  buildDerivs = list(run = list(ignore = 'i'))
)
```

We can see that it gives identical results as above, looking at only the Jacobian to keep the output short.

```
my_derivs_demo2 <- derivs_demo2()
C_my_derivs_demo2 <- compileNimble(my_derivs_demo2)
d <- 1.2
```

```
x <- c(2.1, 2.2)
C_my_derivs_demo2$derivsRun(d, x)$jacobian
```

```
##           [,1]      [,2]      [,3]
## [1,] -0.1689652 -0.09655153  0.00000000
## [2,] -0.1569948  0.00000000 -0.08563352
```

One might think it should be obvious that `i` should not be involved in derivatives, but sometimes math is actually done with a for-loop index. Another way to ensure derivatives won't be tracked for `i` would be to write `i <- 1L`, before the loop. By assigning a definite integer to `i`, it will be established as definitely of integer type throughout the function and thus not have derivatives tracked. (If you are not familiar with the `L` in `R`, try `storage.mode(1L)` vs `storage.mode(1)`.) The only way to mark a double-precision variable to be a regular (not derivative-tracking) type is to name it in the `ignore` set.

Another case where the `ignore` or `integer` methods are useful is in determining the start and/or end values of a for-loop. If you need `for(i in 1:n) {...}` where you previously set `n <- length(some_values)`, you need `n` to be a non-derivative tracking type. You can achieve this either by including `"n"` in the `ignore` set or by doing `n <- 0L` before setting `n <- length(some_values)`, to establish with certainty that `n` is an integer.

Because derivatives are not tracked for integer or logical variables, arguments to `run` that have `integer` or `logical` types will not have derivatives tracked. You can also include an argument name (even for a double-precision argument) in the `ignore` set. In any of these cases, you must then be careful in passing a non-derivative tracking type to another function. If passed as an argument to another function, the argument in that function must also be a non-derivative tracking type (`integer`, `logical`, or in the `ignore` set).

If you need to assign a value from a derivative-tracked variable to a non-derivative tracked variable, you must use `ADbreak` (see the next subsection for more on this). For example if `y` is in the `ignore` set, `y <- ADbreak(x)` assigns the value of `x` (with no derivative tracking) to `y`. You can also do `y <- ADbreak(x)` if `y` is *not* in the `ignore` set, and the result is that `y` has derivative tracking and is initialized to the value of `x`, but all derivatives of `y` with respect to `x` are 0. This severs the derivative (chain-rule) relationship between `x` and any subsequent calculations with `y`, but it does allow `y` to be used where a derivative-tracking variable is expected (e.g. as another function's argument). **Derivative-tracked variables that are severed by `ADbreak` will be “baked in” to the AD tape from the first call.** This means their values will be constants and **will not change on subsequent calls** even if the input values change, unless you use the `reset` feature. See section 17.4.4 to learn more about “baked in” values.

Only some math operations will work with a mix of derivative-tracking and regular variables. For example, most scalar operations will work with one argument of each kind. However, more complicated operations such as matrix multiplication require the arguments to be either both derivative-tracking or both regular.

Note the more elaborate value for `buildDerivs`. In the previous example, this was just a character vector. Here it is a named list, with each element itself being a list of control values. The control value `ignore` is a character vector of variable names to ignore in derivative tracking. `buildDerivs = 'run'` is equivalent to `buildDerivs = list(run = list())`.

17.4.2.2 Holding function arguments (including variables passed from a model) out of derivative tracking

Derivatives can be tracked through function calls, including from a model to a user-defined distribution or function (see 17.2 and 17.4.3). However, if one function is tracking derivatives for `x` and passes `x` as an argument to another function where it is non-derivative tracking type, there will be an error (most likely from `compileNimble`). The problem is that, as noted above, the C++ variable types used for tracking derivatives or not are different and are not automatically interchangeable. Importantly, variables passed from models are always derivative-tracking (even if they will always have integer values).

For this situation, you can use `ADbreak` to stop derivative tracking. For example, if one has a `nimbleFunction` argument `x`, one would do `x_noDeriv <- ADbreak(x)` and then use `x_noDeriv` in subsequent calculations. This “breaks” derivative tracking in the sense that the derivative of `x_noDeriv` with respect to `x` is 0. However, it is important to realize that `x_noDeriv` is still a derivative-tracking type (unless it is in the `ignore` set), so it can be passed to other functions involved in the derivative-tracked operations. And, on the other hand, it can’t be used as a for-loop index or as the start or end of a for-loop extent. As noted above, if the values of `x` or `x_noDeriv` are used in further calculations, their values will be “baked in” to the tape.

Including “`x`” in the `ignore` set won’t work in this case because, if `x` is a function argument, its type will then be changed to be a non-derivative type. If a derivative-tracking type is then passed as the value for `x` there will be a mismatch in types. Since variables passed from models always allow derivative-tracking, one should not `ignore` the corresponding function arguments.

Note that `ADbreak` only works for scalars. If you need to break AD tracking for a non-scalar, you have to write a for-loop to apply `ADbreak` for each element. For example:

```
x_noDeriv <- nimbleNumeric(length(x), init=FALSE)
i <- 1L
for(i in 1:length(x)) x_noDeriv[i] <- ADbreak(x[i])
```

In summary:

- Use the `ignore` set or explicit creation of variables as `integer` or `logical` to create regular variable types.
- If you need an argument to have derivative-tracking (as is always the case when called from a model) but derivatives with respect to it don’t make sense, use `ADbreak` to assign to a local variable that will “bake in” the value and remove it from derivative tracking. The local variable can be regular (if it is in the `ignore` list or was created first as an `integer` or `logical`) or derivative-tracking.

17.4.3 Using AD with multiple nimbleFunctions

Derivatives will be tracked through whatever series of calculations occur in a method, possibly including calls to other methods or `nimbleFunctions` that have `buildDerivs` set. Let’s look at an example where we have a separate function to return the element-wise square root of an input vector. The net calculation for derivatives will be `sqrt(exp(-d*x))`.

```
nf_sqrt <- nimbleFunction(
  run = function(x = double(1)) {
    return(sqrt(x))
```



```

    returnType(double(1))
  },
  buildDerivs = TRUE
)

derivs_demo3 <- nimbleFunction(
  setup = function() {},
  run = function(d = double(), x = double(1)) {
    ans <- exp(-d*x)
    ans <- nf_sqrt(ans)
    return(ans)
    returnType(double(1))
  },
  methods = list(
    derivsRun = function(d = double(), x = double(1)) {
      wrt <- 1:(1 + length(x)) # total length of d and x
      return(derivs(run(d, x), wrt = wrt, order = 0:2))
      returnType(ADNimbleList())
    }
  ),
  buildDerivs = 'run'
)

```

And then let's see it work:

```

my_derivs_demo3 <- derivs_demo3()
C_my_derivs_demo3 <- compileNimble(my_derivs_demo3)
d <- 1.2
x <- c(2.1, 2.2)
C_my_derivs_demo3$derivsRun(d, x)$jacobian

```

```

##           [,1]      [,2]      [,3]
## [1,] -0.2978367 -0.1701924  0.0000000
## [2,] -0.2938488  0.0000000 -0.1602812

```

Note that for a `nimbleFunction` without setup code, one can say `buildDerivs=TRUE`, `buildDerivs = 'run'`, or `buildDerivs = list(run = list())`. One can't take derivatives of `nf_sqrt` on its own, but it can be called by a method of a `nimbleFunction` that can have its derivatives taken (e.g. the `run` method of a `derivs_demo3` object).

17.4.4 Understanding more about how AD works: *taping* of operations

At this point, it will be helpful to grasp more of how AD works and its implementation in `nimble` via the `CppAD` library (Bell, 2022). AD methods work by following a set of calculations multiple times, sometimes in reverse order.

For example, the derivative of $y = \exp(x^2)$ can be calculated (by the chain rule of calculus) as the derivative of $y = \exp(z)$ (evaluated at $z = x^2$) times the derivative of $z = x^2$ (evaluated at x). This can be calculated by first going through the sequence of steps for the value, i.e. (i) $z = x^2$, (ii) $y = \exp(z)$, and then again for the derivatives, i.e. (i) $dz = 2x(dx)$, (ii) $dy = \exp(z)dz$. These steps

determine the instantaneous change dy that results from an instantaneous change dx . (It may be helpful to think of both values as a function of a third variable such as t . Then we are determining dy/dt as a function of dx/dt .) In CppAD, the basic numeric object type (double) is replaced with a special type and corresponding versions of all the basic math operations so that those operations can track not just the values but also derivative information.

The derivative calculations typically need the values. For example, the derivative of $\exp(z)$ is (also) $\exp(z)$, for which the value of z is needed, which in this case is x^2 , which is one of the steps in calculating the value of $\exp(x^2)$. (In fact, in this case the derivative of $\exp(z)$ can more simply use the value of y itself.) Hence, values are calculated before derivatives, which are calculated by stepping through the calculation sequence a *second* time

What was just described is called *forward* mode in AD. The derivatives can also be calculated by working through $\exp(x^2)$ in *reverse* mode. This is often less familiar to graduates of basic calculus courses. In reverse mode, one determines what instantaneous change dx would give an instantaneous change dy . For example, given a value of dy , we can calculate $1/dz = \exp(z)/dy$ and then $(1/dx) = (x^2)/dz$. Again, values are calculated first, followed by derivatives in a second pass through the calculation sequence, this time in reverse order.

In general, choice of forward mode versus reverse mode has to do with the lengths of inputs and outputs and possibly specific operations involved. In nimble, some reasonable choices are used.

In CppAD, the metaphor of pre-digital technology – magnetic tapes – is used for the set of operations in a calculation. When a line with `nimDerivs` is first run, the operations in the function it calls are *taped* and then re-used – *played* – in forward and/or reverse orders to obtain derivatives. Hence we will refer to the AD *tape* in explaining some features below. It is also possible to reset (i.e. re-record) the AD tape used for a particular `nimDerivs` call.

When taped operations are played, the resulting operations can themselves be taped. We call this *meta-taping* or *double-taping*. It is useful because it can sometimes boost efficiency and it can be used to get third and higher order derivatives.

17.4.5 Resetting a `nimDerivs` call

The first time a call to `nimDerivs` runs, it records a tape of the operations in its first argument (`run(d, x)` in the examples here). On subsequent calls, it re-uses that tape without re-recording it. That means subsequent calls are usually much faster. *It also means that subsequent calls **must** use the same sized arguments as the recorded call.* For example, we can use `C_my_derivs_demo$derivsRun` with new arguments of the same size(s):

```
C_my_derivs_demo$derivsRun(-0.4, c(3.2, 5.1))
```

```
## nimbleList object of type NIMBLE_ADCLASS
## Field "value":
## [1] 3.596640 7.690609
## Field "jacobian":
##           [,1]      [,2]      [,3]
## [1,] -11.50925  1.438656  0.000000
## [2,] -39.22211  0.000000  3.076244
## Field "hessian":
## , , 1
```

```
##
##           [,1]      [,2] [,3]
## [1,] 36.829591 -8.2003386    0
## [2,] -8.200339  0.5754624    0
## [3,]  0.000000  0.0000000    0
##
## , , 2
##
##           [,1] [,2]      [,3]
## [1,] 200.03275    0 -23.379452
## [2,]  0.00000    0  0.000000
## [3,] -23.37945    0  1.230497
```

However, if we call `C_my_derivs_demo$derivsRun` with `length(x)` different from 2, the result will be garbage. If we need to change the size of the arguments, we need to re-record the tape. This is done with the `reset` argument.

Here is a slightly more general version of the `derivs_demo` allowing a user to reset the tape. It also takes `order` and `wrt` as arguments instead of hard-coding them.

```
derivs_demo4 <- nimbleFunction(
  setup = function() {},
  run = function(d = double(), x = double(1)) {
    return(exp(-d*x))
    returnType(double(1))
  },
  methods = list(
    derivsRun = function(d = double(), x = double(1),
                        wrt = integer(1), order = integer(1),
                        reset = logical(0, default=FALSE)) {
      return(derivs(run(d, x), wrt = wrt, order = order, reset = reset))
      returnType(ADNimbleList())
    }
  ),
  buildDerivs = 'run'
)
```

Now we will illustrate the use of `reset`. To make shorter output, we'll request only the Jacobian.

```
my_derivs_demo4 <- derivs_demo4()
C_my_derivs_demo4 <- compileNimble(my_derivs_demo4)
d <- 1.2
x <- c(2.1, 2.2)
# On the first call, reset is ignored because the tape must be recorded.
C_my_derivs_demo4$derivsRun(d, x, 1:3, 1)
```

```
## nimbleList object of type NIMBLE_ADCLASS
## Field "value":
## numeric(0)
## Field "jacobian":
```

```
##           [,1]      [,2]      [,3]
## [1,] -0.1689652 -0.09655153  0.00000000
## [2,] -0.1569948  0.00000000 -0.08563352
## Field "hessian":
## <0 x 0 x 0 array of double>
##
```

```
# On the second call, reset=FALSE, so the tape is re-used.
```

```
C_my_derivs_demo4$derivsRun(-0.4, c(3.2, 5.1), 1:3, 1)
```

```
## nimbleList object of type NIMBLE_ADCLASS
## Field "value":
## numeric(0)
## Field "jacobian":
##           [,1]      [,2]      [,3]
## [1,] -11.50925  1.438656  0.000000
## [2,] -39.22211  0.000000  3.076244
## Field "hessian":
## <0 x 0 x 0 array of double>
##
```

```
# If we need a longer X, we need to say reset=TRUE
```

```
C_my_derivs_demo4$derivsRun(1.2, c(2.1, 2.2, 2.3), 1:4, 1, reset=TRUE)
```

```
## nimbleList object of type NIMBLE_ADCLASS
## Field "value":
## numeric(0)
## Field "jacobian":
##           [,1]      [,2]      [,3]      [,4]
## [1,] -0.1689652 -0.09655153  0.00000000  0.00000000
## [2,] -0.1569948  0.00000000 -0.08563352  0.00000000
## [3,] -0.1455711  0.00000000  0.00000000 -0.07595012
## Field "hessian":
## <0 x 0 x 0 array of double>
##
```

```
# Now we can use with reset=FALSE with X of length 3
```

```
C_my_derivs_demo4$derivsRun(-0.4, c(3.2, 5.1, 4.5), 1:4, 1)
```

```
## nimbleList object of type NIMBLE_ADCLASS
## Field "value":
## numeric(0)
## Field "jacobian":
##           [,1]      [,2]      [,3]      [,4]
## [1,] -11.50925  1.438656  0.000000  0.000000
## [2,] -39.22211  0.000000  3.076244  0.000000
## [3,] -27.22341  0.000000  0.000000  2.419859
## Field "hessian":
## <0 x 0 x 0 array of double>
##
```

You could also make multiple instances of `derivs_demo4` and keep track of the argument sizes currently recorded for each one.

Other important situations where you need to reset a tape are when values that are otherwise “baked in” to an AD tape need to be changed. By “baked in”, we mean that some values are permanently part of a tape until it is reset.

17.4.5.1 What gets baked into AD tapes until `reset=TRUE`

Specifically, you need to reset a tape when:

- the extent of any for-loops change. For-loops are not natively recorded, but rather each operation done in execution of the for-loop is recorded. (This is sometimes called “loop unrolling”.) Therefore, in `for(i in start:end)`, the values of `start` and `end` are in effect baked into a tape.
- the outcomes of any if-then-else conditions in a `nimbleFunction` are changed.
- values of any `integer` or `logical` inputs change.
- values of any variables with derivative tracking severed `ADbreak` change. (See 17.4.2.)
- any other member data used in calculations changes. It is possible to create a variable in the `setup` function and use it in methods such as `run`. Such variables become member data in the terminology of object-oriented programming. Such variables will not have derivatives tracked if used in a function while it is being taped, so the values will be baked into the tape. This can be useful if you understand what is happening or confusing if not.

Uncompiled execution of `derivs` ignores the reset argument.

17.4.6 A note on performance benchmarking

If you are interested in measuring the performance of AD in nimble, please remember that the first call to `nimDerivs` and any subsequent call with `reset=TRUE` will be slower, often much slower, than subsequent calls.

17.5 Advanced uses: double taping

Suppose you are interested only in the Jacobian, not in the value, and/or want only some elements of the Jacobian. You might still need the value, but obtaining the value alone from an AD tape (i.e. from `derivs`) will be slower than obtaining it by simply calling the function. On the other hand, AD methods need to calculate the value before calculating first order derivatives, so the value will be calculated anyway. However, in some cases, some of the steps of value calculations aren’t really needed if one only wants, say, first-order derivatives. In addition, if not all elements of the Jacobian are wanted, then some unnecessary calculations will be done internally that one might want to avoid.

A way to cut out unnecessary calculations is to record a tape of a tape, which we call double taping. Let’s see an example before explaining further.

```
derivs_demo5 <- nimbleFunction(
  setup = function() {},
  run = function(d = double(), x = double(1)) {
    return(exp(-d*x))
```

```

    returnType(double(1))
  },
  methods = list(
    jacobian_run_wrt_d = function(d = double(), x = double(1),
                                wrt = integer(1),
                                reset = logical(0, default=FALSE)) {
      ans <- derivs(run(d, x), wrt = wrt, order = 1, reset = reset)
      jac <- ans$jacobian[,1] # derivatives wrt 'd' only
      return(jac)
      returnType(double(1))
    },
    derivsJacobian = function(d = double(), x = double(1),
                              wrt = integer(1),
                              order = integer(1),
                              reset = logical(0, default = FALSE)) {
      innerWrt <- nimInteger(value = 1, length = 1)
      ans <- nimDerivs(jacobian_run_wrt_d(d, x, wrt = innerWrt, reset = reset),
                      wrt = wrt, order = order, reset = reset)

      return(ans)
      returnType(ADNimbleList())
    }
  ),
  buildDerivs = c('run', 'jacobian_run_wrt_d')
)

```

What is happening in this code? `jacobian_run_wrt_d` is a method that returns part of the Jacobian of `run(d, x)`, specifically the first column, which contains derivatives with respect to `d`. It happens to use AD to do it, but otherwise it is just some function with inputs `d` and `x` and a vector output. (Arguments that are integer or logical do not have derivatives tracked.) `derivsJacobian` calculates derivatives of `jacobian_run_wrt_d`. This means that the value returned by `derivsJacobian` will be the value of `jacobian_run_wrt_d`, which comprises the first derivatives of `run` with respect to `d`. The jacobian returned by `derivsJacobian` will contain second derivatives of `run` with respect to `d` and each of `d`, `x[1]`, and `x[2]`. And the hessian returned by `derivsJacobian` will contain some *third* derivatives of `run`. Notice that `buildDerivs` now includes `jacobian_run_wrt_d`.

Let's see this in use.

```

my_derivs_demo5 <- derivs_demo5()
C_my_derivs_demo5 <- compileNimble(my_derivs_demo5)
d <- 1.2
x <- c(2.1, 2.2)
C_my_derivs_demo5$derivsJacobian(d, x, wrt = 1:3, order = 0:2)

## nimbleList object of type NIMBLE_ADCLASS
## Field "value":
## [1] -0.1689652 -0.1569948
## Field "jacobian":
##           [,1]      [,2]      [,3]
## [1,] 0.3548269 0.1222986 0.0000000

```

```
## [2,] 0.3453885 0.0000000 0.1170325
## Field "hessian":
## , , 1
##
##          [,1]      [,2] [,3]
## [1,] -0.74513642 -0.08786189  0
## [2,] -0.08786189 -0.05020679  0
## [3,]  0.00000000  0.00000000  0
##
## , , 2
##
##          [,1] [,2]      [,3]
## [1,] -0.7598548  0 -0.10047667
## [2,]  0.0000000  0  0.00000000
## [3,] -0.1004767  0 -0.05480546
```

We can compare these results to those shown above from the same values of `d` and `x`. The value here is the same as `jacobian[1:2, 1]` above. The `jacobian` here is the same as `t(hessian[1, 1:3, 1:2])` or (by symmetry of second derivatives) `t(hessian[1:3, 1, 1:2])` above. And the `hessian[i,j,k]` here contains the third derivative of the `k`-th output of `run(d, x)` with respect to `d`, the `i`-th input, and the `j`-th input.

It is also possible to call `derivs` on (i.e. tape) a function containing a `derivs` call from which some 0-th and/or 2nd order derivatives are extracted. It is even possible to triple tape by calling `derivs` on a function calling `derivs` on a function calling `derivs`, and so on.

17.6 Derivatives involving model calculations

Obtaining derivatives involving model calculations takes some special considerations, and there are two ways to do it.

We will use the Poisson GLMM above as an example model for which we want derivatives. We can obtain derivatives with respect to any variables for all or any subset of model calculations.

17.6.1 Method 1: `nimDerivs` of `model$calculate`

Recall that `model$calculate(nodes)` returns the sum of the log probabilities of all stochastic nodes in `nodes`. Deterministic calculations are also executed; they contribute 0 to the sum of probabilities but may be needed for inputs to subsequent calculations. Calculations are done in the order of `nodes`, which should be a valid order for the model, often obtained from `model$getDependencies`.

The simplest way to get derivatives for model calculations is to use `model$calculate` as the function taped by `derivs` (here shown by its alternative name `nimDerivs` for illustration).

```
derivs_nf <- nimbleFunction(
  setup = function(model, with_respect_to_nodes, calc_nodes) {},
  run = function(order = integer(1),
                  reset = logical(0, default=FALSE)) {
    ans <- nimDerivs(model$calculate(calc_nodes), wrt = with_respect_to_nodes,
                     order = order, reset = reset)
```

```

    return(ans)
    returnType(ADNimbleList())
  }
)

```

In `derivs_nf`:

- The mere presence of `model`, `with_respect_to_nodes`, and `calc_nodes` as `setup` arguments makes them available as member data for `run` or other methods.
- `model` will be a model object returned from `nimbleModel`.
- `with_respect_to_nodes` will be the names of nodes we want derivatives with respect to.
- `calc_nodes` will be the nodes to be calculated, in the order given.
- `order` can contain any of 0 (value), 1 (1st order), or 2 (2nd order) derivatives requested, as above.
- `reset` should be TRUE if the AD tape should be reset (re-recorded), as above. There are additional situations when a tape for `model$calculate` should be reset, discussed below.

Thus, `nimDerivs(model$calculate(calc_nodes), wrt = with_respect_to_nodes, <other args>)` takes derivatives of a function whose inputs are the values of `with_respect_to_nodes` (using their values in the model object) and whose output is the summed log probability returned by `model$calculate(calc_nodes)`. The internal handling of this case is distinct from other calls to `nimDerivs`.

Now we can make an instance of `derivs_nf`, compile the model and `nimbleFunction` instance, and look at the results. We will use the model from the HMC and Laplace examples.

```

model_code <- nimbleCode({
  # priors
  intercept ~ dnorm(0, sd = 100)
  beta ~ dnorm(0, sd = 100)
  sigma ~ dhalfflat()
  # random effects and data
  for(i in 1:10) {
    # random effects
    ran_eff[i] ~ dnorm(0, sd = sigma)
    for(j in 1:5) {
      # data
      y[i,j] ~ dpois(exp(intercept + beta*X[i,j] + ran_eff[i]))
    }
  }
})
set.seed(123)
X <- matrix(rnorm(50), nrow = 10)
model <- nimbleModel(model_code, constants = list(X = X), calculate = FALSE,
                     buildDerivs = TRUE) # Here is the argument needed for AD.

model$intercept <- 0
model$beta <- 0.2
model$sigma <- 0.5

```

```

model$calculate() # This will return NA because the model is not fully initialized.

## [1] NA

model$simulate(model$getDependencies('ran_eff'))
model$calculate() # Now the model is fully initialized: all nodes have valid values.

## [1] -78.44085

model$setData('y') # Now the model has y marked as data, with values from simulation.

wrt_nodes <- c('intercept','beta', 'sigma')
calc_nodes <- model$getDependencies(wrt_nodes)
derivs_all <- derivs_nf(model, wrt_nodes, calc_nodes)
cModel <- compileNimble(model)
cDerivs_all <- compileNimble(derivs_all, project = model)
derivs_result <- cDerivs_all$run(order = 0:2)
derivs_result

## nimbleList object of type NIMBLE_ADCLASS
## Field "value":
## [1] -78.44085
## Field "jacobian":
##           [,1]      [,2]      [,3]
## [1,] -9.358104 -0.3637619 -5.81414
## Field "hessian":
## , , 1
##
##           [,1]      [,2]      [,3]
## [1,] -62.35820 -16.21705  0.00000
## [2,] -16.21705 -69.47074  0.00000
## [3,]  0.00000  0.00000 -45.11516

```

As above, using `order = 0:2` results in the the value (0th order), Jacobian (1st order), and Hessian (2nd order). The function `model$calculate` is organized here to have inputs that are the current values of `intercept`, `beta`, and `sigma` in the model. It has output that is the summed log probability of the `calc_nodes`. The Jacobian columns are first derivatives with respect to `intercept`, `beta` and `sigma`, respectively. The first and second indices of the Hessian array follow the same ordering. For example `derivs_result$hessian[2,1,1]` is the second derivative with respect to `beta` (first index = 2) and `intercept` (second index = 1).

In the case of `model$calculate`, the first index of the Jacobian and the last index of the Hessian are always 1 because derivatives are of the first (and only) output value.

The ordering of inputs is similar to that used above, such as for the arguments `d` and `x`, but in this case the inputs are model nodes. When non-scalar nodes such as matrix or array nodes are used as `with_respect_to_nodes`, the resulting elements of Jacobian columns and Hessian first and second indices will follow column-major order. For example, for a 2x2 matrix `m`, the element order would be `m[1, 1]`, `m[2, 1]`, `m[1, 2]`, `m[2, 2]`. This will usually be the same ordering as the names returned by `model$expandNodeNames("m", returnScalarElements=TRUE)`.

The actual values used as inputs are the current values in the compiled model object. These would typically be set by code like `values(model, with_respect_to_nodes) <- my_values` before the call to `nimDerivs` in the above example.

As above, derivatives can also be calculated in uncompiled execution, but they will be much slower and less accurate. Uncompiled execution is mostly useful for checking that compiled derivatives are working correctly. This will use values in the *uncompiled* model object.

```
derivs_all$run(order = 0:2)

## nimbleList object of type NIMBLE_ADCLASS
## Field "value":
## [1] -78.44085
## Field "jacobian":
##           [,1]      [,2]      [,3]
## [1,] -9.358104 -0.3637619 -5.81414
## Field "hessian":
## , , 1
##
##           [,1]      [,2]      [,3]
## [1,] -62.35821 -1.621705e+01  0.000000e+00
## [2,] -16.21705 -6.947075e+01  2.384186e-07
## [3,]  0.00000  2.384186e-07 -4.511517e+01
```

We can see that all the results clearly match the compiled results to a reasonable numerical precision.

17.6.2 Method 2: `nimDerivs` of a method that calls `model$calculate`

Sometimes one needs derivatives of calculations done in a `nimbleFunction` as well as in a model. And sometimes one needs to change values in a model before and/or after doing model calculations and have those changes recorded in the derivative tape.

For these reasons, it is possible to take derivatives of a method that includes a call to `model$calculate`. Continuing the above example, say we want to take derivatives with respect to the log of `sigma`, so the input vector will be treated as `intercept`, `beta`, and `log(sigma)`, in that order. We will convert `log(sigma)` to `sigma` before using it in the model, and we want the derivative of that transformation included in the tape (i.e. using the chain rule).

(Using `log(sigma)` instead of `sigma` is useful so an algorithm such as optimization or HMC can use an unconstrained parameter space. A constraint such as `sigma > 0` can be difficult for many algorithms. See below for NIMBLE's automated parameter transformation system if you need to do this systematically.)

Here is a `nimbleFunction` to do that.

```
derivs_nf2 <- nimbleFunction(
  setup = function(model, wrt_nodes, calc_nodes) {
    derivsInfo <- makeModelDerivsInfo(model, wrt_nodes, calc_nodes)
    updateNodes <- derivsInfo$updateNodes
    constantNodes <- derivsInfo$constantNodes
    n_wrt <- length(wrt_nodes)
    # If wrt_nodes might contain non-scalar nodes, the more general way
```

```

    # to determine the length of all scalar elements is:
    # length(model$expandNodeNames(wrt_nodes, returnScalarComponents = TRUE))
  },
  run = function(x = double(1)) {
    x_trans <- x                # x[1:2] don't need transformation
    x_trans[3] <- exp(x[3])     # transformation of x[3]
    values(model, wrt_nodes) <- x_trans # put inputs into model
    ans <- model$calculate(calc_nodes) # calculate model
    return(ans)
    returnType(double(0))
  },
  methods = list(
    derivsRun = function(x = double(1),
                        order = integer(1),
                        reset = logical(0, default=FALSE)) {

      wrt <- 1:n_wrt
      ans <- nimDerivs(run(x), wrt = wrt, order = order, reset = reset,
                      model = model, updateNodes = updateNodes,
                      constantNodes = constantNodes)

      return(ans)
      returnType(ADNimbleList())
    }
  ),
  buildDerivs = list(run = list()) # or simply 'run' would work in this case
)

```

Let's see how this can be used.

```

derivs_all12 <- derivs_nf2(model, wrt_nodes, calc_nodes)
cDerivs_all12 <- compileNimble(derivs_all12, project = model)
params <- values(model, wrt_nodes)
params[3] <- log(params[3])
cDerivs_all12$derivsRun(params, order = 0:2)

```

```

## nimbleList object of type NIMBLE_ADCLASS
## Field "value":
## [1] -78.44085
## Field "jacobian":
##           [,1]      [,2]      [,3]
## [1,] -9.358104 -0.3637619 -2.90707
## Field "hessian":
## , , 1
##
##           [,1]      [,2]      [,3]
## [1,] -62.35820 -16.21705  0.00000
## [2,] -16.21705 -69.47074  0.00000
## [3,]  0.00000  0.00000 -14.18586

```

Notice that these results are the same as we saw above except for third elements, which represent

derivatives with respect to `sigma` above and to `log(sigma)` here.

There are several important new points here:

- The only valid way to get values into the model that will be recorded on the AD tape is with `values(model, nodes) <-> some_values` as shown. Other ways such as `nimCopy` or `model[[node]] <-> some_value` are not currently supported to work with AD.
- The call to `nimDerivs` of `run` must be told some information about the model calculations that will be done inside of `run`. `model` is of course the model that will be used. `constantNodes` is a vector of node names whose values are needed for the calculations but are not expected to change until you use `reset = TRUE`. Values of these nodes will be baked into the AD tape until `reset = TRUE`. `updateNodes` is a vector of node names whose values are needed for the calculations, might change between calls even with `reset = FALSE`, and are neither part of `wrt_nodes` nor a deterministic part of `calc_nodes`. The function `makeModelDerivsInfo`, as shown in the `setup` code, determines what is usually needed for `constantNodes` and `updateNodes`.
- In Method 1 above, the NIMBLE compiler automatically determines `constantNodes` and `updateNodes` using `makeModelDerivsInfo` based on the inputs to `nimDerivs(model$calculate(...), ...)`.
- One can use double-taping, but if so *both* calls to `nimDerivs` need the `model`, `updateNodes`, and `constantNodes` arguments, which should normally be identical.

17.6.2.1 Advanced topic: more about `constantNodes` and `updateNodes`

In most cases, you can obtain `constantNodes` and `updateNodes` from `makeModelDerivsInfo` without knowing exactly what they mean. But for advanced uses and possible debugging needs, let's explore these arguments in more detail. The purpose of `constantNodes` and `updateNodes` is to tell the AD system about all nodes that will be needed for taped model calculations. Specifically:

- `updateNodes` and `constantNodes` are vectors of nodes names that together comprise any nodes that are necessary for `model$calculate(calc_nodes)` but are:
 - not in the `wrt` argument to `nimDerivs` (only relevant for Method 1), and
 - not assigned into the model using `values(model, nodes) <-> some_values` prior to `model$calculate` (only relevant for Method 2), and
 - not a deterministic node in `calc_nodes`.
- `updateNodes` includes node names satisfying those conditions whose values might change between uses of the tape regardless of the `reset` argument.
- `constantNodes` includes node names satisfying those conditions whose values will be baked into the tape (will not change) until the next call with `reset=TRUE`.

To fix ideas, say that we want derivatives with respect to `ran_eff[1]` for calculations of it and the data that depend on it, including any deterministic nodes. In other words, `wrt` will be `ran_eff[1]`, and `calc_nodes` will be:

```
model$getDependencies('ran_eff[1]')
```

```
## [1] "ran_eff[1]"
## [2] "lifted_exp_oPintercept_plus_beta_times_X_oBi_comma_j_cB_plus_ran_eff_oBi_cB_cP_L7[1, ]"
## [3] "lifted_exp_oPintercept_plus_beta_times_X_oBi_comma_j_cB_plus_ran_eff_oBi_cB_cP_L7[1, ]"
```

```
## [4] "lifted_exp_oPintercept_plus_beta_times_X_oBi_comma_j_cB_plus_ran_eff_oBi_cB_cP_L7[1, 3]"
## [5] "lifted_exp_oPintercept_plus_beta_times_X_oBi_comma_j_cB_plus_ran_eff_oBi_cB_cP_L7[1, 4]"
## [6] "lifted_exp_oPintercept_plus_beta_times_X_oBi_comma_j_cB_plus_ran_eff_oBi_cB_cP_L7[1, 5]"
## [7] "y[1, 1]"
## [8] "y[1, 2]"
## [9] "y[1, 3]"
## [10] "y[1, 4]"
## [11] "y[1, 5]"
```

(The “lifted” nodes here are for the `exp(intercept + beta*X[i,j] + ran_eff[i])`, i.e. the inputs to `dpois`. See chapters 2 and 14 to learn about lifted nodes.)

In this case, the log probability of `ran_eff[1]` itself requires `sigma`. Since `sigma` is not in `wrt_nodes`, it is not assigned into the model by the line `values(model, wrt_nodes) <- x_trans`. It is also not a deterministic node (or any node) in `calc_nodes`, so it must be included in `updateNodes` or `constantNodes`. Since it might change between calls, it should be included in `updateNodes`.

Next, notice that the stochastic node `y[1, 1]` in `calc_nodes` means that the log probability of `y[1, 1]` will be calculated, and this requires the actual value of `y[1, 1]`. This node is part of `calc_nodes` but is not a deterministic part of it, so it must be provided in either `updateNodes` or `constantNodes`. When data values will not be changed often, it is better to put those nodes in `constantNodes`, because that will be more efficient than putting them in `updateNodes`. (If the data values are changed, use `reset=TRUE`, which will reset values of all `constantNodes` in the tape.)

The function `makeModelDerivsInfo` inspects the model and determines the usual needs for `updateNodes` and `constantNodes`. By default, data nodes are put in `constantNodes`. Use `dataAsConstantNodes = FALSE` in `makeModelDerivsInfo` if you want them put in `updateNodes`.

Note that a deterministic node in `calc_nodes` will have its value calculated as part of the operations recorded in the AD tape, so it does not need to be included in `updateNodes` or `constantNodes`.

As usual in model-generic programming in NIMBLE, be aware of lifted nodes and their implications. Suppose in the Poisson GLMM we had used a precision parameterization for the random effects, with the changes in this code snippet:

```
precision ~ dgamma(0.01, 0.01)
# <other lines>
ran_eff[i] ~ dnorm(0, tau = precision)
```

This would result in a lifted node for the standard deviation, calculated as `1/sqrt(precision)`. That lifted node is what would actually be used in `dnorm` for each `ran_eff[i]`. Now if `ran_eff[1]` is in `wrt_nodes`, the *lifted node* (but *not* `precision`) will be in `updateNodes` (as determined by `makeModelDerivsInfo`). If you then change the value of `precision`, you must be sure to calculate the lifted node before obtaining derivatives. Otherwise the value of the lifted node will correspond to the old value of `precision`. These considerations are not unique to AD but rather are part of model-generic programming (see chapter 16).

An example of model-generic programming to update any lifted nodes that depend on `precision` would be:

```
model$calculate(model$getDependencies('precision', determOnly=TRUE))
```

In Method 1 above, NIMBLE automatically uses `makeModelDerivsInfo` based on the code `nimDerivs(model$calculate(<args>), <args>)`. However, in Method 2, when `model$calculate(<args>)` is used in a method such as `run`, then a call to `nimDerivs(run(<args>), <args>)` requires the `model`, `updateNodes` and `constantNodes` to be provided. Hence, the two functions (`run` and `derivsRun`) must be written in coordination.

17.6.2.2 When do you need to reset a tape with model calculations?

The additional rule for when tapes need to be reset based on model calculations is:

- when values of any `constantNodes` are changed. Usually these will only be data nodes.

17.7 Parameter transformations

Many algorithms that in some way explore a parameter space are best used in an unconstrained parameter space. For example, there are a bunch of optimization methods provided in R's `optim`, but only one (L-BFGS-B) allows constraints on the parameter space. Similarly, HMC is implemented to work in an unconstrained parameter space.

NIMBLE provides a `nimbleFunction` to automatically create a transformation from original parameters to unconstrained parameters and the inverse transformation back to the original parameters. Denoting $\mathbf{x}$ as an original parameter and $\mathbf{g}(\mathbf{x})$ as the transformed parameter, the cases handled include:

- If scalar $x \in (0, \infty)$, then $g(x) = \log(x)$. For example, $\mathbf{x} \sim \text{dweibull}$ in a model means $x \in (0, \infty)$.
- If scalar $x \in (0, 1)$, then $g(x) = \text{logit}(x)$. For example, $\mathbf{x} \sim \text{dbeta}$ in a model means $x \in (0, 1)$.
- If scalar $x \in (a, \infty)$, then $g(x) = \log(x - a)$.
- If scalar $x \in (-\infty, b)$, then $g(x) = -\log(b - x)$.
- If scalar $x \in (a, b)$, then $g(x) = \text{logit}((x - a)/(b - a))$. For example $\mathbf{x} \sim \text{dunif}(\mathbf{a}, \mathbf{b})$ in a model means $x \in (a, b)$.
- If matrix $\mathbf{x}[1:n, 1:n] \sim \text{dwishart}$ or $\mathbf{x}[1:n, 1:n] \sim \text{dinvwishart}$ in a model, then $g(x) = \text{chol}(x)$ for non-diagonal elements of $\text{chol}(x)$ and $g(x) = \log(\text{chol}(x))$ for diagonal elements of $\text{chol}(x)$, where $\text{chol}(x)$ is the Cholesky decomposition of x . Note that x in these cases follows a Wishart or inverse Wishart distribution, respectively, and thus is a random precision or covariance matrix. That means it must be positive definite and thus have positive diagonal elements of $\text{chol}(x)$.
- If vector $\mathbf{x}[1:n] \sim \text{ddirch}$ in a model, then $g(x[1]) = \text{logit}(x[1])$ and $g(x[i]) = \text{logit}(x[i]/(1 - \sum_{k=1}^{i-1} x[k]))$ for $i > 1$. In this case, $\mathbf{x}$ follows a Dirichlet distribution and thus has the simplex constraint that $\sum_{i=1}^n x[i] = 1$.
- If vector $\mathbf{x}[1:n] \sim \text{dlkj_corr_cholesky}$, $g(x)$ is the LKJ Cholesky transformation. This provides the LKJ prior for a covariance matrix.

Note that the scalar transformations are all monotonic increasing functions of the original parameter.

The worked example next will illustrate use of the parameter transformation system.

Chapter 18

Example: maximum likelihood estimation using `optim` with gradients from `nimDerivs`.

In this example, we will obtain maximum likelihood estimates of a Poisson GLM. It will be similar to the GLMM above, but without the random effects. Maximization of a likelihood function can be done with methods that don't require gradients, but it is much faster to use methods that do use gradients.

To illustrate the parameter transformation system, we will give `p` a uniform prior, constrained to values between 0 and 1. The maximum likelihood estimation below will not use the prior probability, but nevertheless the prior will be interpreted for its constraints on valid values in this case. This need for parameter transformation is somewhat artificial in the interest of simplicity.

```
model_code <- nimbleCode({
  # priors (will be ignored for MLE)
  p ~ dunif(0, 1)
  log_p <- log(p)
  beta ~ dnorm(0, sd = 100)
  # random effects and data
  for(i in 1:50) {
    # data
    y[i] ~ dpois(exp(log_p + beta*X[i]))
  }
})

set.seed(123)
X <- rnorm(50) # Simulate values for X

model <- nimbleModel(model_code, constants = list(X = X), calculate = FALSE,
  buildDerivs = TRUE) # Create nimble model object
```

As in the first example above, we will simulate data using the model itself. One can instead provide data as an argument to `nimbleModel`.

```

model$p <- 0.5      # Assign parameter values and simulate data (y).
model$beta <- 0.2
model$calculate()  # We expect this to be NA because there are no y values yet.

## [1] NA

model$simulate(model$getDependencies(c('beta', 'p'), self = FALSE)) # Simulate y values.
model$calculate() # Now we expect a valid result.

## [1] -53.33015

model$setData('y') # Now the model has y marked as data, with values from simulation.
Cmodel <- compileNimble(model) # Make compiled version.

```

Now that the example model is set up, we'll write a `nimbleFunction` to provide the negative log likelihood and its gradient using an unconstrained parameter space. The default behavior of `optim` below is to do a minimization problem, so returning the negative log likelihood means that `optim` will find the maximum likelihood parameters.

```

logLikelihood_nf <- nimbleFunction(
  setup = function(model, paramNodes) {
    # Determine nodes for calculating the log likelihood for parameters given by
    # paramNodes, ignoring any priors.
    calcNodes <- model$getDependencies(paramNodes, self = FALSE)
    # Set up the additional arguments for nimDerivs involving model$calculate
    derivsInfo <- makeModelDerivsInfo(model, paramNodes, calcNodes)
    updateNodes <- derivsInfo$updateNodes
    constantNodes <- derivsInfo$constantNodes
    # Create a parameter transformation between original and unconstrained
    # parameter spaces.
    transformer <- parameterTransform(model, paramNodes)
  },
  methods = list(
    neg_logLikelihood_p = function(p = double(1)) {
      # Put values in model and calculate negative log likelihood.
      values(model, paramNodes) <- p
      return(-model$calculate(calcNodes))
      returnType(double())
    },
    neg_logLikelihood = function(ptrans = double(1)) {
      # Objective function for optim,
      # using transformed parameter space.
      p <- transformer$inverseTransform(ptrans)
      return(neg_logLikelihood_p(p))
      returnType(double())
    },
    gr_neg_logLikelihood = function(ptrans = double(1)) {
      # Gradient of neg log likelihood
      p <- transformer$inverseTransform(ptrans)

```



```

    d <- derivs(neg_logLikelihood_p(p), wrt = 1:length(p), order = 1,
               model = model, updateNodes = updateNodes,
               constantNodes = constantNodes)
    return(d$jacobian[1,])
    returnType(double(1))
  },
  transform = function(p = double(1)) {
    # Give user access to the transformation ...
    return(transformer$transform(p))
    returnType(double(1))
  },
  inverseTransform = function(ptrans = double(1)) { # ... and its inverse.
    return(transformer$inverseTransform(ptrans))
    returnType(double(1))
  }
),
buildDerivs = 'neg_logLikelihood_p'
)

```

Sometimes you might want a call to `transformer$transform` or `transformer$inverseTransform` to be included in a derivative calculation. In this case, we don't.

Next we build and compile an instance of `logLikelihood_nf` specialized (by the ‘setup’ function) to our model and its parameters.

```

ll_glm <- logLikelihood_nf(model, c("p", "beta"))
Cll_glm <- compileNimble(ll_glm, project = model)

```

Now let's see how the parameter transformation works for this model. For `beta`, it already has an unconstrained range of valid values, so it has no transformation (i.e. it has an identity transformation). For `p`, it is constrained between 0 and 1, so it will be logit-transformed.

```
Cll_glm$transform(c(0.2, 1))
```

```
## [1] -1.386294  1.000000
```

```

# The inverse transformation goes back to the original parameters, within rounding:
Cll_glm$inverseTransform(c(-1.386294, 1))

```

```
## [1] 0.2000001 1.0000000
```

Now we are ready to use the methods provided in `Cll_glm` in a call to R's `optim` to find the MLE.

```

MLE <- optim(c(0,0), # initial values
            fn = Cll_glm$neg_logLikelihood, # function to be minimized
            gr = Cll_glm$gr_neg_logLikelihood, # gradient of function to be minimized
            method = "BFGS") # optimization method to use
MLE$par

```

```
## [1] 0.08257249 0.25577134
```

```
-MLE$value
```

```
## [1] -47.73243
```

Those outputs show the MLE in the parameters `logit(p)`, and `beta`, following by the maximum log likelihood value.

Now let's confirm that it's working correctly by comparing to results from `glm`.

```
glm_fit <- glm(I(model$y) ~ X, family = poisson)
coef(glm_fit)
```

```
## (Intercept)          X
## -0.6527680    0.2557751
```

```
logLik(glm_fit)
```

```
## 'log Lik.' -47.73243 (df=2)
```

We see that the coefficient for `X` matches `beta` (within numerical tolerance), and the maximum log likelihoods match. To understand the intercept, we need to be careful about the parameter transformation used in `nimble`. The inverse transformation will give `p`, and the log of that should match the intercept estimated by `glm`.

```
C11_glm$inverseTransform(MLE$par)
```

```
## [1] 0.5206314 0.2557713
```

```
log(C11_glm$inverseTransform(MLE$par)[1]) # This should match glm's intercept
```

```
## [1] -0.652713
```

It looks like it worked.

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